

Data Science and Machine Learning

From Data to Knowledge

Color Images of the Book

MICHELE DI NUZZO

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1. DATA SCIENCE AND MACHINE LEARNING

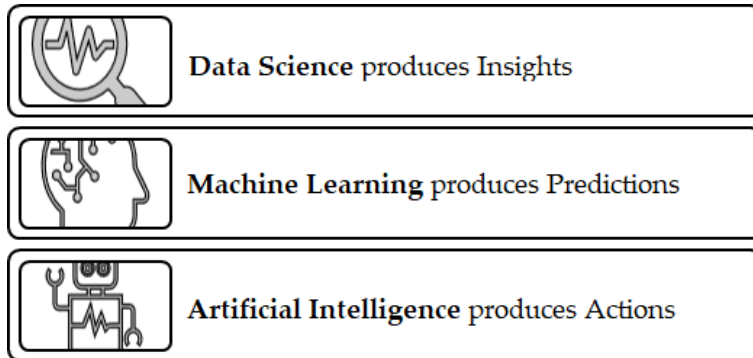


Figure 1.1 - Data Science, Machine Learning and AI

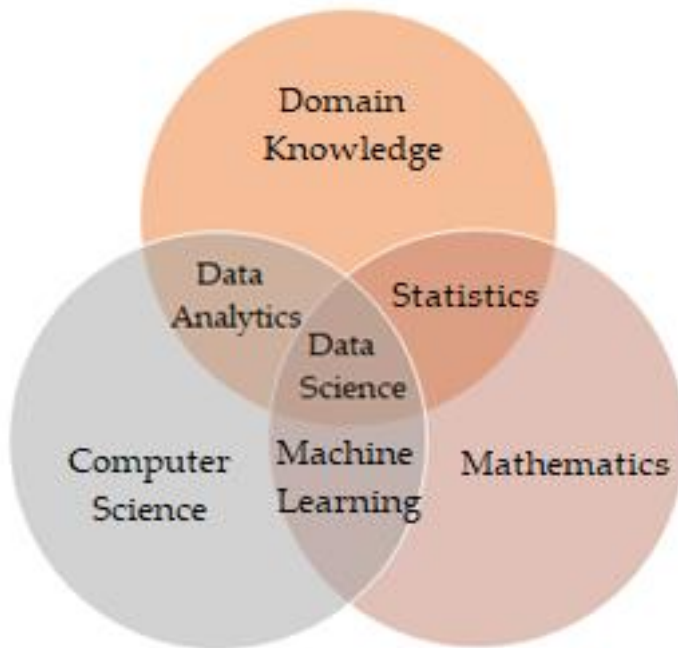


Figure 1.2 - Data Scientist's skills

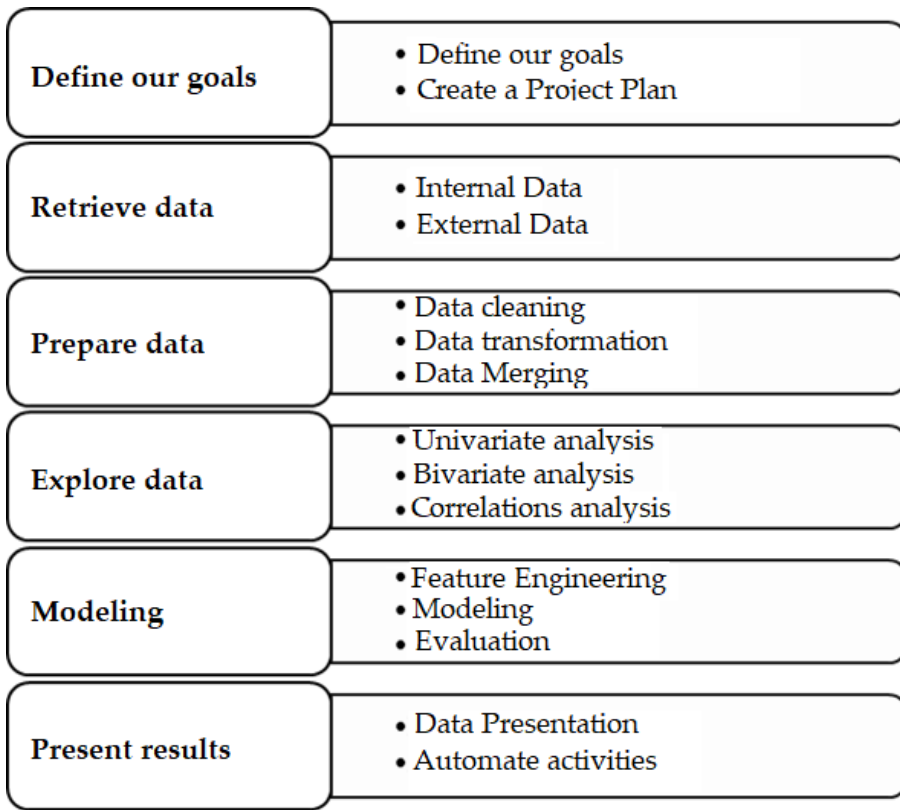


Figure 1.3 - The process of a data science project

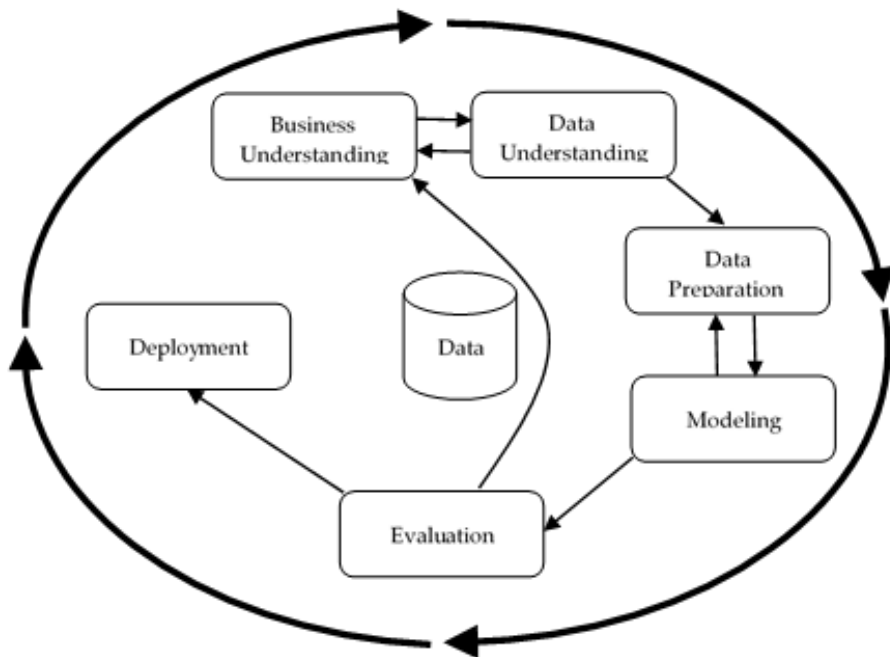


Figure 1.4 - The steps of the CRISP-DM model

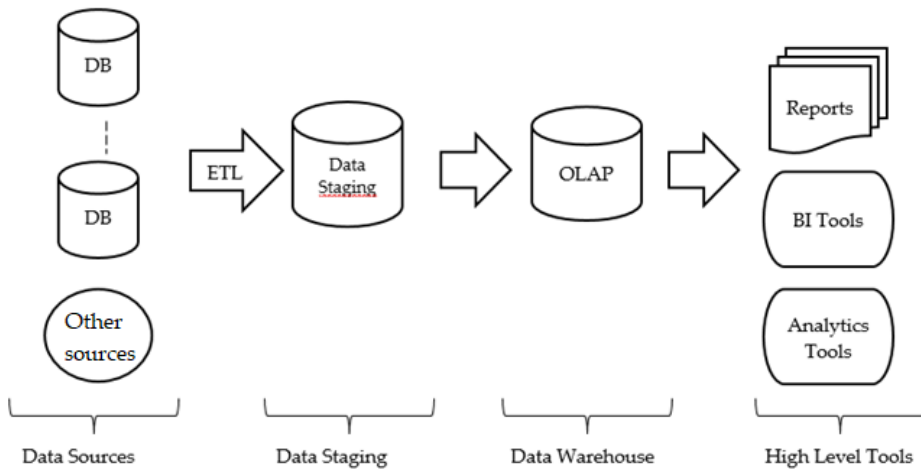


Figure 1.5 - A Business Intelligence system

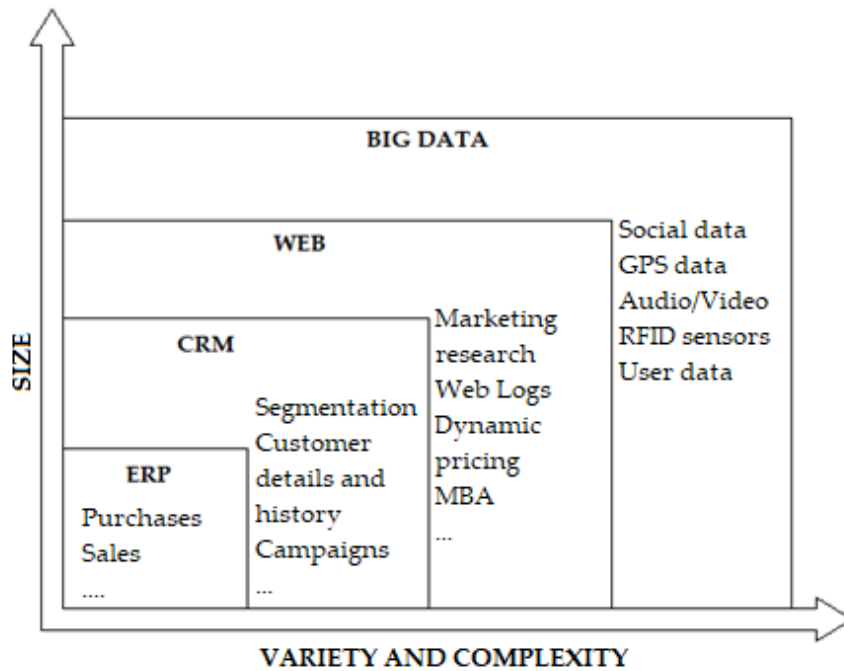


Figure 1.6 – The evolution of Big Data

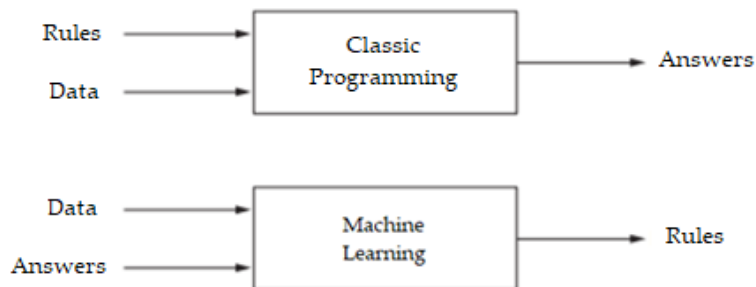


Figure 1.7 – Classic programming vs Machine Learning

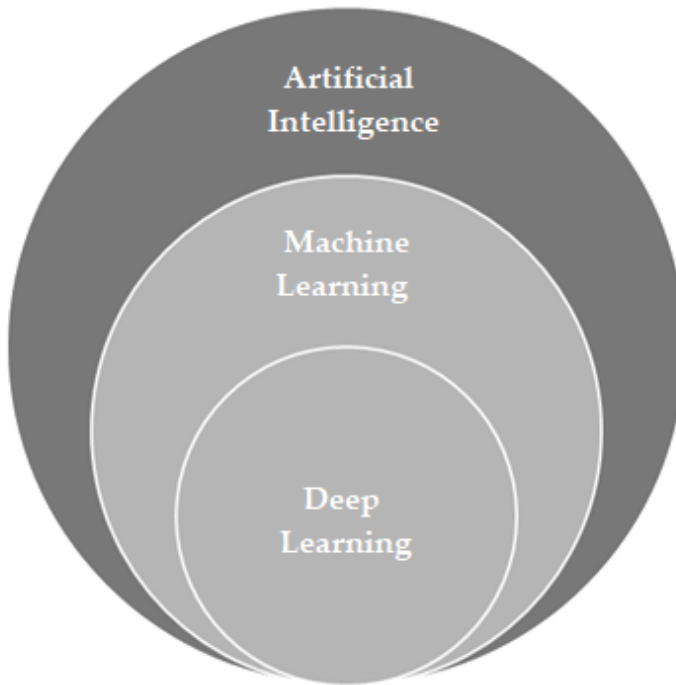


Figure 1.8 – Artificial Intelligence, Machine Learning e Deep Learning

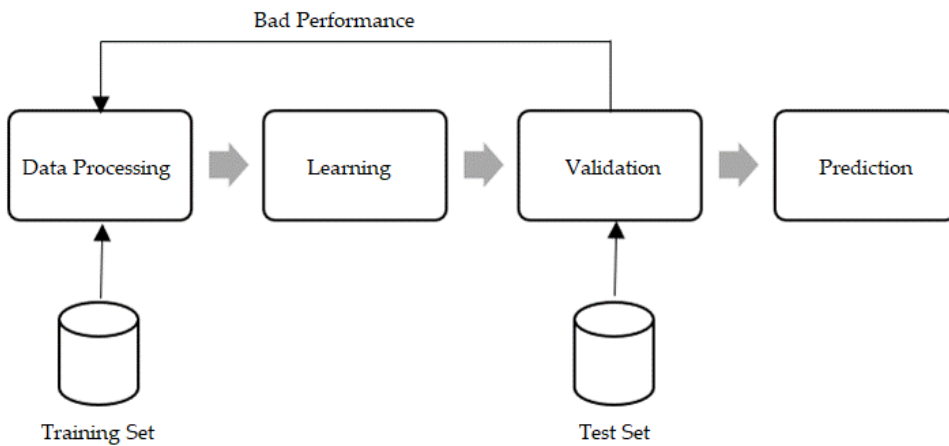


Figure 1.9 – A Machine Learning project

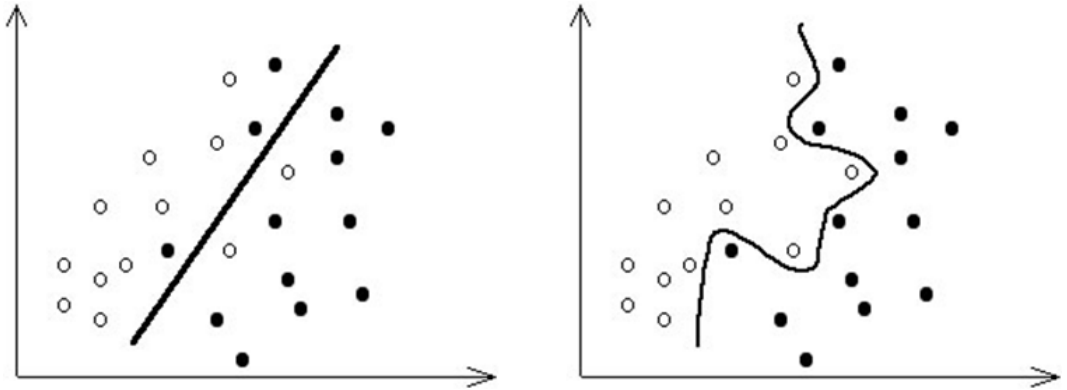


Figure 1.10 – Classifiers with underfitting and with overfitting

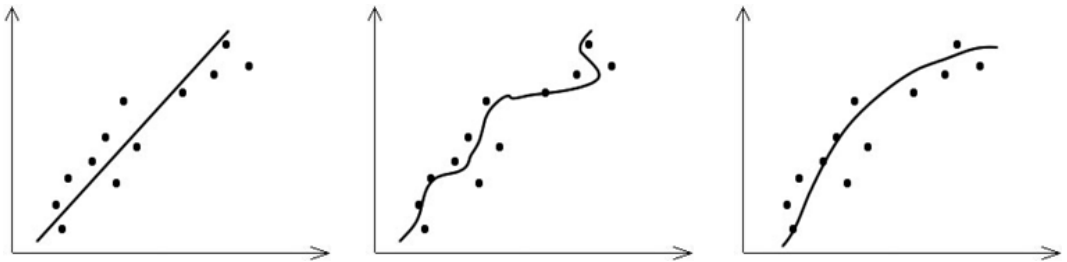


Figure 1.11 - Regressors with underfitting, overfitting and balanced

2. R FOR DATA SCIENCE

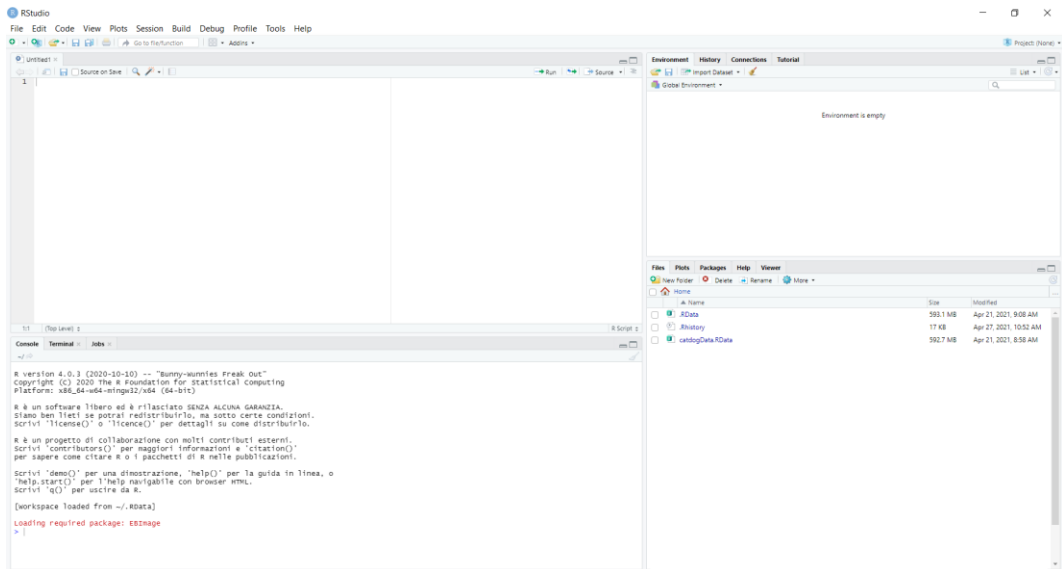


Figure 2.1 - R Studio

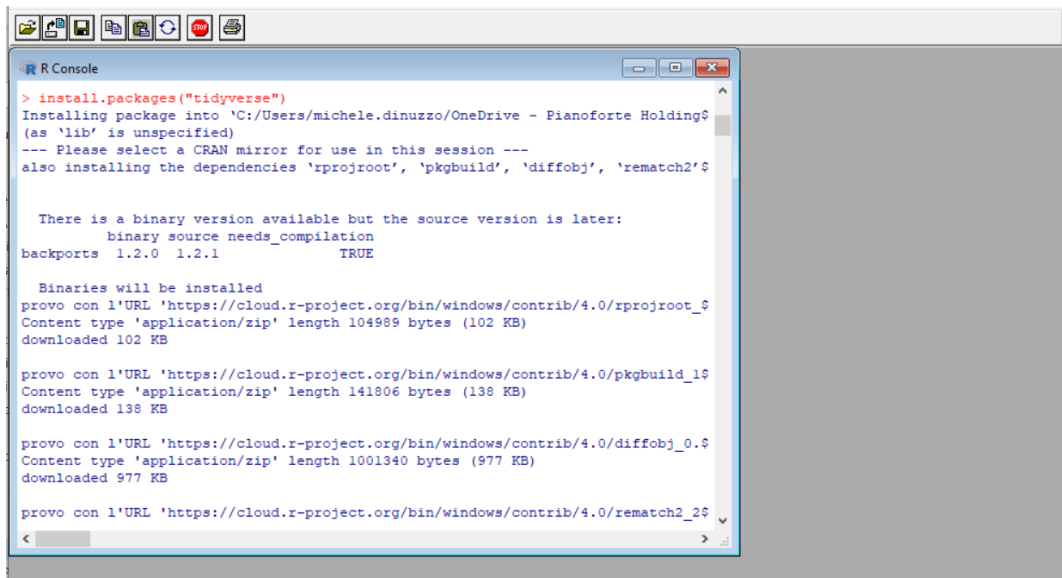


Figure 2.2 – Installing the tidyverse package

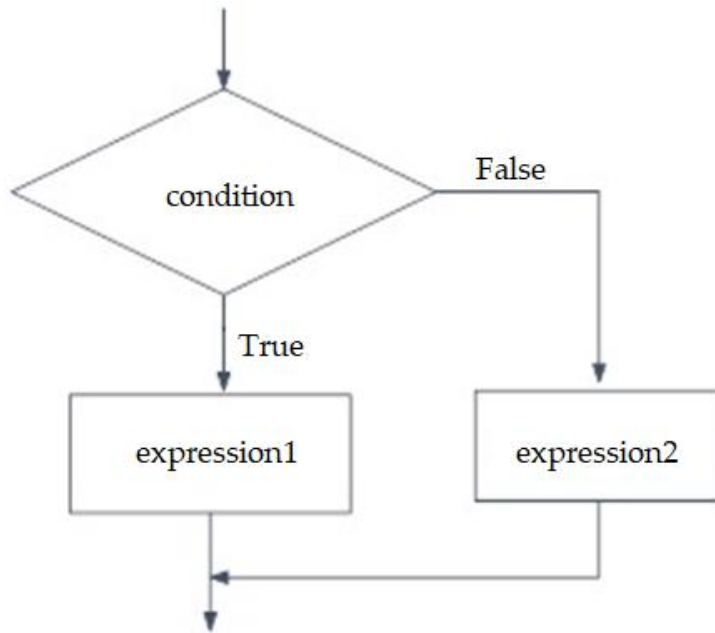


Figure 2.3 - if-else flow diagram

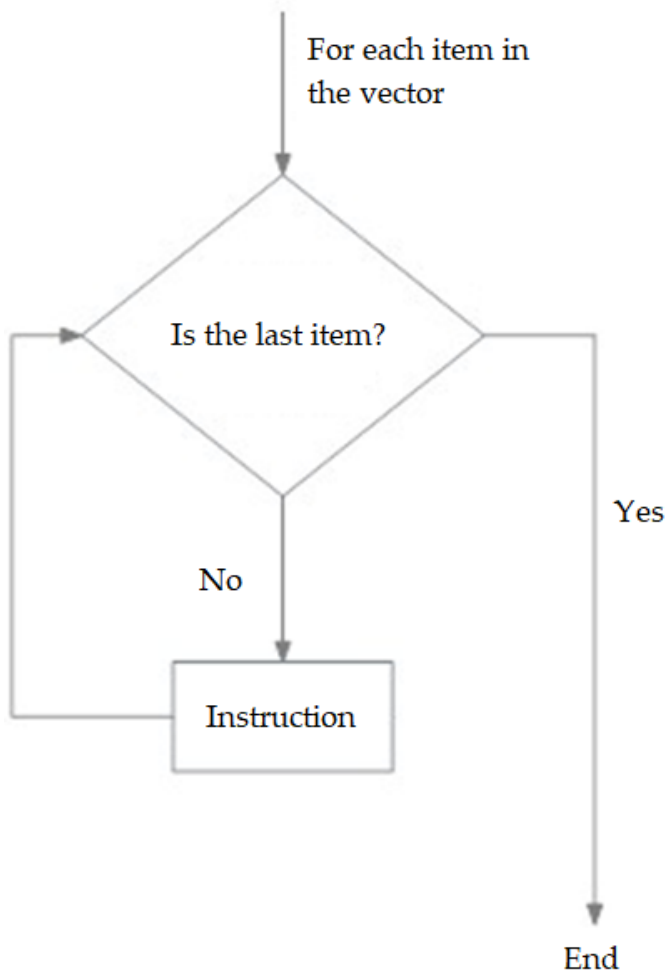


Figure 2.4 – for flow diagram

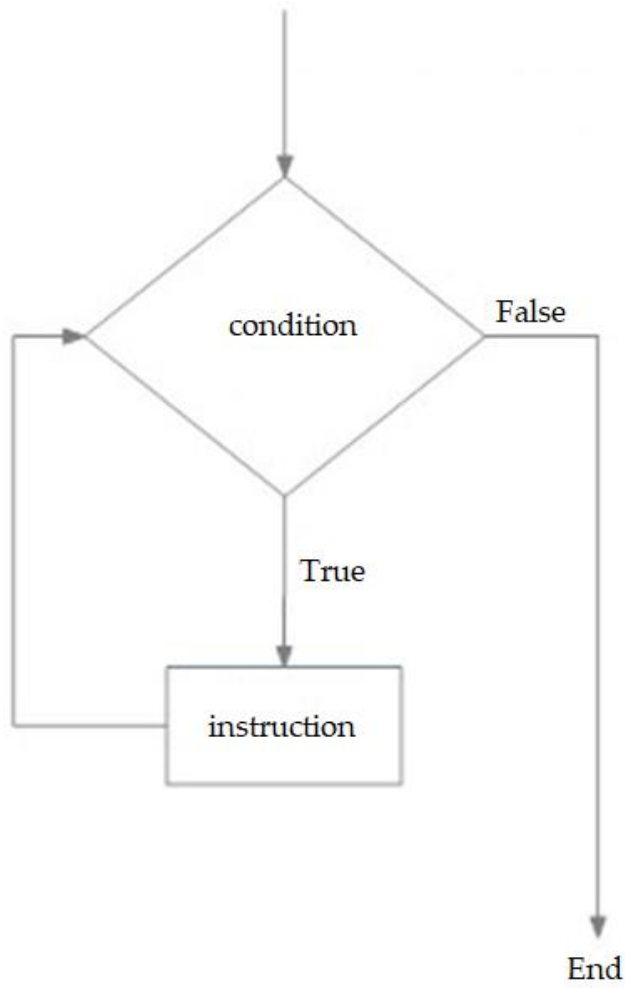


Figure 2.5 – The while flow diagram

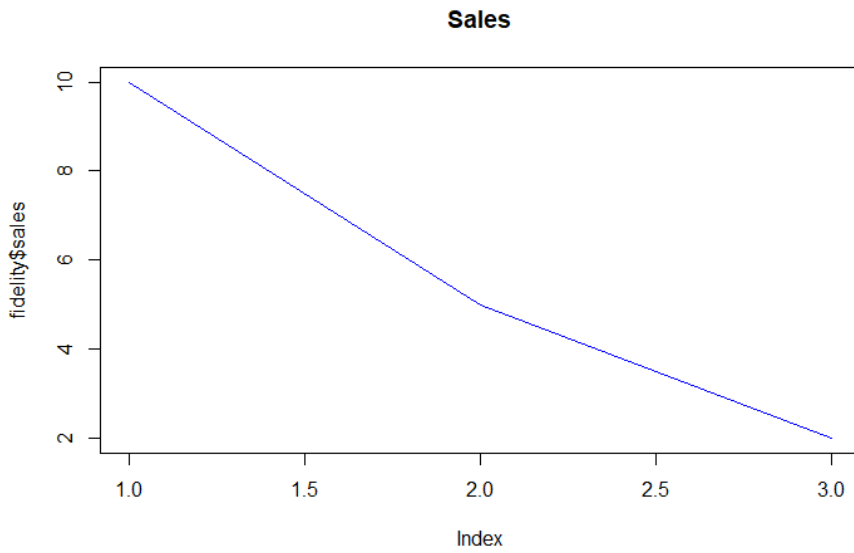


Figure 2.6 - A chart generated with the plot function

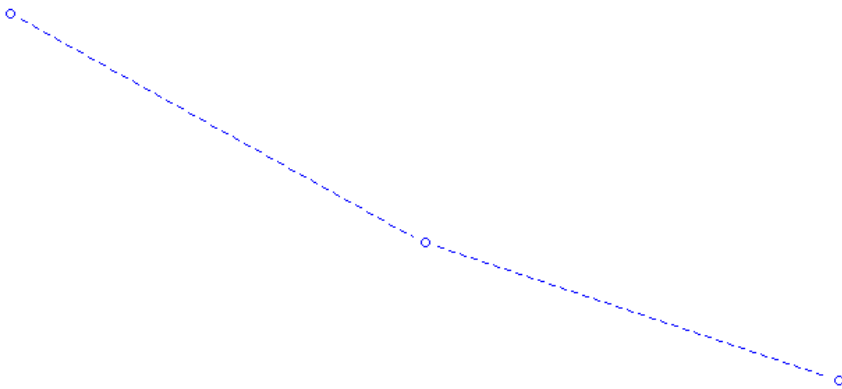


Figure 2.7 - Our new chart

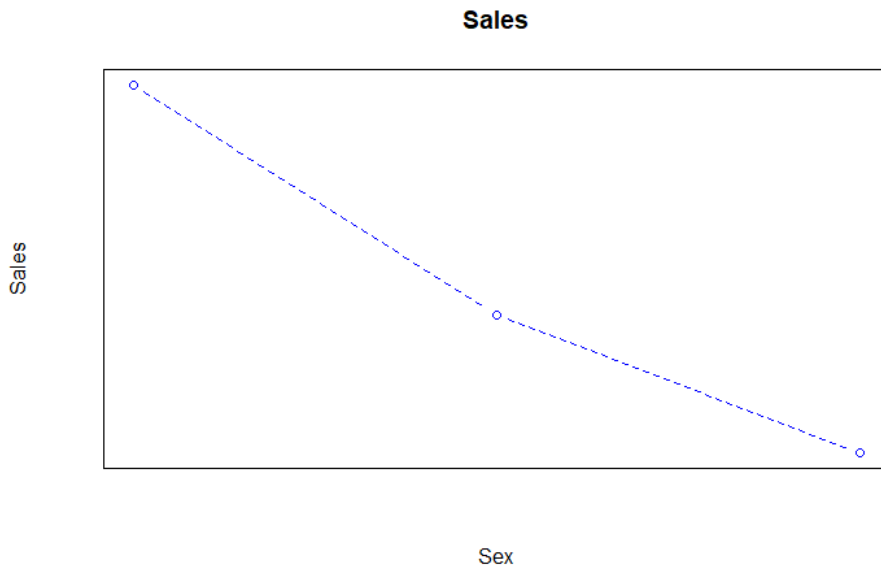


Figure 2.8 - The chart with titles and box

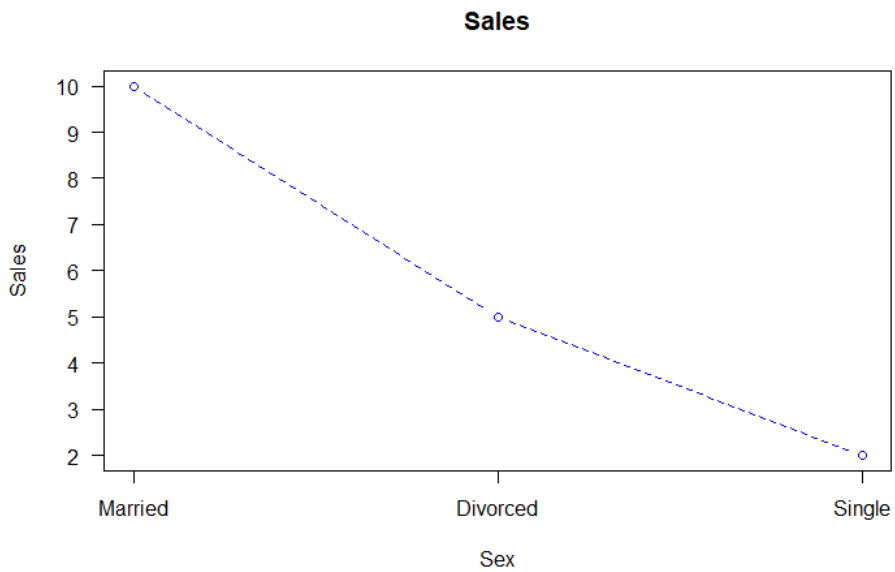


Figure 2.9 - Our final chart

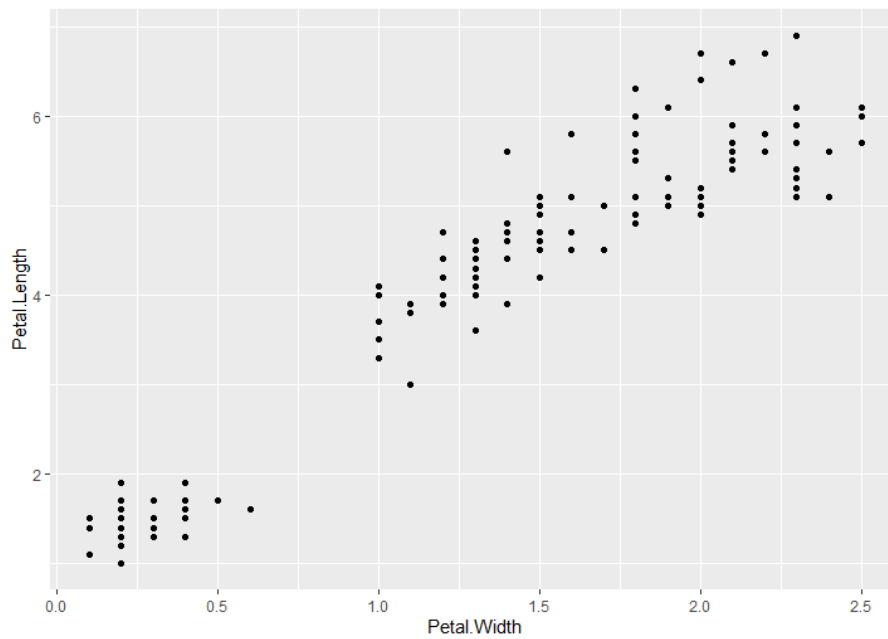


Figure 2.10 - Plot iris data with the `qplot()` function

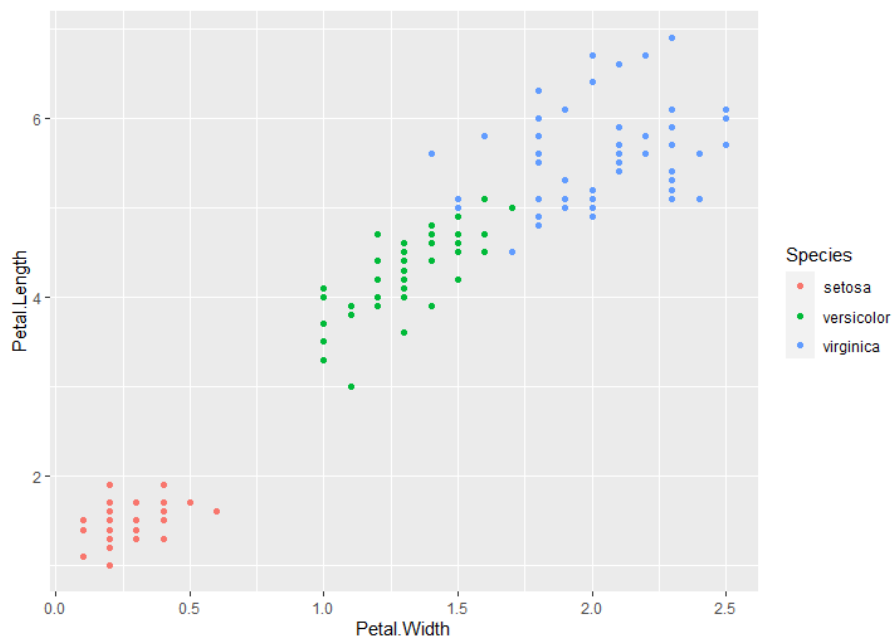


Figure 2.11 - Data grouped by color

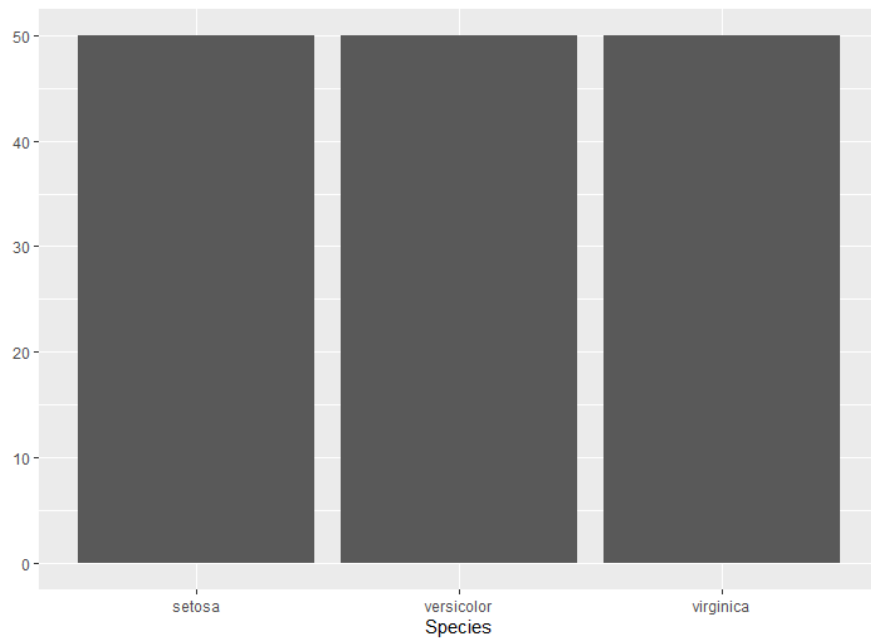


Figure 2.12 - *qplot with one variable*

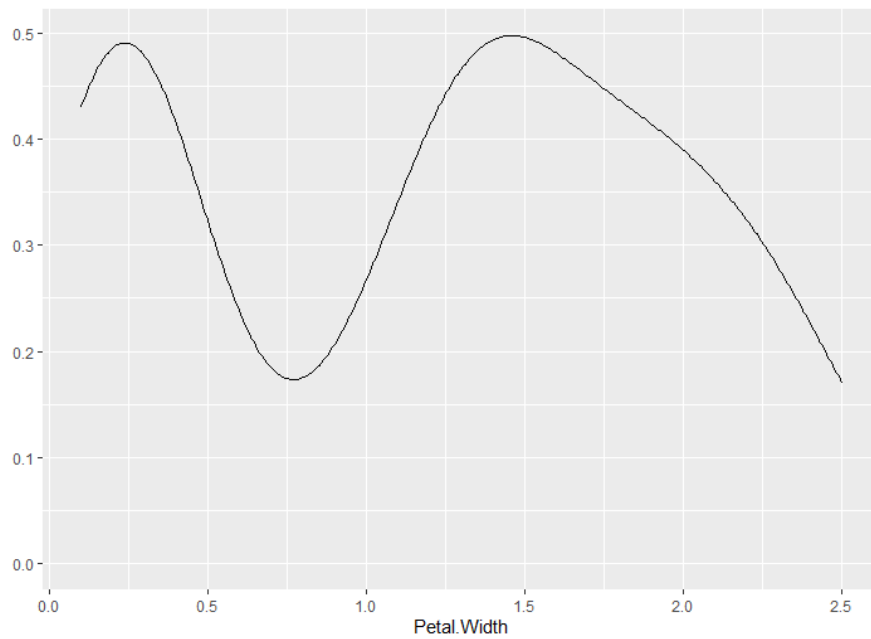


Figure 2.13 - *Chart of a variable with geom = density*

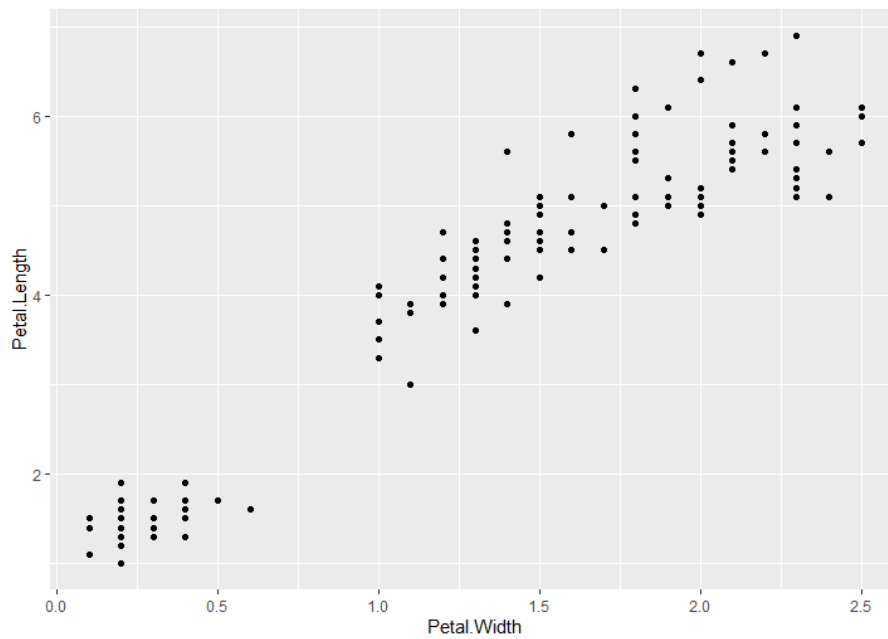


Figure 2.14 - Scatterplot of two variables

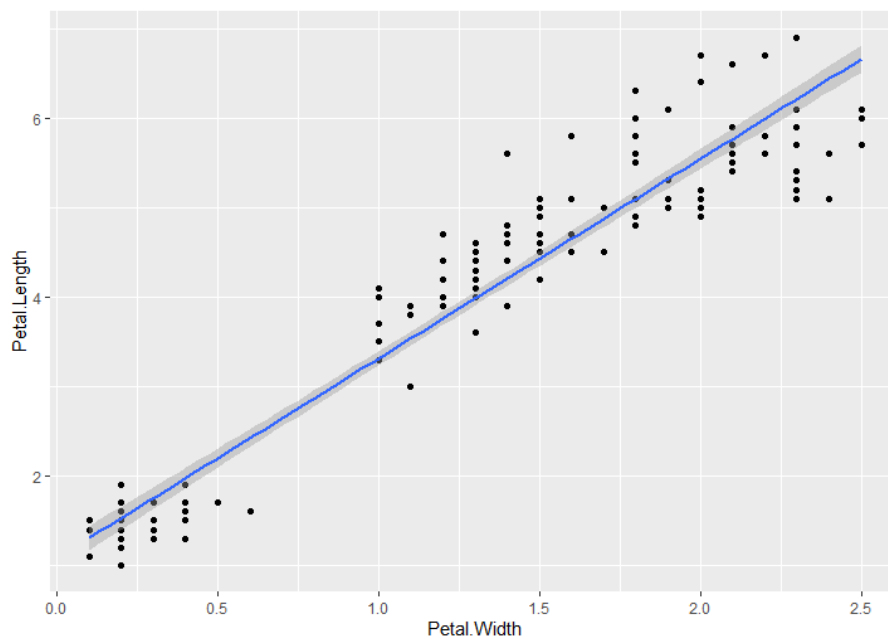


Figure 2.15 - Adding a regression line

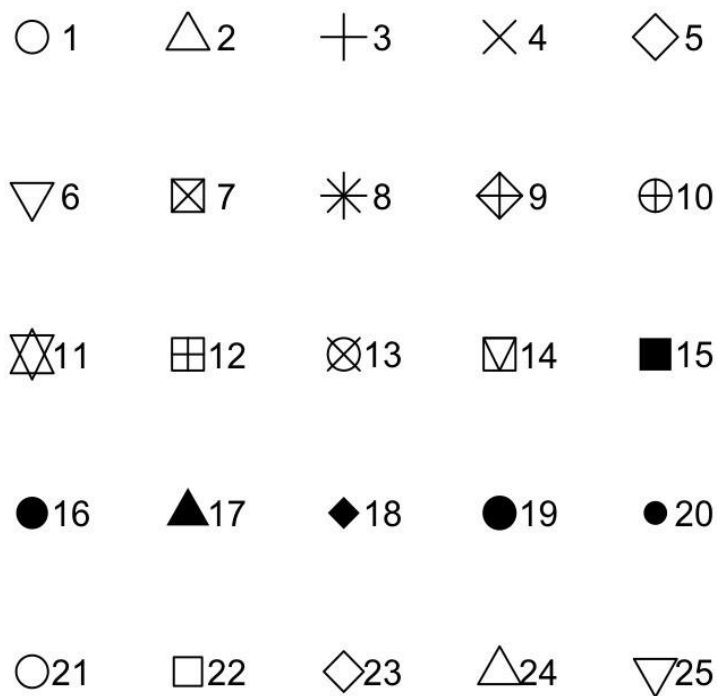


Figure 2.16 - Possible shapes of the points in R

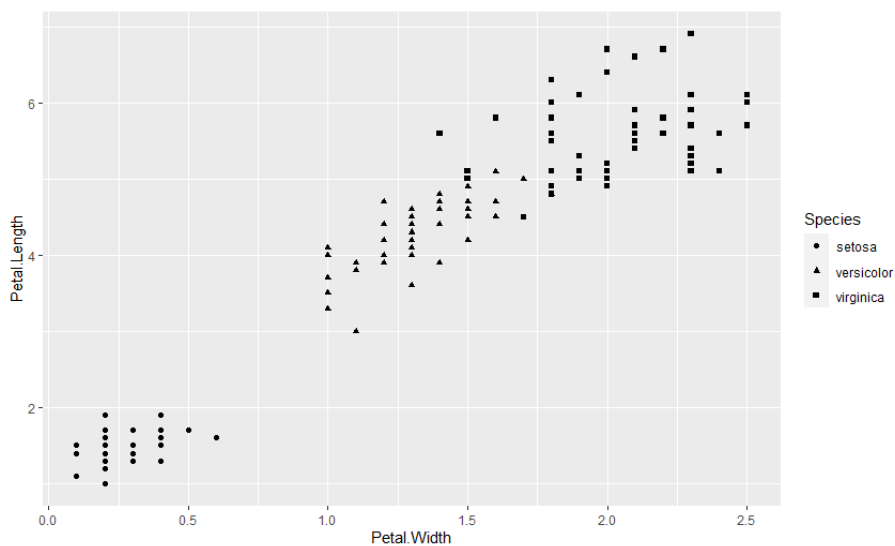


Figure 2.17 – Point shapes by class

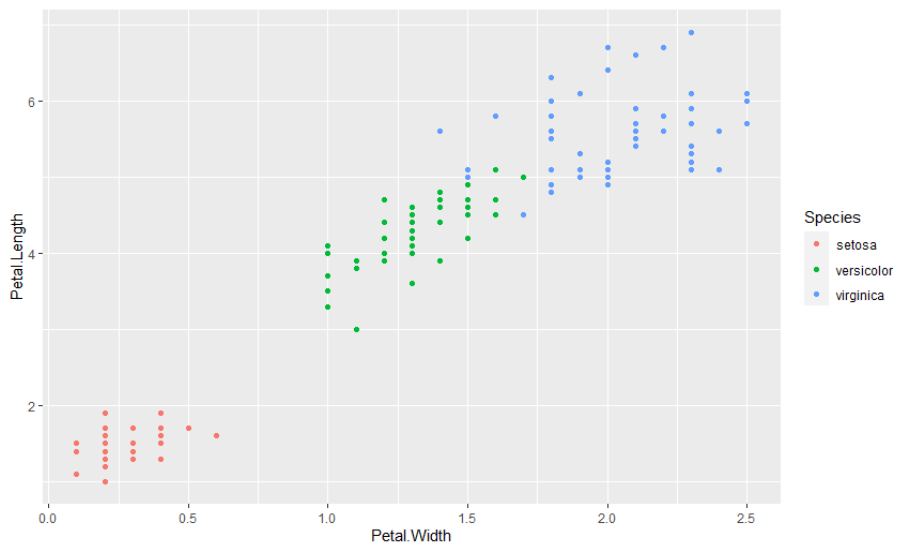


Figure 2.18 – Classes by color

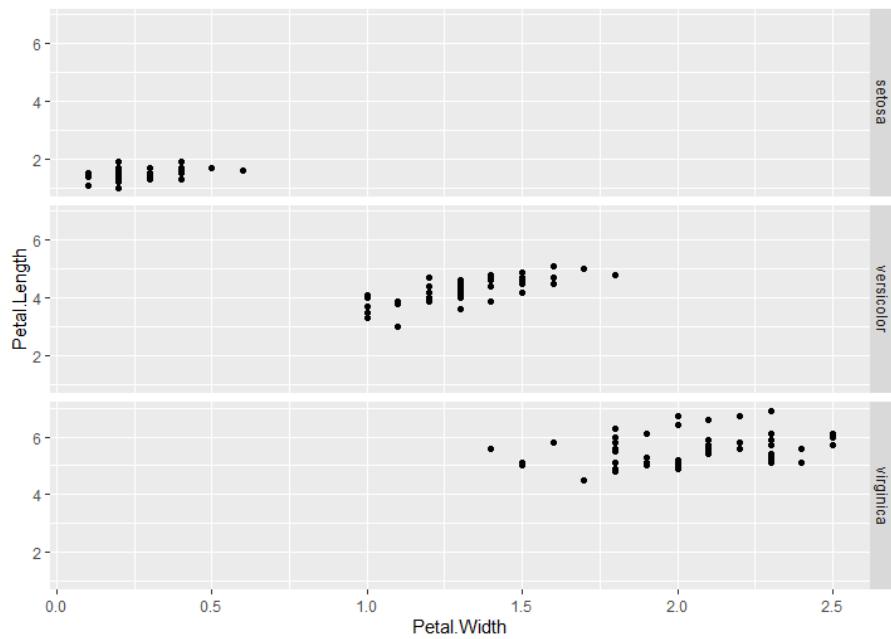


Figure 2.19 – Chart divided by rows

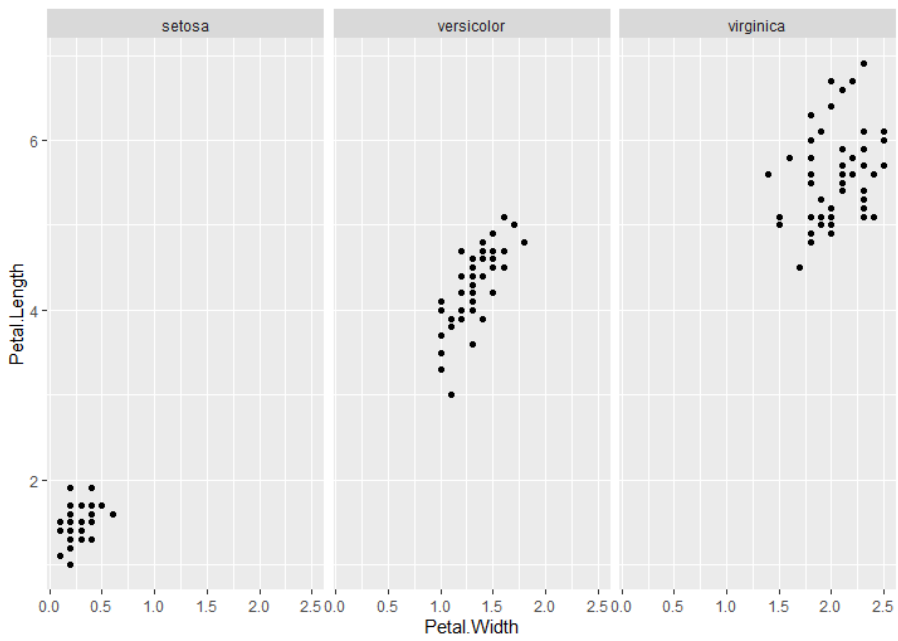


Figure 2.20 – Chart divided by columns

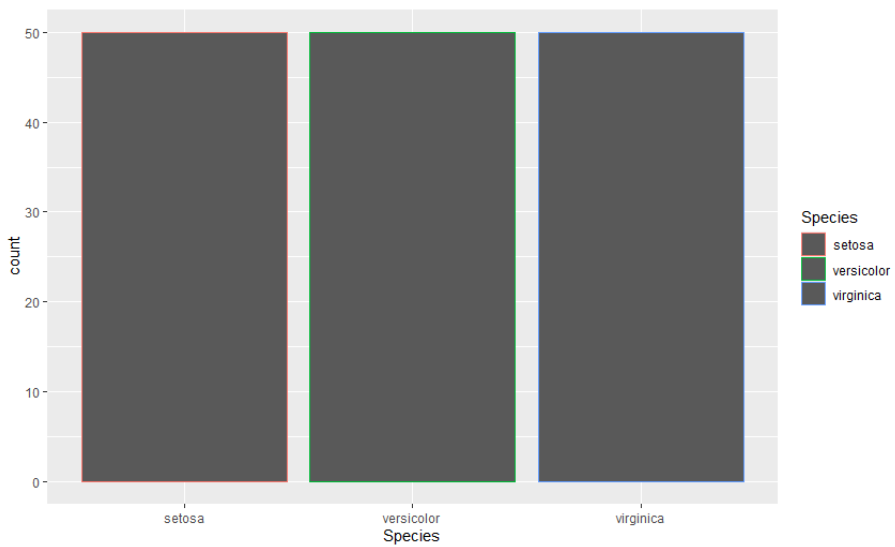


Figure 2.21 - The color parameters

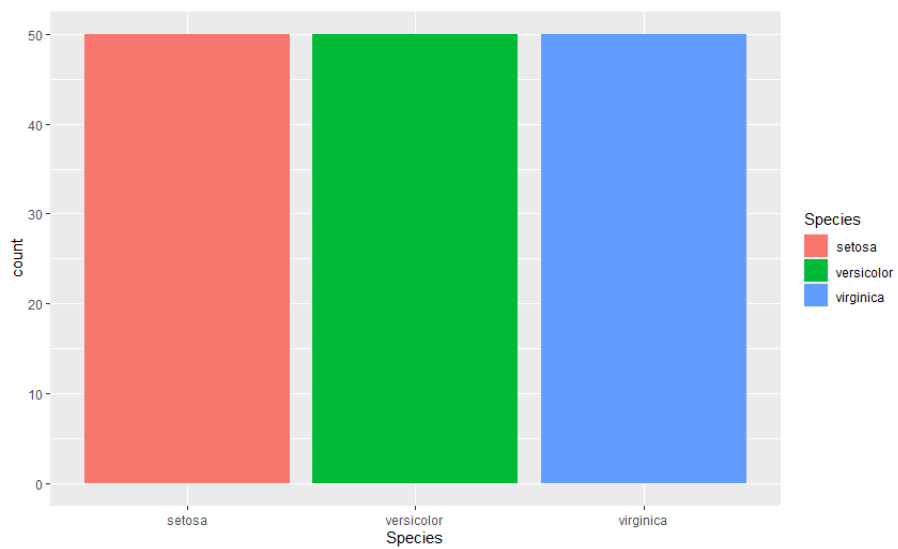


Figure 2.22 – The fill parameter

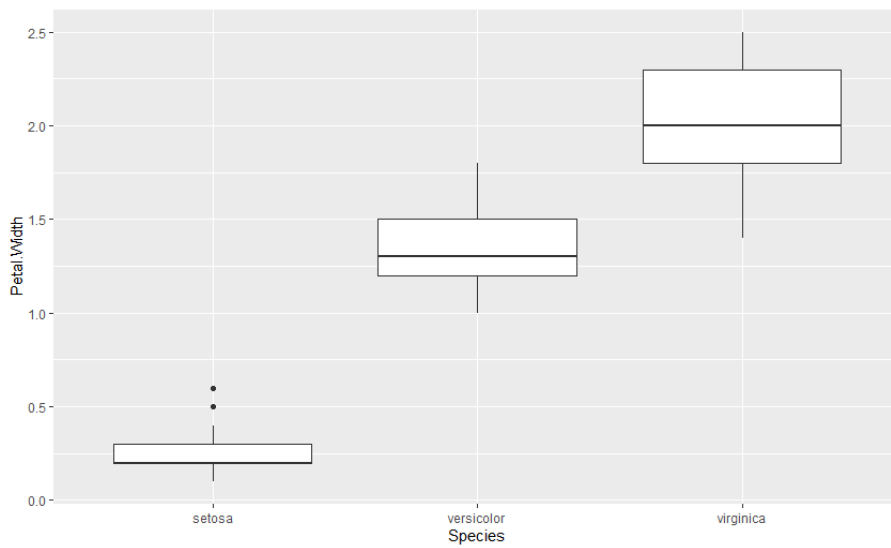


Figure 2.23 - Boxplot of Petal.Width

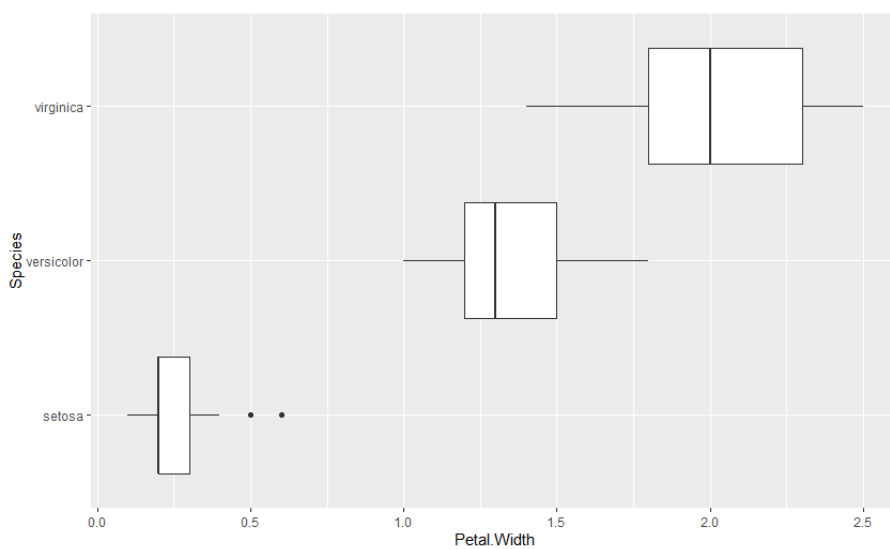


Figure 2.24 - Boxplot of *Petal.Width* with inverted axes

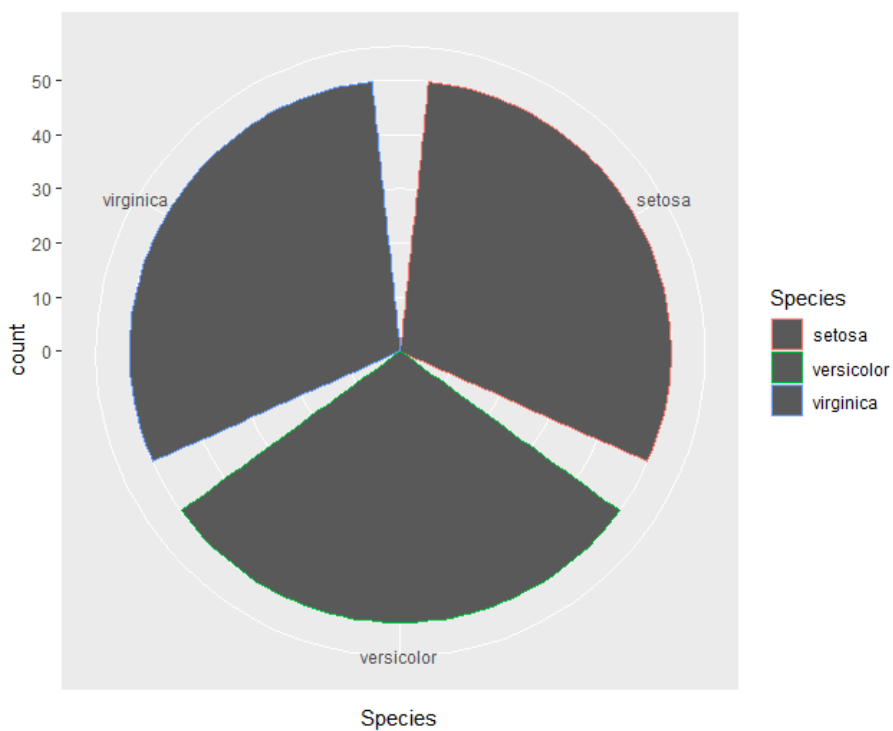


Figure 2.25 – Polar coordinates

```
5.1,3.5,1.4,0.2,Iris-setosa  
4.9,3.0,1.4,0.2,Iris-setosa  
4.7,3.2,1.3,0.2,Iris-setosa  
4.6,3.1,1.5,0.2,Iris-setosa  
5.0,3.6,1.4,0.2,Iris-setosa  
5.4,3.9,1.7,0.4,Iris-setosa
```

Figure 2.26 - A data file with comma separated fields

3. MATHEMATICS AND ALGEBRA

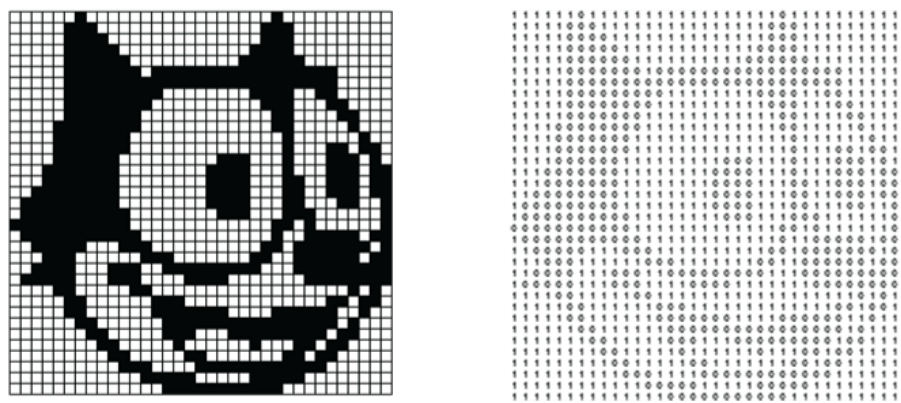


Figure 3.1 - Matrix representation of the image

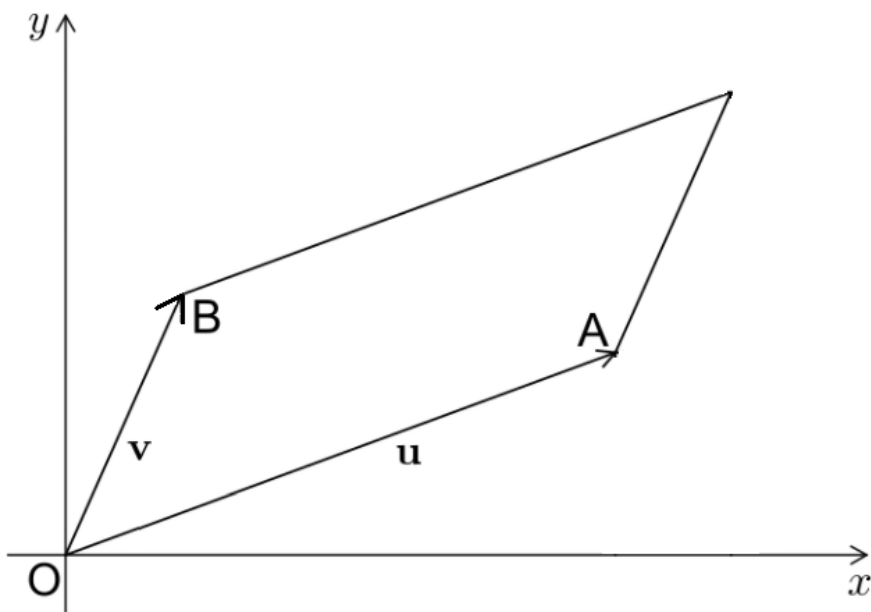


Figure 3.2 - Vector representation

$$\begin{array}{|c|} \hline A \\ \hline m * n \end{array} = \begin{array}{|c|} \hline U \\ \hline m * p \end{array} \begin{array}{|c|} \hline D \\ \hline p * p \end{array} \begin{array}{|c|} \hline V^T \\ \hline p * n \end{array}$$

Figure 3.3 - SVD

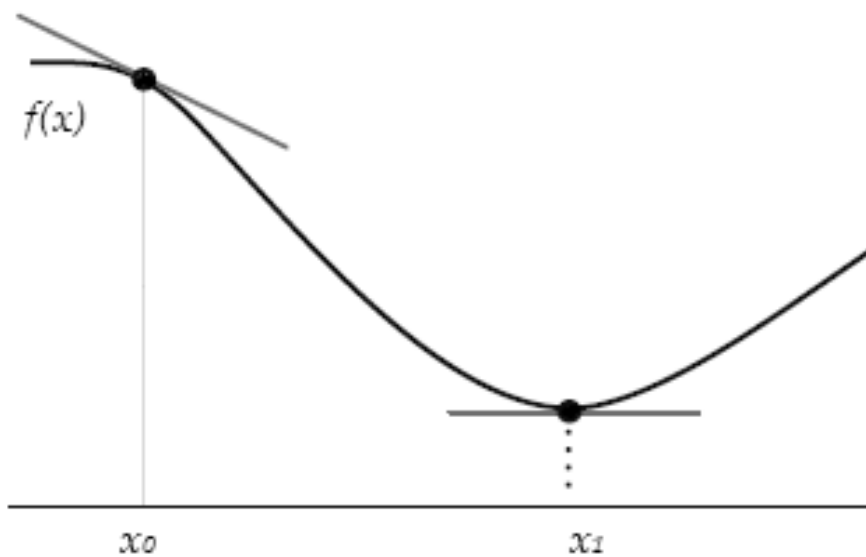
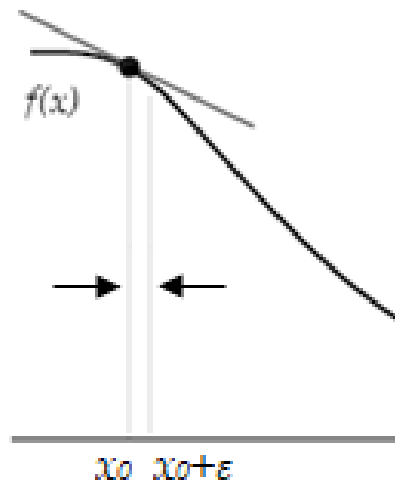


Figure 3.4 – A generic function



4. DESCRIPTIVE STATISTICS

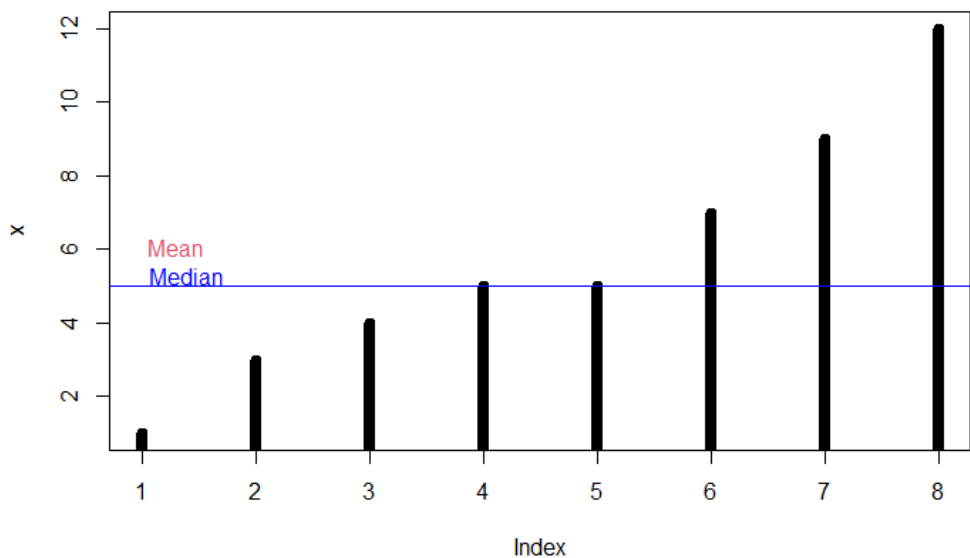


Figure 4.1 – Mean and median

Family	Revenue
Family 1	32.000
Family 2	43.000
Family 3	28.000
Family 4	29.500
Family 5	34.000
Family 6	42.000
Family 7	47.000
Family 8	-
Mean	36.500

Figure 4.2 – Mean example

Family	Revenue
Family 1	32.000
Family 2	43.000
Family 3	28.000
Family 4	29.500
Family 5	34.000
Family 6	42.000
Family 7	47.000
Family 8	1.300.000
Mean	194.438

Figure 4.3 – Mean Example

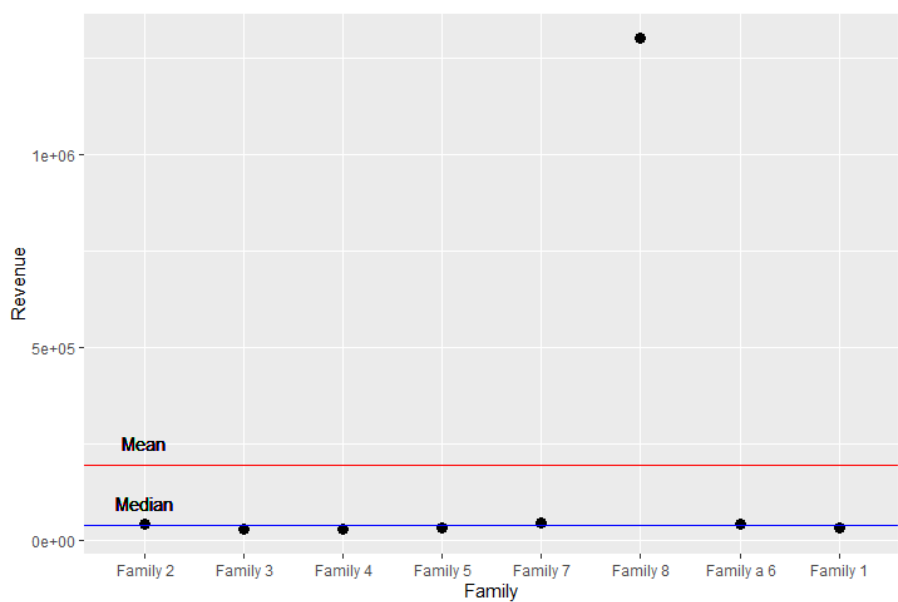


Figure 4.4 - Chart and median

Obs.	Sample A children	Sample B children	Sample C children
1	2	1	1
2	2	2	5
3	2	3	2
4	2	3	2
5	2	2	0
6	2	1	1
7	2	2	3
Mean	2	2	2
Std. Dev.	0	0,76	1,51
Variance	0	0,57	2,29

Figure 4.5 - Three samples with the same mean

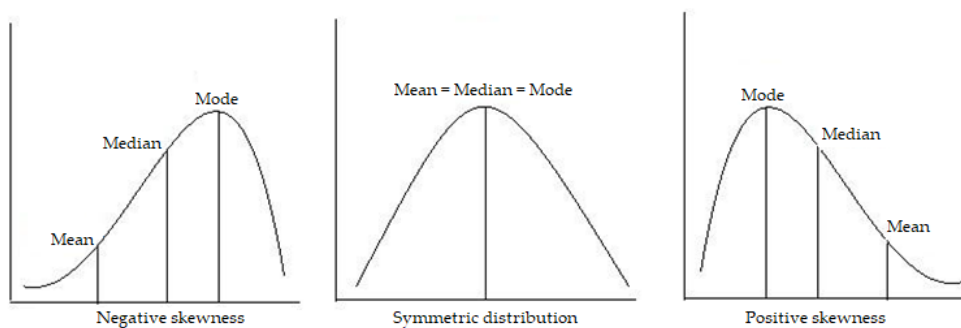


Figure 4.6 - Skewness

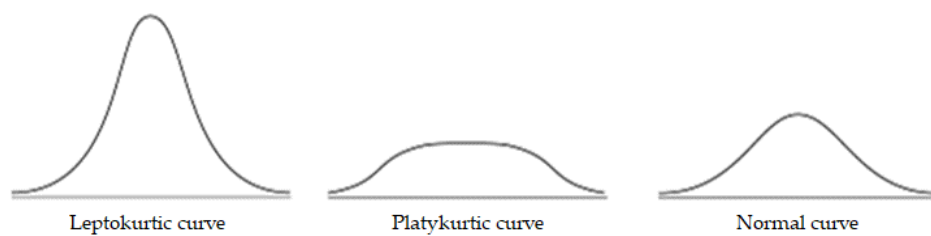


Figure 4.7 – Kurtosis

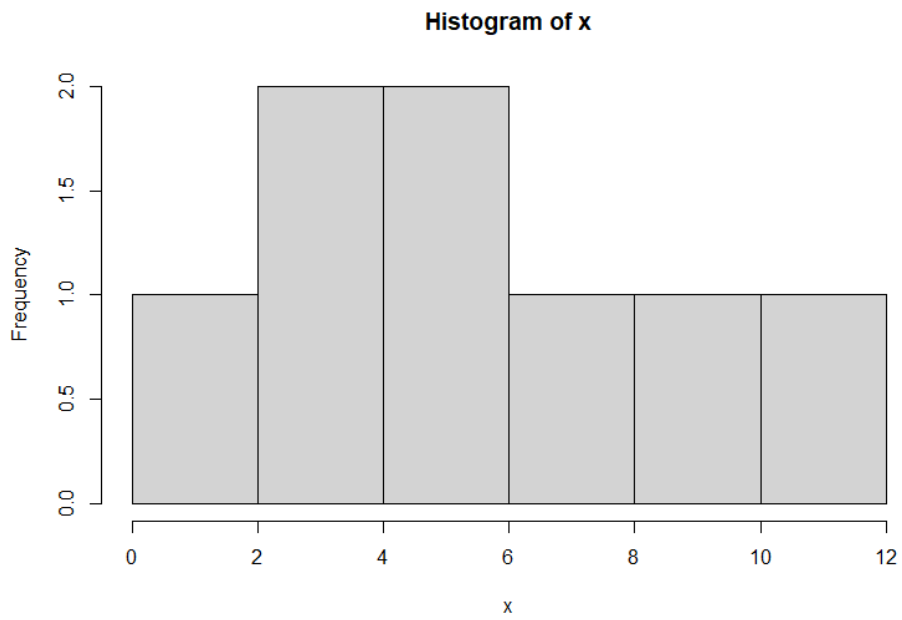


Figure 4.8 – Histogram for our variable

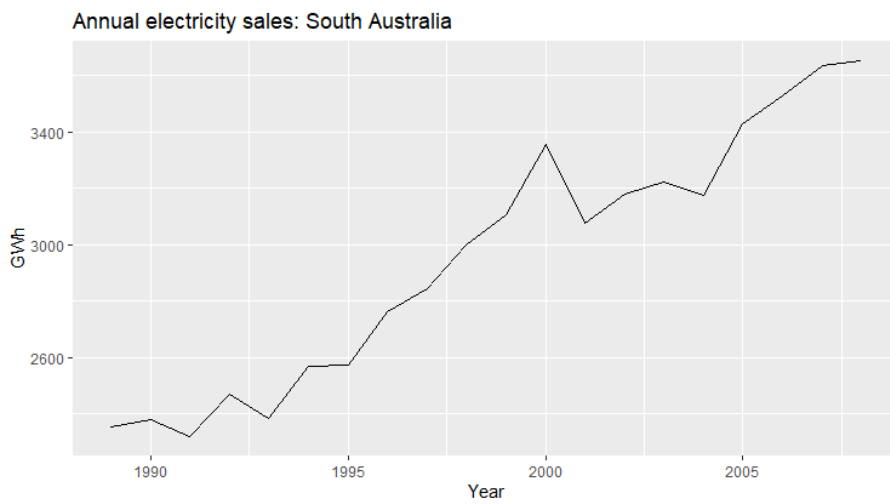


Figure 4.9 – elecsales dataset

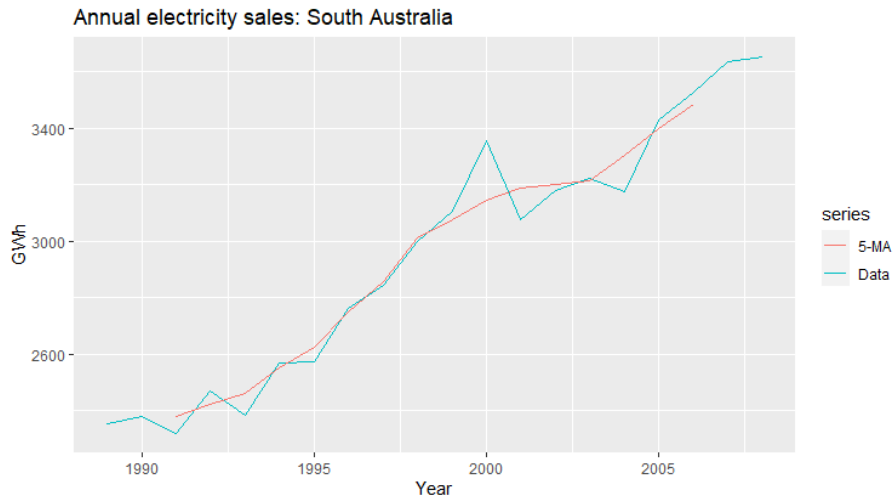


Figure 4.10 - Plot of the 5-MA

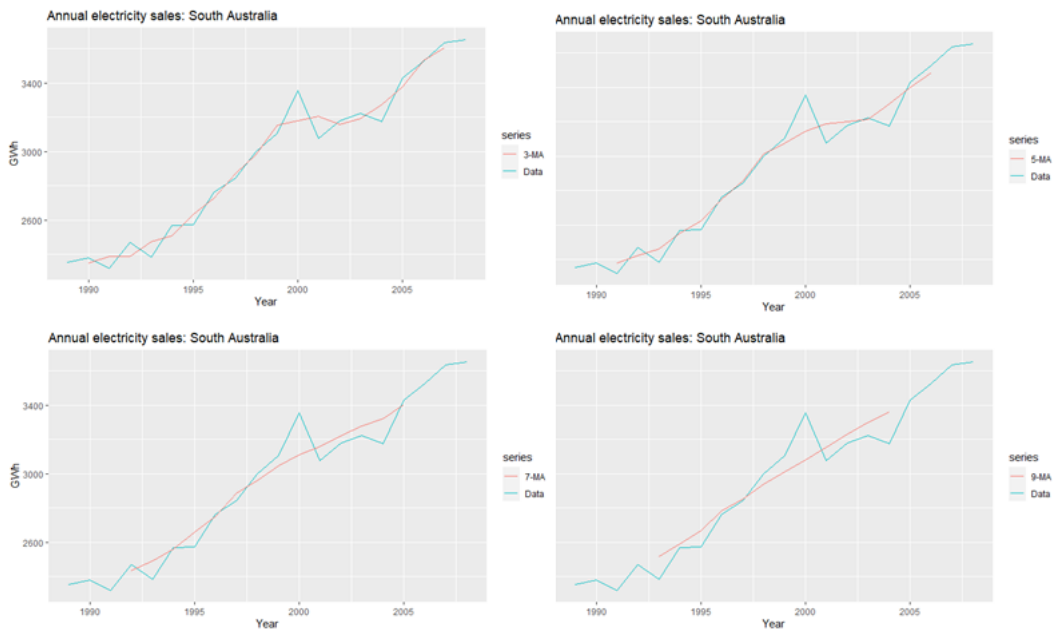


Figure 4.11 – Moving average for different m

In

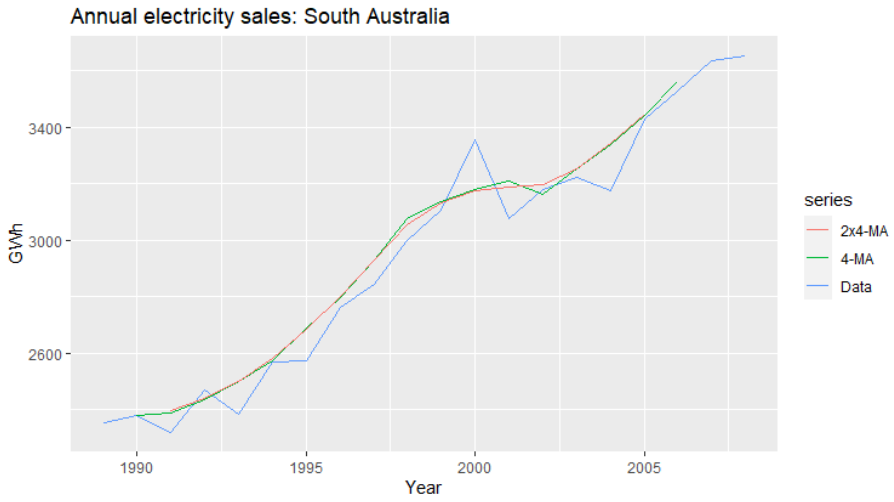


Figure 4.12 - 2x4MA

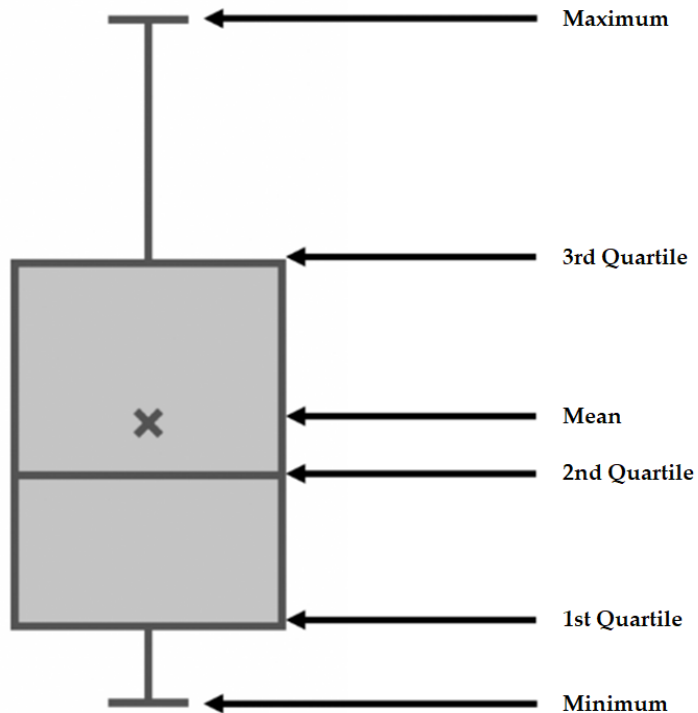


Figure 4.13 - Boxplot

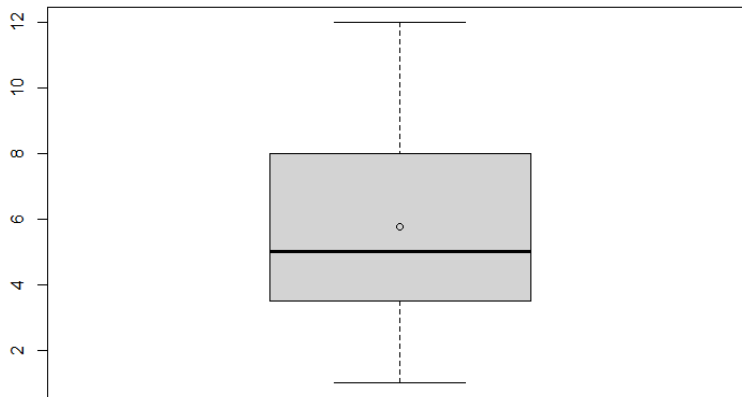


Figure 4.14 - A boxplot

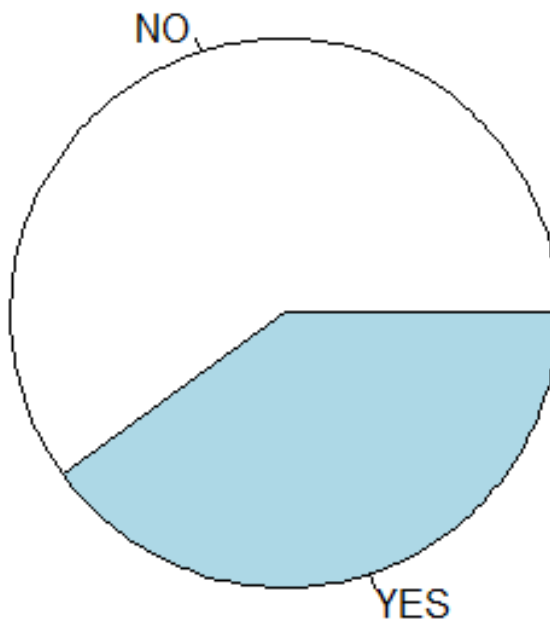


Figure 4.15 – Pie chart of a discrete variable

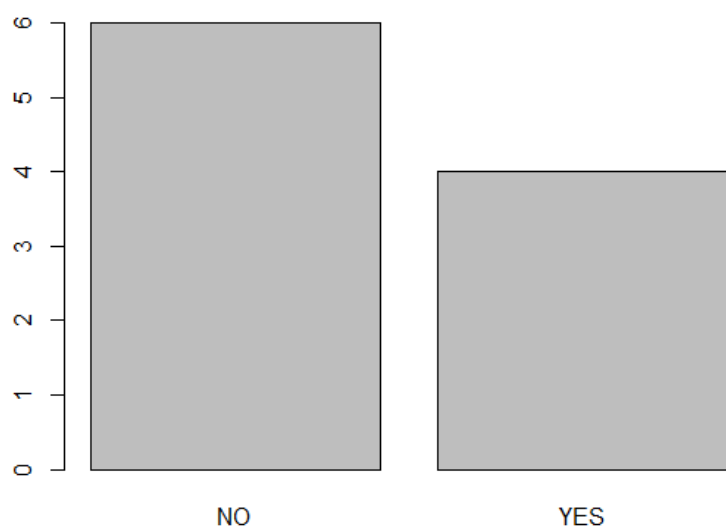


Figure 4.16 - Barplot

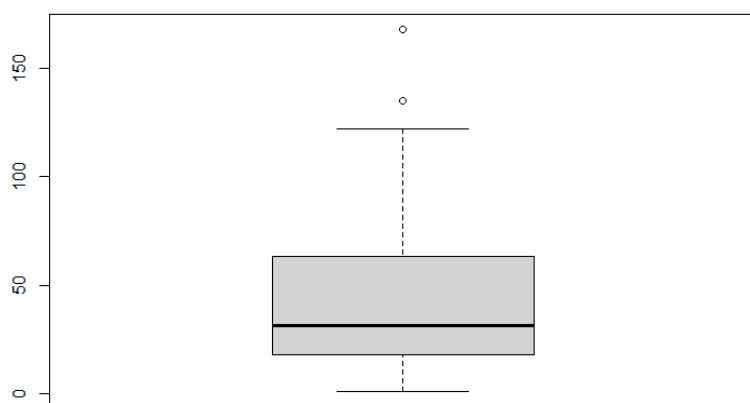


Figure 4.17 – Ozone variable boxplot

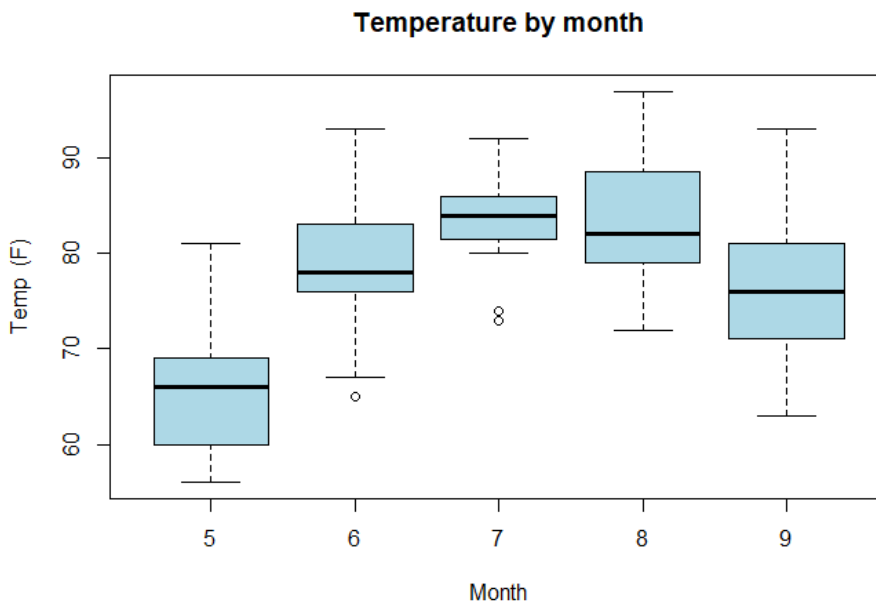


Figure 4.18 - Boxplot of Temp by month

Loyal	Man	Woman	Total
YES	$f_{y,m}$	$f_{y,w}$	f_{yes}
NO	$f_{n,m}$	$f_{n,w}$	f_{no}
TOTAL	f_m	f_w	f

Figure 4.19 – Contingency table

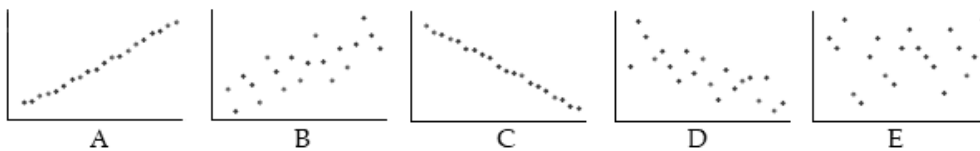


Figure 4.20 – Scatter plot examples

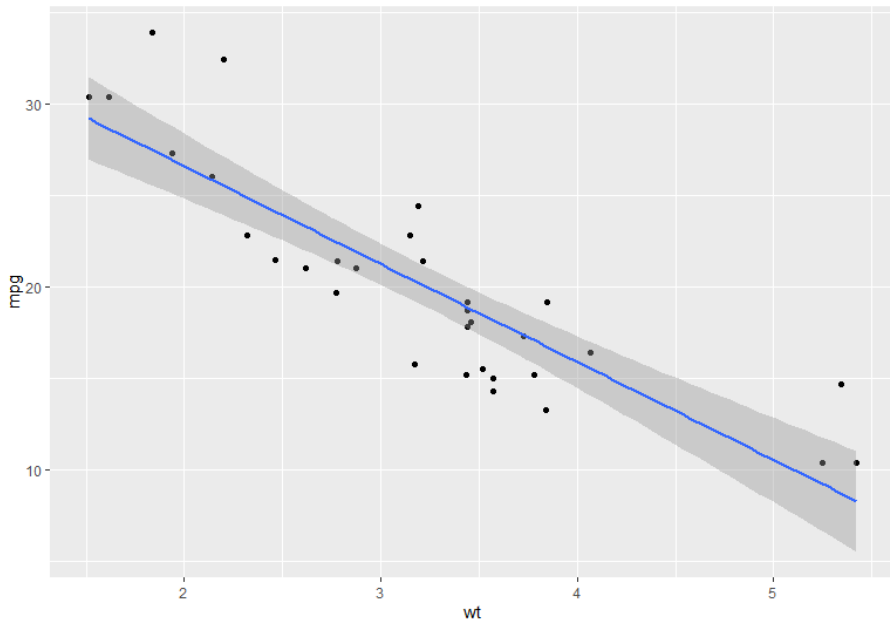


Figure 4.21 - Scatterplot of mpg e wt

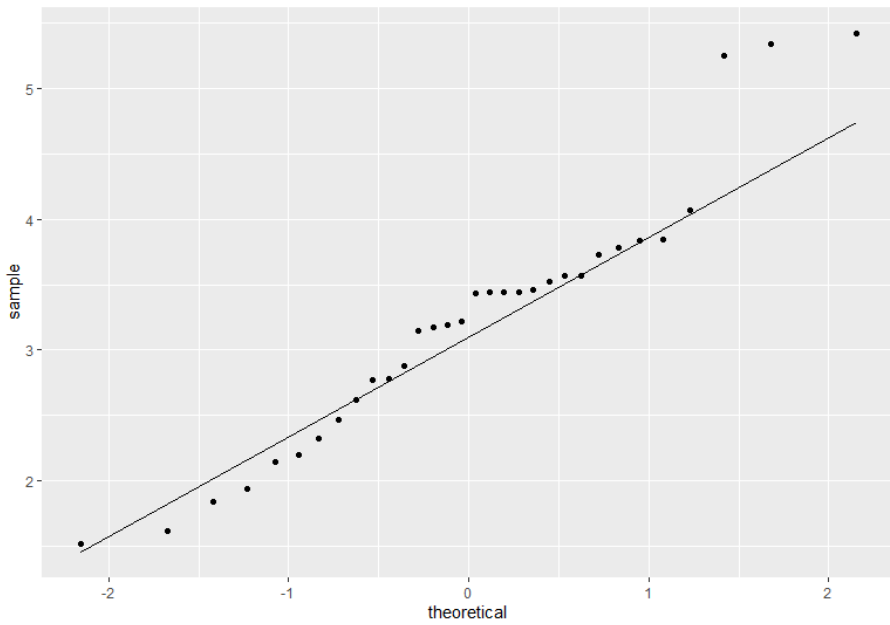


Figure 4.22 - Q-Q chart of wt

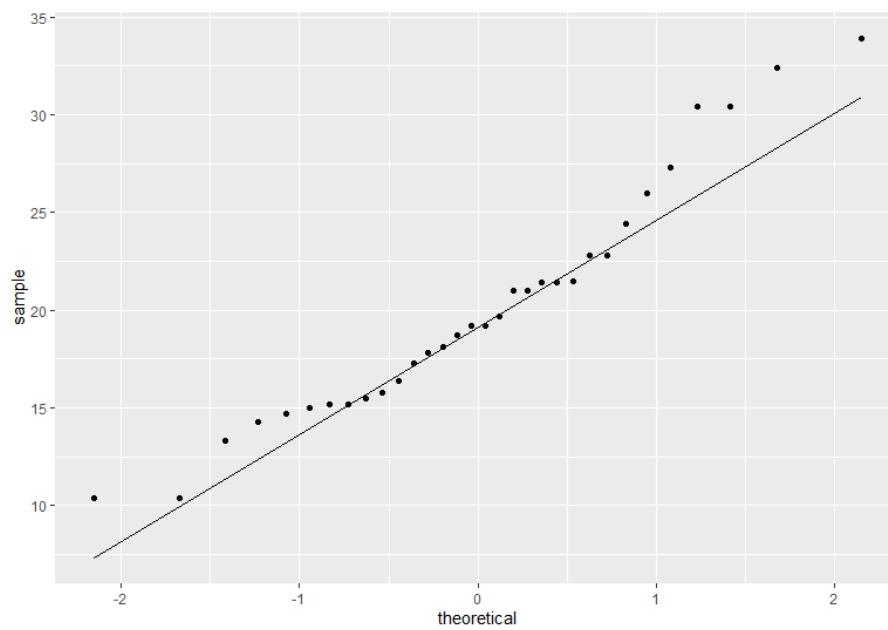


Figure 4.23 - Q-Q chart of mpg

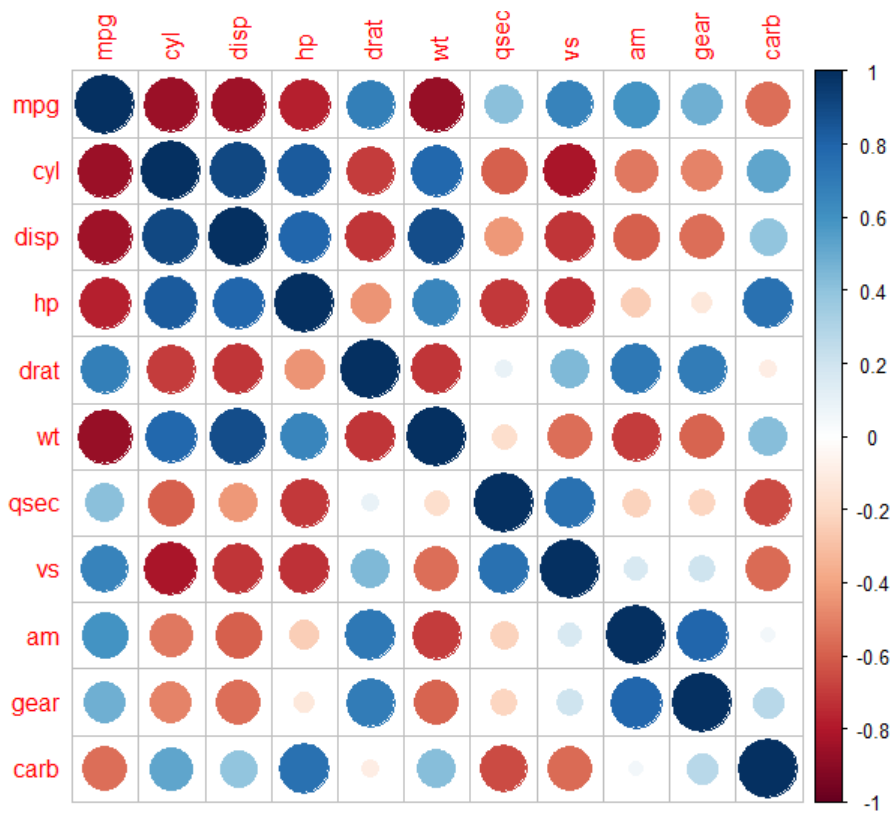


Figure 4.24 – Correlation Chart

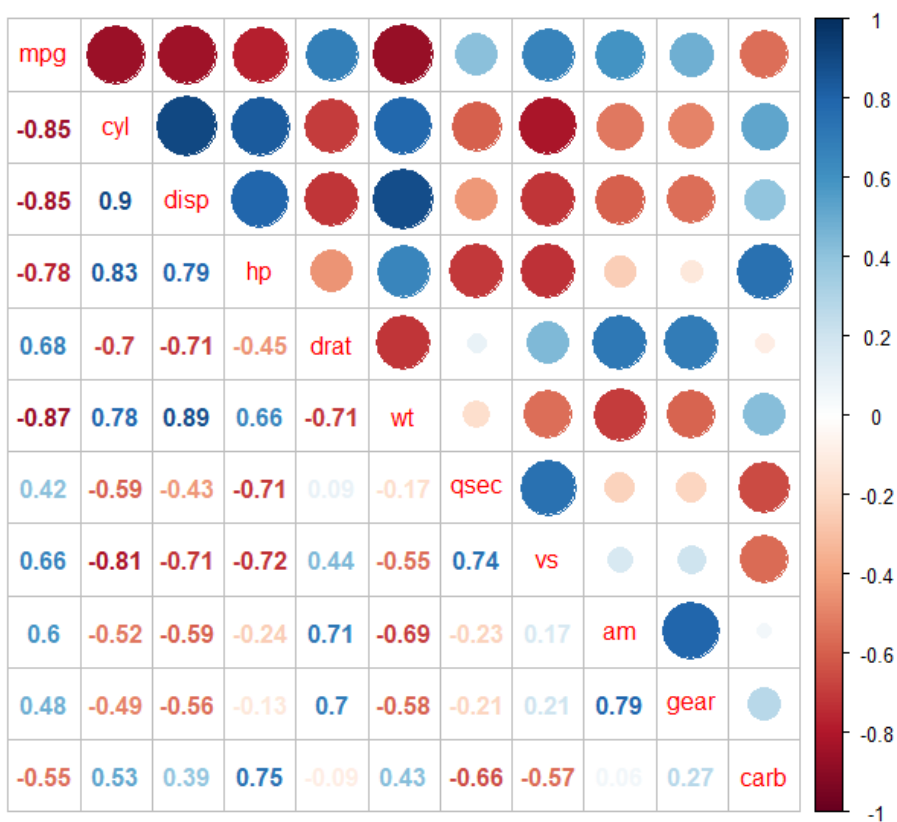


Figure 4.25 – Correlation values

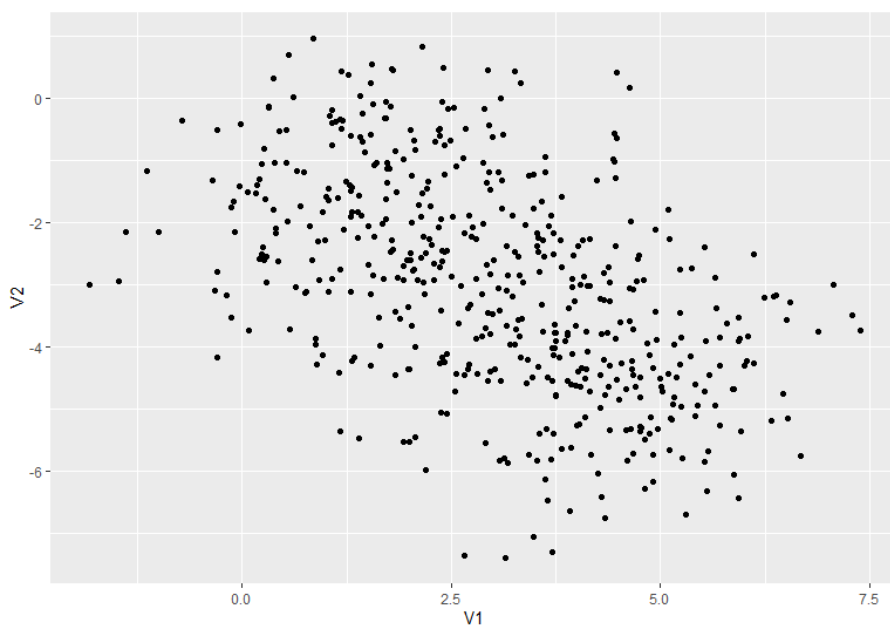


Figure 4.26 – Data generated by simulation

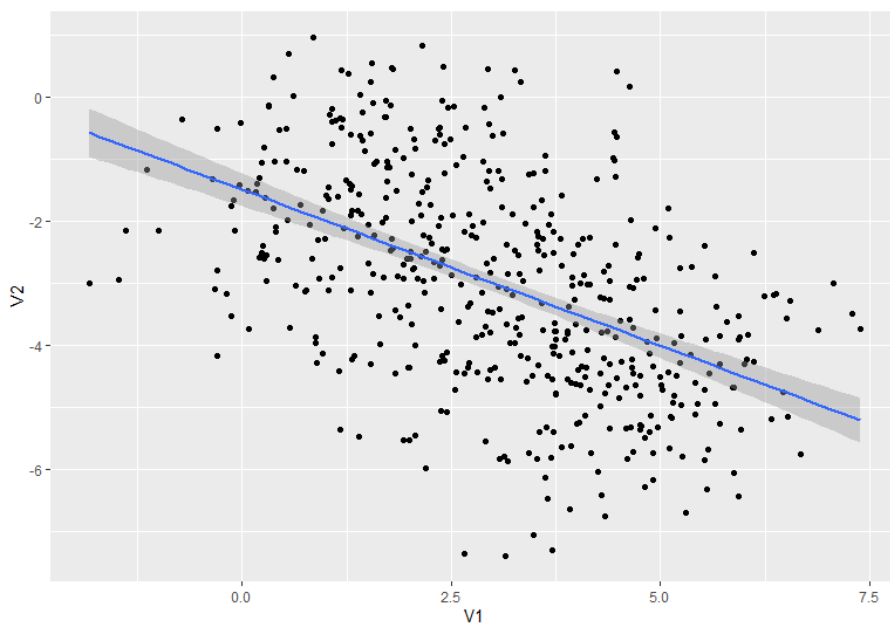


Figure 4.27 – Regression line

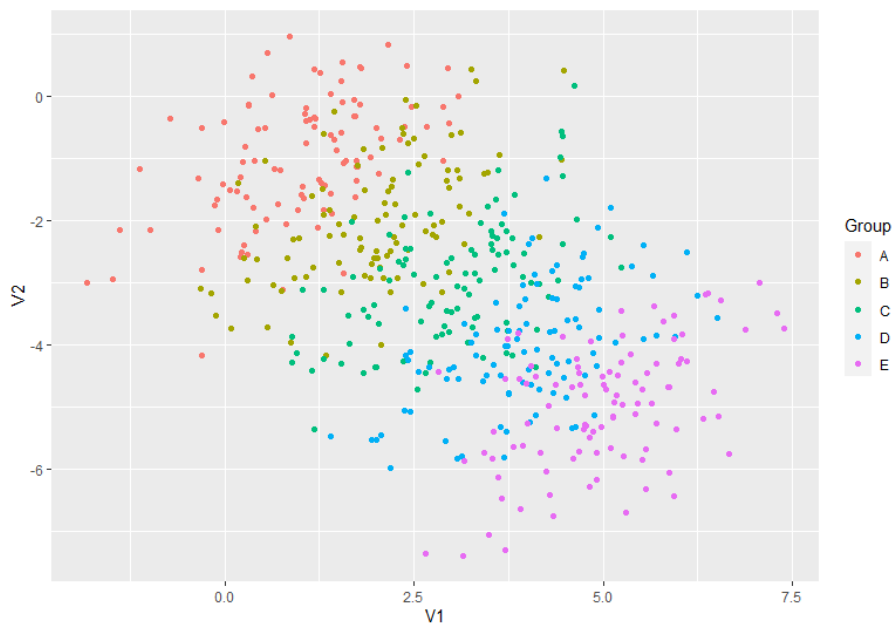


Figure 4.28 – Divide by Group

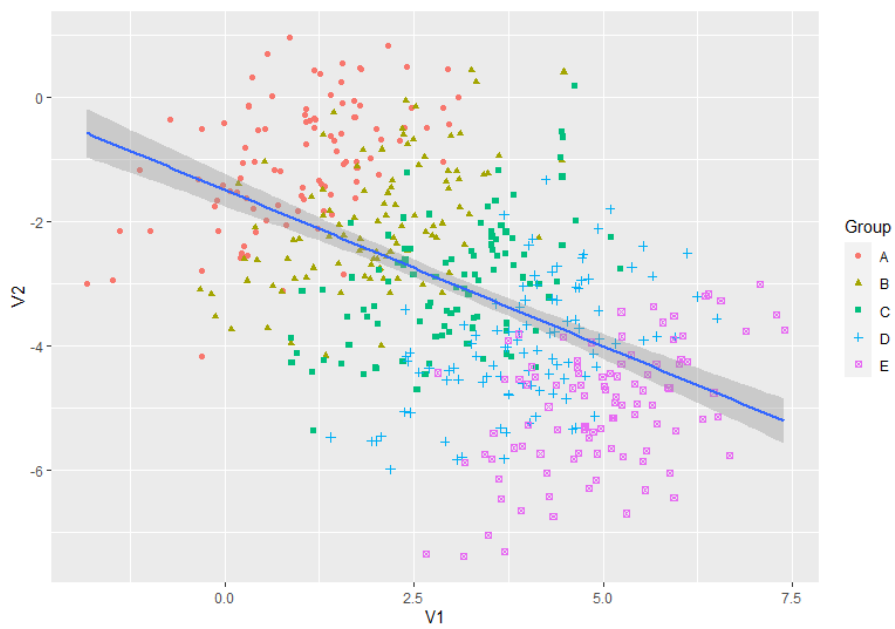


Figure 4.29 – Regression Line

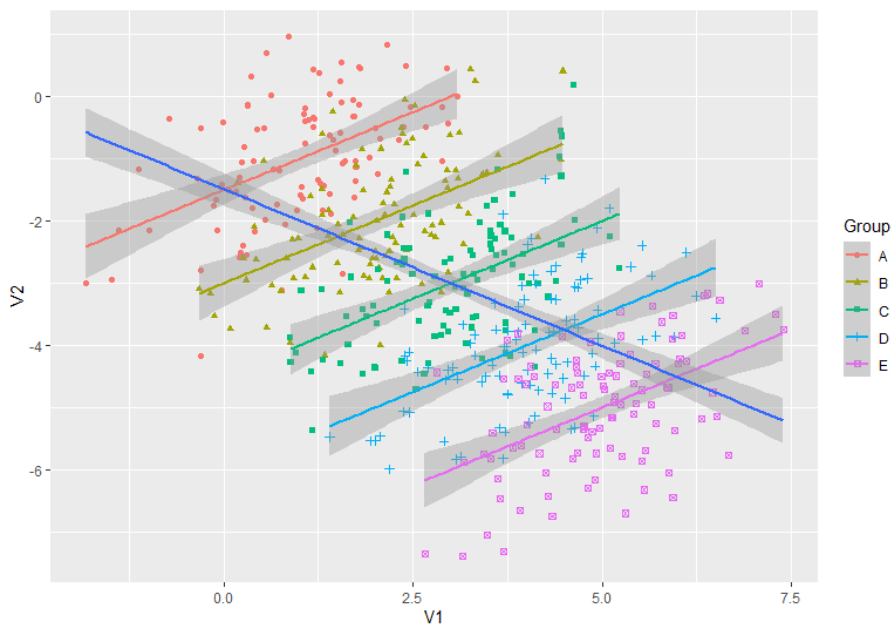


Figure 4.30 – Correlations between groups

5. PROBABILITY

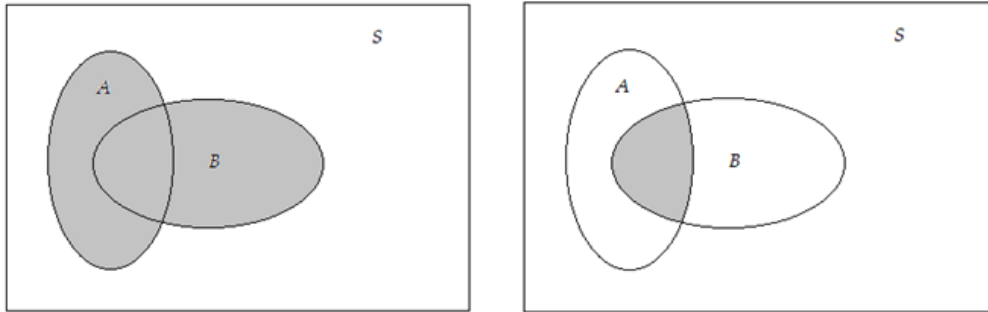


Figure 5.1 – Union and intersection

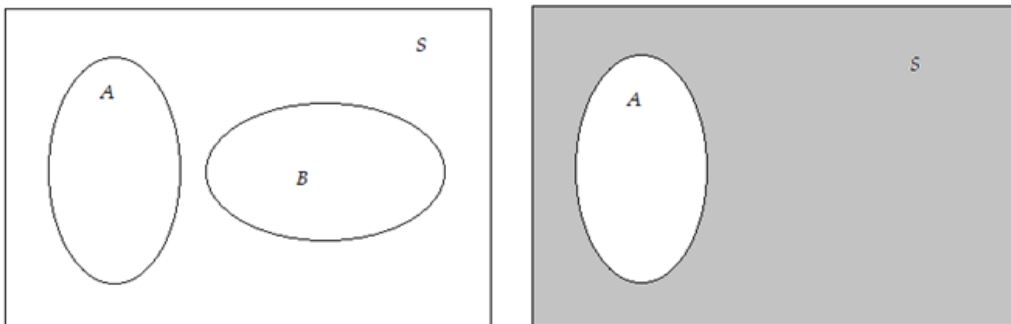


Figure 5.2 – Disjoint events and complementary events

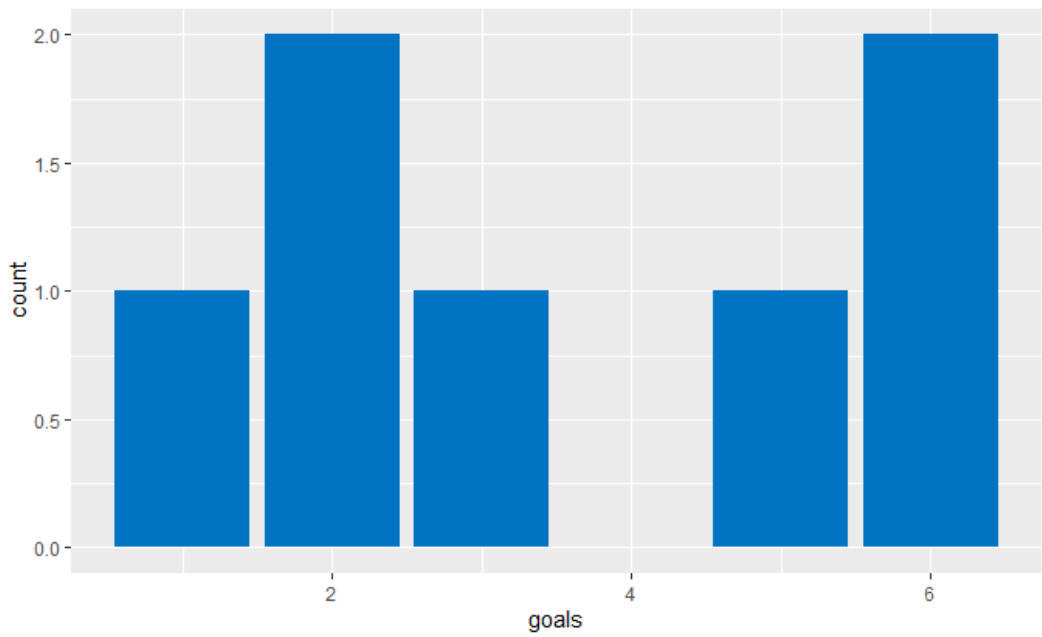


Figure 5.3 – The Goals variable

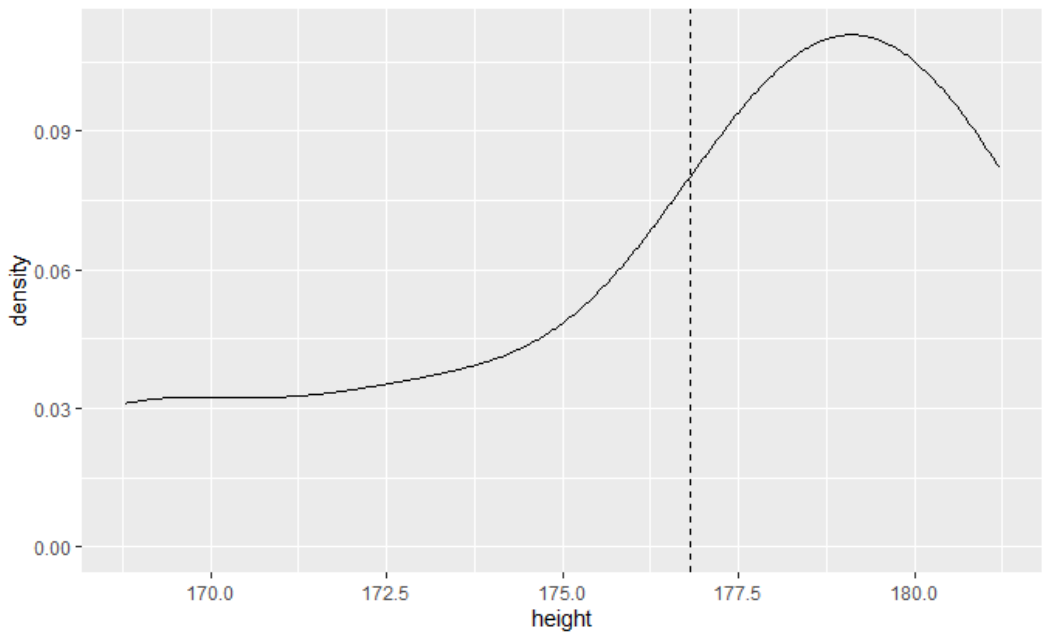


Figure 5.4 – Height variable distribution

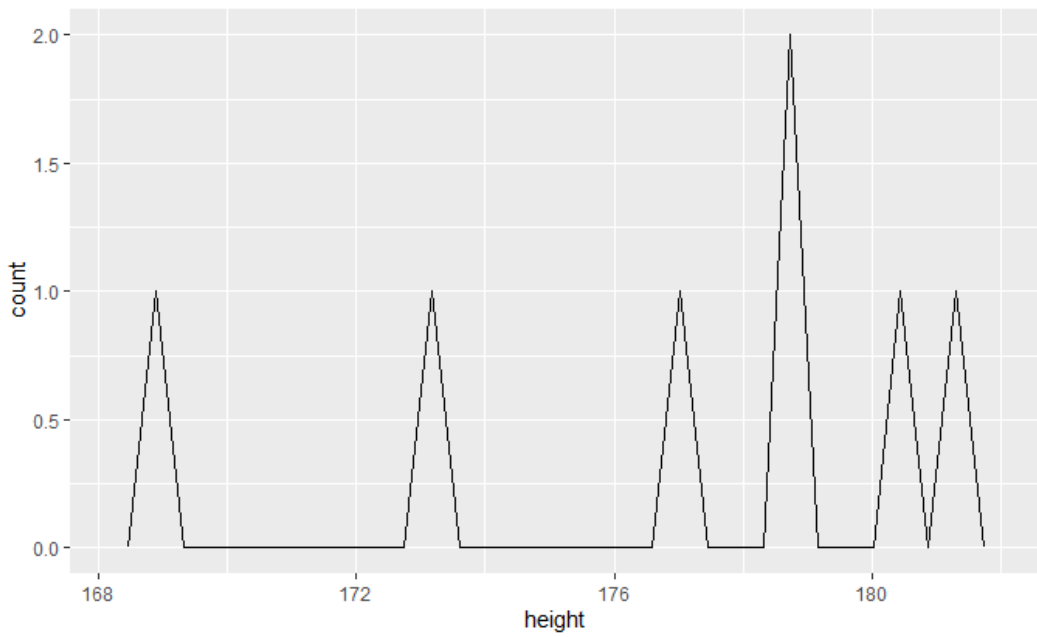


Figure 5.5 – Frequency plot

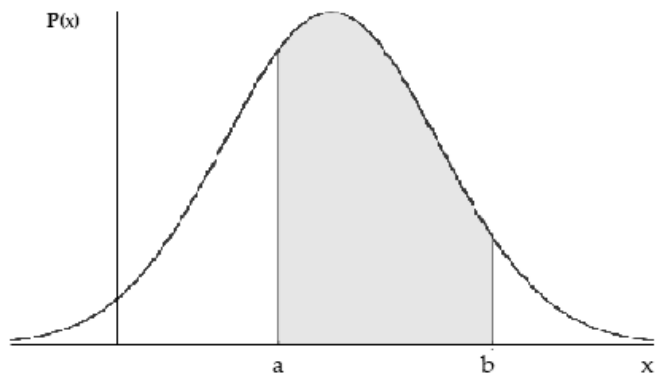


Figure 5.6 - Probability Density Function

X	P(X)	F(X)
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6

Figure 5.7 – CDF

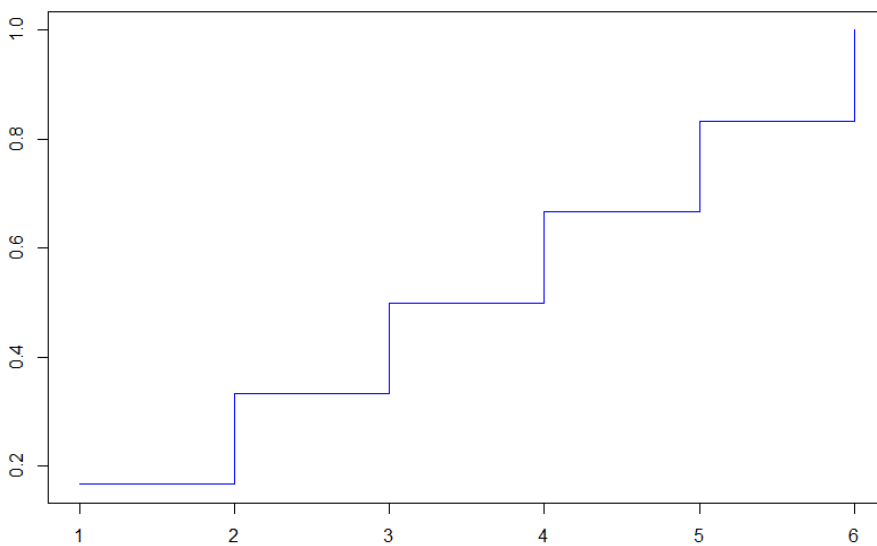


Figure 5.8 – CDF chart

6. DISTRIBUTIONS

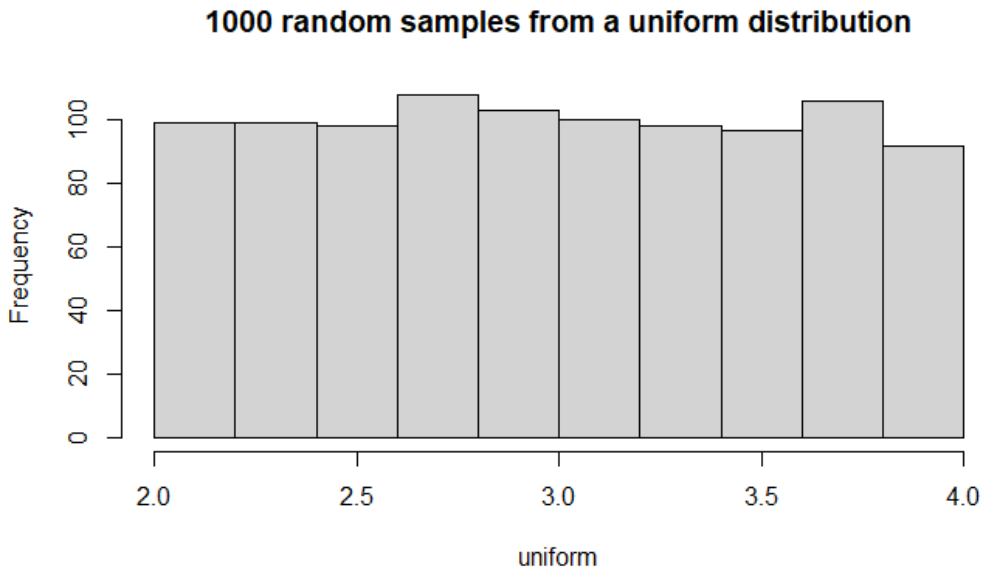


Figure 6.1 – Samples from a uniform distribution

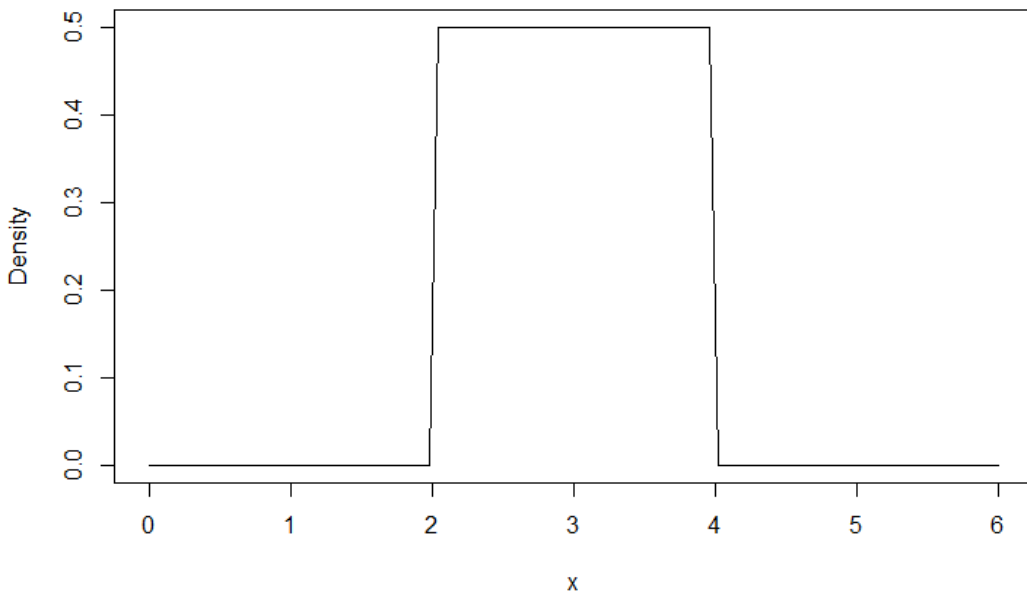


Figure 6.2 – Uniform distribution

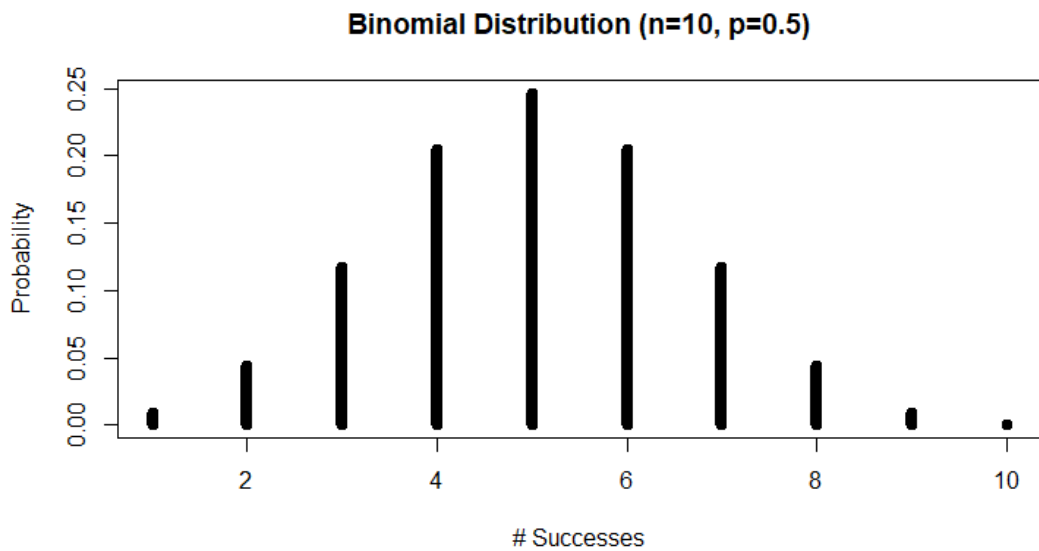


Figure 6.3- Binomial Distribution $B(10,0.5)$

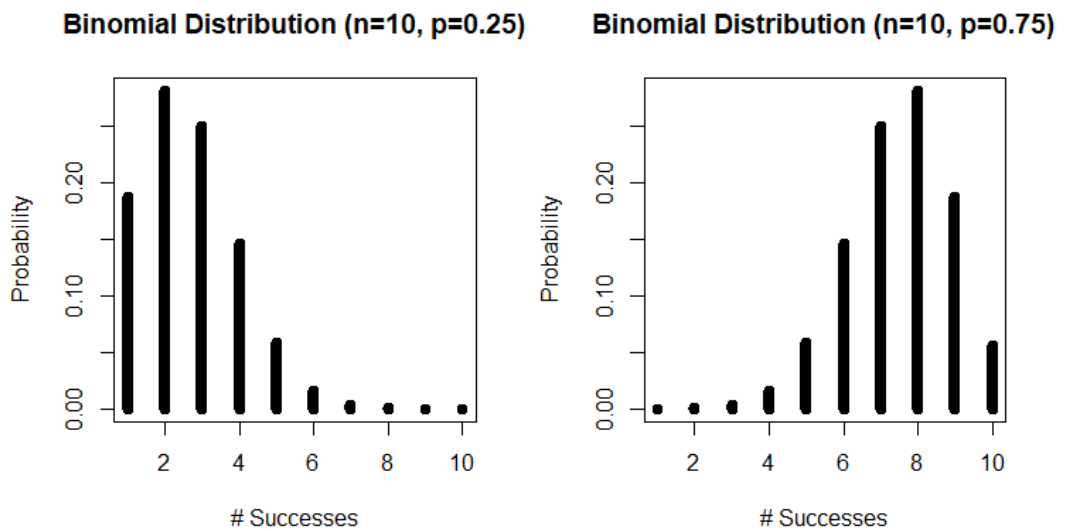


Figure 6.4 – Binomial Distributions $B(10,0.25)$ e $B(10,0.75)$

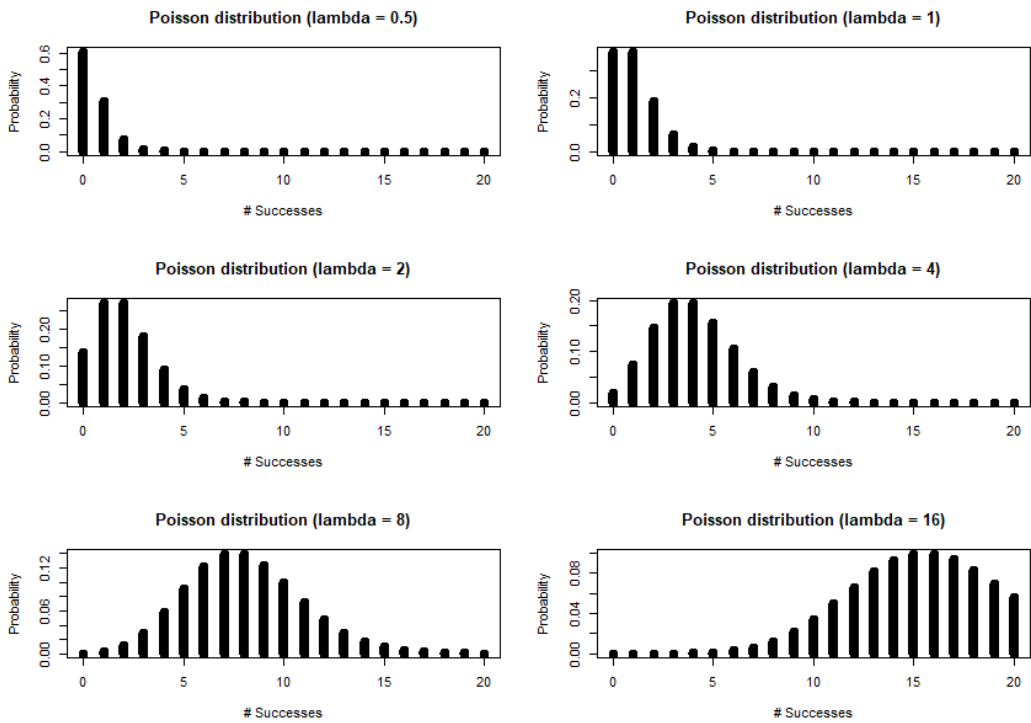


Figure 6.5 – Poisson distribution for some λ

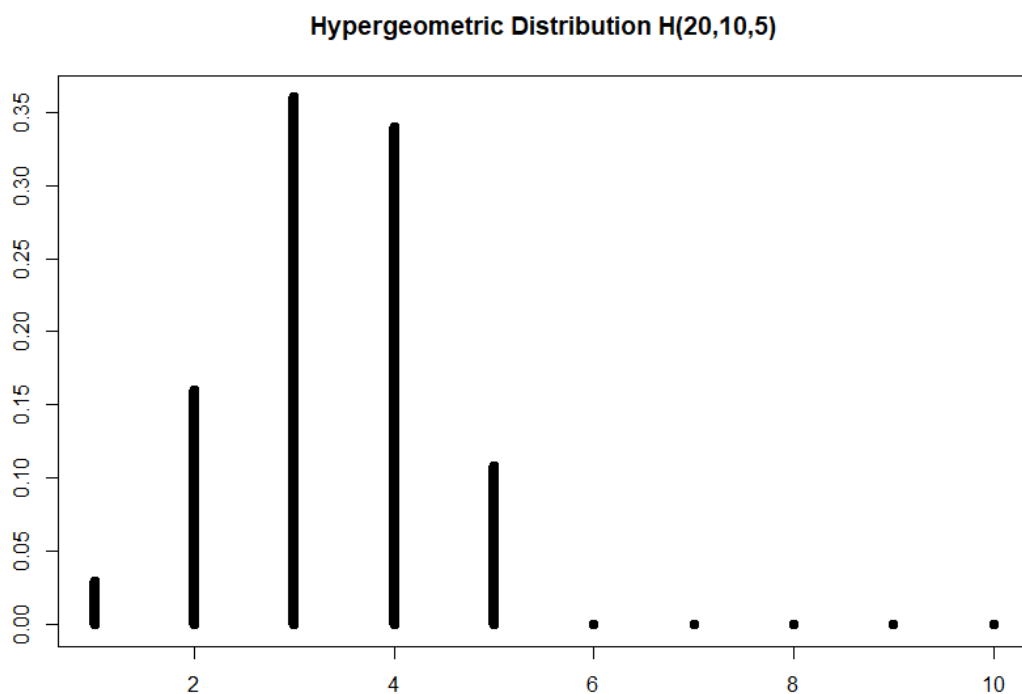


Figure 6.6 - $H(20,10,5)$

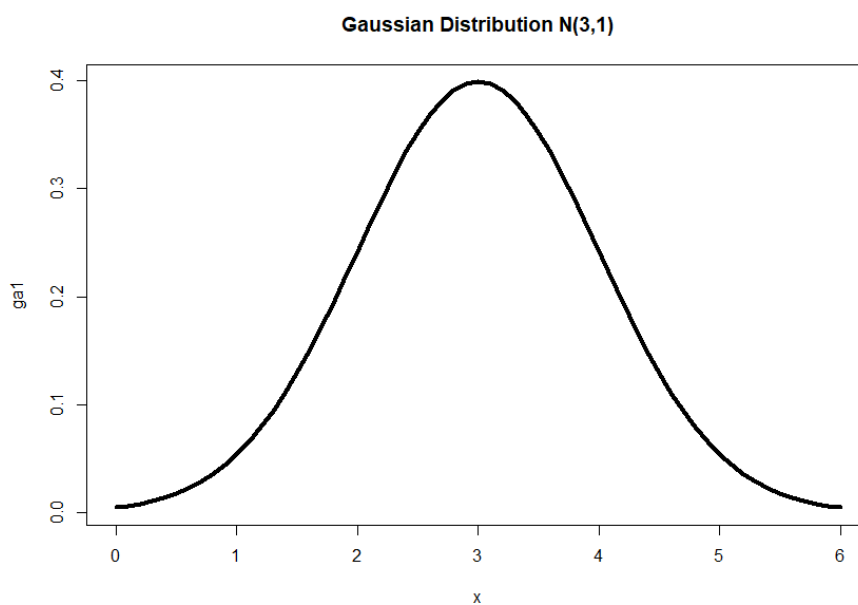


Figure 6.7 – Gaussian distribution with $\mu = 3$ and $\sigma = 1$

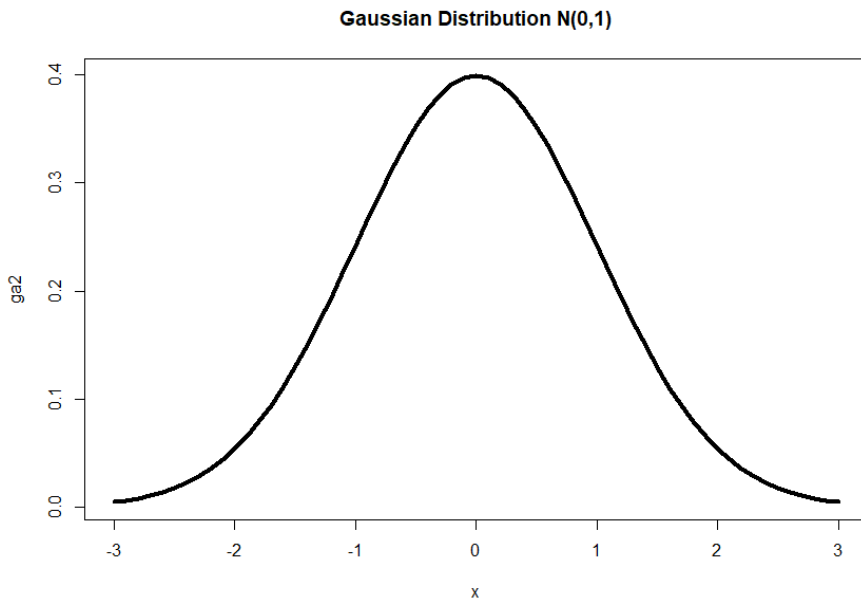


Figure 6.8 – Gaussian distribution with $\mu=0$ and $\sigma = 1$

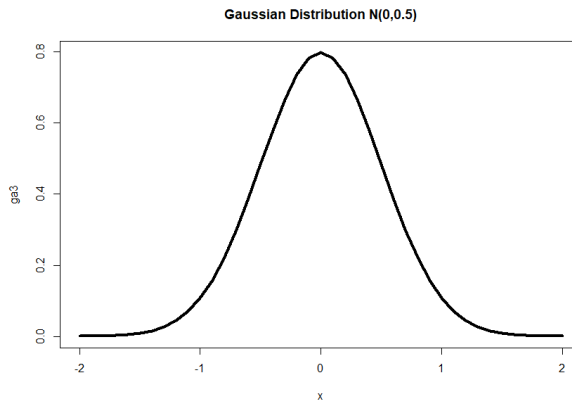


Figure 6.9 – Gaussian distribution with $\mu=0$ and $\sigma = 0,5$

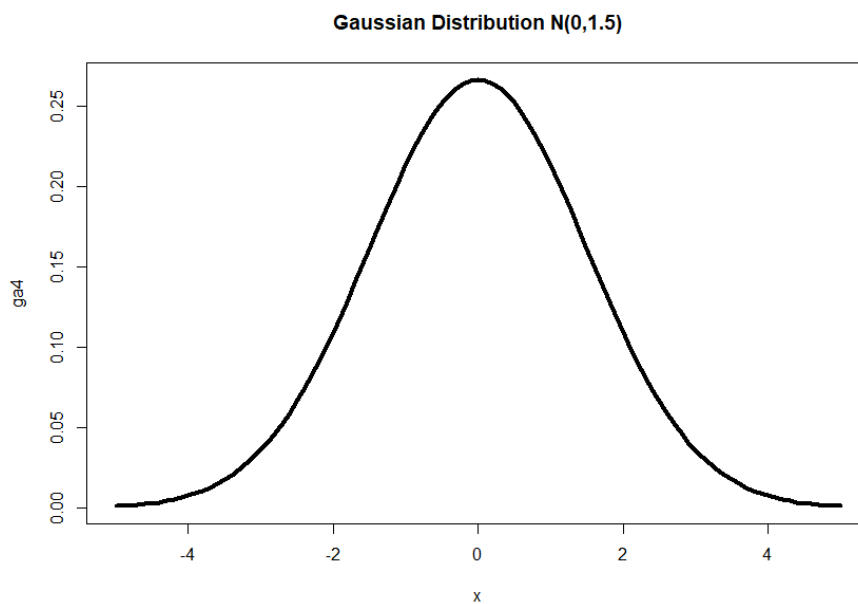


Figure 6.10 – Gaussian distribution with $\mu=0$ and $\sigma = 1,5$

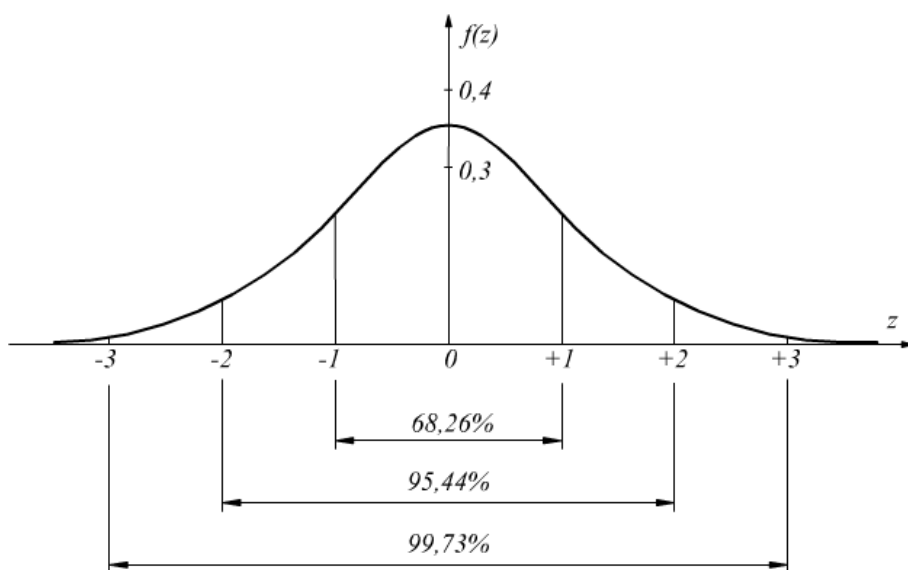


Figure 6.11 - Rule 68-95-99

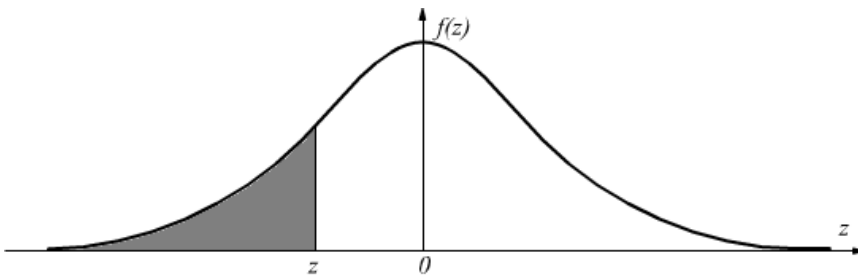


Figure 6.12 – Standard normal curve per negative values of z

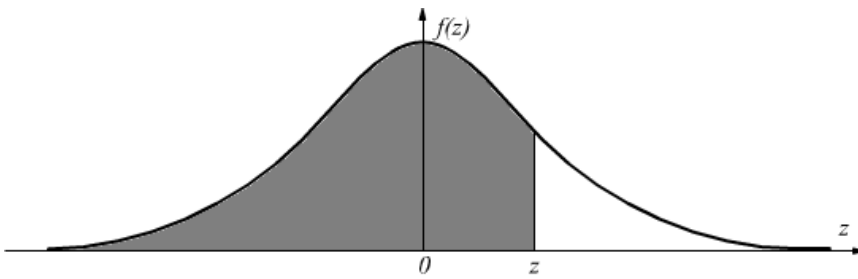


Figure 6.13 - Standard normal curve for positive values of z

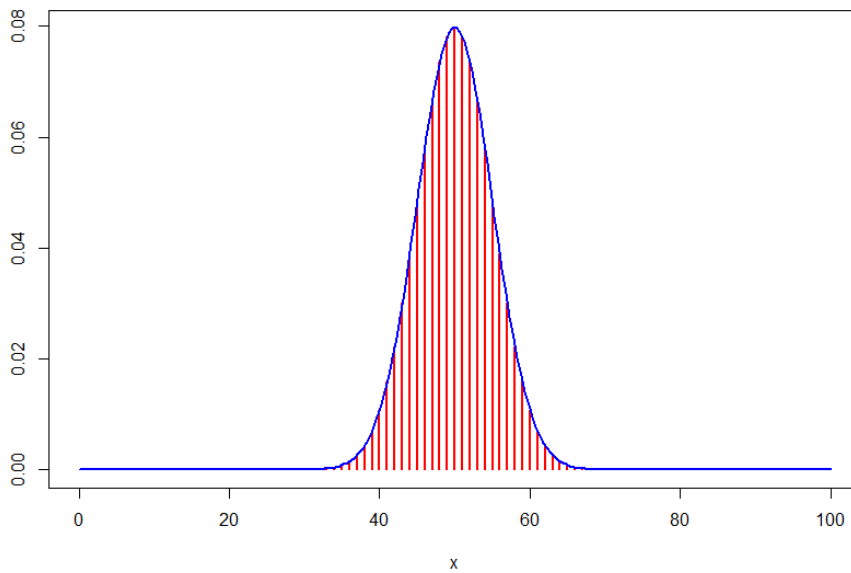


Figure 6.14 – Binomial to normal approximation

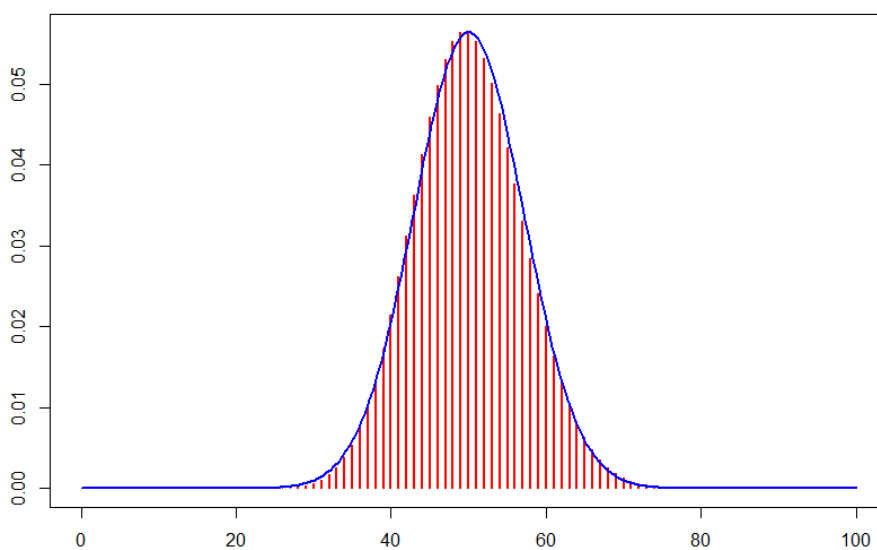


Figure 6.15 – Poisson to normal approximation

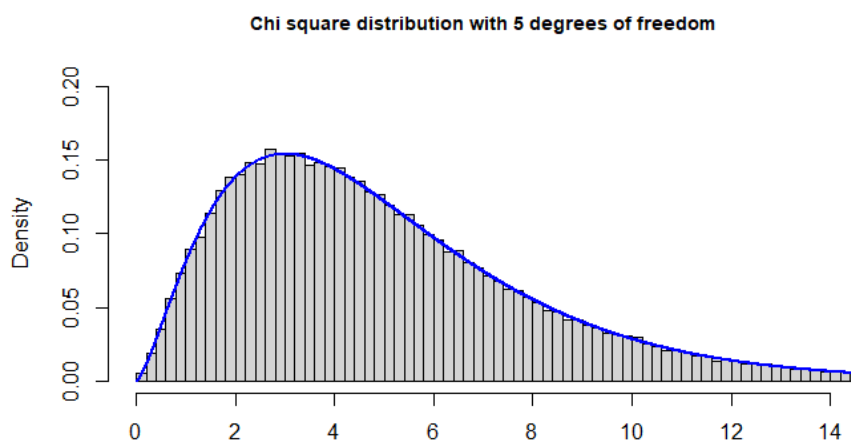


Figure 6.16 - Chi square distribution with 5 degrees of freedom

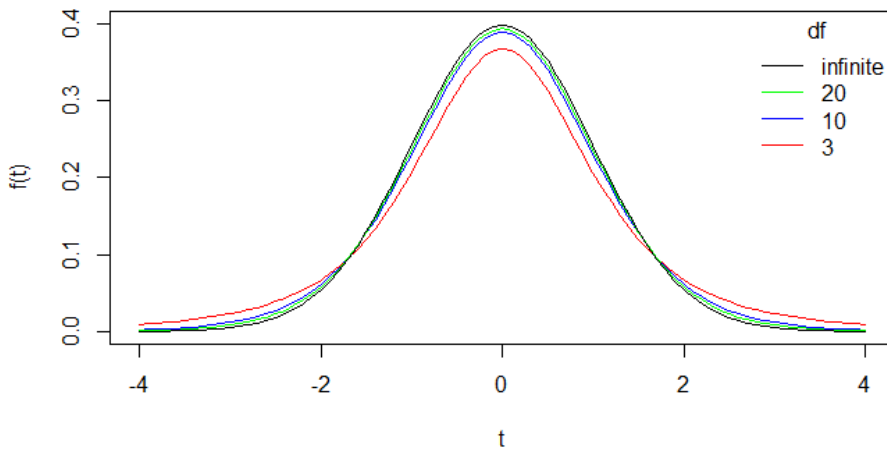


Figure 6.17 – Student t distribution

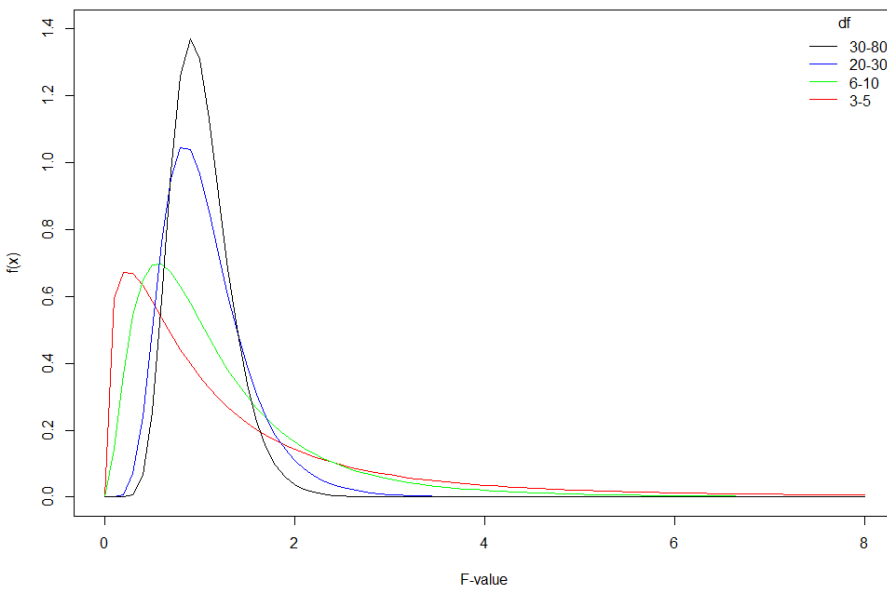


Figure 6.18 – F distribution with many degrees of freedom

7. INFERENCE STATISTICS

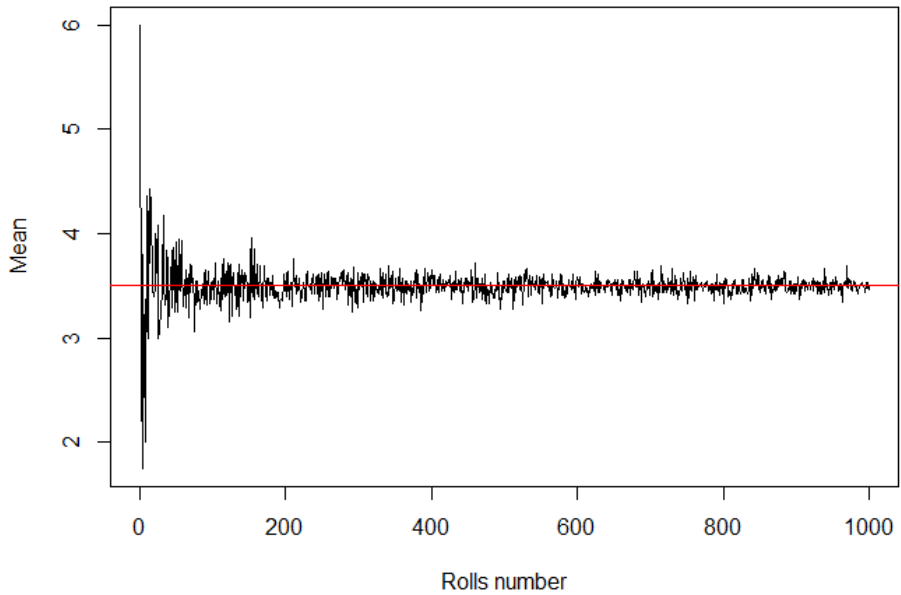


Figure 7.1 - By increasing the number of samples, the average approaches the expected value.

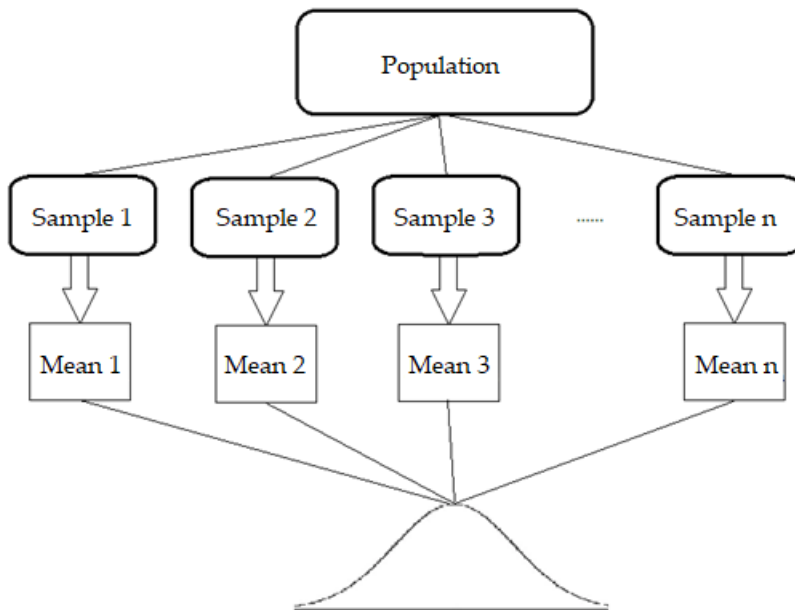


Figure 7.2 – Central Limit Theorem

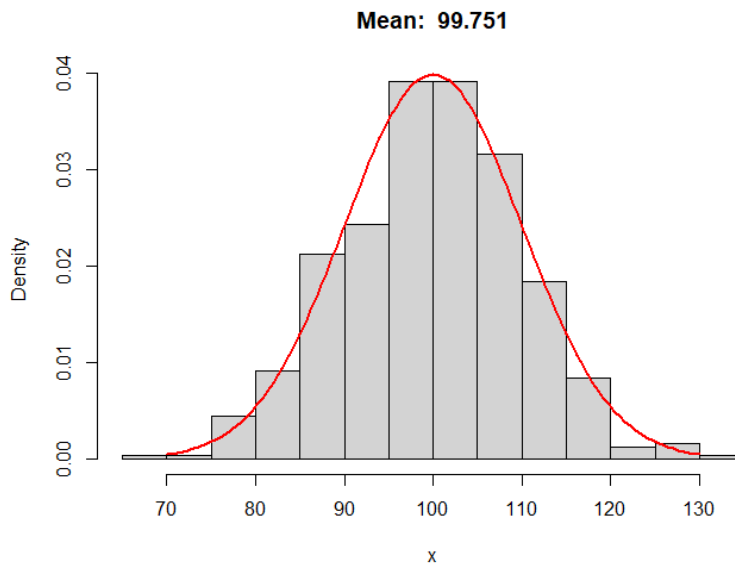


Figure 7.3 - Distribution of the 500 extracted elements

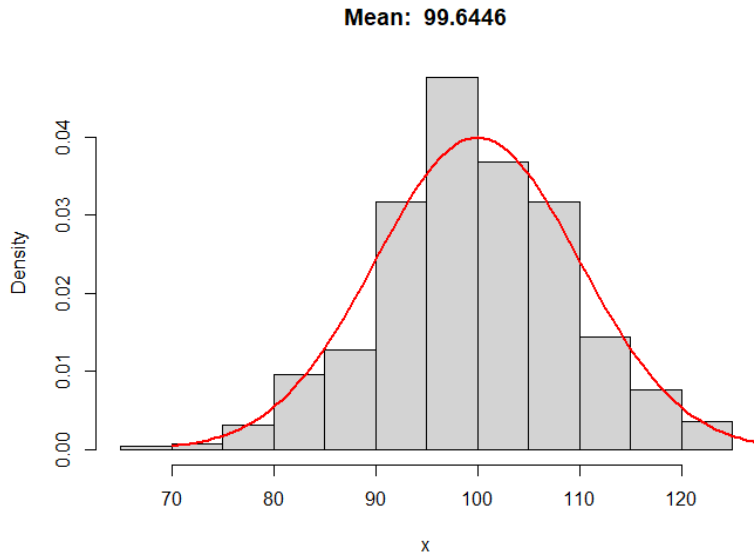


Figure 7.4 – Other 500 items

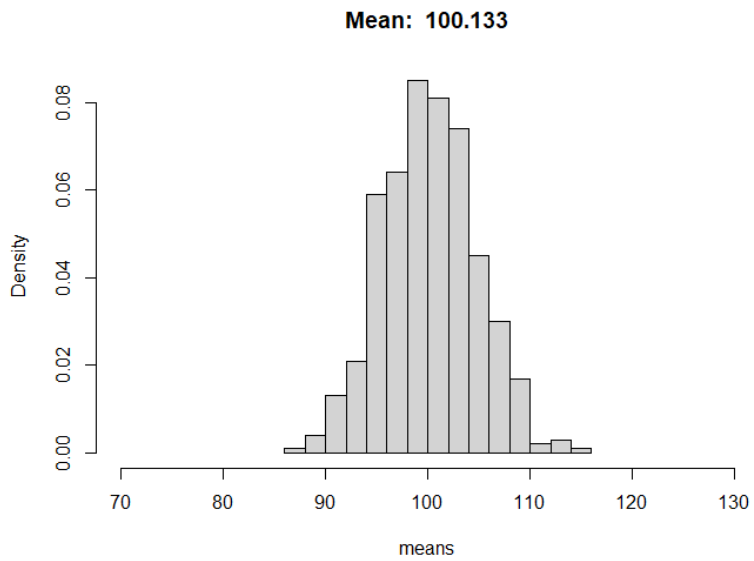


Figure 7.5 – Means distribution for $n = 5$

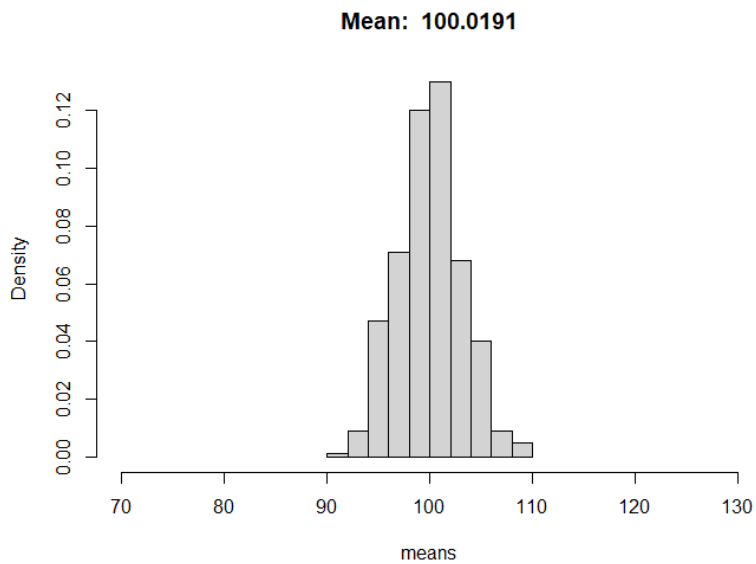


Figure 7.6 – Means distribution with $n = 10$

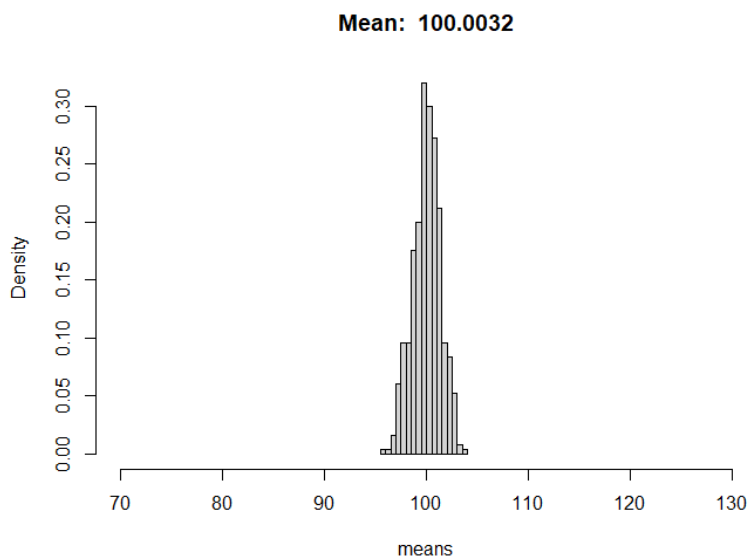


Figure 7.7 – Means distribution with $n = 50$

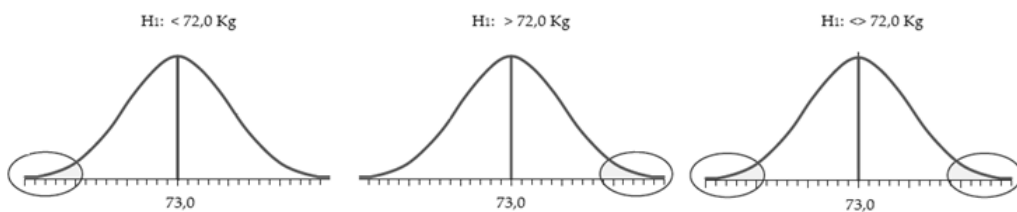


Figure 7.8 - Using the Alternative Hypothesis to Determine Test Tails

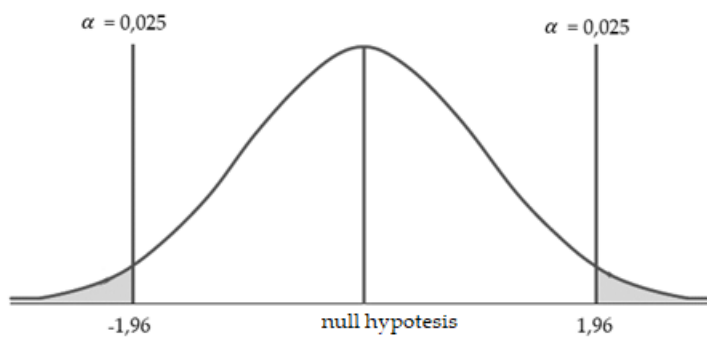


Figure 7.9 - Accept and reject region for the two-tailed test

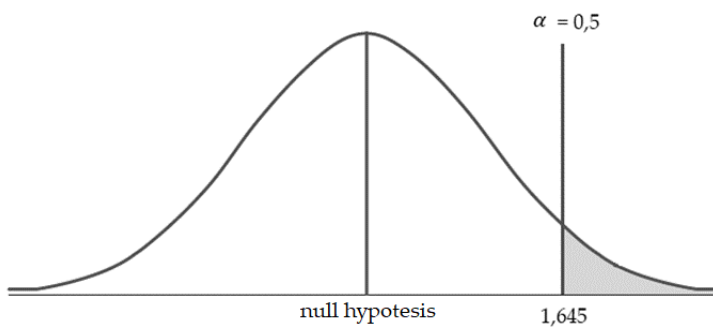


Figure 7.10 - Acceptance and rejection region for one-tailed tests

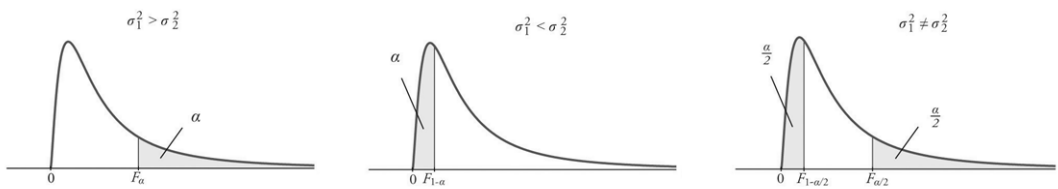


Figure 7.11 – F Distribution

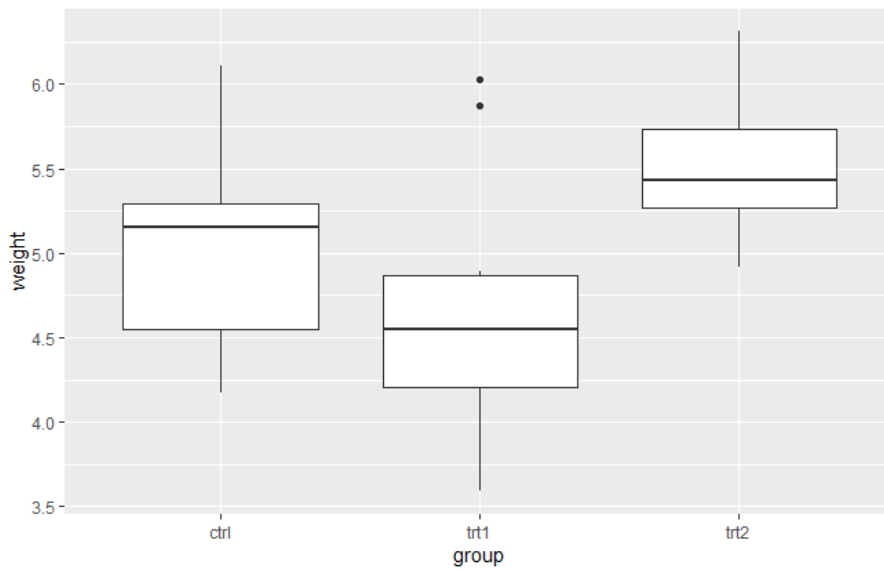


Figure 7.12 - Boxplot of weight variable

		SEX		
		Loyal		
		Woman	Man	Total
TYPE	Not Loyal	105	15	120
	Loyal	80	300	380
Total		185	315	500

Figure 7.13 – Two discrete variables

		SEX		
	Loyal	Woman	Man	Total
TYPE	Not Loyal	$120 \cdot 185 / 500$	$120 \cdot 315 / 500$	120
	Loyal	$380 \cdot 185 / 500$	$380 \cdot 315 / 500$	380
	Total	185	315	500

Figure 7.14 – Creating the expected frequency table

		SEX		
	Loyal	Woman	Man	Total
TYPE	Not Loyal	44,4	75,6	120
	Loyal	140,6	239,4	380
	Total	185	315	500

Figure 7.15 – Expected frequency table

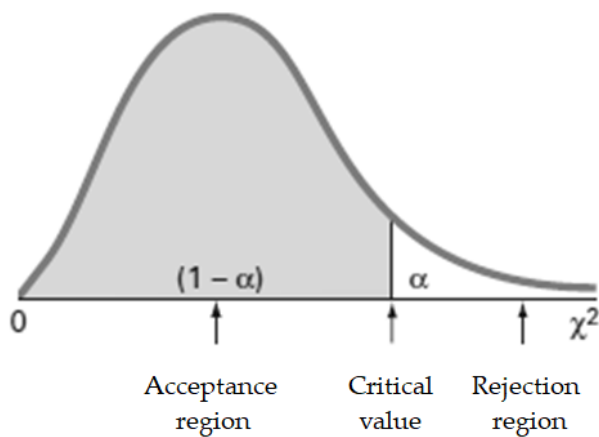


Figure 7.16 – Chi Square distribution

<i>gl/alfa</i>	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.6424	2.7055	3.8415	5.0239	6.6349	7.8794	10.8274	12.1153	15.1343
2	3.2189	4.6052	5.9915	7.3778	9.2104	10.5965	13.8150	15.2014	18.4247
3	4.6416	6.2514	7.8147	9.3484	11.3449	12.8381	16.2660	17.7311	21.1040
4	5.9886	7.7794	9.4877	11.1433	13.2767	14.8602	18.4662	19.9977	23.5064
5	7.2893	9.2363	11.0705	12.8325	15.0863	16.7496	20.5147	22.1057	25.7507
6	8.5581	10.6446	12.5916	14.4494	16.8119	18.5475	22.4575	24.1016	27.8527
7	9.8032	12.0170	14.0671	16.0128	18.4753	20.2777	24.3213	26.0179	29.8814
8	11.0301	13.3616	15.5073	17.5345	20.0902	21.9549	26.1239	27.8674	31.8268
9	12.2421	14.6837	16.9190	19.0228	21.6660	23.5893	27.8767	29.6669	33.7247
10	13.4420	15.9872	18.3070	20.4832	23.2093	25.1881	29.5879	31.4195	35.5572
11	14.6314	17.2750	19.6752	21.9200	24.7250	26.7569	31.2635	33.1382	37.3647
12	15.8120	18.5493	21.0261	23.3367	26.2170	28.2997	32.9092	34.8211	39.1306
13	16.9848	19.8119	22.3620	24.7356	27.6882	29.8193	34.5274	36.4768	40.8735
14	18.1508	21.0641	23.6848	26.1189	29.1412	31.3194	36.1239	38.1085	42.5752
15	19.3107	22.3071	24.9958	27.4884	30.5780	32.8015	37.6978	39.7173	44.2596
16	20.4651	23.5418	26.2962	28.8453	31.9999	34.2671	39.2518	41.3077	45.9255
17	21.6146	24.7690	27.5871	30.1910	33.4087	35.7184	40.7911	42.8808	47.5591
18	22.7595	25.9894	28.8693	31.5264	34.8052	37.1564	42.3119	44.4337	49.1853
19	23.9004	27.2036	30.1435	32.8523	36.1908	38.5821	43.8194	45.9738	50.7873
20	25.0375	28.4120	31.4104	34.1696	37.5663	39.9969	45.3142	47.4977	52.3832

Figure 7.17 - Table with expected values of chi square distribution

8. SAMPLING

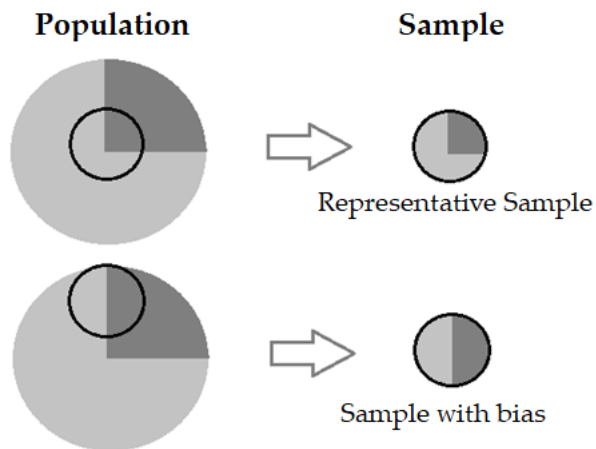


Figure 8.1 – Good sampling e biased sampling

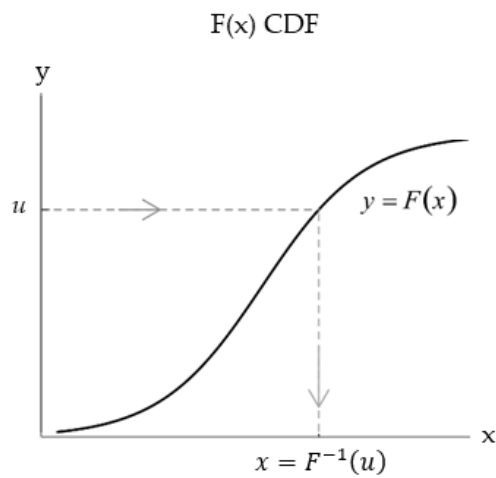


Figure 8.2 - Inverse Transform Sampling

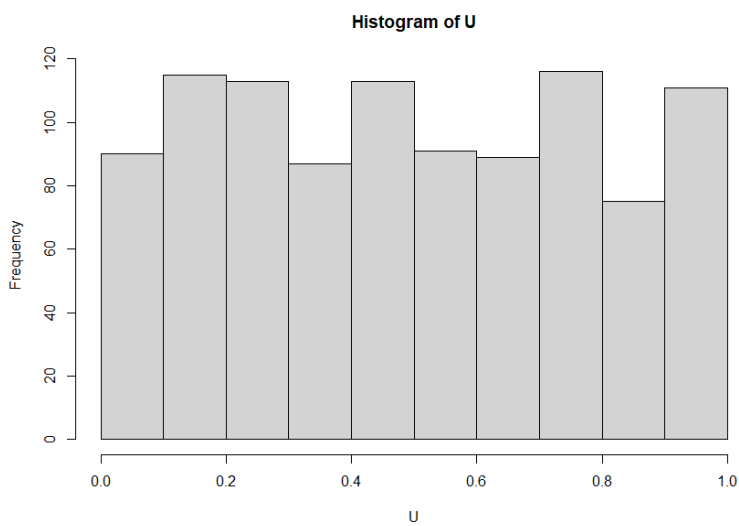


Figure 8.3 – Histogram of uniform distribution

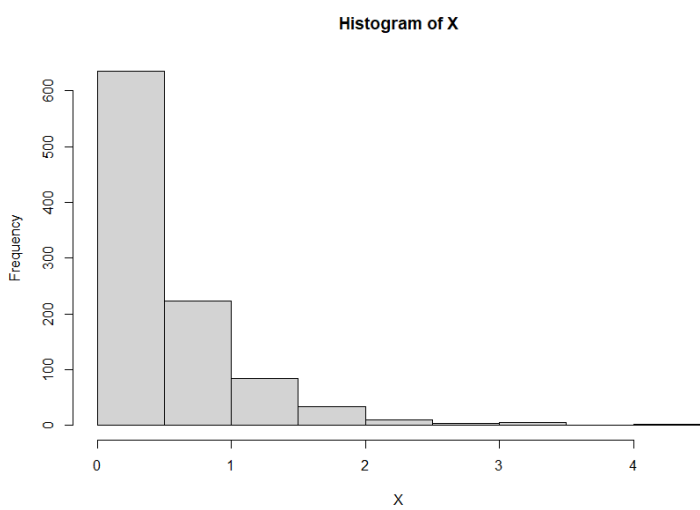


Figure 8.4 – The new distribution

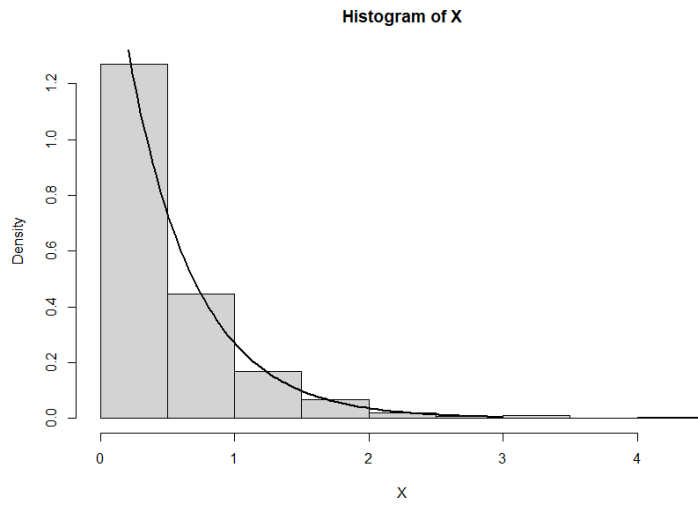


Figure 8.5 – Exponential trend of the distribution

9. DATA PREPARATION

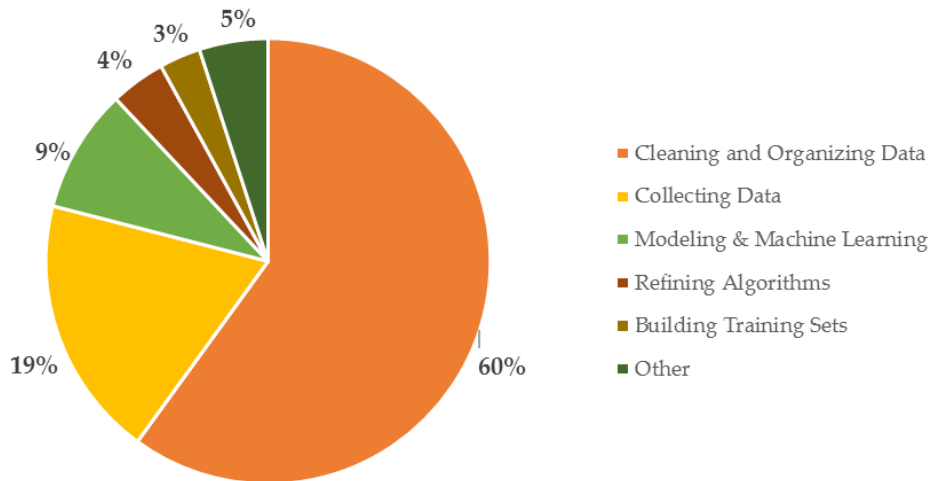


Figure 9.1 – Time distribution in a project

Customer	Sex	Marital Status	Work	Purchase	Amount	Loyal
1	Woman	Married	Employee	10	1.000	YES
2	Man	Divorced	Freelancer	5	500	YES
3	Man	Single	Employee	2	200	NO
4	Woman	Single	Employee	1	100	NO
5	Woman	Single	Manager	1	100	NO
6	Man	Divorced	Employee	8	800	YES
7	Woman	Divorced	Employee	0	0	NO
8	Man	Married	Employee	13	1.300	YES
9	Man	Married	Manager	2	200	NO
10	Man	Married	Freelancer	7	700	YES

Figure 9.2 Sample customer dataset

Customer	Data	Product	Purchase	Amount
1	06/01/2021	Product A	1	100
1	08/02/2021	Product B	1	100
1	15/03/2021	Product A	1	100
1	16/03/2021	Product C	1	100
1	03/05/2021	Product C	1	100
1	04/05/2021	Product B	1	100
1	06/06/2021	Product A	1	100
1	31/07/2021	Product A	1	100
1	24/09/2021	Product C	1	100
1	15/12/2021	Product A	1	100

Figure 9.3 – Transaction example

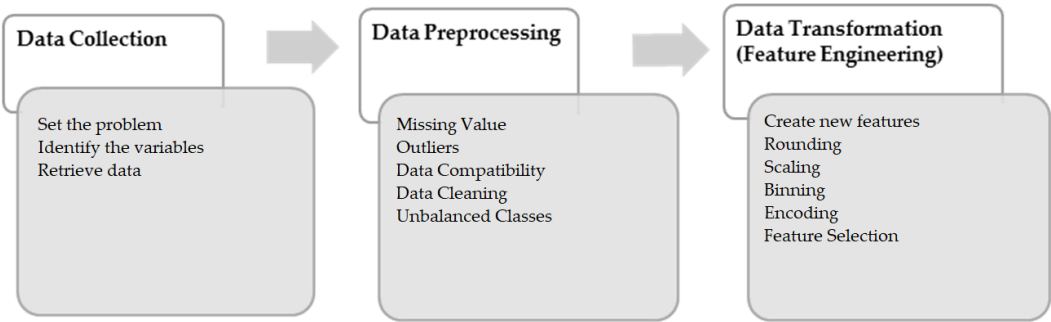


Figure 9.4 – Data preparation steps

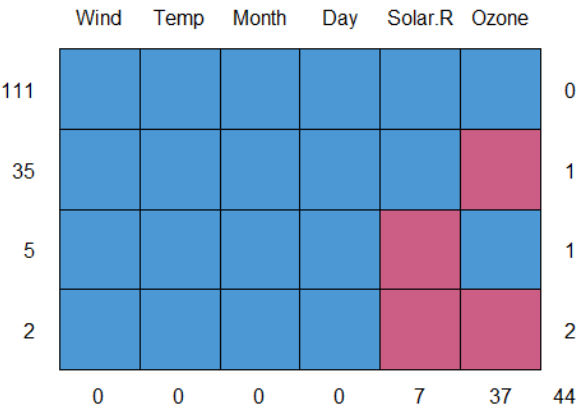


Figure 9.5 – Map of missing values

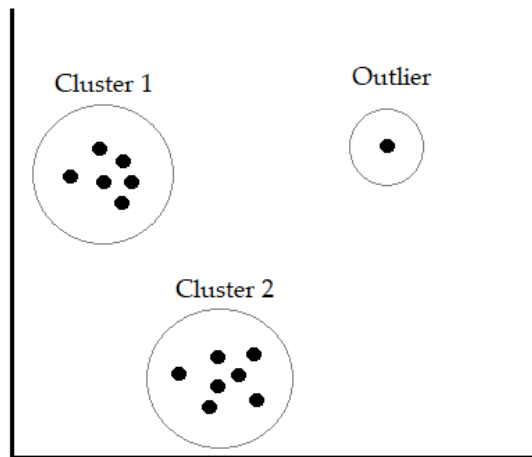


Figure 9.6 - Clustering for identifying outliers

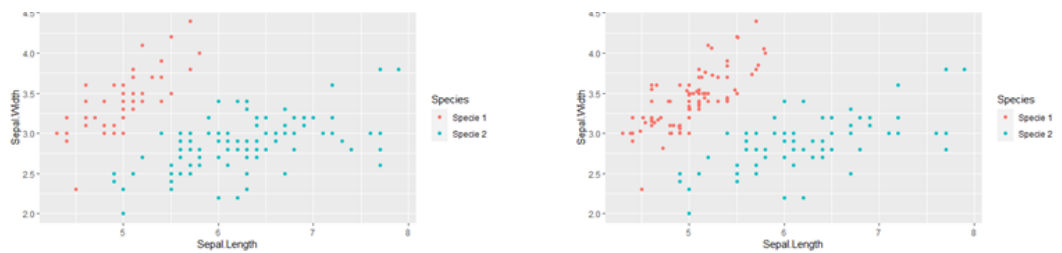


Figure 9.7 – The dataset before and after balancing

10. FEATURE ENGINEERING

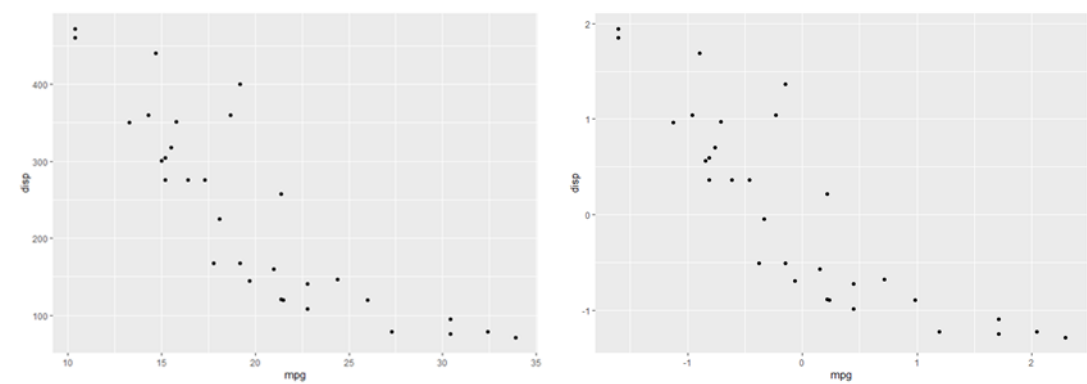


Figure 10.1 – Variables before and after scaling

Age	Bin
12	15
16	15
18	15
22	26
25	26
30	26
35	41
42	41
45	41
48	51
50	51
55	51

Figure 10.2 - Equi-depth binning

Age	Bin
12	[12,22]
16	[12,22]
18	[12,22]
22	[12,22]
25	[23,33]
30	[23,33]
35	[34,44]
42	[34,44]
45	[45,55]
48	[45,55]
50	[45,55]
55	[45,55]

Figure 10.3 - Equi-width binning

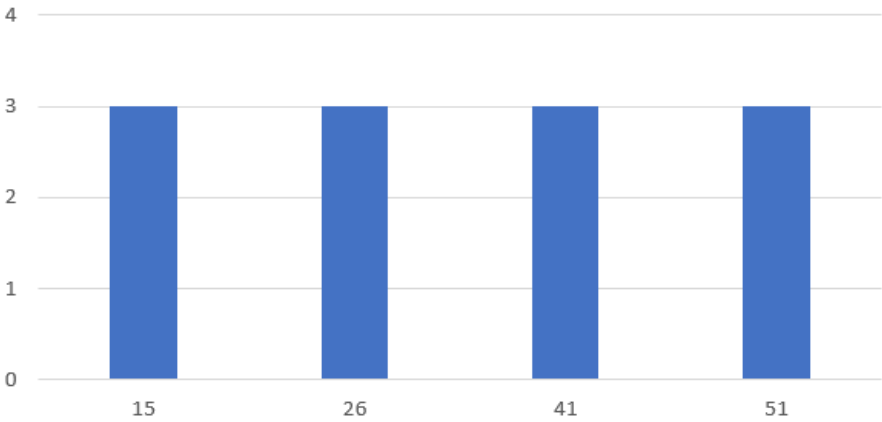


Figure 10.4 - Equi-depth binning

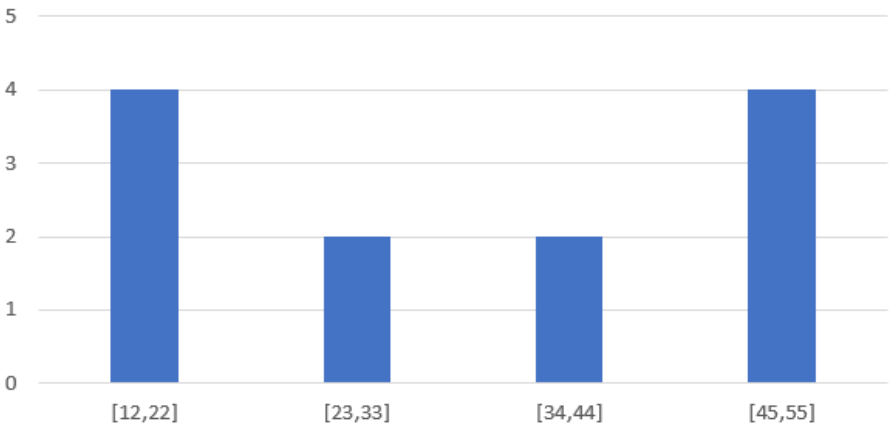


Figure 10.5 - Equi-width binning

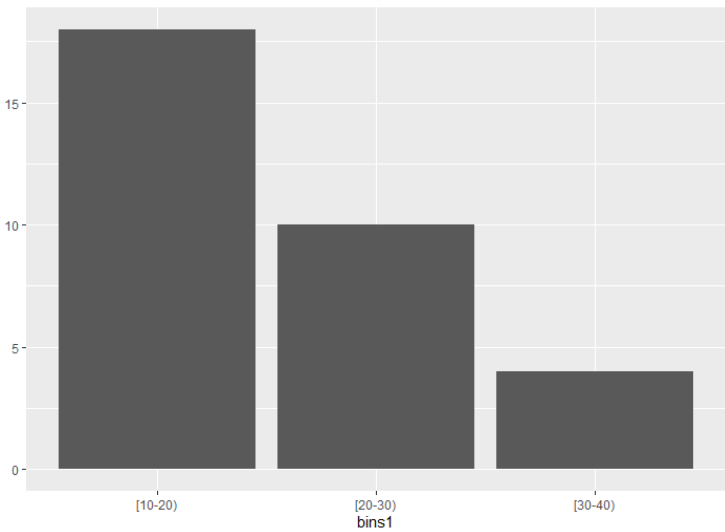


Figure 10.6 - Plot of the bins

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb	bins
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4	[20-30)
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4	[20-30)
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1	[20-30)
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1	[20-30)
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2	[10-20)
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1	[10-20)
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4	[10-20)
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2	[20-30)
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2	[20-30)
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4	[10-20)
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4	[10-20)

Figure 10.7 - The dataset with the bins column

11. MACHINE LEARNING

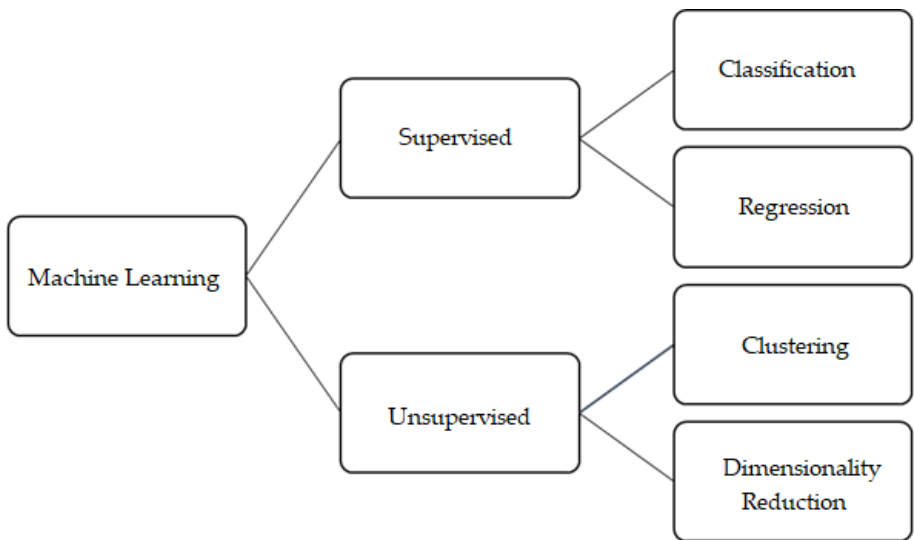


Figure 11.1 – Some ML algorithms

Number	Color	Shape	Size	Result
1	Red	Round	Small	Eatable
2	Green	Round	Medium	Eatable
3	Yellow	Long	Medium	Eatable

Example

Feature

Class

Figure 11.2 - Items needed for the training step

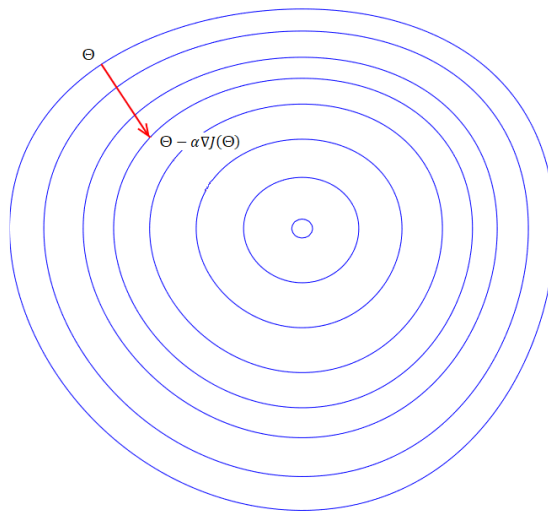


Figure 11.3 – Gradient Descent

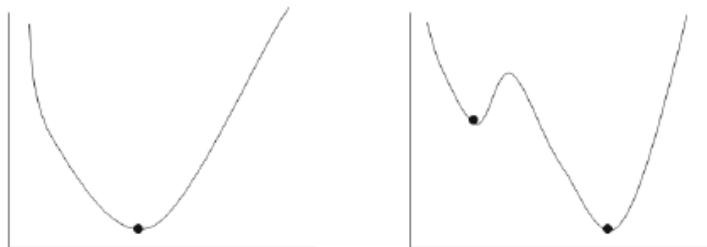


Figure 11.4 - Convex function and non-convex function

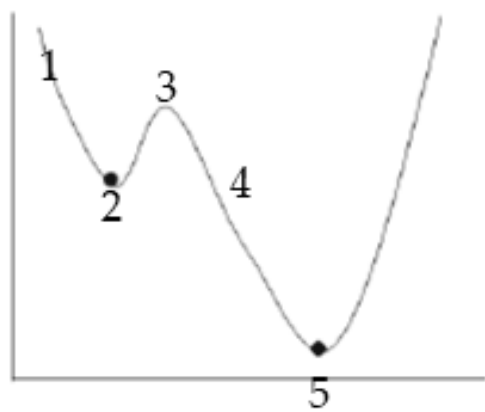


Figure 11.5 – Non-convex function

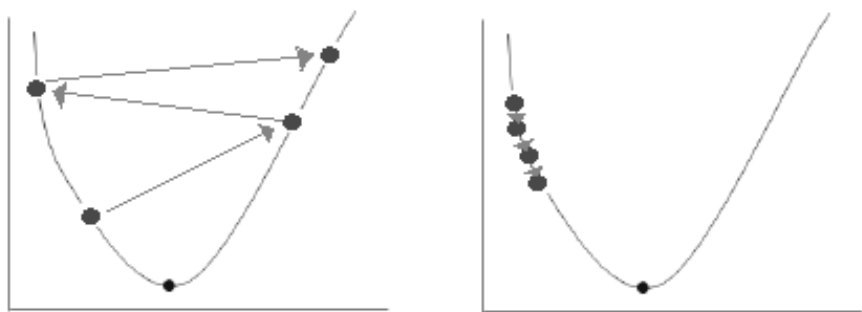


Figure 11.6 – Learning rate effect

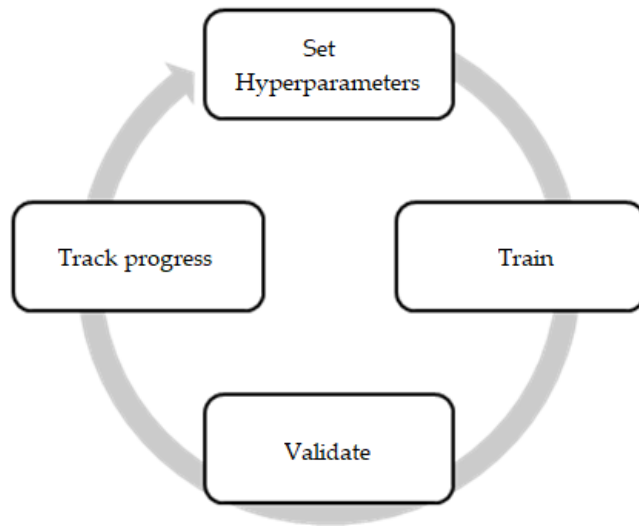


Figure 11.7 – Hyperparameter tuning

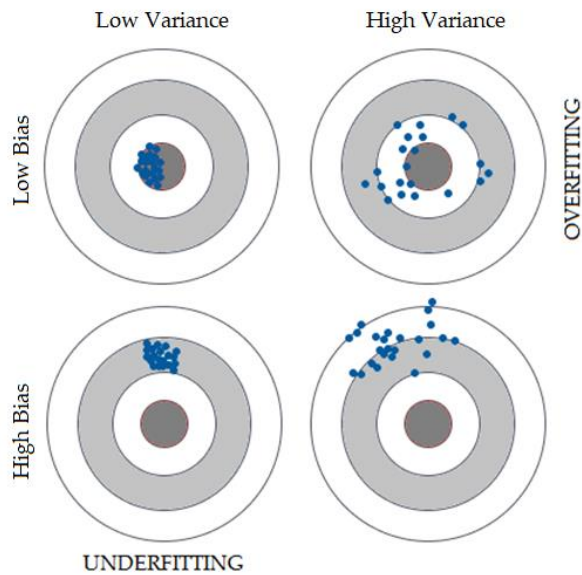


Figure 11.8 – Bias and variance effect

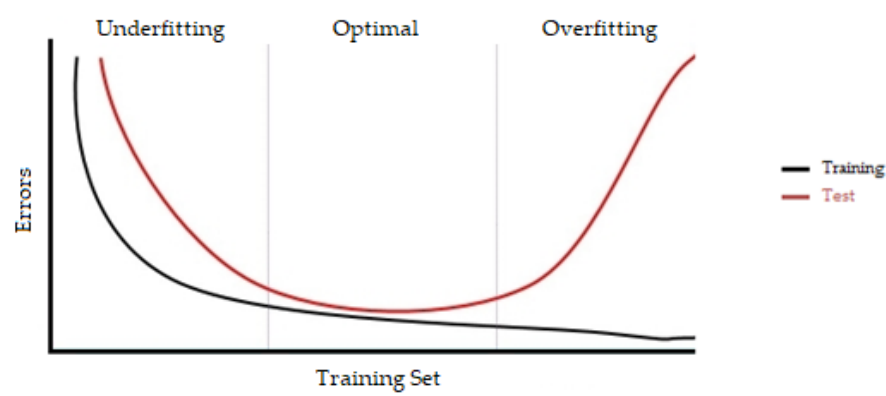


Figure 11.9 – Learning curve

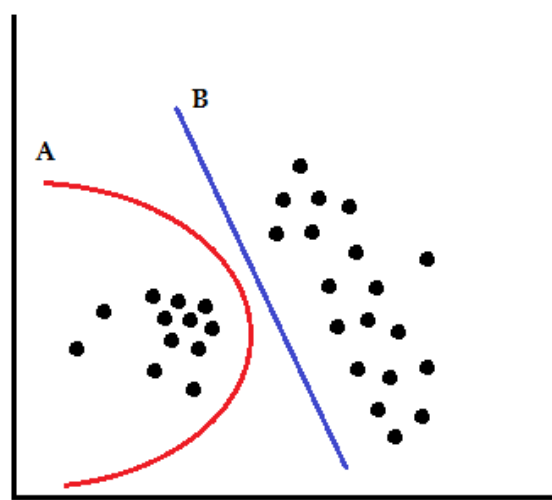


Figure 11.10 - Occam's Razor: solution B is preferred because it is the simplest

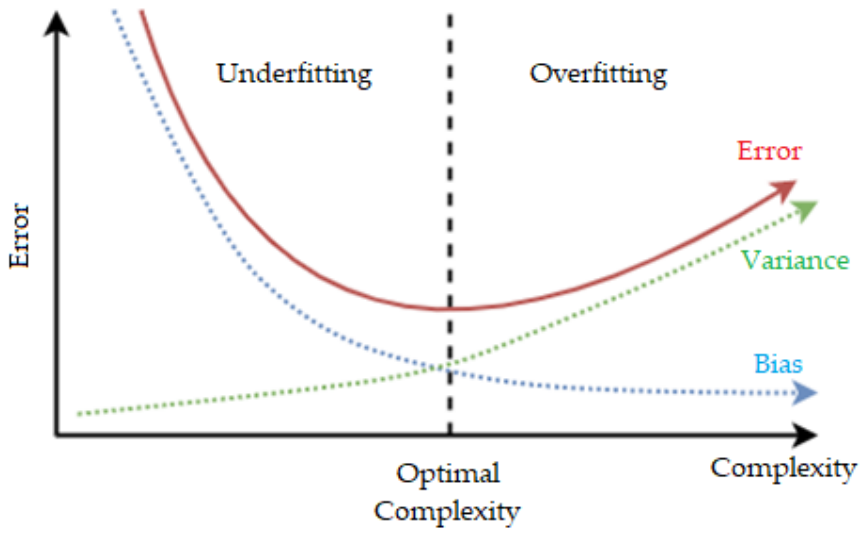


Figure 11.11 - Bias-Variance Trade-Off

12. VALIDATION

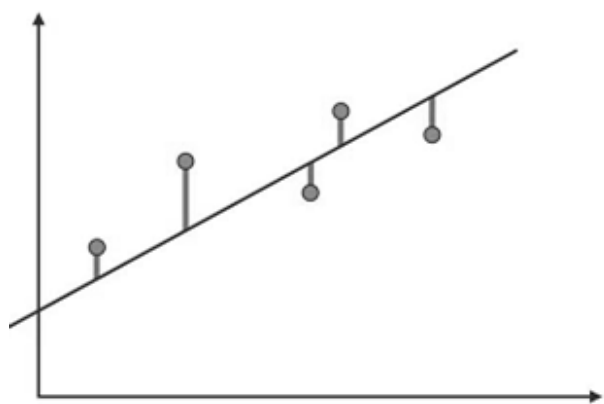


Figure 12.1 – Error in regression

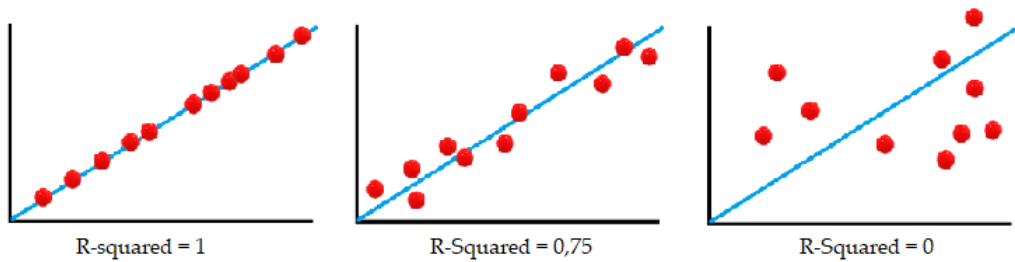


Figure 12.2 - R-squared

		Predictions		Tot
		YES	NO	
Actual	YES	TP	FN	TP+FN
	NO	FP	TN	FP+TN
Tot		TP+FP	TN+FN	P+N

Figure 12.3 – Confusion Matrix

Decile (% of elements)	TPR Base Model	TPR Our Model	Lift
0	0	0	0,00
10	10	60	6,00
20	20	75	3,75
30	30	82	2,73
40	40	88	2,20
50	50	92	1,84
60	60	95	1,58
70	70	98	1,40
80	80	100	1,25
90	90	100	1,11
100	100	100	1,00

Figure 12.4 – Table for the Gain Chart

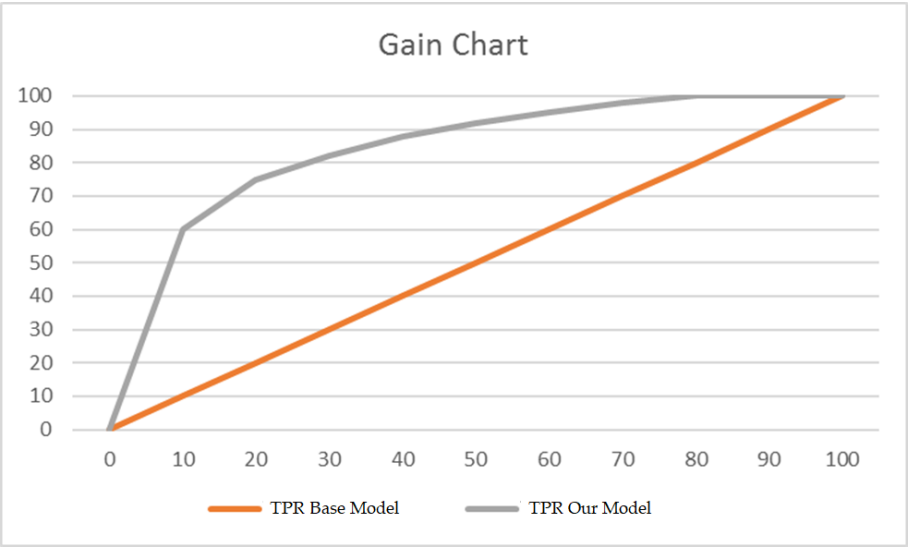


Figure 12.5 - Gain Chart

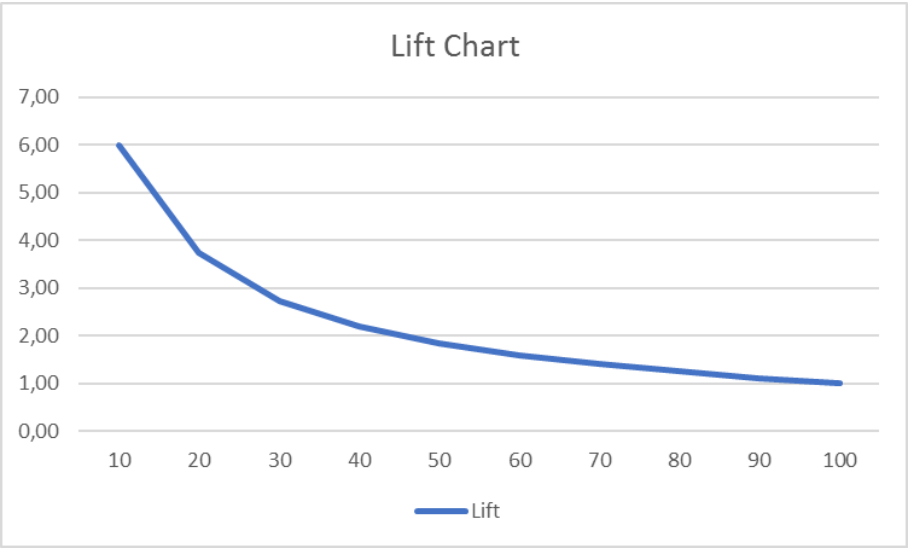


Figure 12.6 - Lift Chart

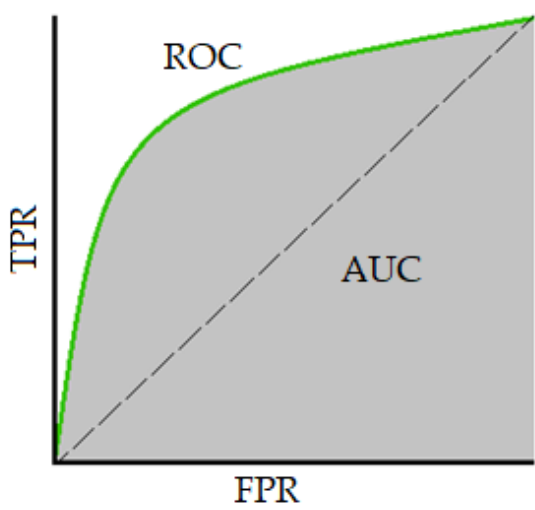


Figure 12.7 – ROC chart

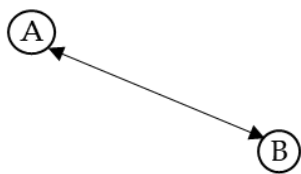


Figure 12.8 – Euclidean Distance

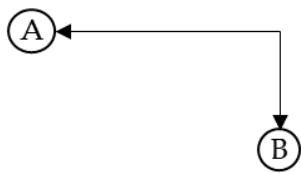


Figure 12.9 – Manhattan distance

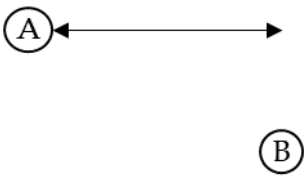


Figure 12.10 – Lagrange distance

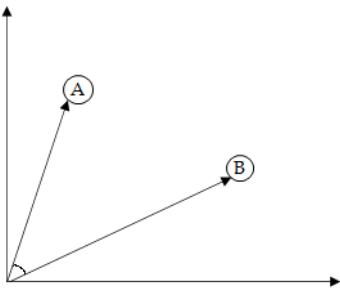


Figure 12.11 – Cosine distance

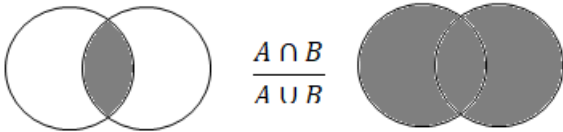


Figure 12.12 – Jaccard distance

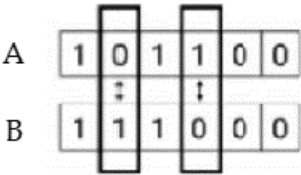


Figure 12.13 – Hamming distance

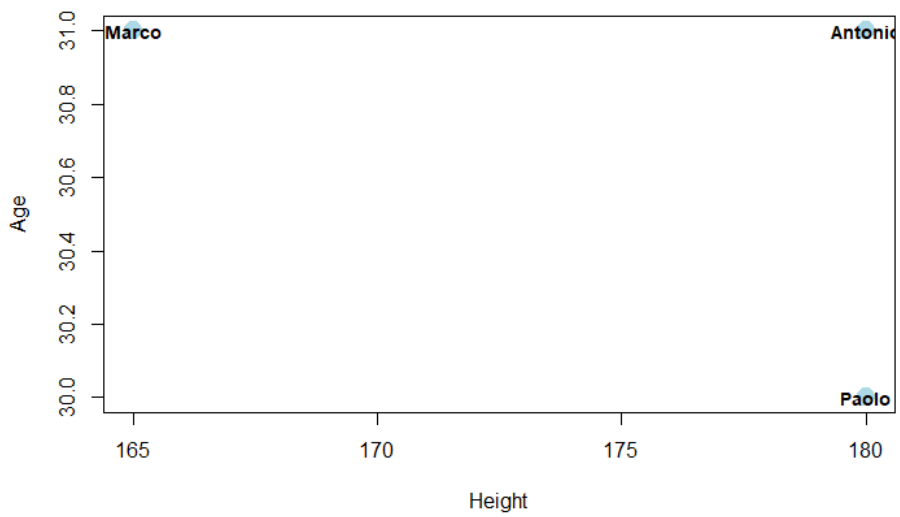


Figure 12.14 – Chart of the x matrix

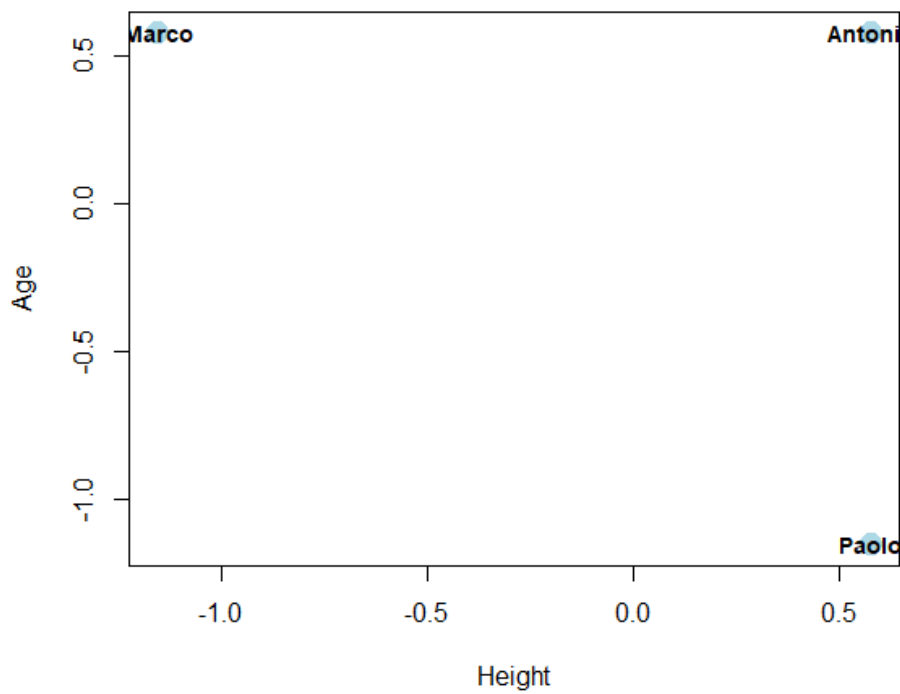


Figure 12.15 – Chart of the scaled variables



Figure 12.16 - Resubstitution Validation



Figure 12.17 - Hold-out Validation

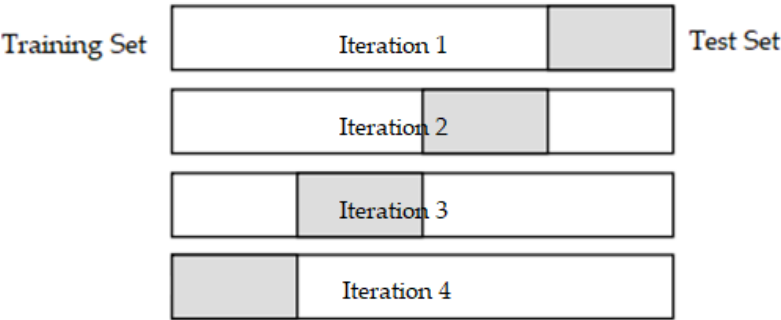


Figure 12.18 - K-fold Cross Validation

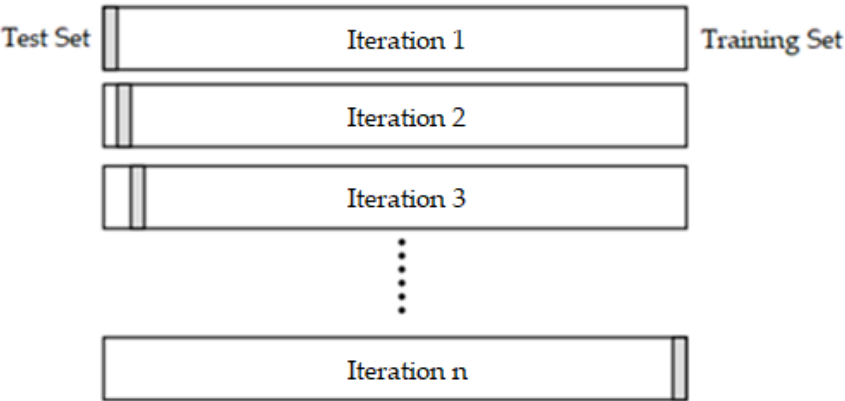


Figure 12.19 - Leave One Out Cross Validation

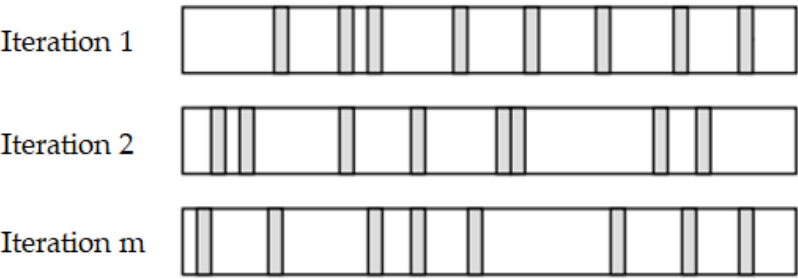


Figure 12.20 - Random Subsampling Validation

13. REGRESSION ALGORITHMS

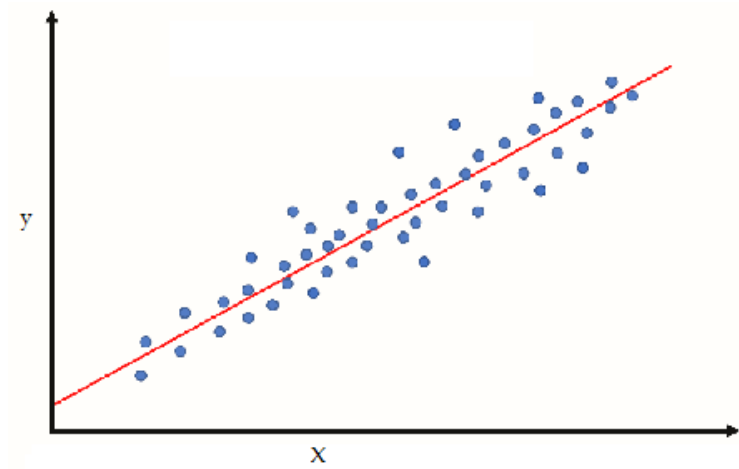


Figure 13.1 – Linear regression example

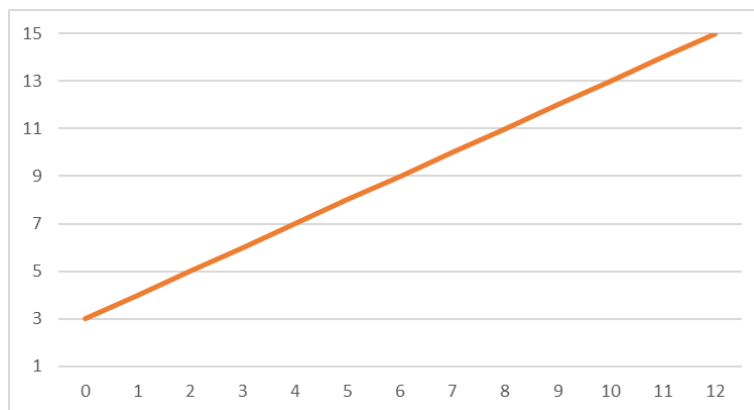


Figure 13.2 - Chart of a newborn's weight per month

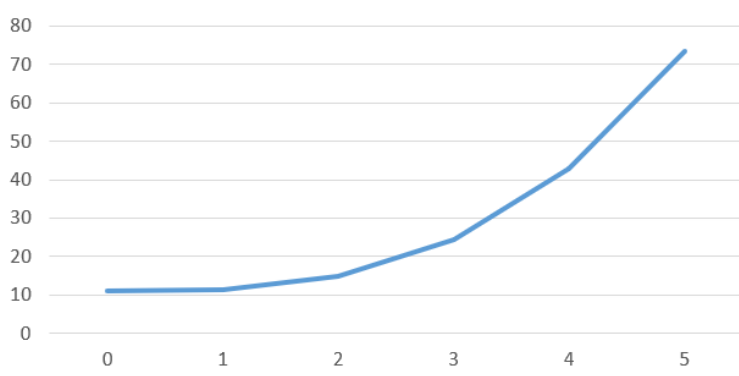


Figure 13.3 - Example of non-linear relationship

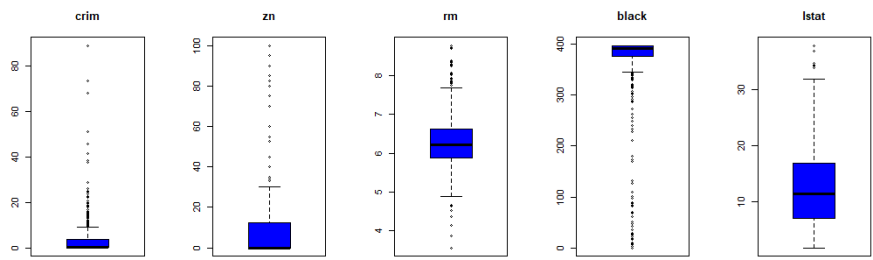


Figure 13.4 - Boxplot of some variables

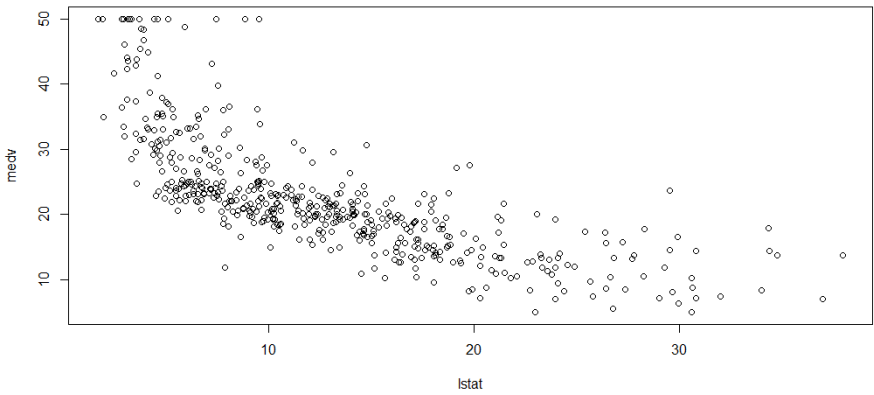


Figure 13.5 – Relationship between lstat and medv

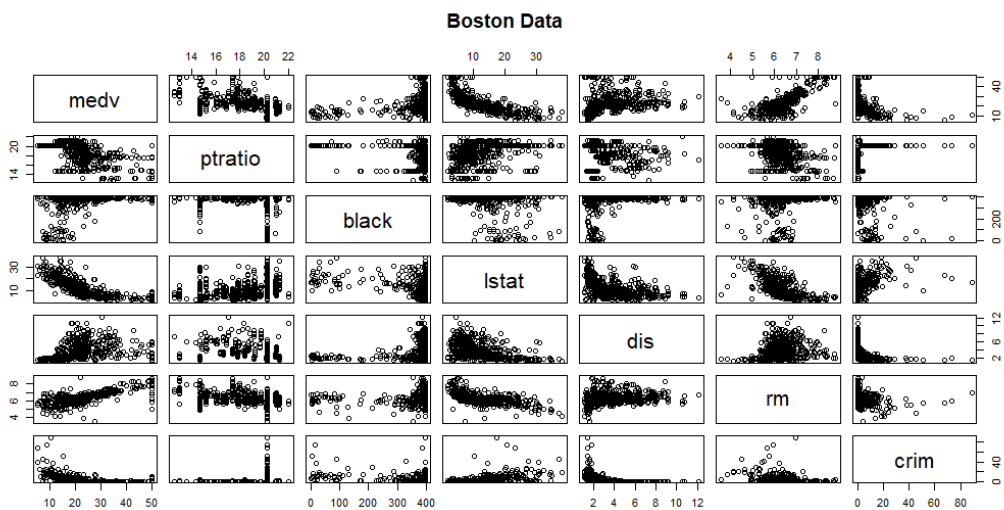


Figure 13.6 – Relationship between all variables

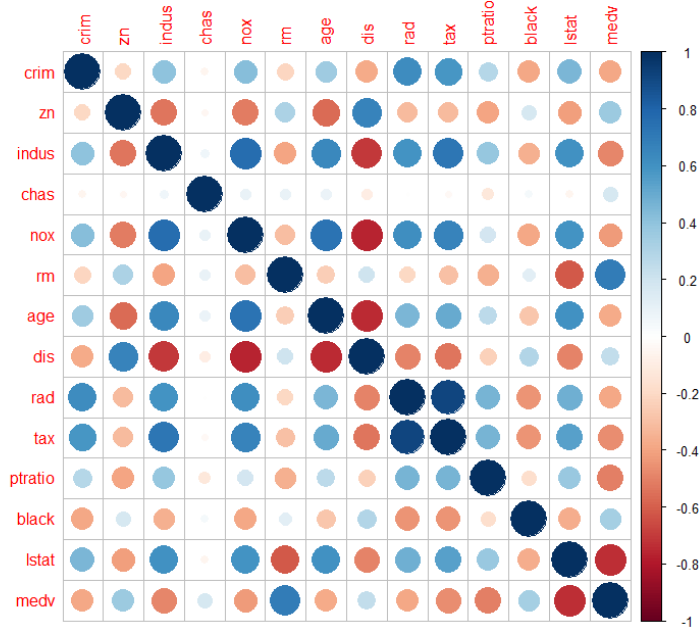


Figure 13.7 – Correlations chart

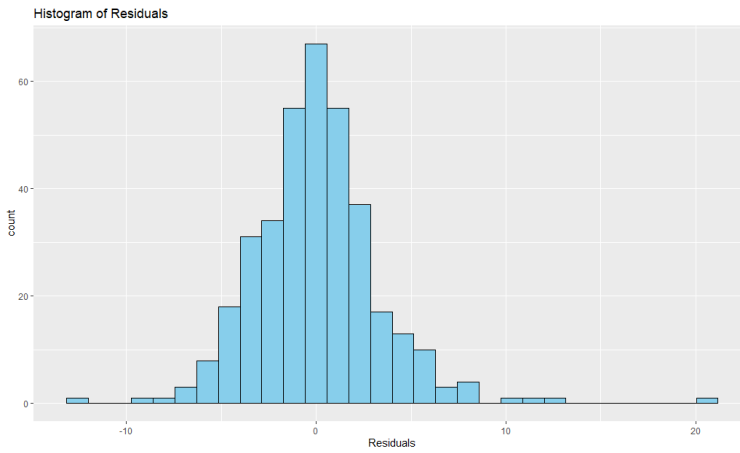


Figure 13.8 – Residuals distribution

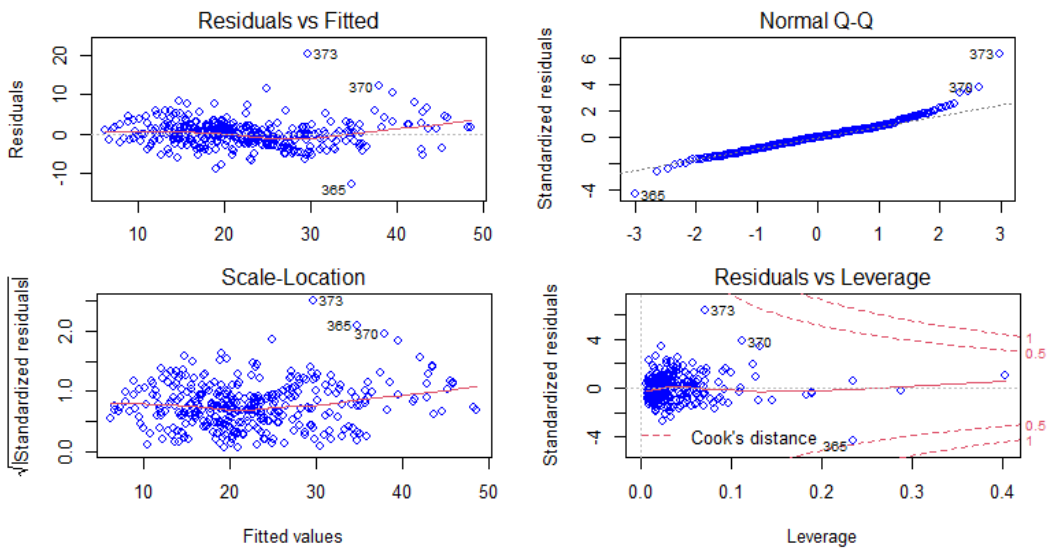


Figure 13.9 – Residuals plot

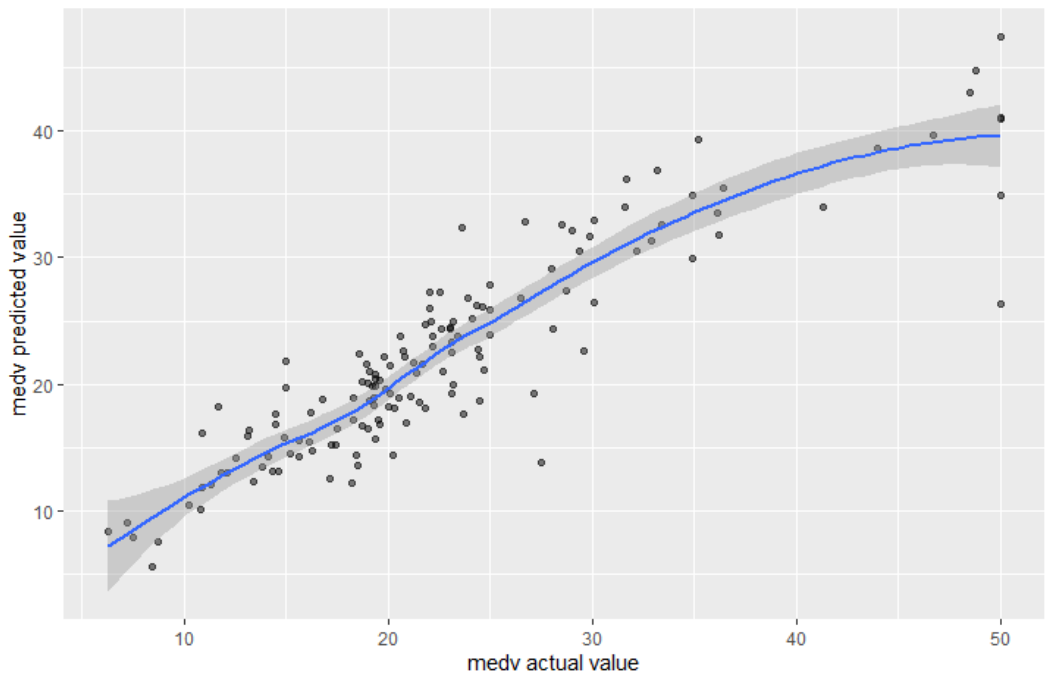


Figure 13.10 - Comparison between the predicted and the actual variable

	AtBat	Hits	HmRun	Runs	RBI	Walks	Years	CatBat	CHits	ChmRun	CRuns	CRBI	CWalks	League	Division	PutOuts	Assists	Errors	Salary	NewLeague
-Andy Atkinson	293	66	1	30	29	14	1	293	66	1	30	29	14	A	E	446	33	20	N/A	A
-Alan Ashby	315	81	7	24	38	39	14	3449	835	69	321	414	375	N	W	632	43	10	475,000	N
-Alvin Davis	479	130	18	66	72	76	3	1624	457	63	224	266	263	A	W	880	82	14	480,000	A
-Andre Dawson	496	141	20	65	78	37	11	5628	1575	225	828	838	354	N	E	200	11	3	500,000	N
-Andres Galaraga	321	87	10	39	42	30	2	396	101	12	48	46	33	N	E	805	40	4	91,500	N
-Alfredo Griffin	594	169	4	74	51	35	11	4408	1133	19	501	336	194	A	W	282	421	25	750,000	A
-Al Newman	185	37	1	23	8	21	2	214	42	1	30	9	24	N	E	76	127	7	70,000	A
-Argenis Salazar	298	73	0	24	24	7	3	509	108	0	41	37	12	A	W	121	283	9	100,000	A
-Andres Thomas	323	81	6	26	32	8	2	341	86	6	32	34	8	N	W	143	290	19	75,000	N
-Andre Thornton	401	92	17	49	66	65	13	5206	1332	253	784	890	866	A	E	0	0	0	1100,000	A
-Alan Trammell	574	159	21	107	75	59	10	4631	1300	90	702	504	488	A	E	238	445	22	517,143	A
-Alex Trevino	202	53	4	31	26	27	9	1876	467	15	192	186	161	N	W	304	45	11	512,500	N
-Andy VanSlyke	418	113	13	48	61	47	4	1512	392	41	205	204	203	N	E	211	11	7	550,000	N
-Alan Wiggins	239	60	0	30	11	22	6	1941	510	4	309	103	207	A	E	121	151	6	700,000	A
-Bill Almon	196	43	7	29	27	30	13	3231	825	36	376	290	238	N	E	80	45	8	240,000	N
-Billy Beane	183	39	3	20	15	11	3	201	42	3	20	16	11	A	W	118	0	0	N/A	A
-Buddy Bell	568	158	20	89	75	73	15	8068	2273	177	1045	993	732	N	W	105	290	10	775,000	N

Figure 13.11 – Hitters dataset

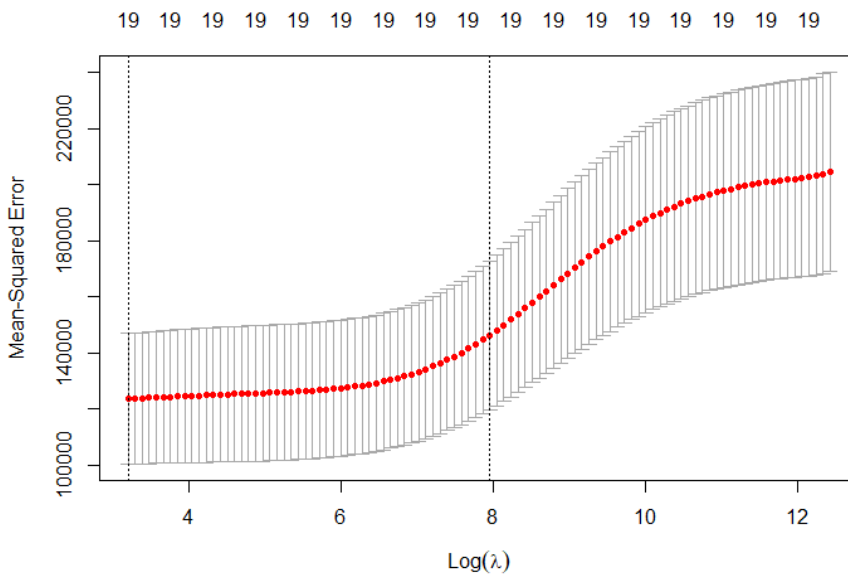


Figure 13.12 - MSE by lambda

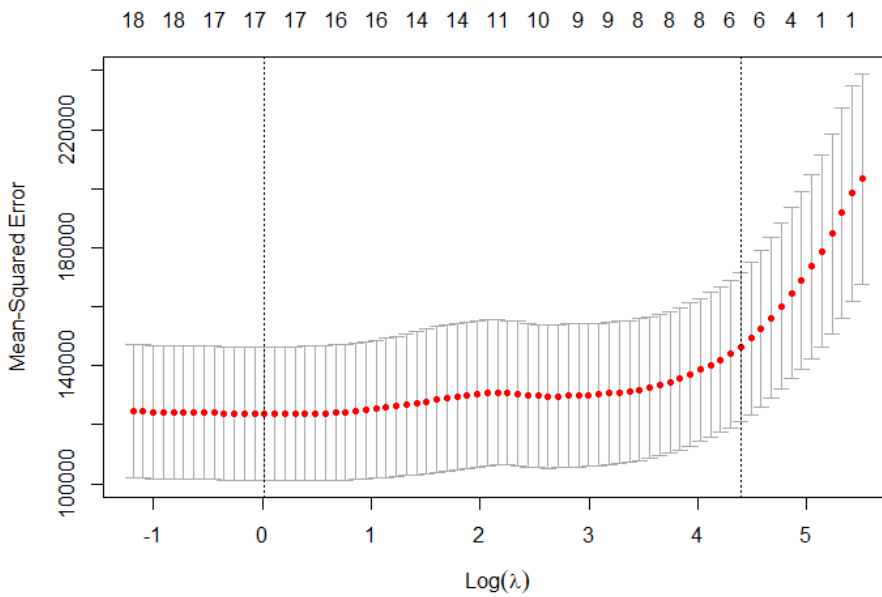


Figure 13.13 – MSE by lambda for LASSO

14. CLASSIFICATION ALGORITHMS

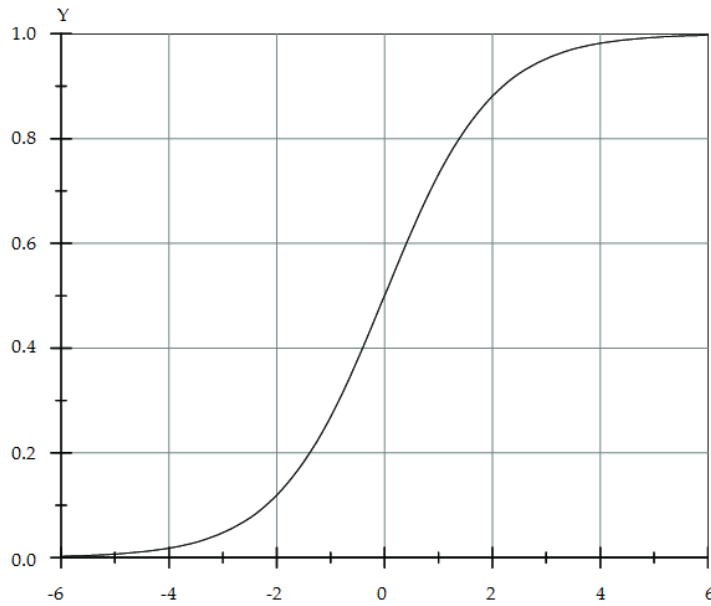


Figure 14.1 – Sigmoid function

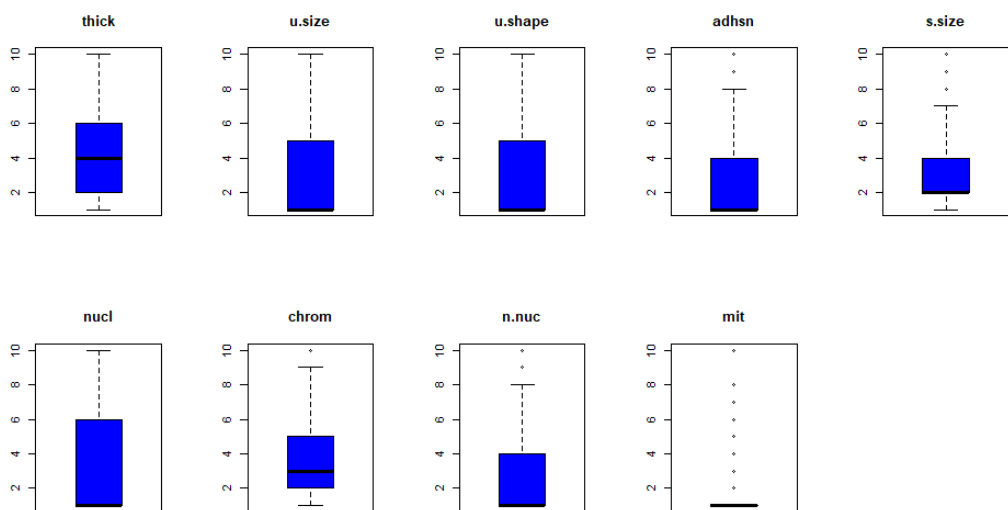


Figure 14.2 – Boxplots of the features

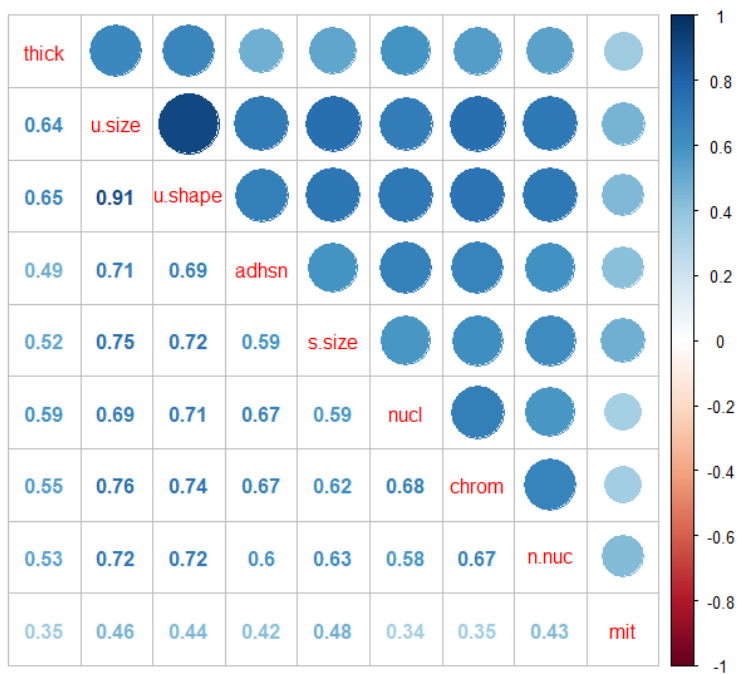


Figure 14.3 – Correlations between variables

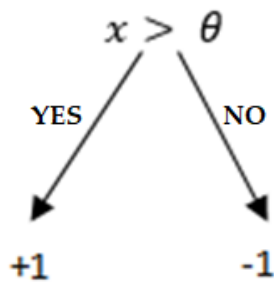


Figure 14.4 - A node of a classification tree

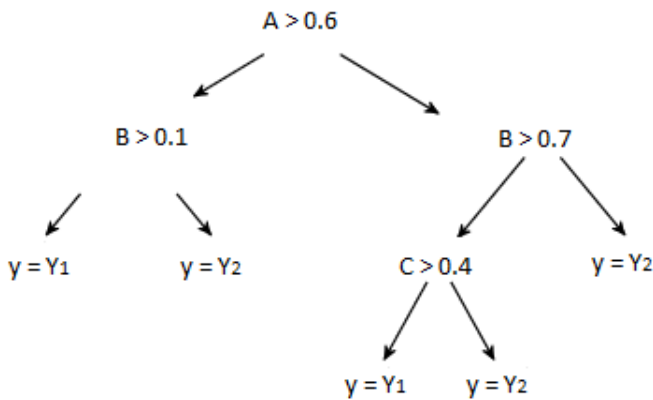


Figure 14.5 – Binary nodes

	X2	X30	X2.1	X1	X2.2	X2.3	X2.4	X2.5	X1.1	X2.6	X2.7	X2.8	X2.9	X2.10	X1.00	X85	X18	X4.0	X	X1.2
1	2	50	1	1	2	1	2	2	1	2	2	2	2	2	0.9	135	42	3.5	NA	1
2	2	78	1	2	2	1	2	2	2	2	2	2	2	2	0.7	96	32	4.0	NA	1
3	2	31	1	NA	1	2	2	2	2	2	2	2	2	2	0.7	46	52	4.0	80	1
4	2	34	1	2	2	2	2	2	2	2	2	2	2	2	1.0	NA	200	4.0	NA	1
5	2	34	1	2	2	2	2	2	2	2	2	2	2	2	0.9	95	28	4.0	75	1
6	1	51	1	1	2	1	2	1	2	2	1	1	2	2	NA	NA	NA	NA	NA	1
7	2	23	1	2	2	2	2	2	2	2	2	2	2	2	1.0	NA	NA	NA	NA	1
8	2	39	1	2	2	1	2	2	2	1	2	2	2	2	0.7	NA	48	4.4	NA	1
9	2	30	1	2	2	2	2	2	2	2	2	2	2	2	1.0	NA	120	3.9	NA	1
10	2	39	1	1	1	2	2	2	1	1	2	2	2	2	1.3	78	30	4.4	85	1
11	2	32	1	2	1	1	2	2	2	1	2	1	2	2	1.0	59	249	3.7	54	1
12	2	41	1	2	1	1	2	2	2	1	2	2	2	2	0.9	61	60	3.9	52	1
13	2	30	1	2	2	1	2	2	2	1	2	2	2	2	2.2	57	144	4.9	78	1
14	2	47	1	1	1	2	2	2	2	2	2	2	2	2	NA	NA	50	NA	NA	1

Figure 14.6 – Hepatitis dataset

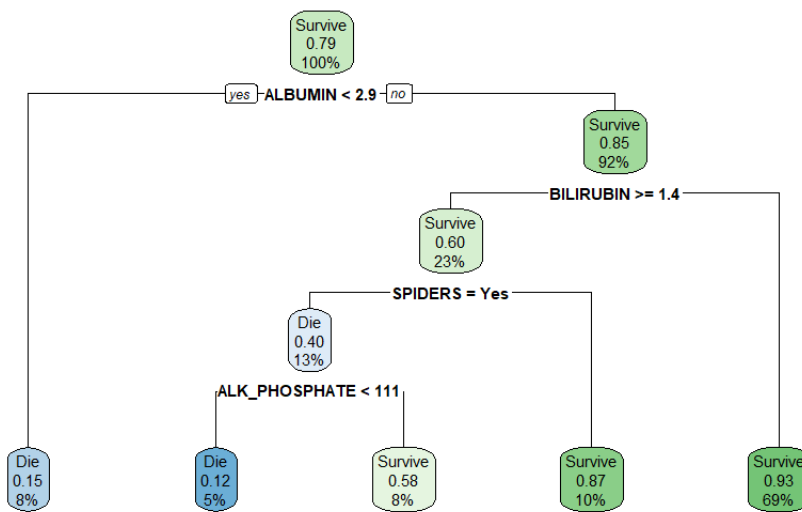


Figure 14.7 – The hepatitis dataset tree

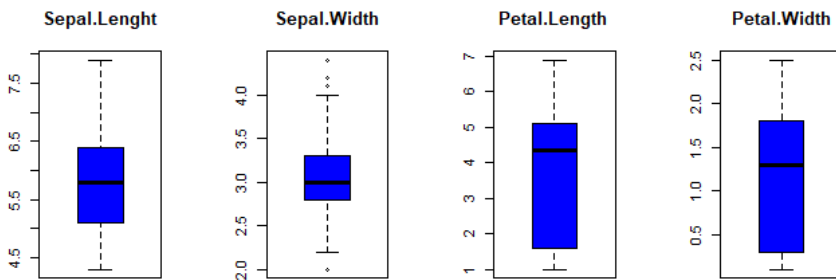


Figure 14.8 – Features boxplots

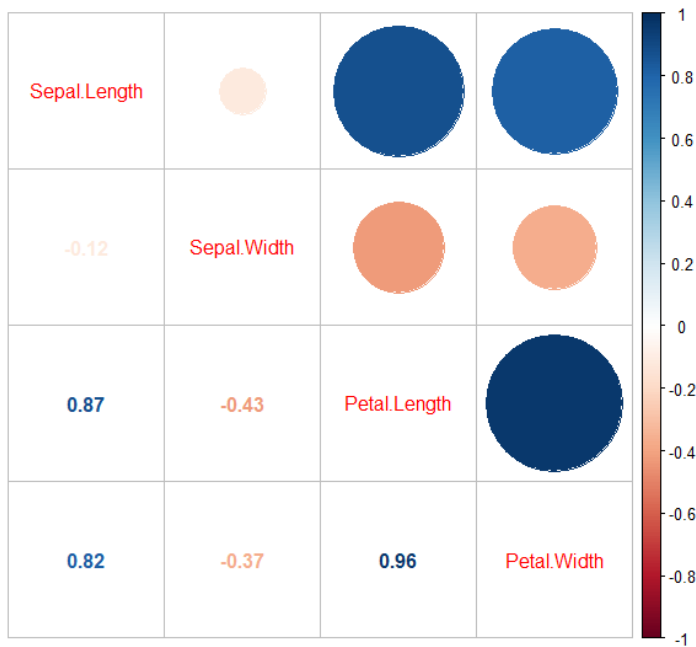


Figure 14.9 - Features corrplot

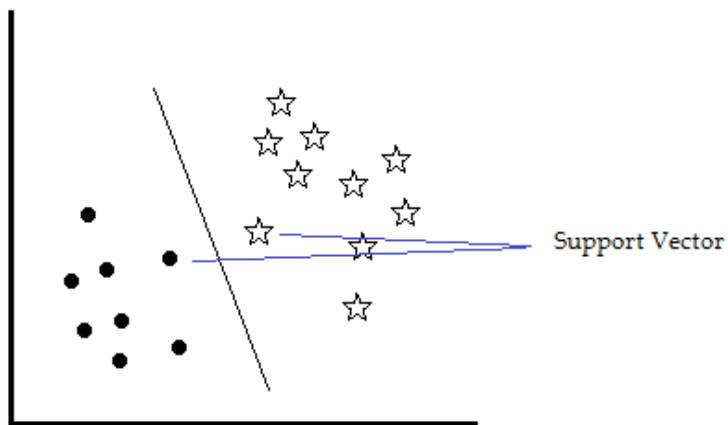


Figure 14.10 - Support Vectors

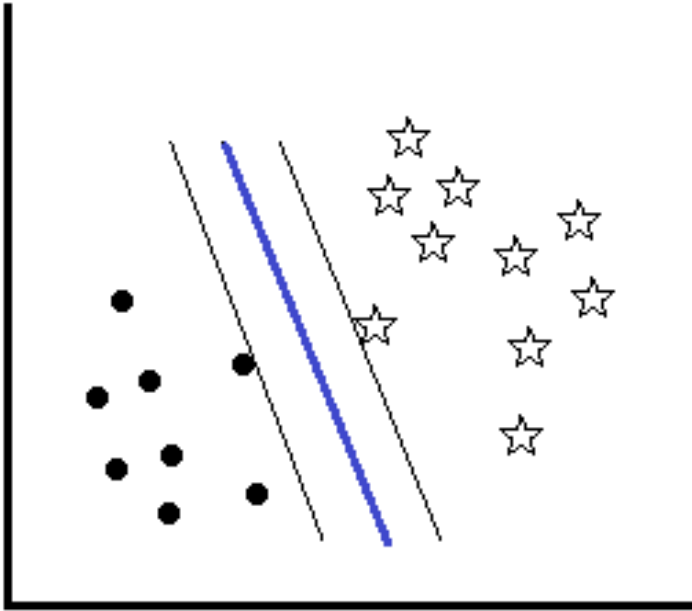


Figure 14.11 – Maximizing distance between hyperplane and support vectors

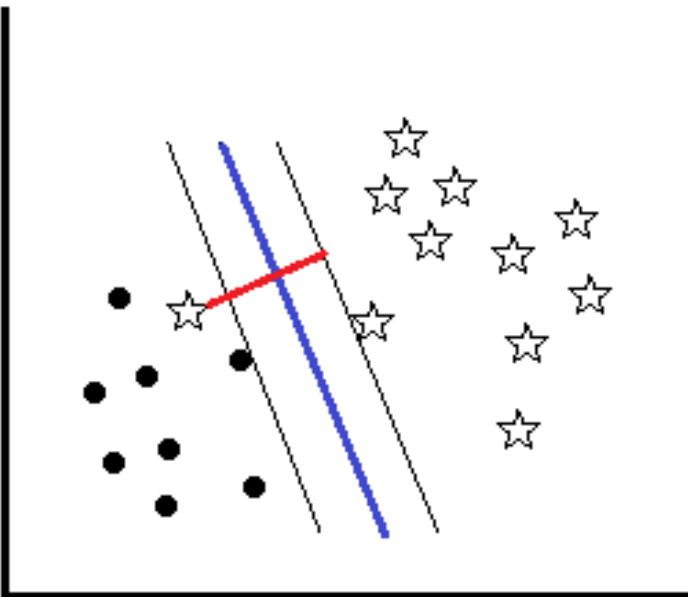


Figure 14.12 – Nonlinearly separable classes

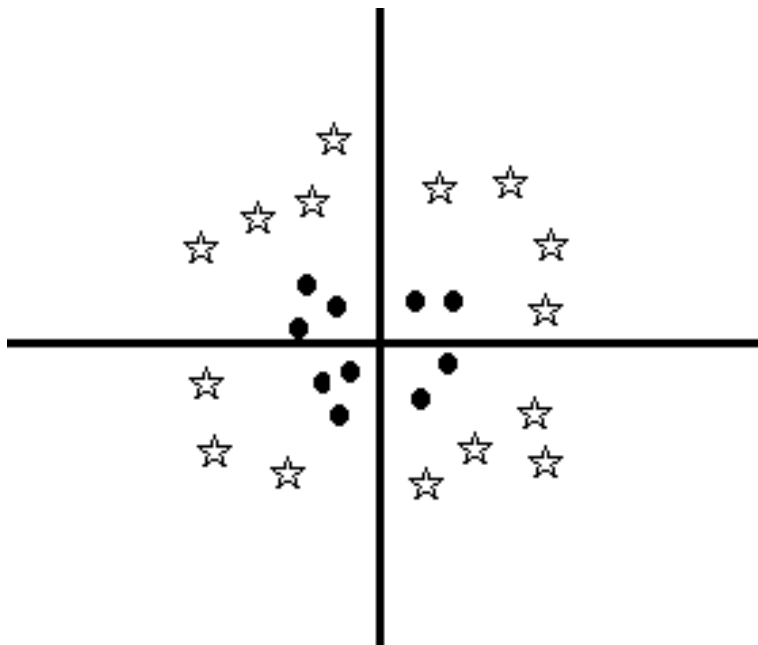


Figure 14.13 – Nonlinearly separable class

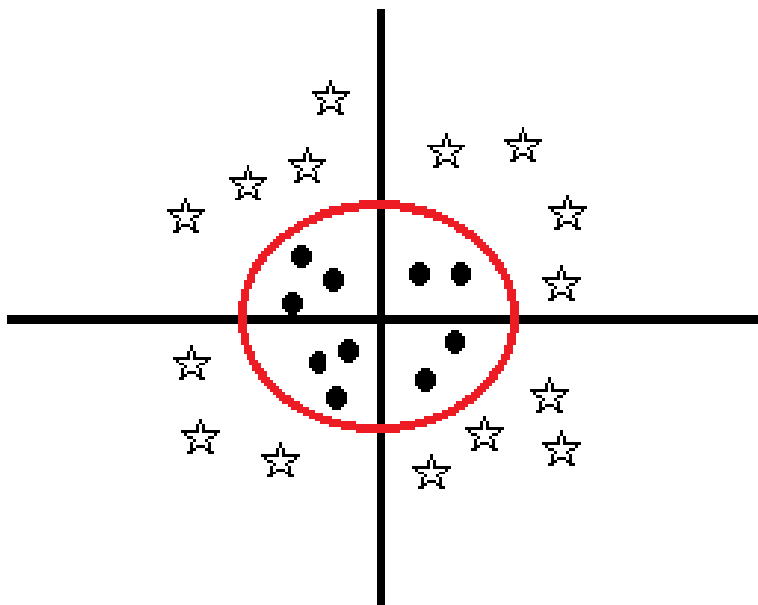


Figure 14.14 – Separation with a higher dimension model

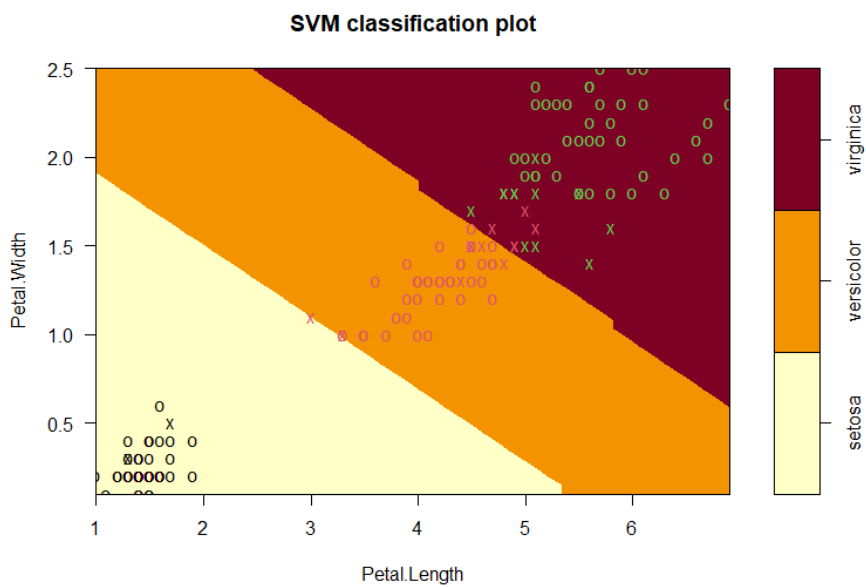


Figure 14.15 - Chart of the first SVM model

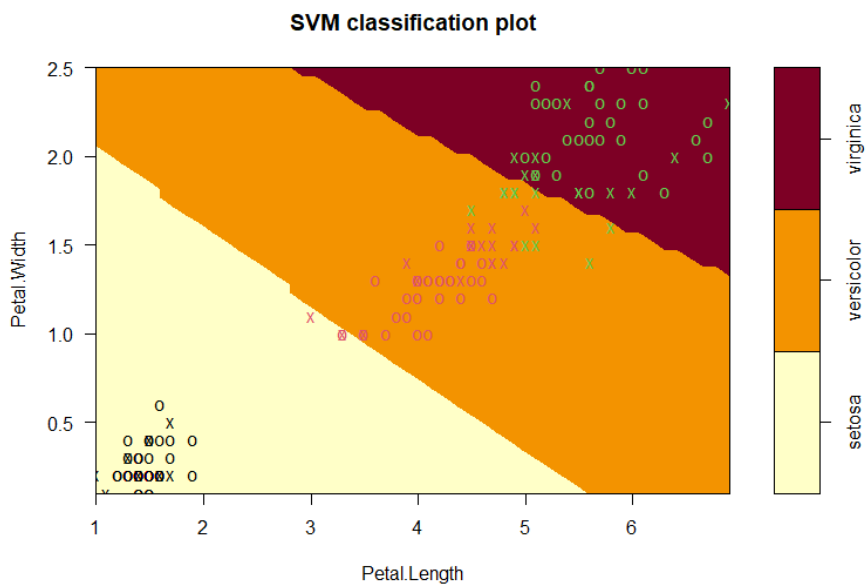


Figure 14.16 – Second SVM model chart

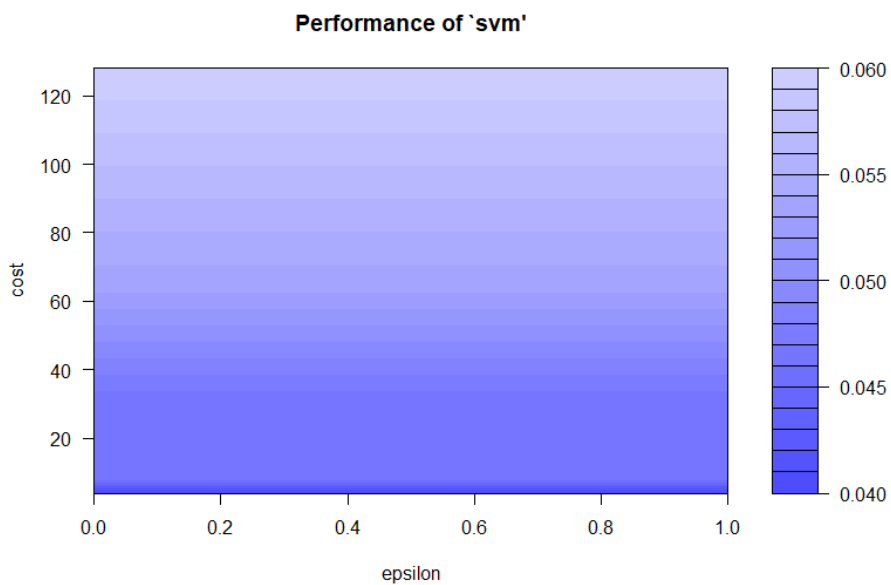


Figure 14.17 – SVM performance

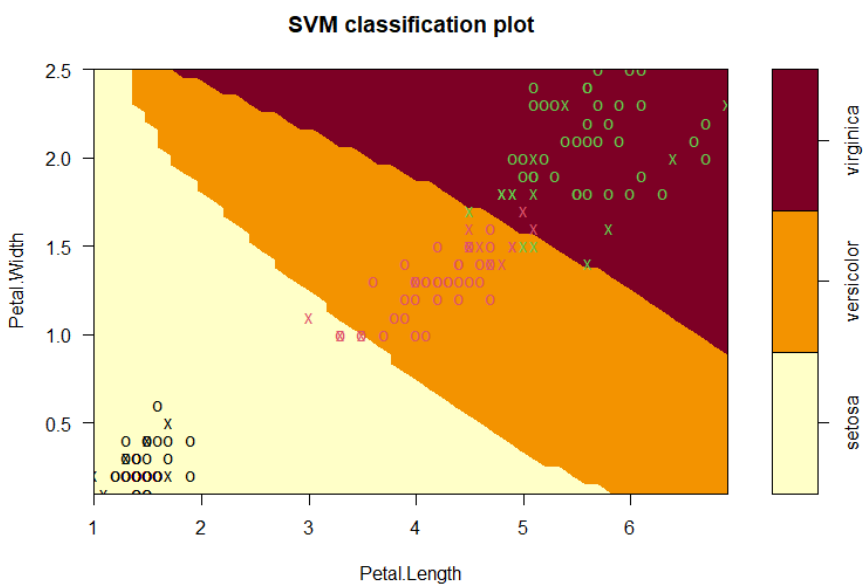


Figure 14.18 – Best SVM model

15. CLUSTERING ALGORITHMS



Figure 15.1- Clustering example

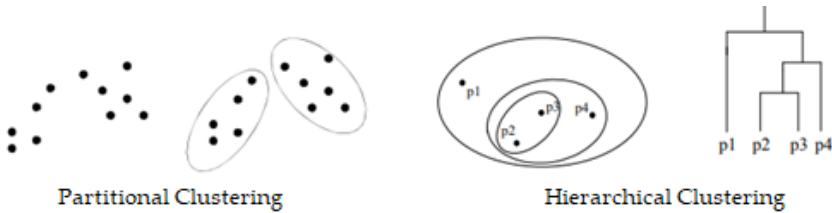


Figure 15.2 – Clustering Algorithms

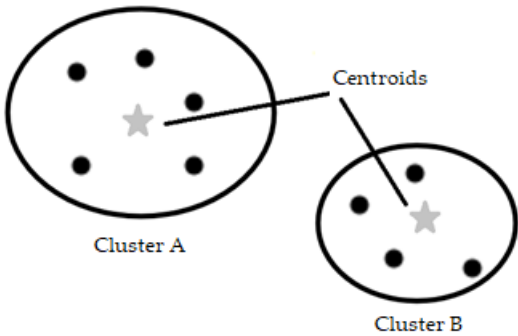


Figure 15.3 - Clusters and their centroids

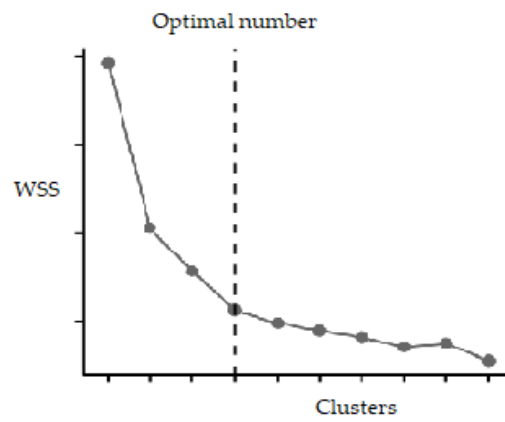


Figure 15.4 – The Elbow method

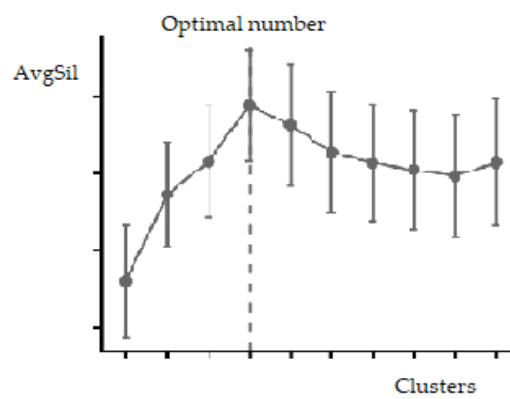


Figure 15.5 – Silhouette method

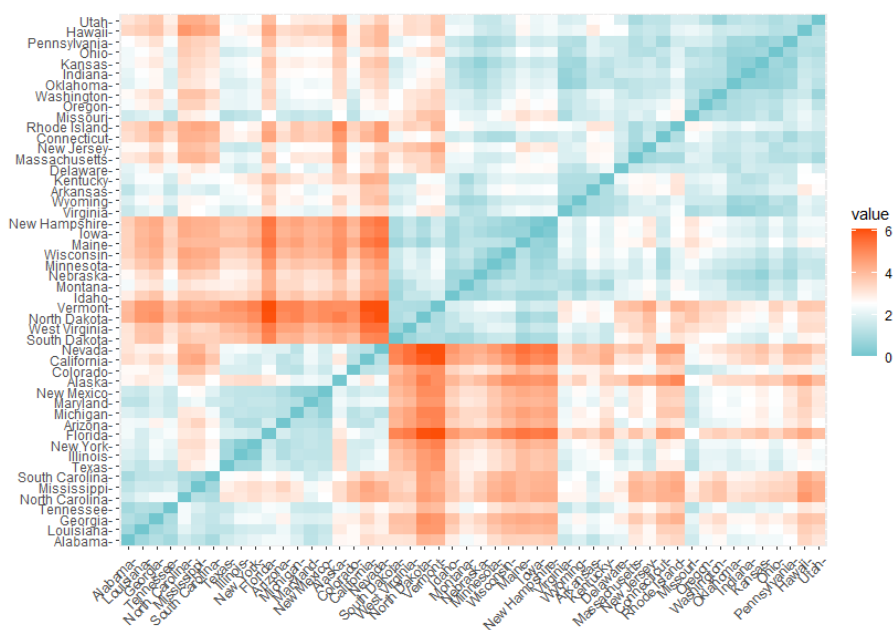


Figure 15.6 – Distance matrix

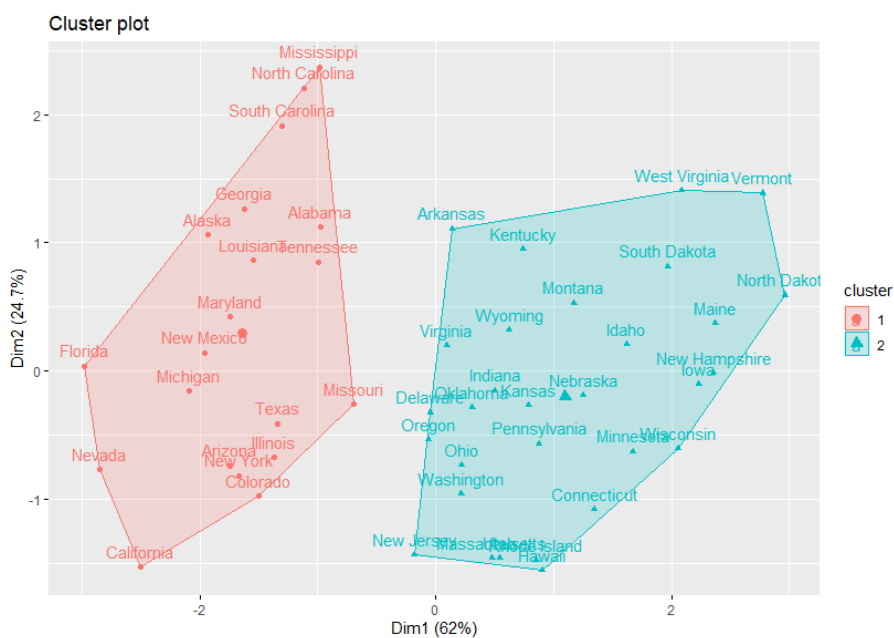


Figure 15.7 - Clustering with $k = 2$

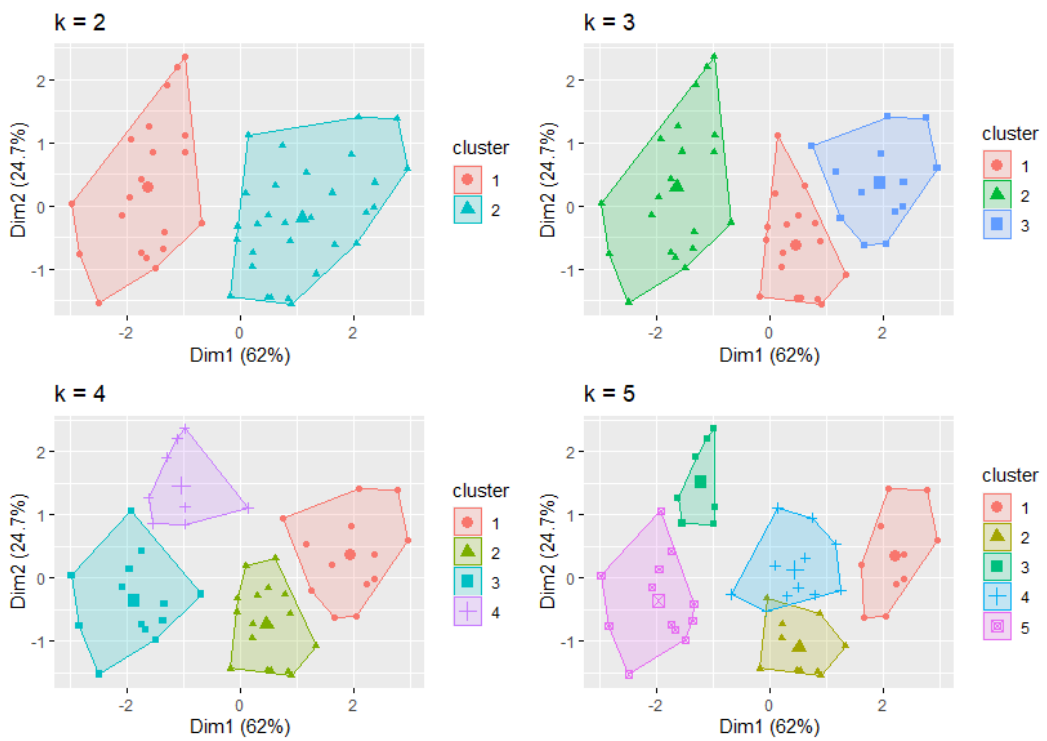


Figure 15.8 - K-means for different k

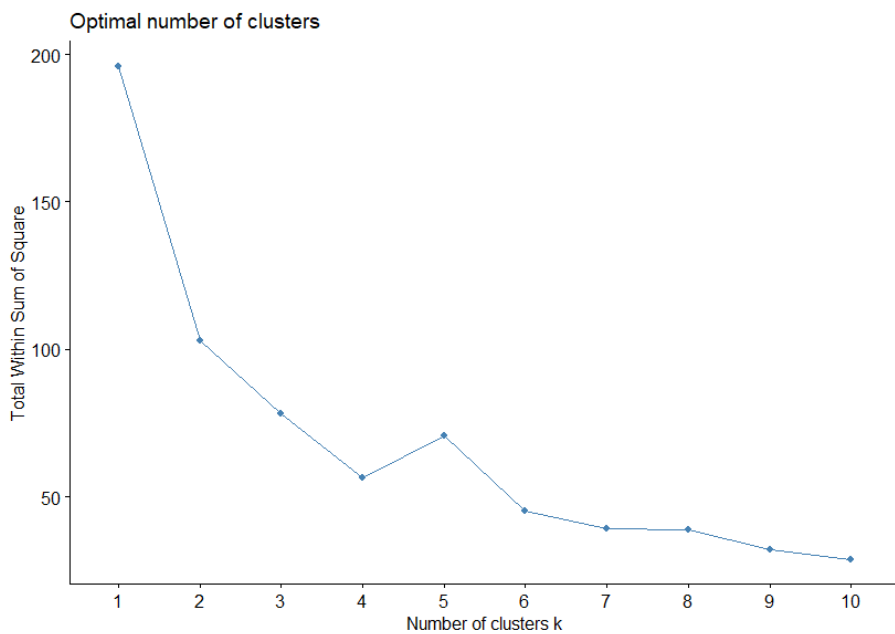


Figure 15.9 – Elbow method

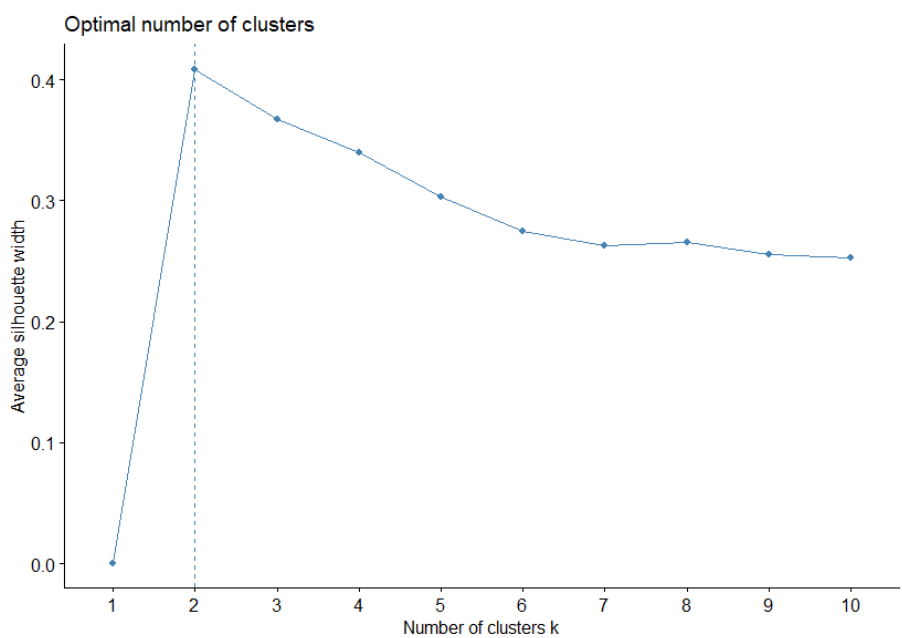


Figure 15.10 – Silhouette method

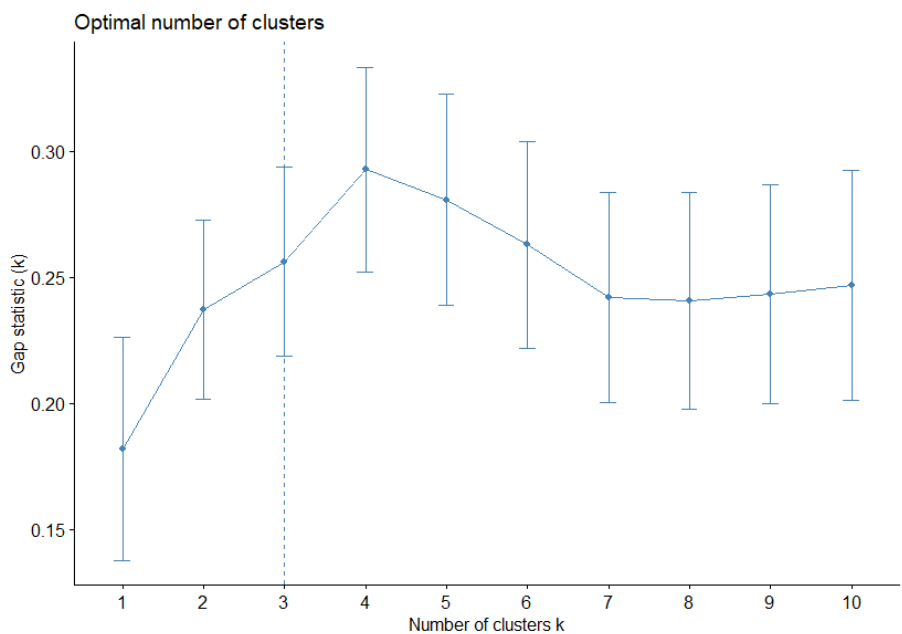


Figure 15.11 - Gap statistics method

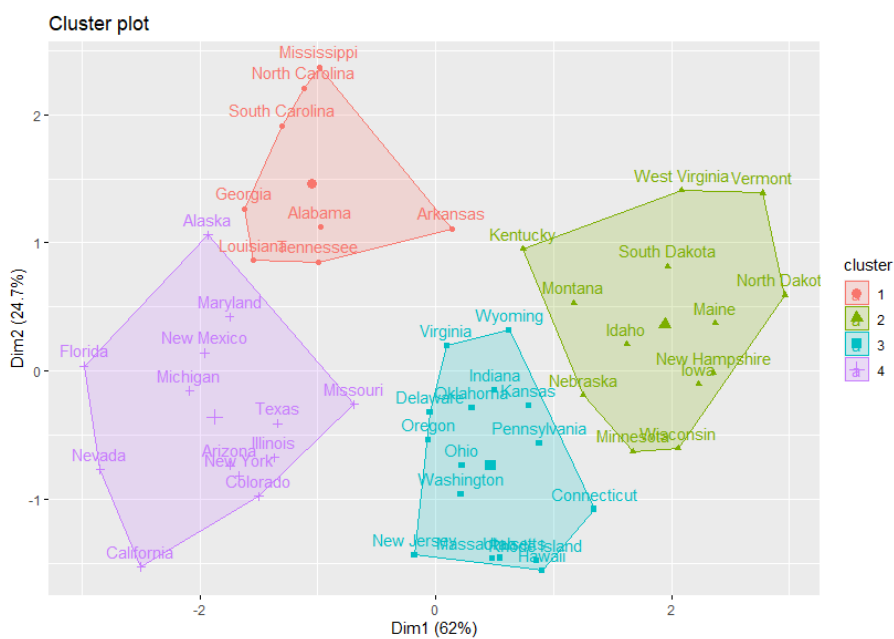


Figure 15.12 - Clustering with $k = 4$

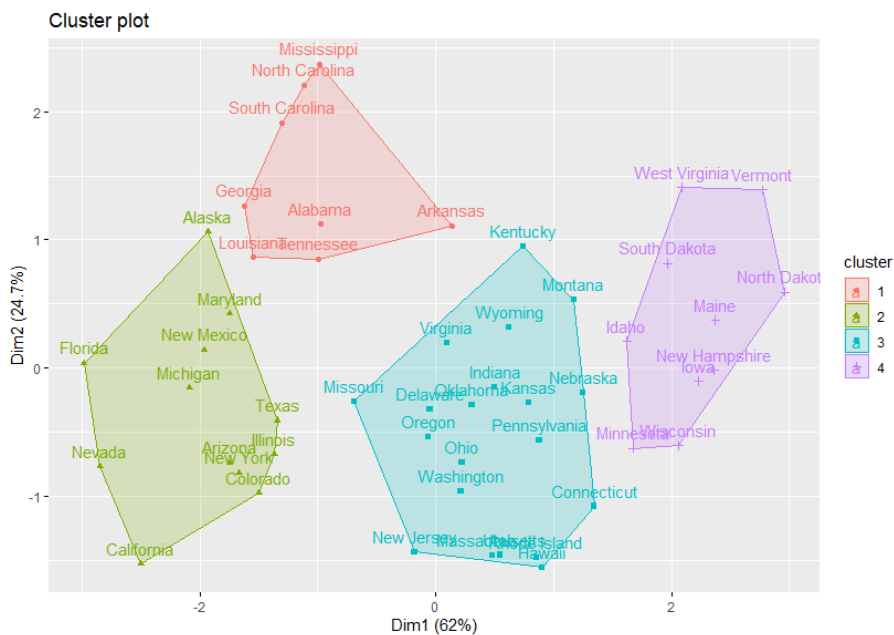


Figure 15.13 – PAM Clustering

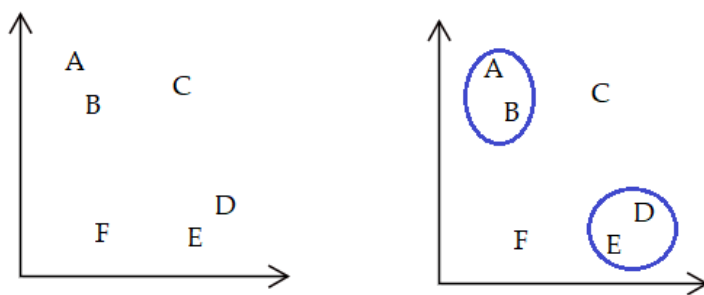


Figure 15.14 – Initial configuration and first selection



Figure 15.15 – First iteration

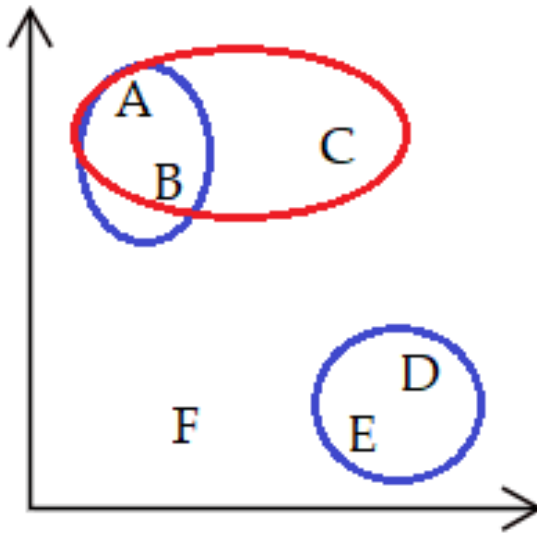


Figure 15.16 - The cluster C is the closest to the cluster (A,B)

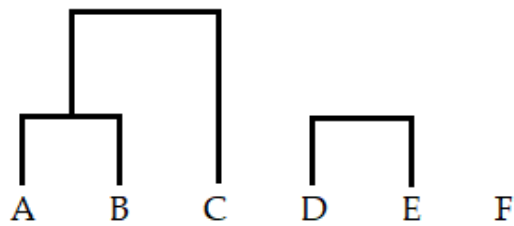


Figure 15.17 - Next iteration

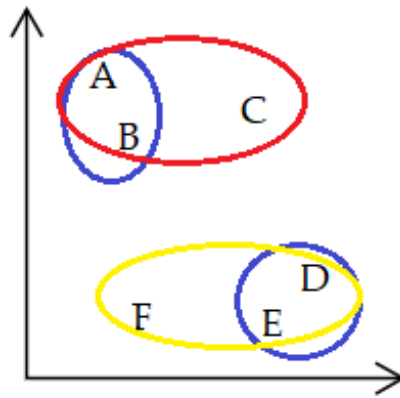


Figure 15.18 – The F cluster is the closest to the cluster (D,E)

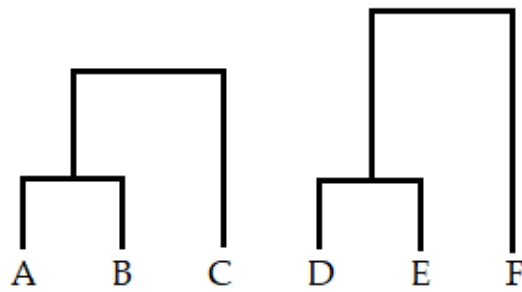


Figure 15.19 – Next iteration

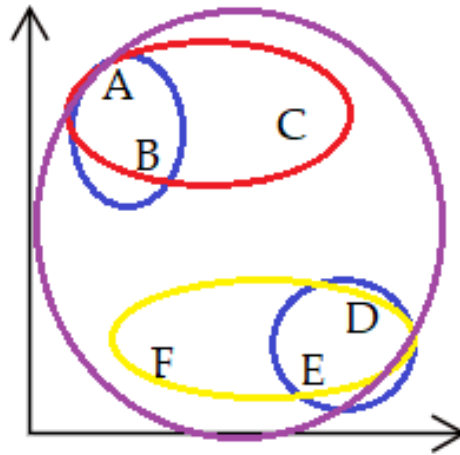


Figure 15.20 – The final cluster

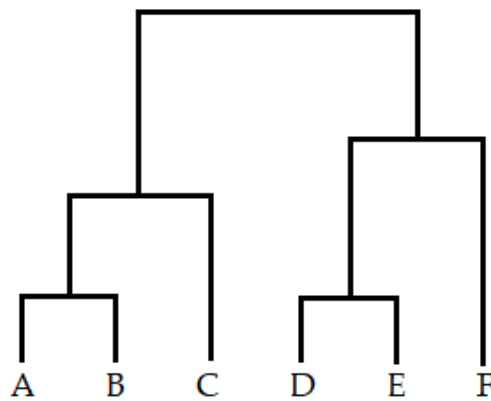


Figure 15.21 – Hierarchical Clustering

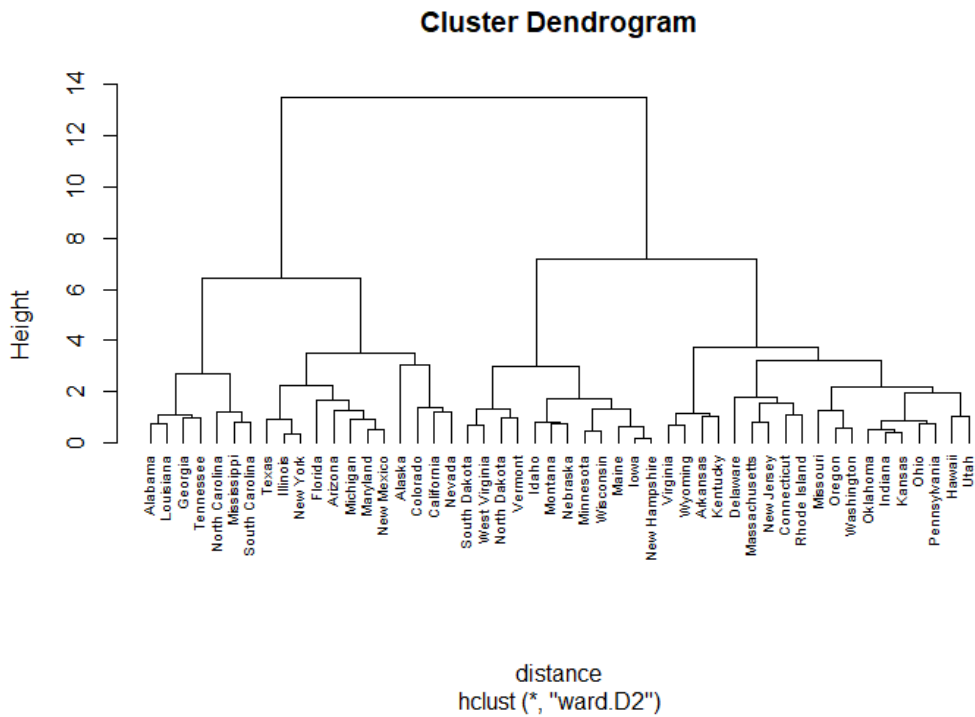


Figure 15.22 - Dendrogram of hierarchical clustering

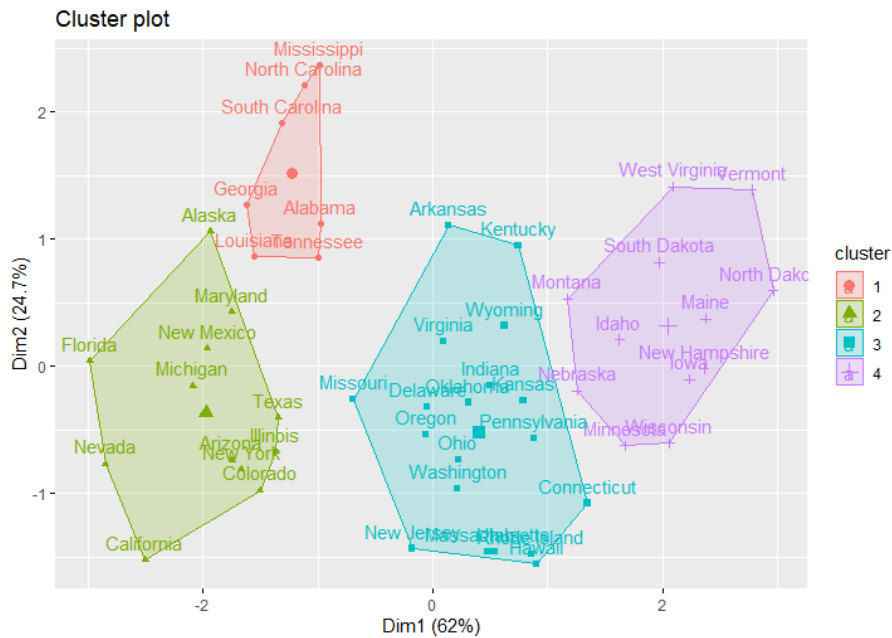


Figure 15.23 – Identified clusters

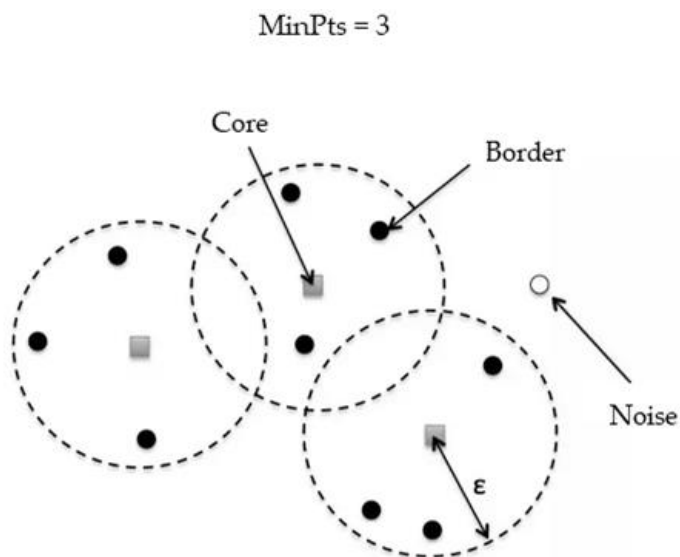


Figure 15.24 – Points classification in DBSCAN

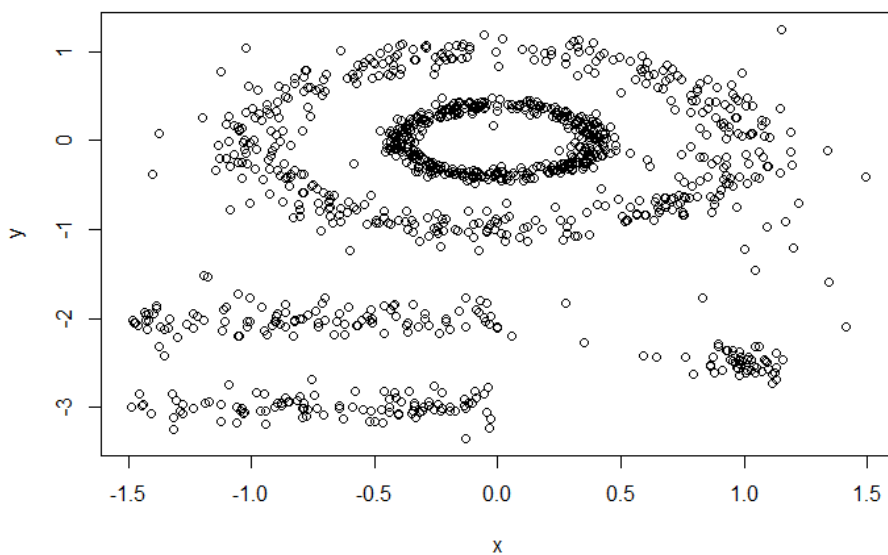


Figure 15.25 – Multishapes data

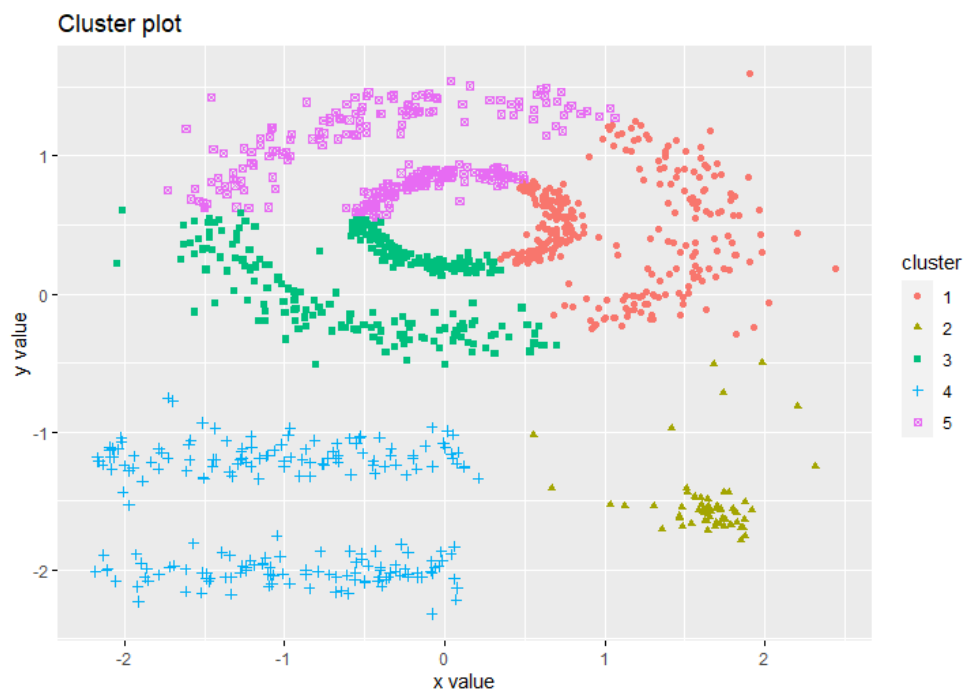


Figure 15.26 - *k*-means clustering results

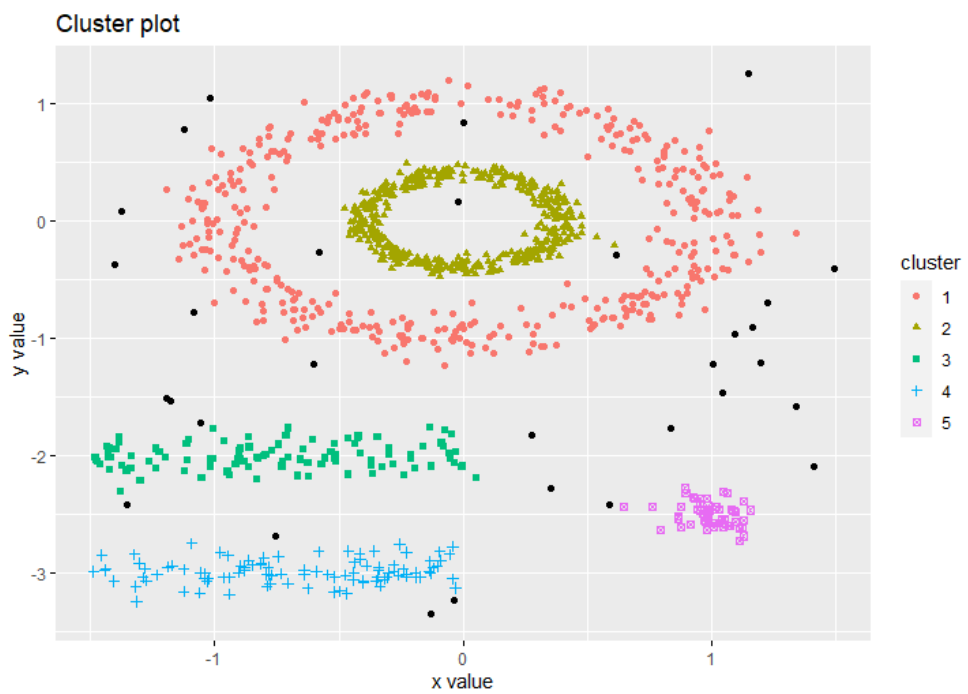


Figure 15.27 - Clustering with DBScan

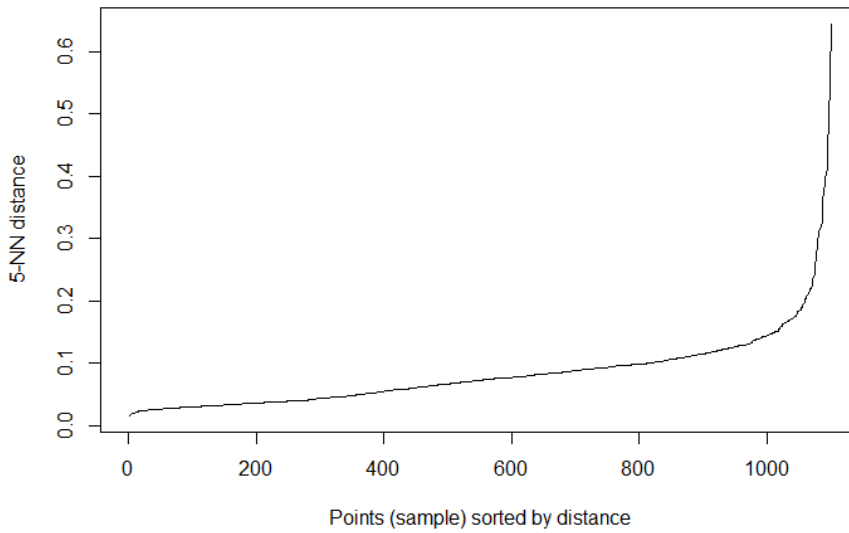


Figure 15.28 – Chart of average distance for epsilon

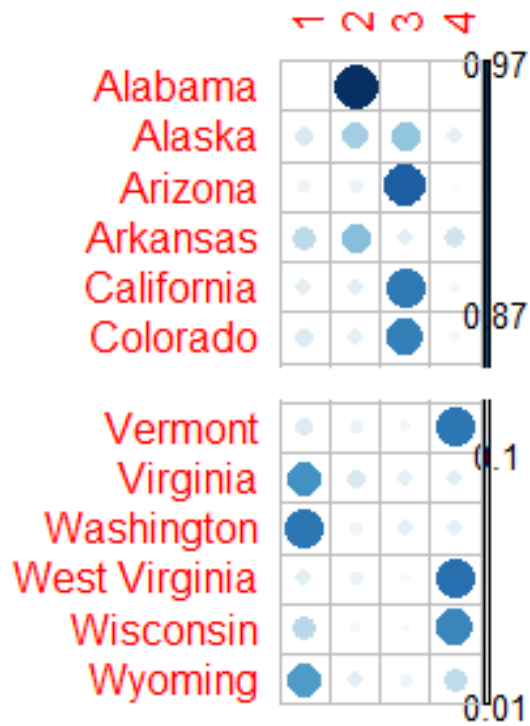


Figure 15.29 – Membership chart

16. DIMENSIONALITY REDUCTION

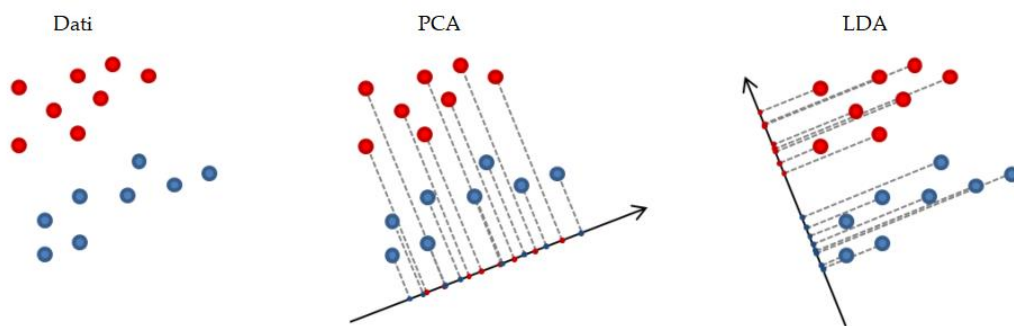


Figure 16.1 - PCA vs LDA

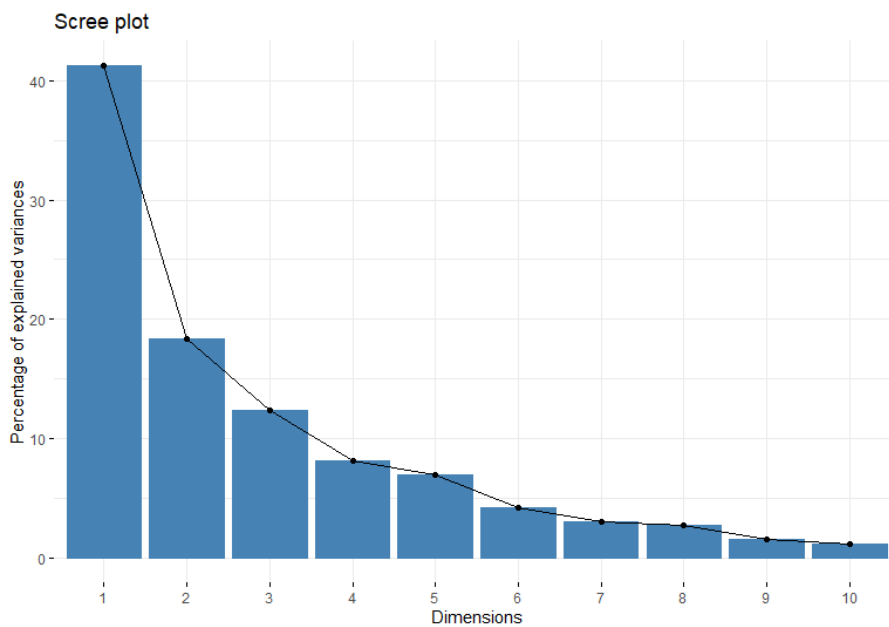


Figure 16.2 - Chart of variance explained by component

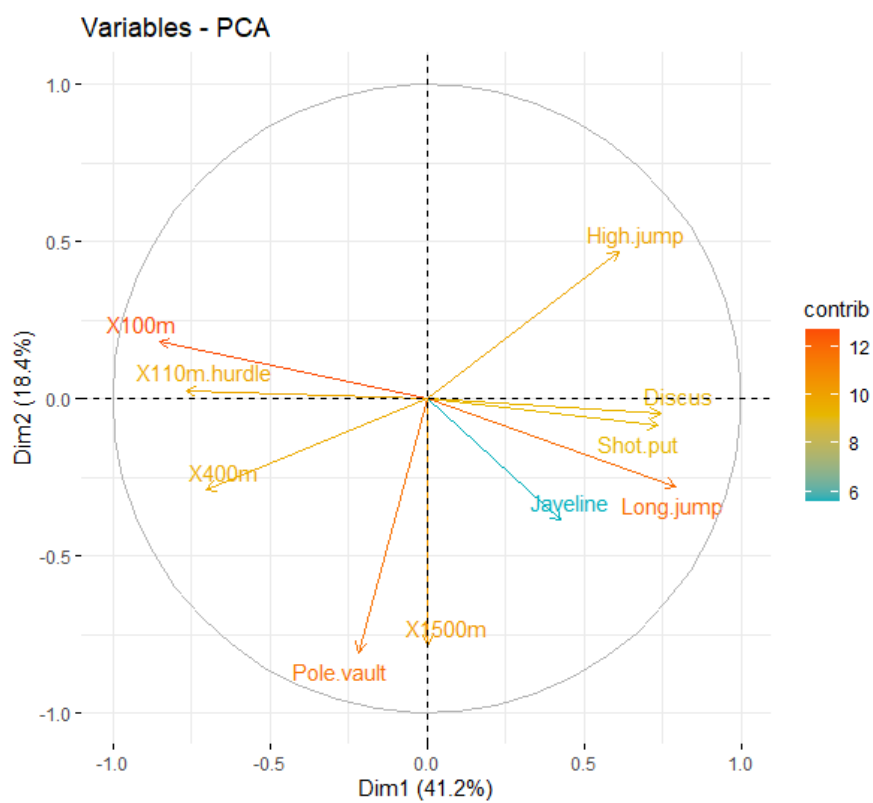


Figure 16.3 – Correlations chart

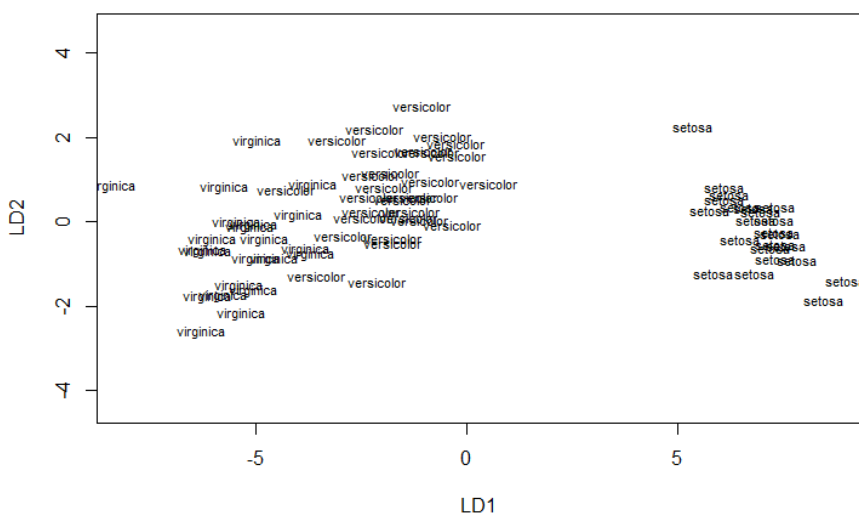


Figure 16.4 – Classification with LDA

17. SEMI-SUPERVISED ALGORITHMS

Feature 1	Feature 2	Feature 3	Feature 4	Class	
6	27	47	9	YES	Supervised
10	43	26	13	YES	
24	17	42	43	YES	
35	41	21	1	YES	
45	27	19	25	-	Unsupervised
46	5	22	36	-	
46	3	23	38	-	
42	33	4	41	-	
45	26	49	6	-	
17	14	35	47	-	
46	39	15	43	-	
22	33	6	28	-	
50	7	43	14	-	
32	8	2	37	-	
18	20	16	40	-	
7	40	1	47	-	
43	1	47	16	-	
1	10	50	9	-	
14	13	2	37	-	

Figure 17.1 – Example of Semi-Supervised problem

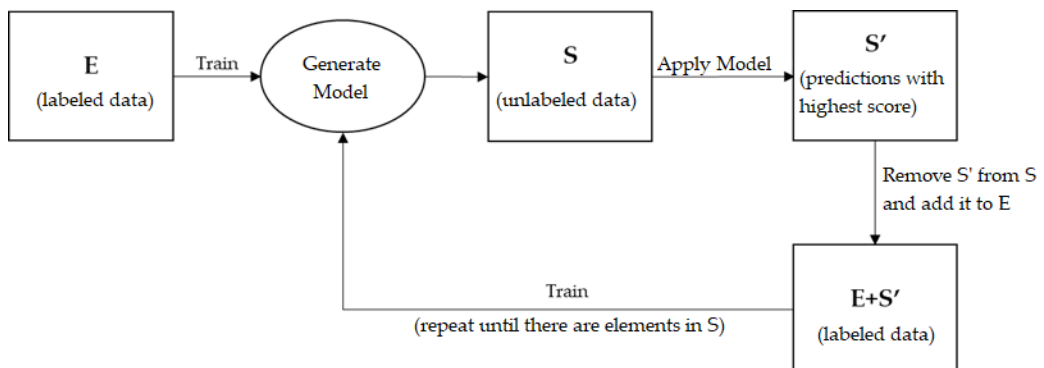


Figure 17.2 - Self training method

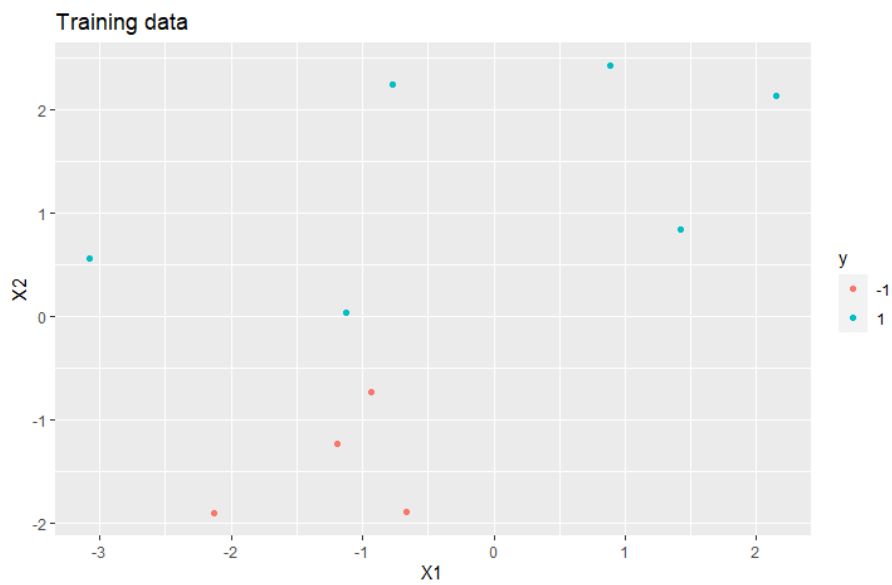


Figure 17.3 – Labeled training data

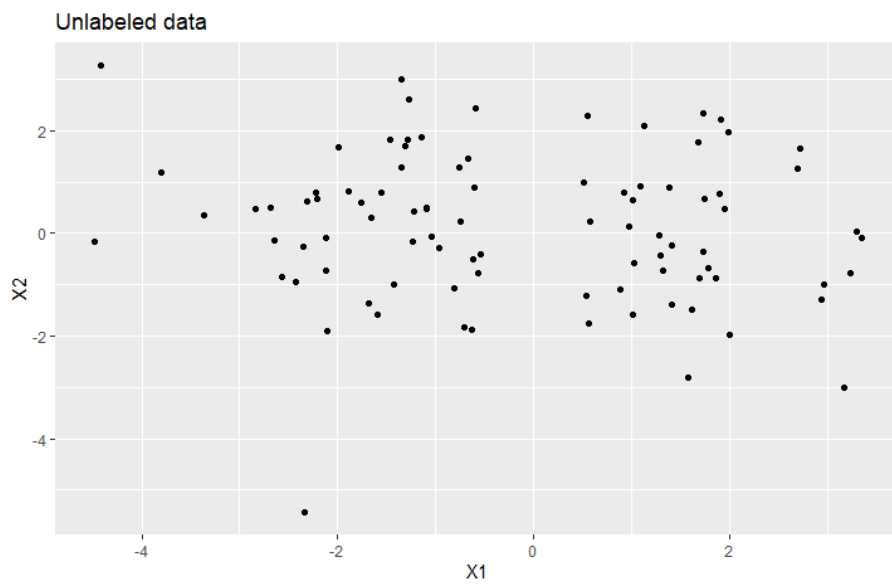


Figure 17.4 – Unlabeled data

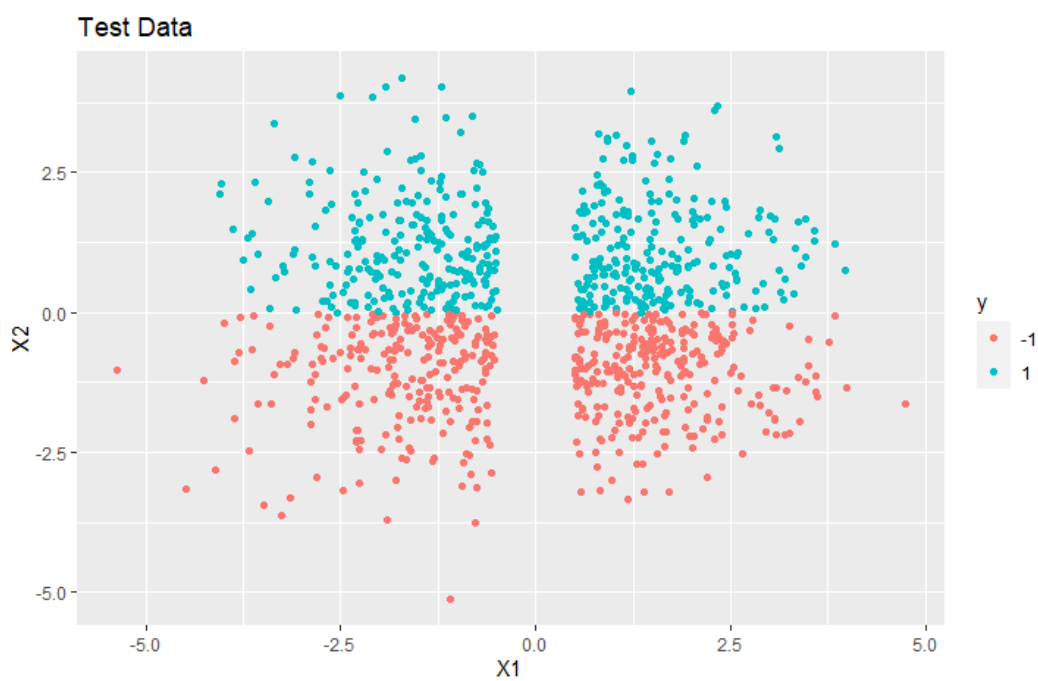


Figure 17.5 – Test data

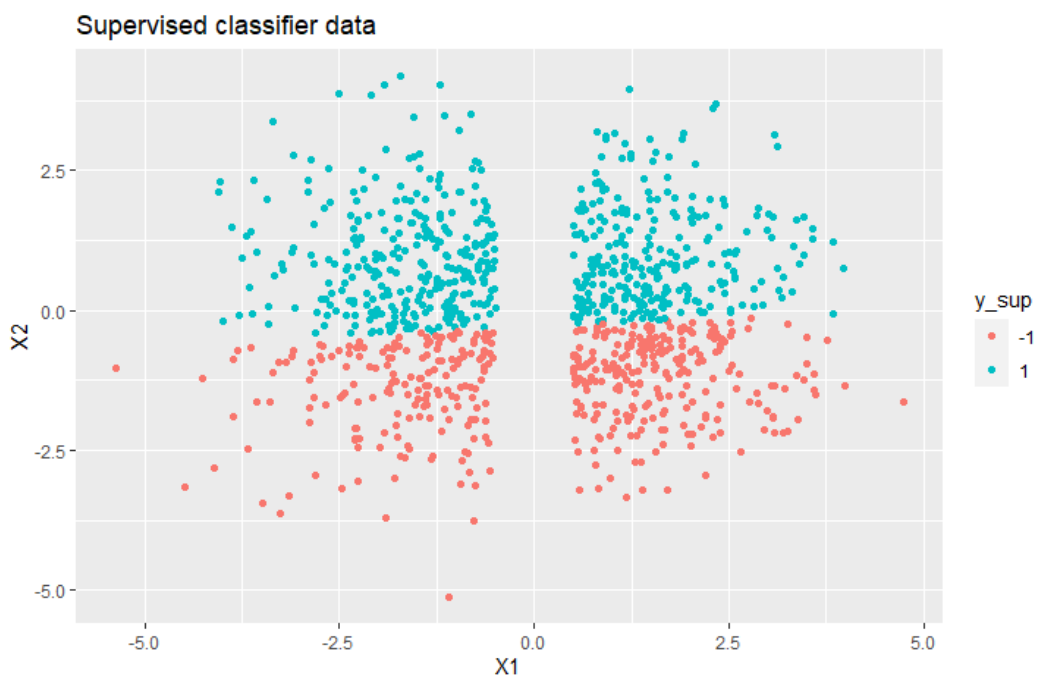


Figure 17.6 – Supervised classifier results

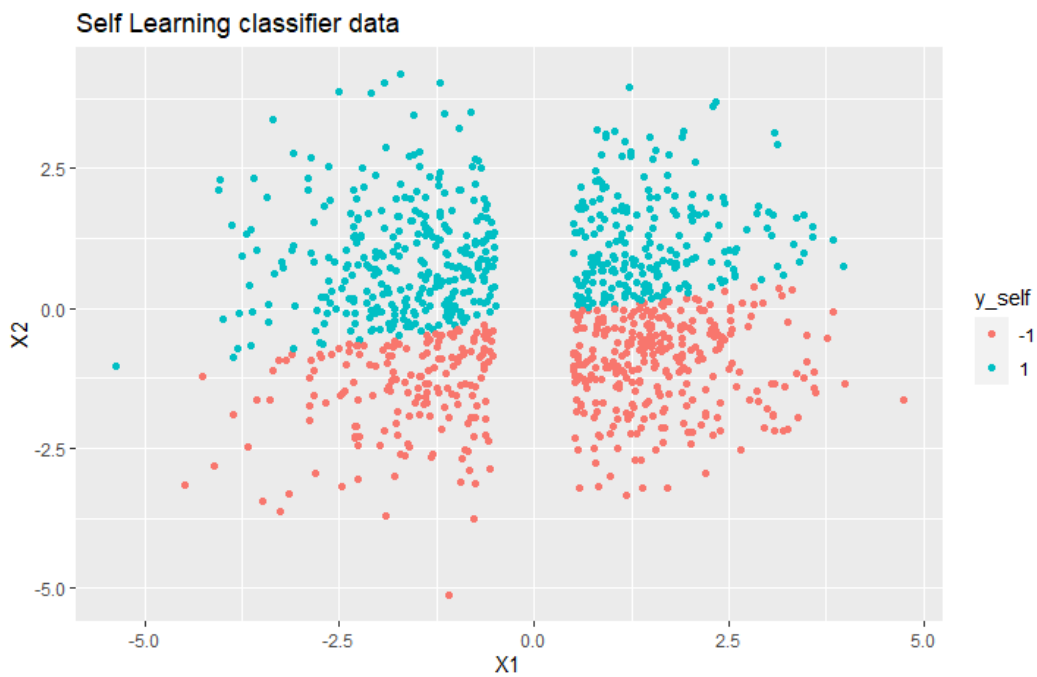


Figure 17.7 – Self-learning classifier

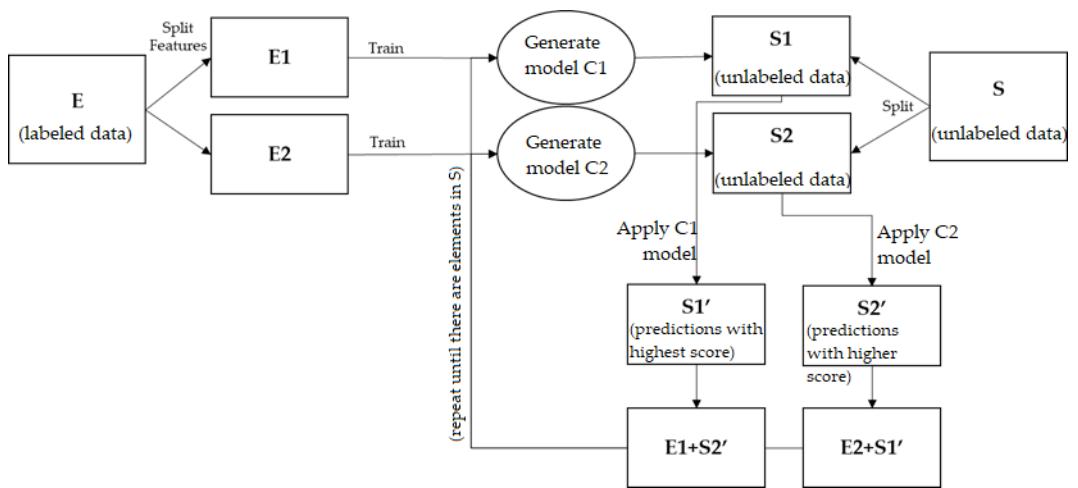


Figure 17.8 - Co-training

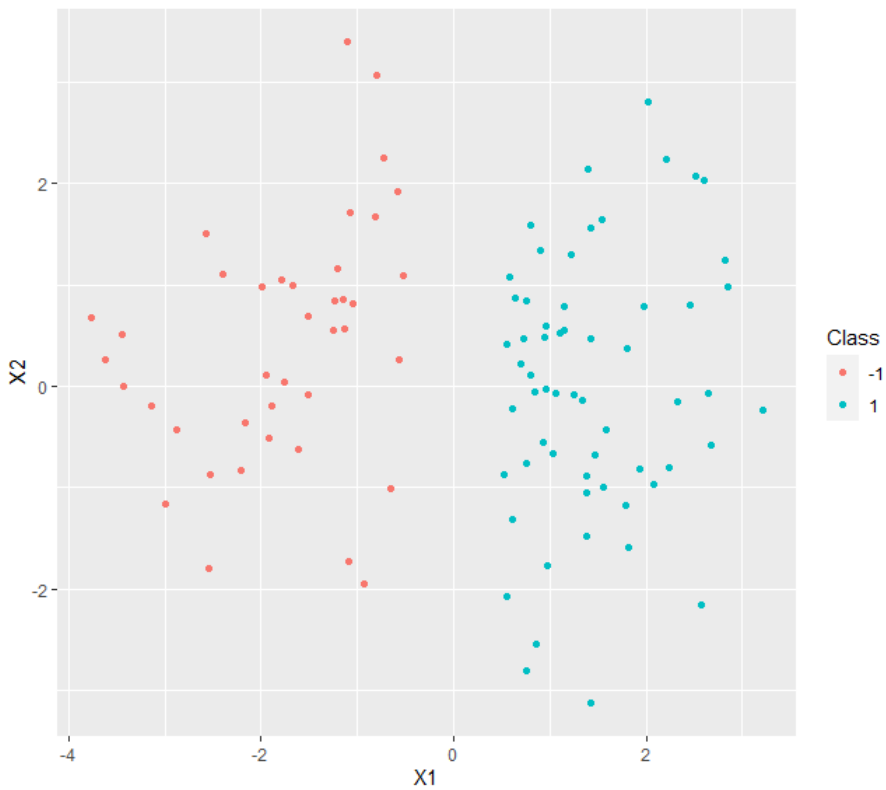


Figure 17.9 – Generated dataset

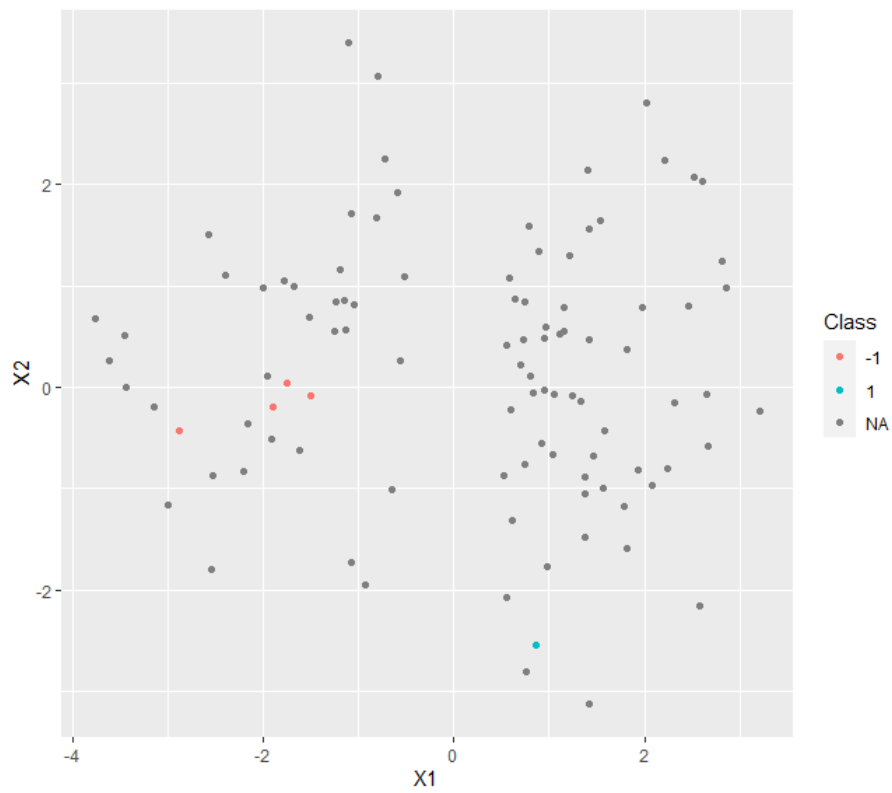


Figure 17.10 – Dataset with missing labels

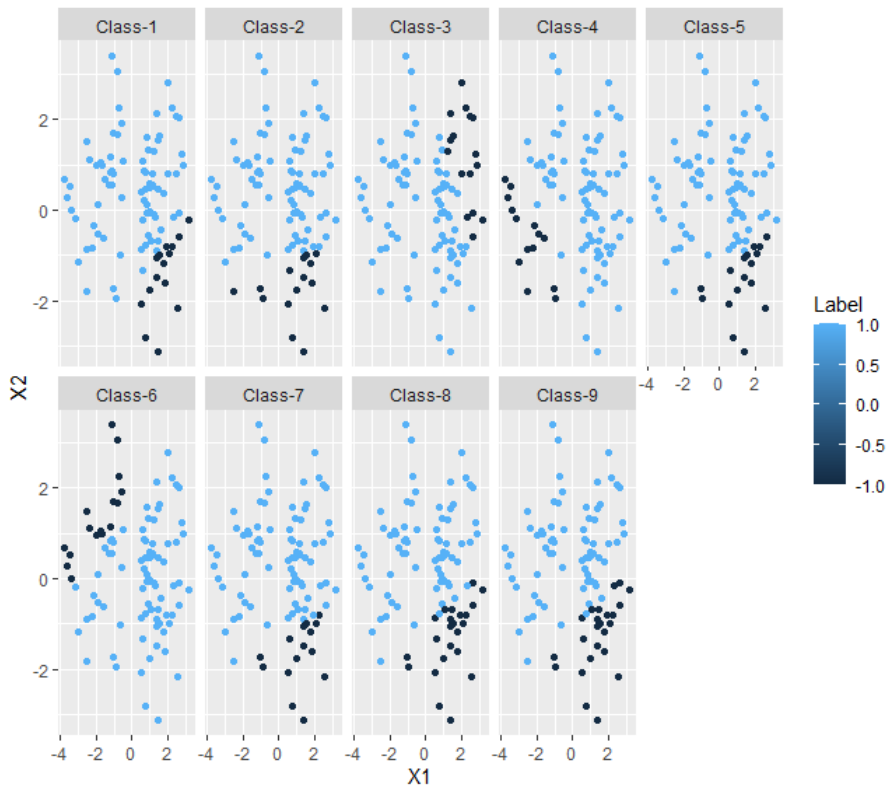


Figure 17.11 – Considered Labels

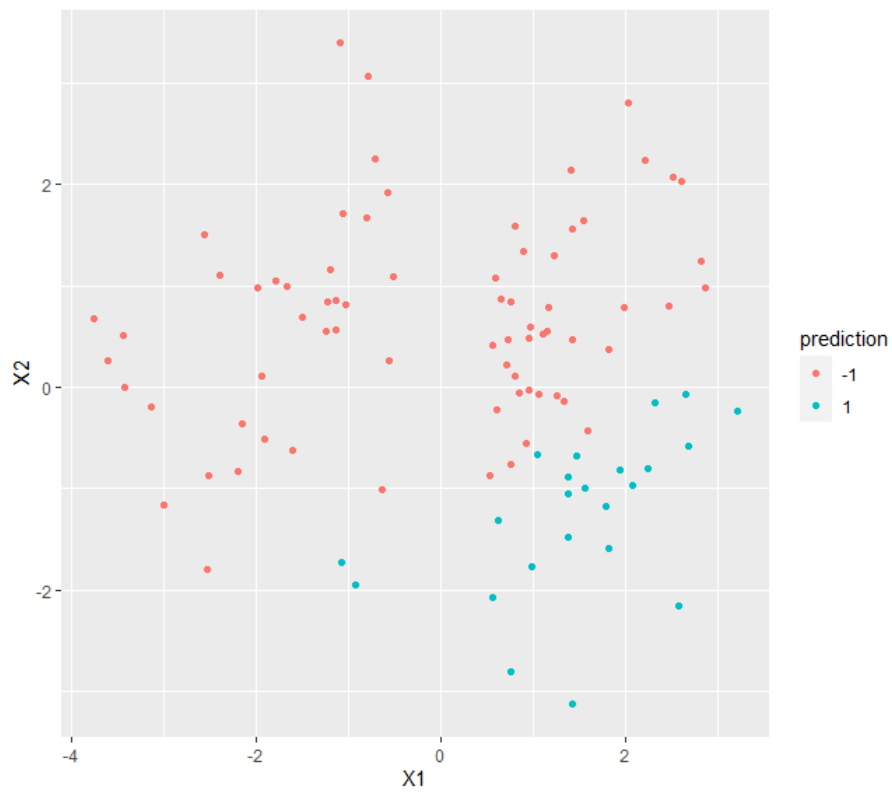


Figure 17.12 – Final labeling

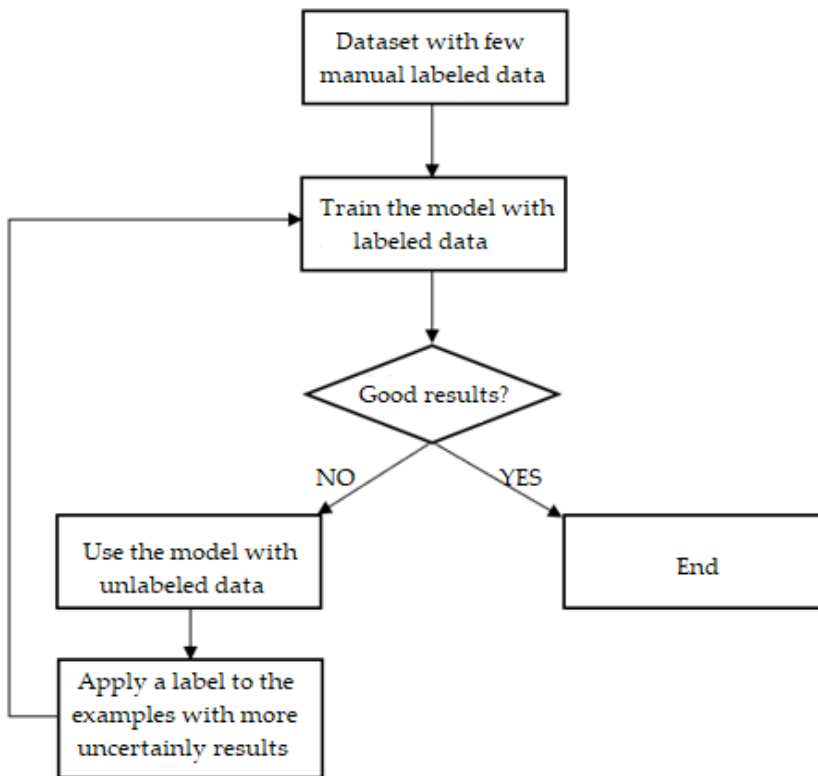


Figure 17.13 – Iterative process in Active Learning

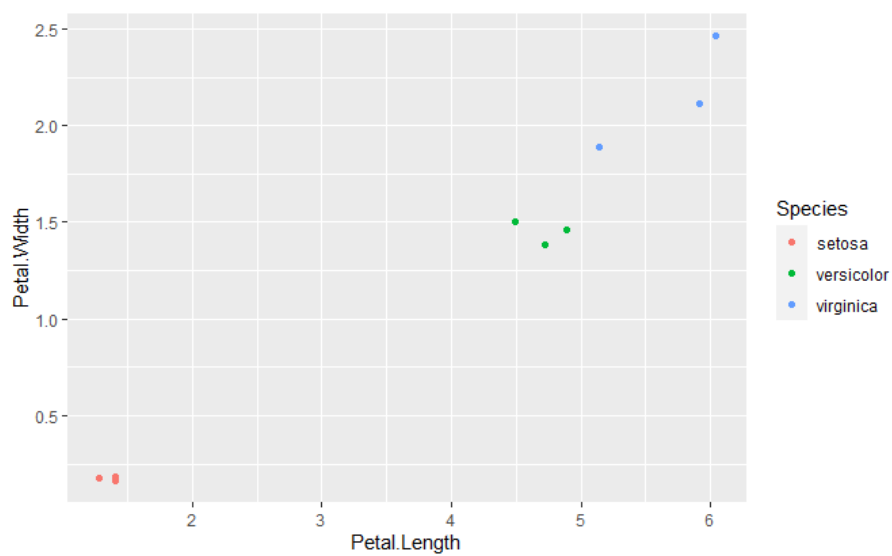


Figure 17.14 – Labeled data

18. LAZY LEARNING

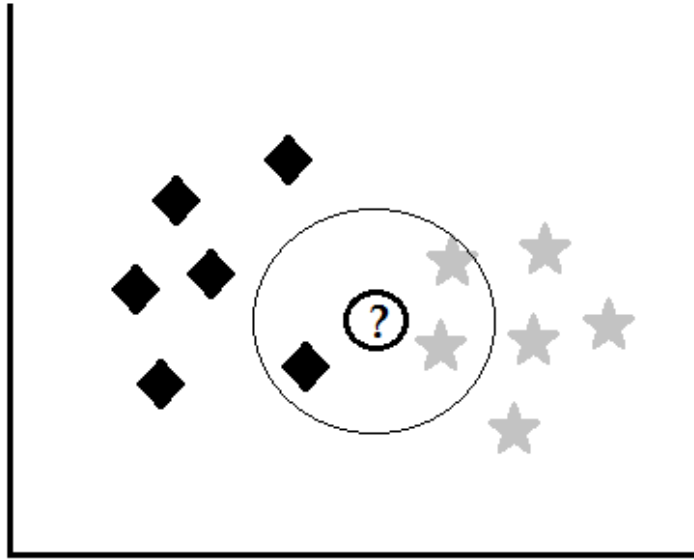


Figure 18.1 – KNN example

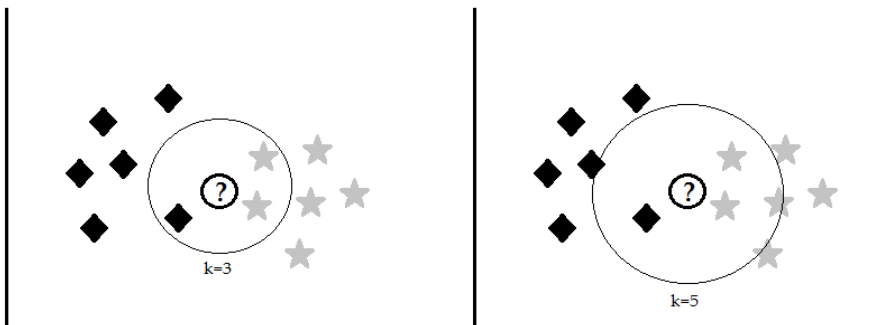


Figure 18.2 – Choosing k in KNN

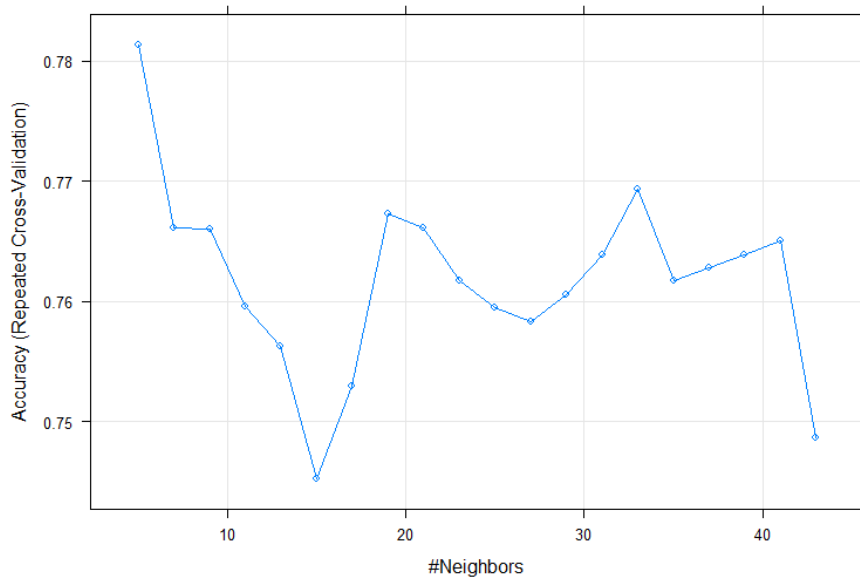


Figure 18.3 - Model performance for different K values

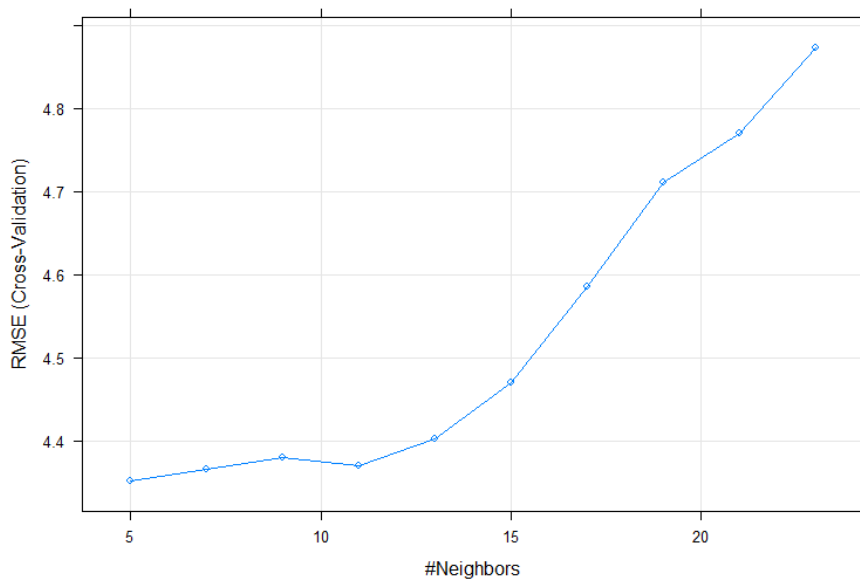


Figure 18.4 – RMS chart for different k values

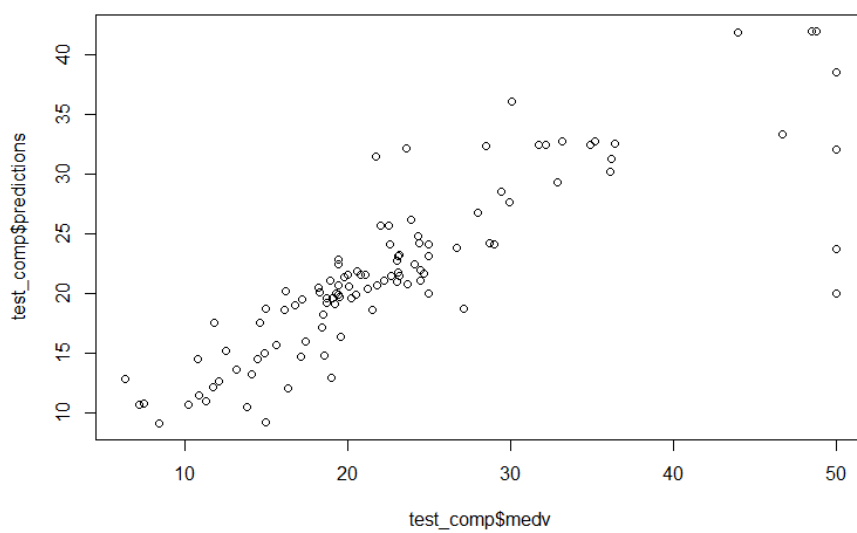


Figure 18.5 - Expected vs actual value

19. ENSEMBLE METHODS

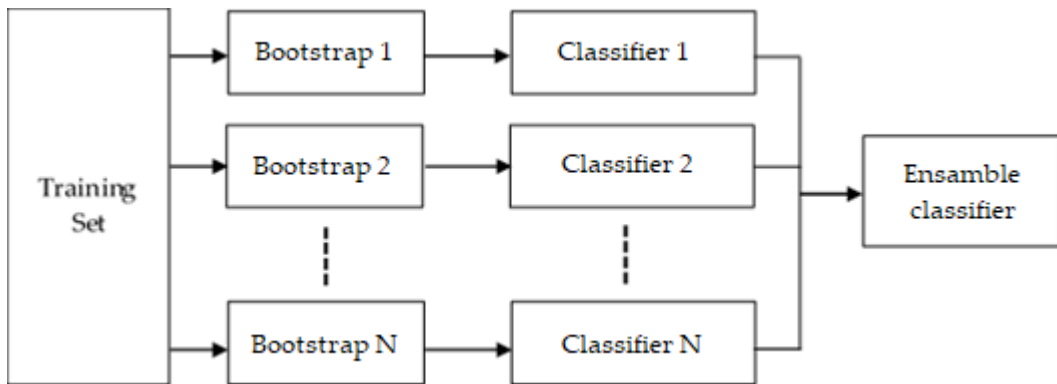


Figure 19.1 - Bagging

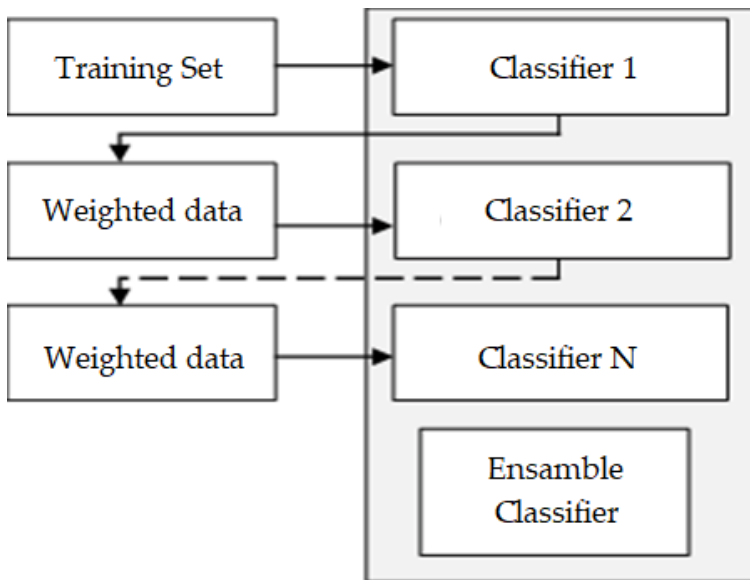


Figure 19.2 – Boosting

20. ASSOCIATION RULES

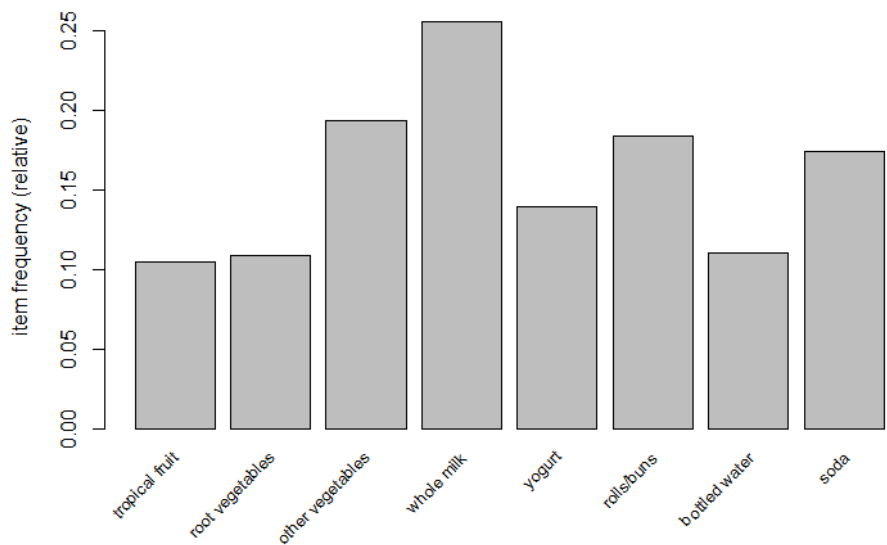


Figure 20.1 - Itemset with support greater than 0.1

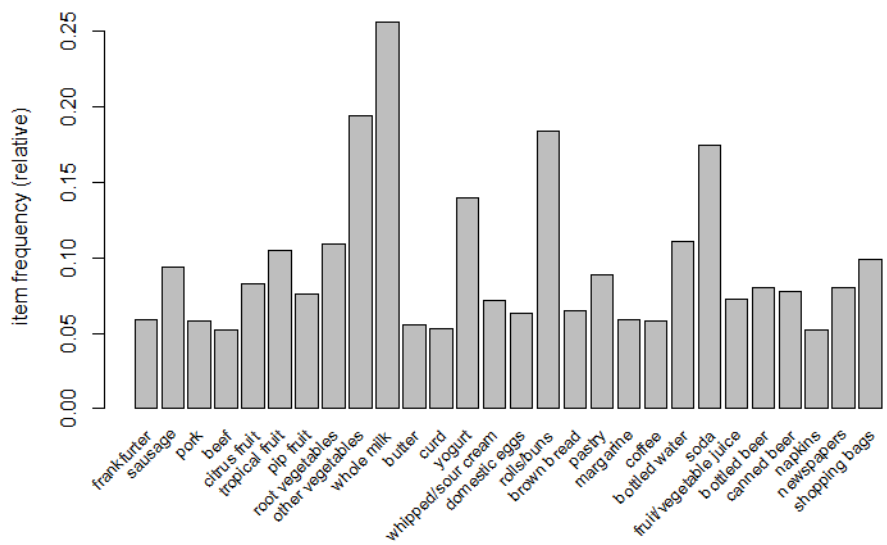


Figure 20.2 - Itemset with support greater than 0.05

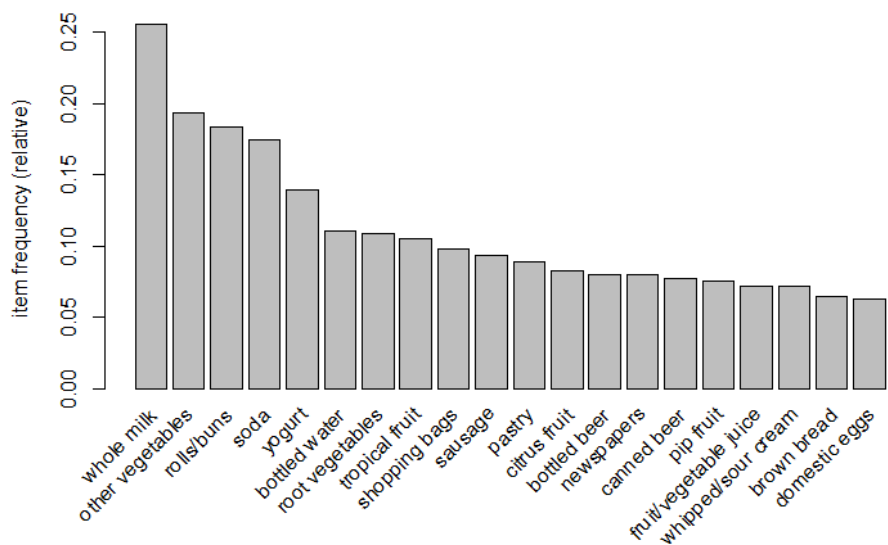


Figure 20.3 – Top 20 frequency itemset

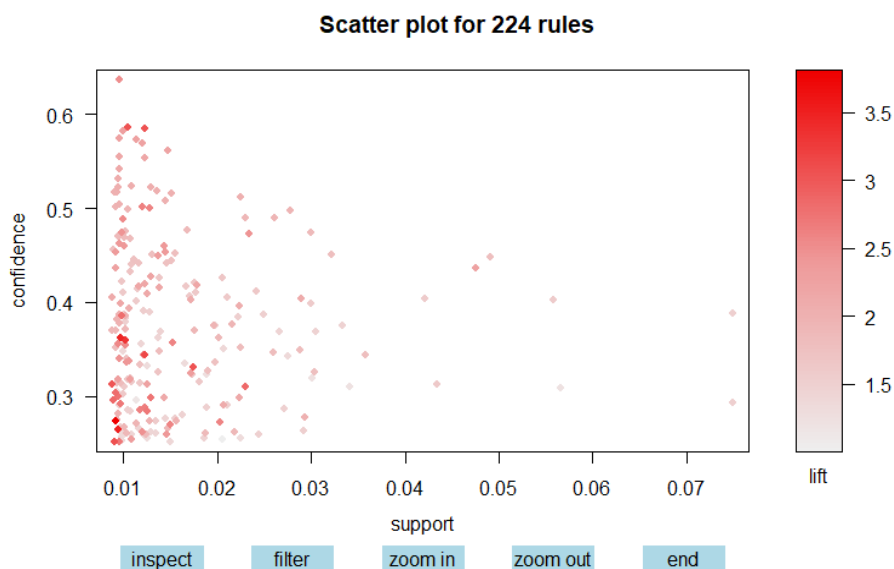


Figure 20.4 – Support and confidence chart

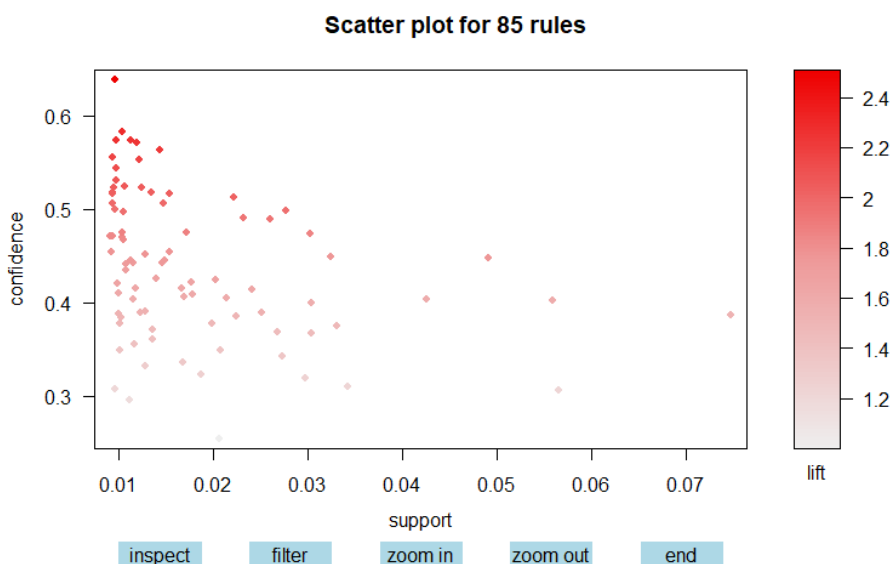


Figure 20.5 - Rules for milk-related products

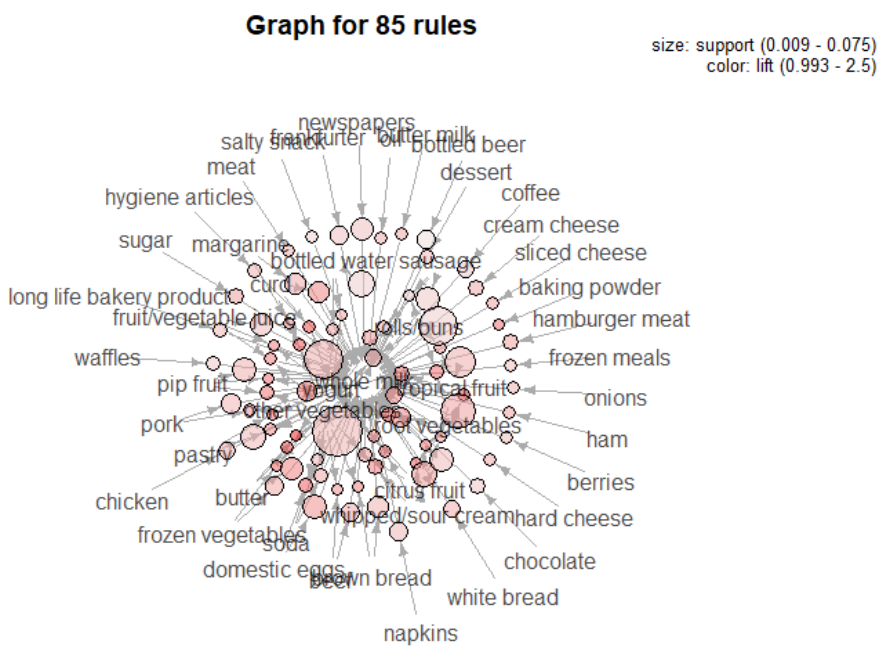


Figure 20.6 – Rules chart

21. TIME SERIES

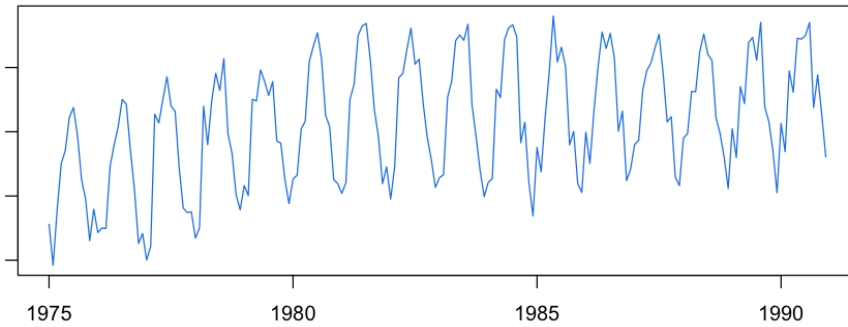


Figure 21.1 – Time series example

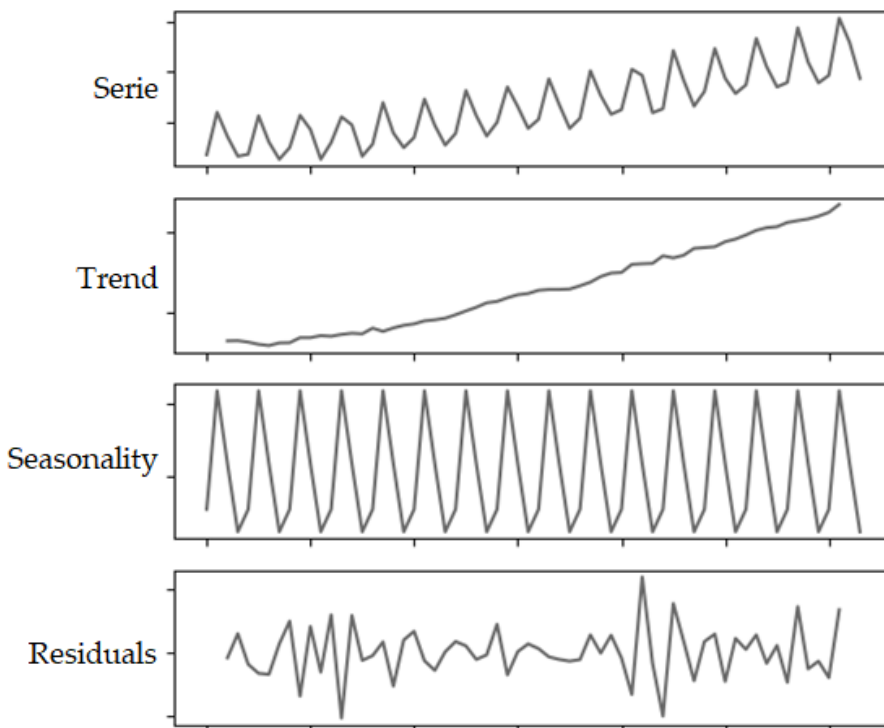


Figure 21.2 - Time series decomposition

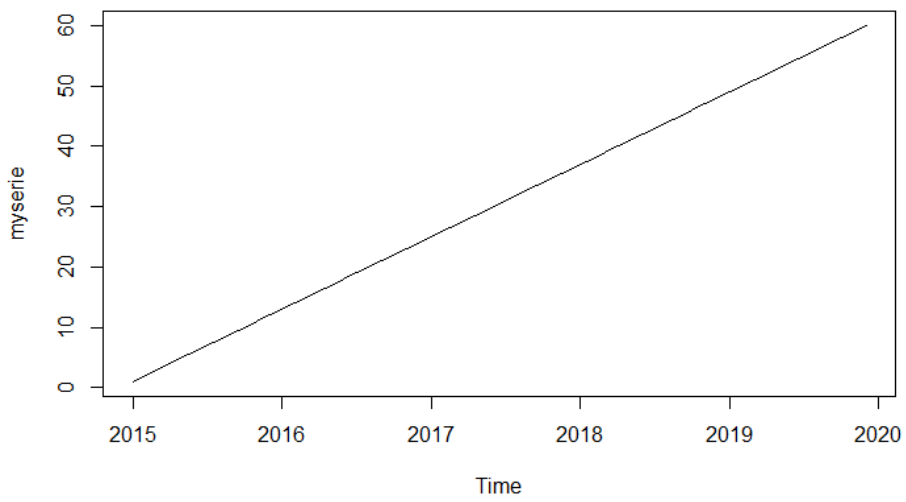


Figure 21.3 - Our simple time series

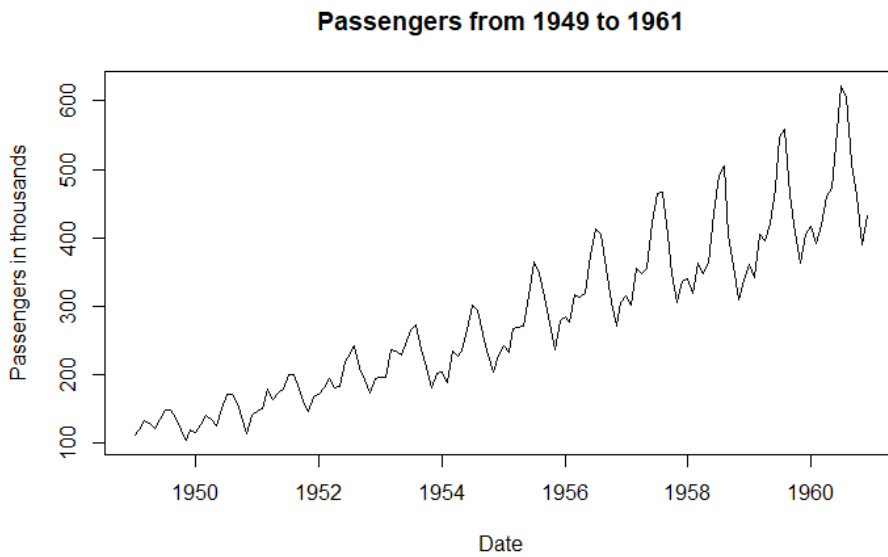


Figure 21.4 – AirPassengers time series

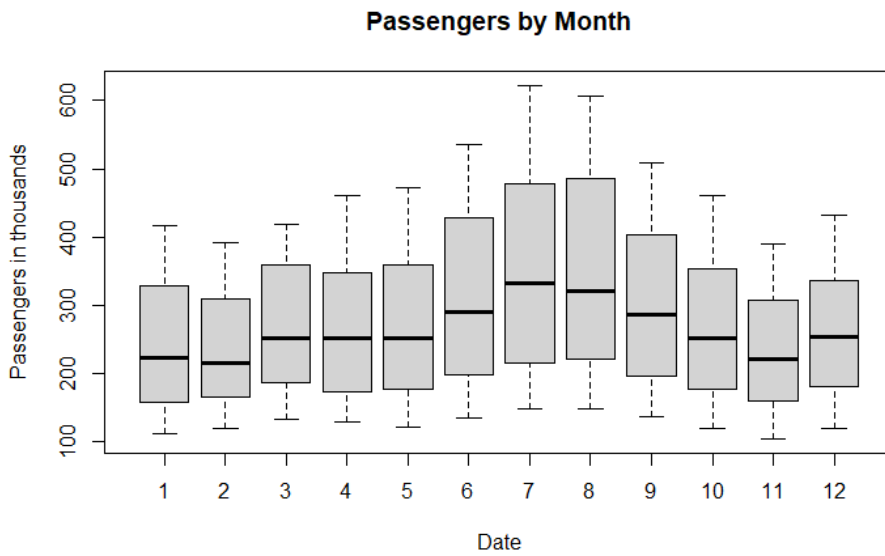


Figure 21.5 - Boxplot by month

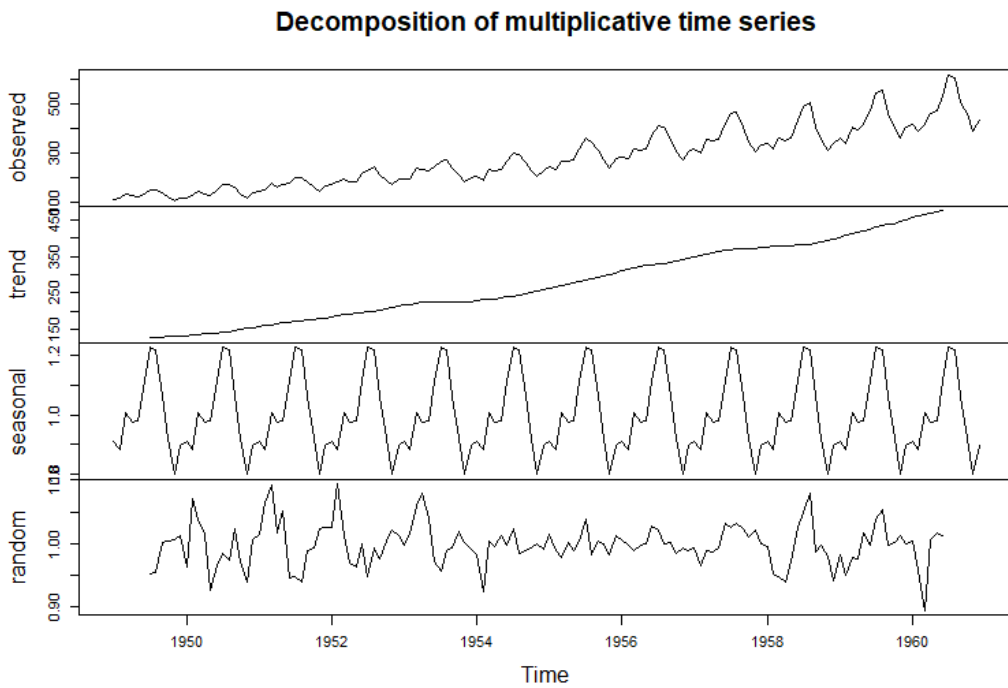


Figure 21.6 – The components of the time series



Figure 21.7 – Residuals analysis

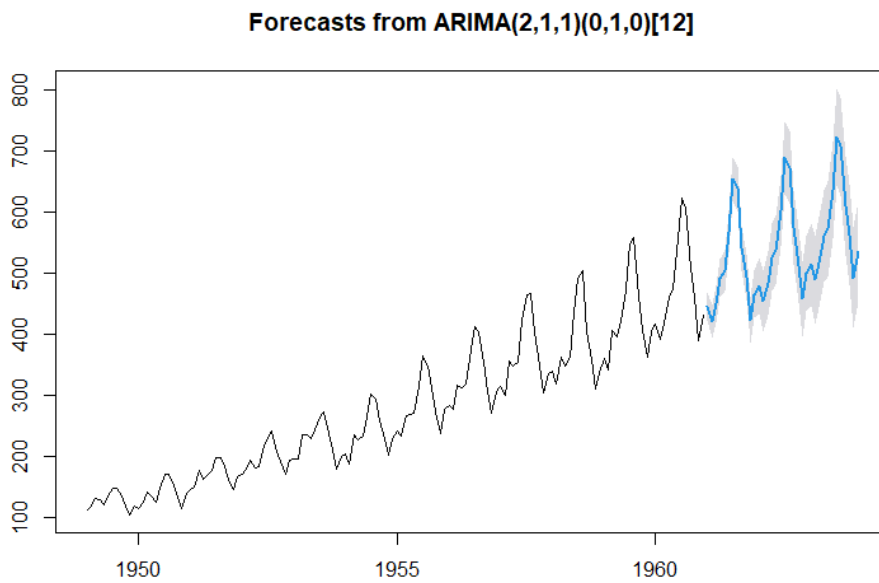


Figure 21.8 - Forecast

22. NEURAL NETWORK

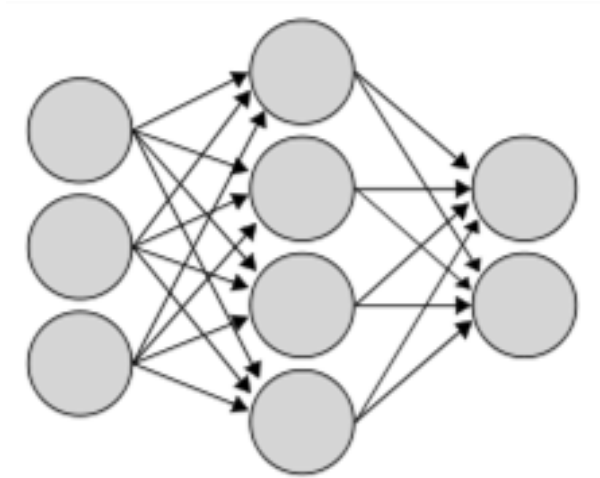


Figure 22.1 – Neural Network representation

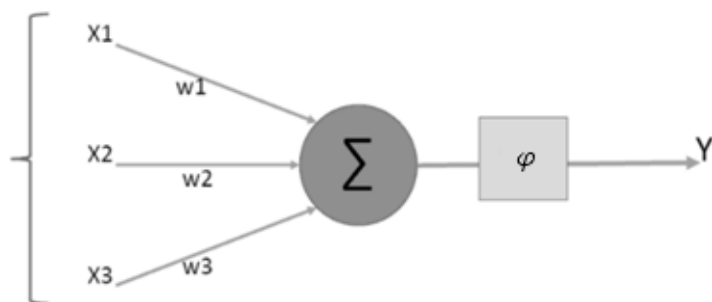


Figure 22.2 -Neuron structure

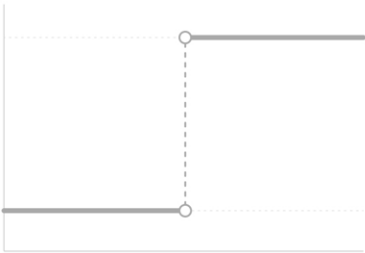


Figure 22.3 – Threshold function

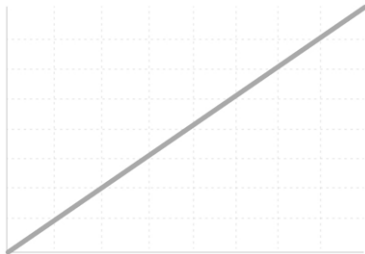


Figure 22.4 – Linear function

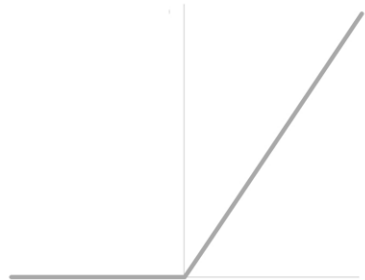


Figure 22.5 – ReLu Function

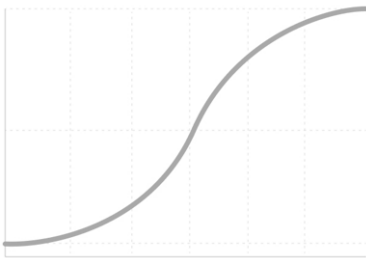


Figure 22.6 – Logistic function

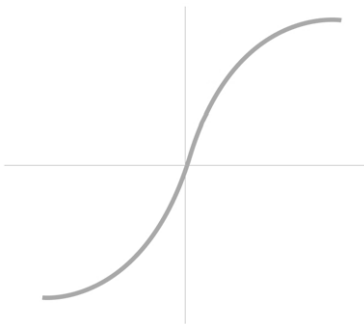


Figure 22.7 – Hyperbolic tangent function

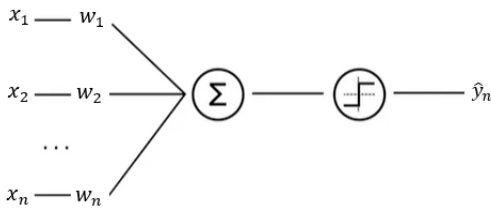


Figure 22.8 - Perceptron

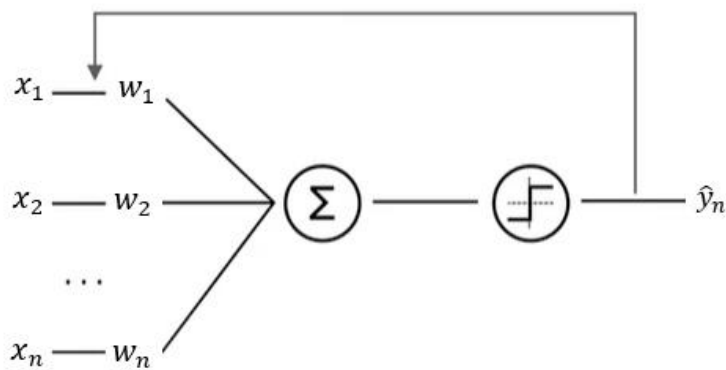


Figure 22.9 – Perceptron algorithm

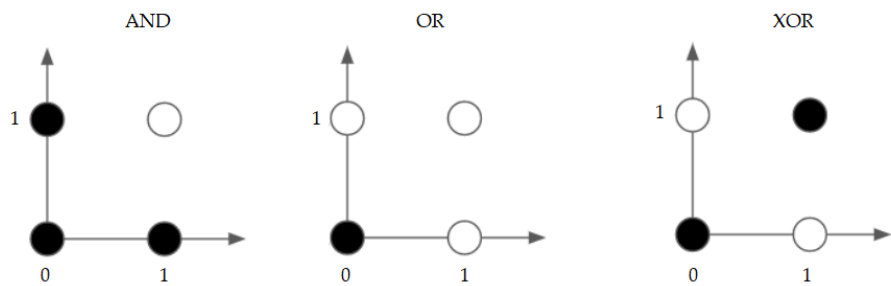


Figure 22.10 - Functions AND, OR e XOR

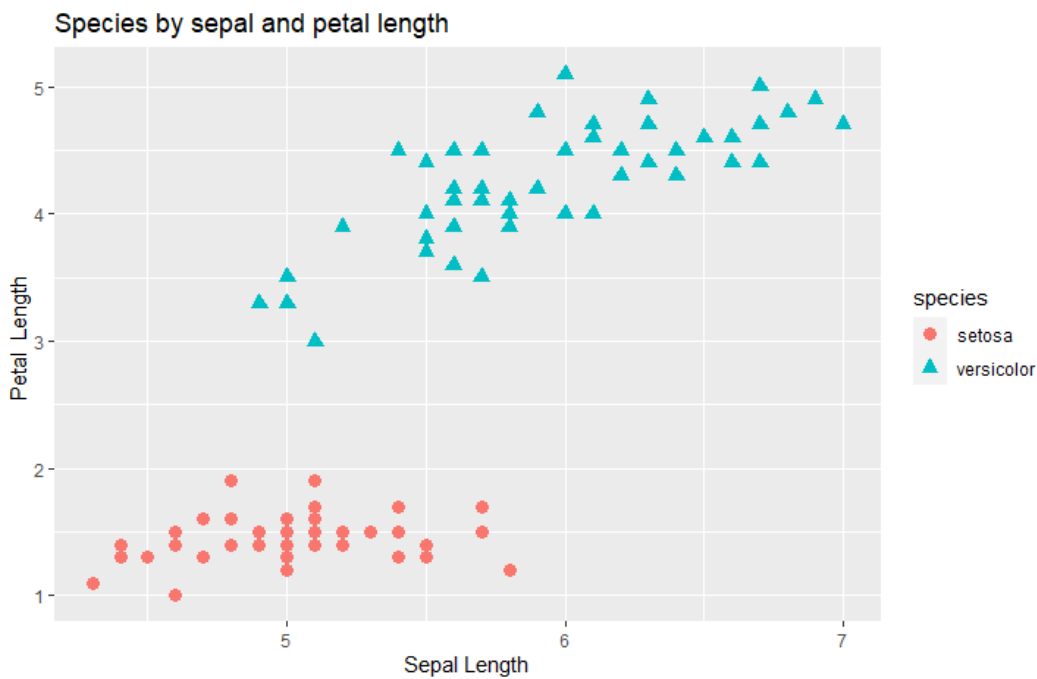


Figure 22.11 - Linearly separable classes

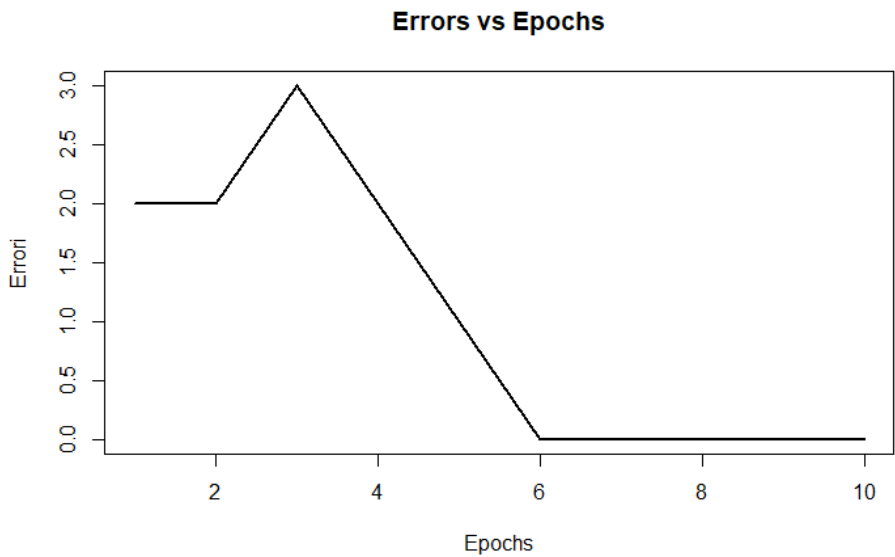


Figure 22.12 – Errors chart

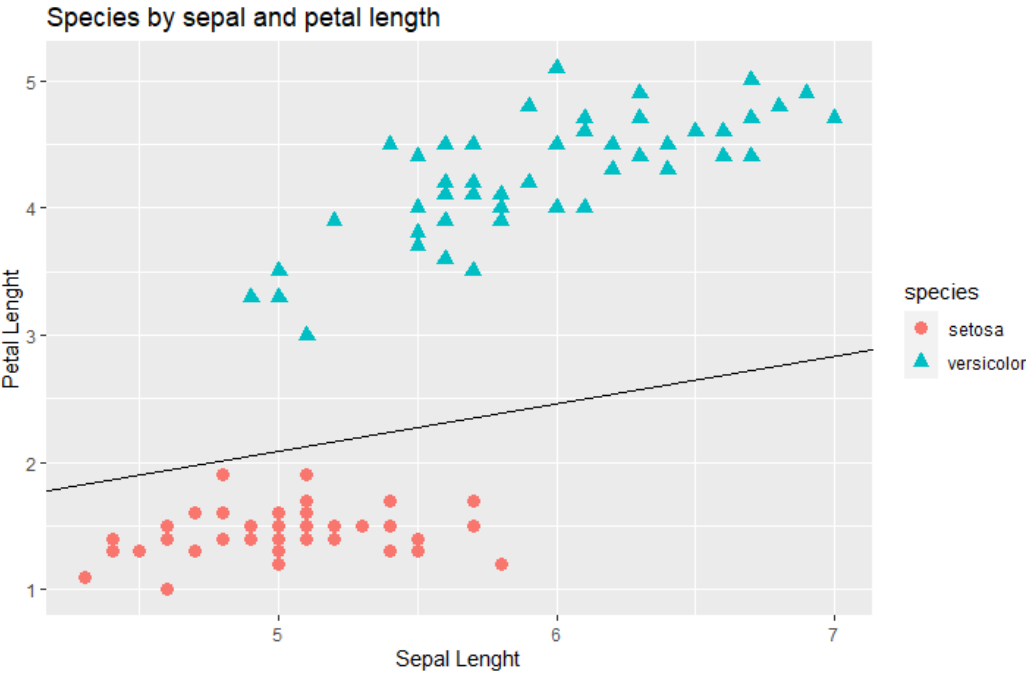


Figure 22.13 – The final perceptron

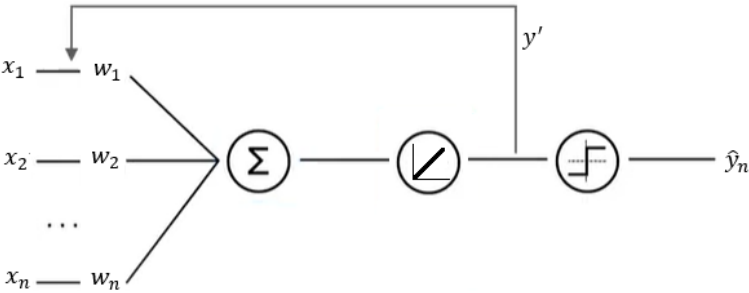


Figure 22.14 – Adaline algorithm

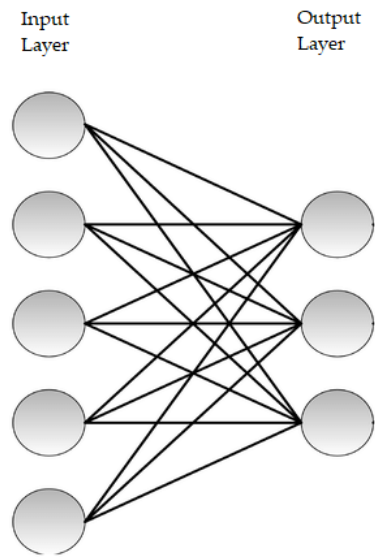


Figure 22.15 – SLP Neural Network

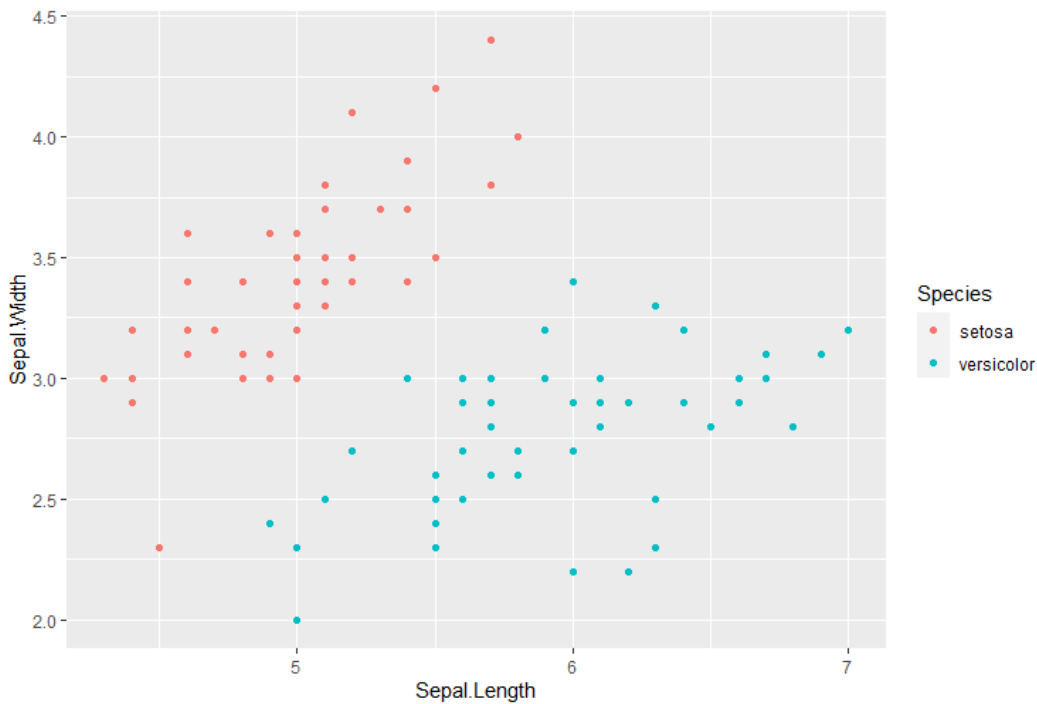
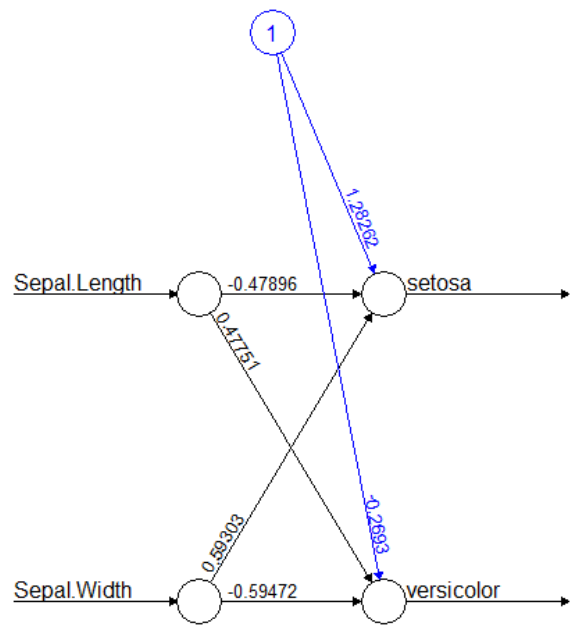


Figure 22.16 – Linear separable classes



Error: 4.107052 Steps: 2023

Figure 22.17 – The SLP of our example

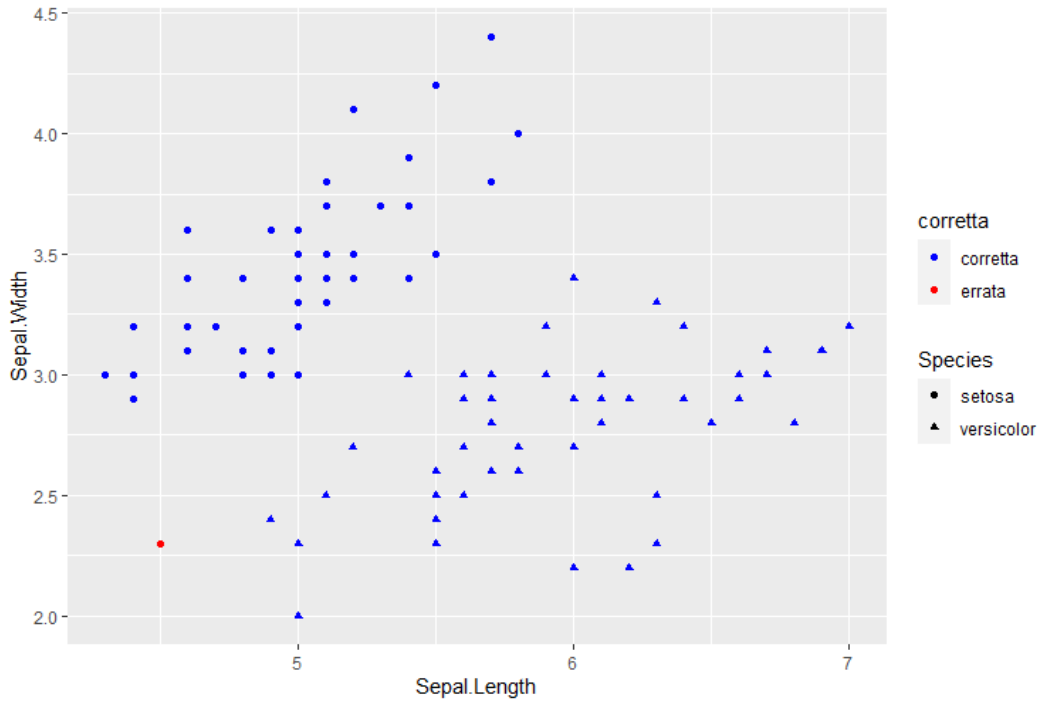


Figure 22.18 - Correctly identified classes

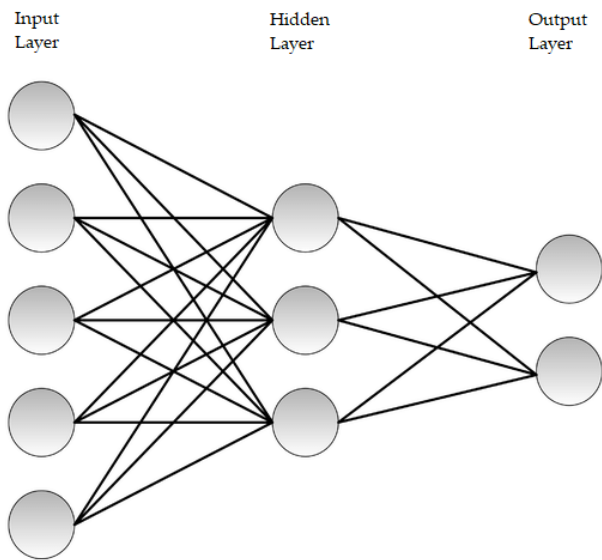


Figure 22.19 - MLP

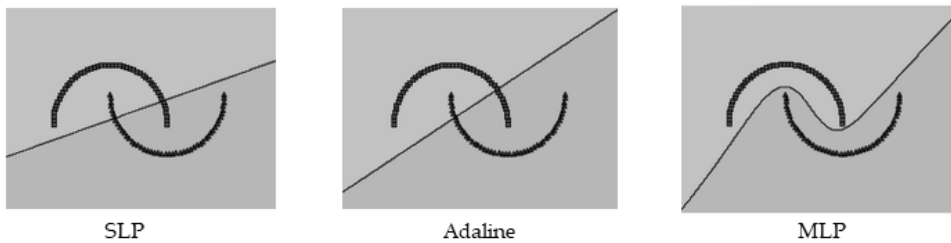


Figure 22.20 - Solving nonlinear problems with MLP

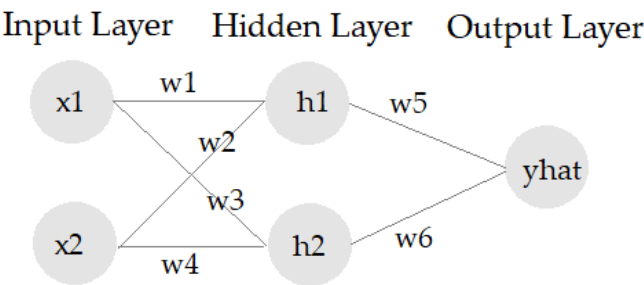


Figure 22.21 – The network of our example

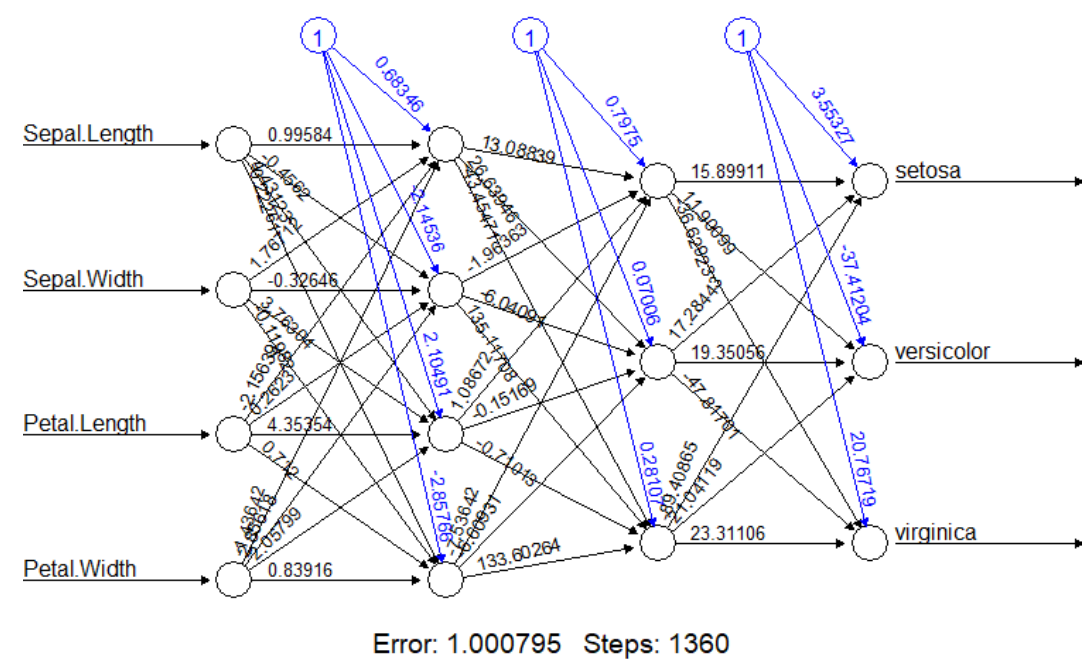


Figure 22.22 – The neural network model

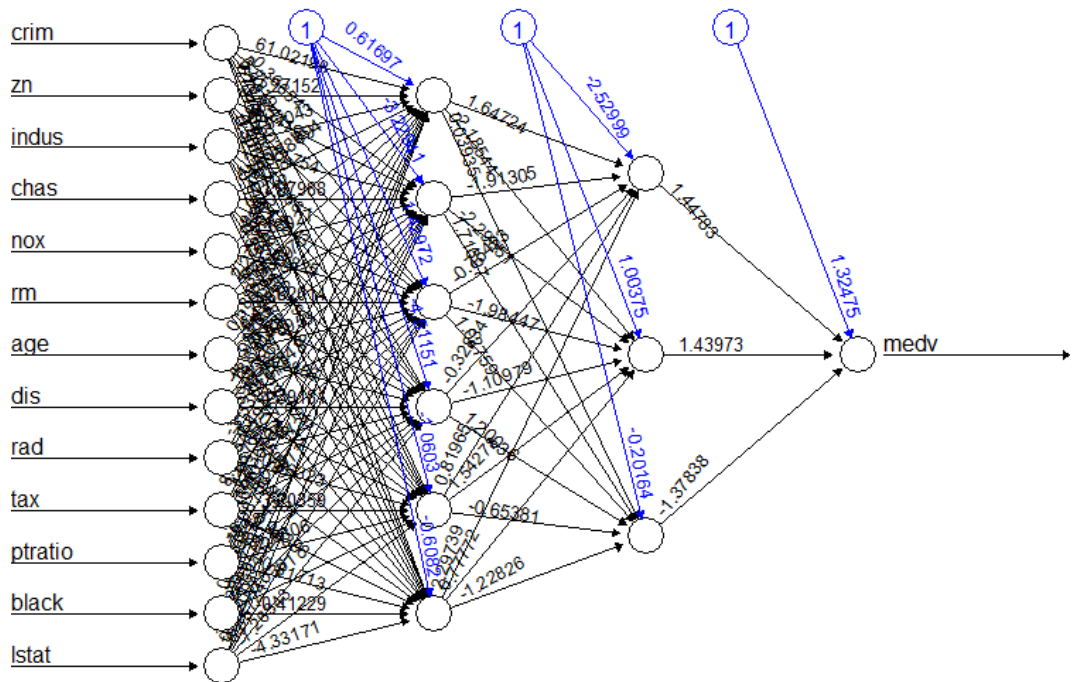


Figure 22.23 – Neural network for the boston dataset3t

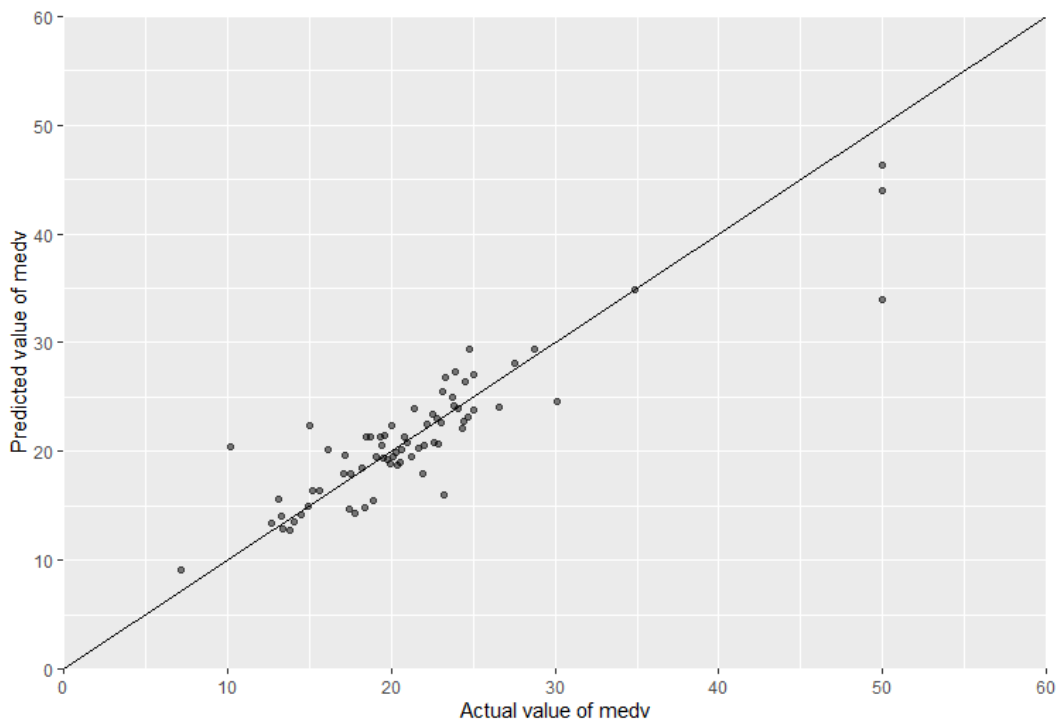


Figure 22.24 – Prediction vs Actual

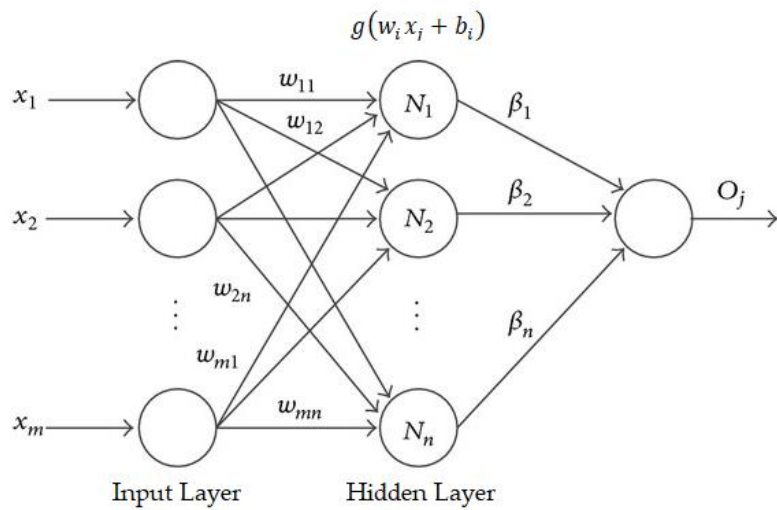


Figure 22.25 – ELM structure

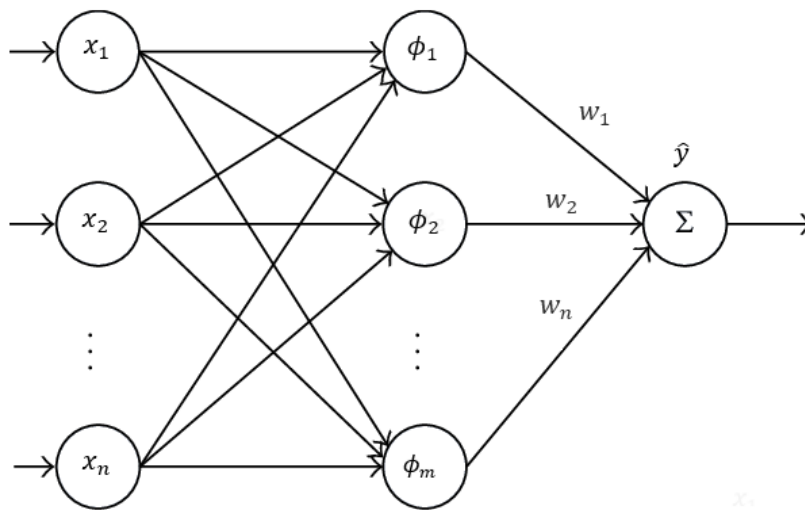


Figure 22.26 – RBF Network

23. DEEP LEARNING

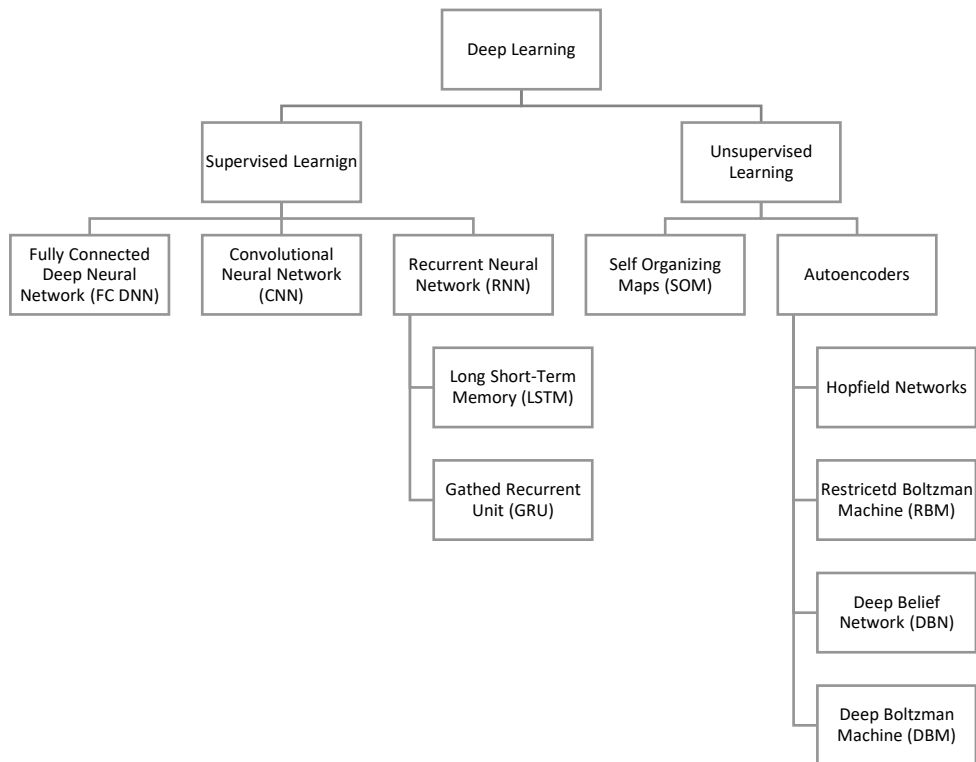


Figure 23.1 - Some Deep Learning models

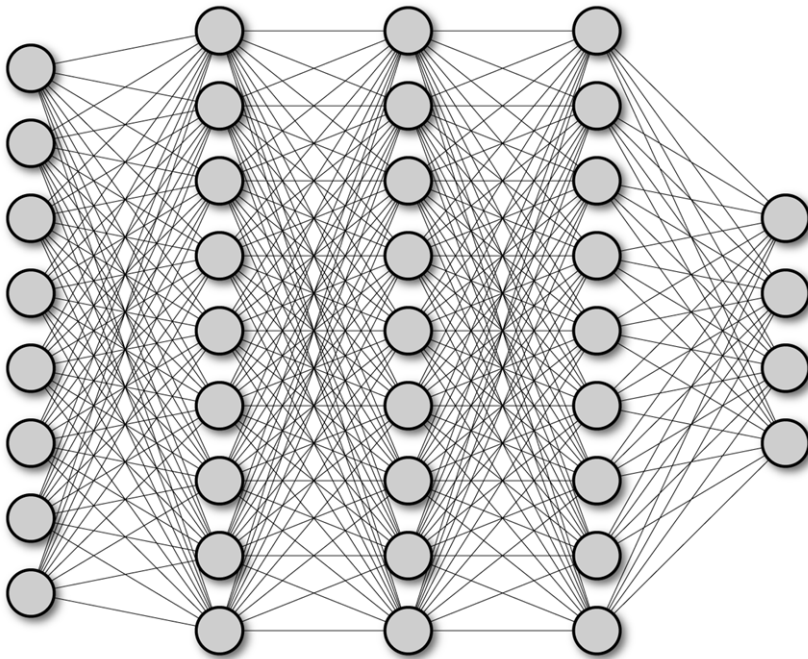


Figure 23.2 – Fully connected neural network



Figure 23.3 – Some examples from the MNIST dataset

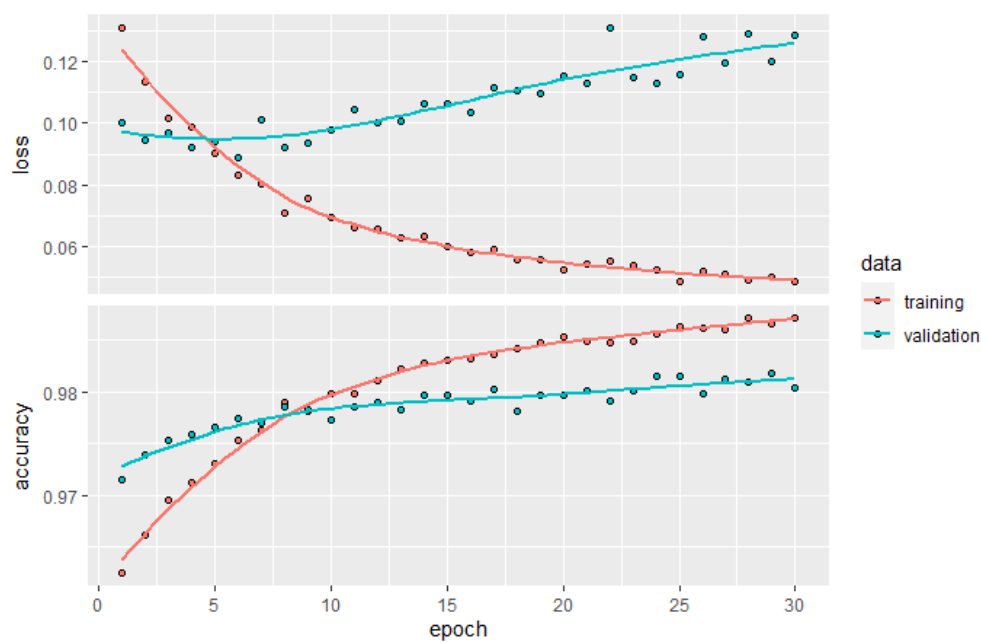


Figure 23.4 - loss and accuracy

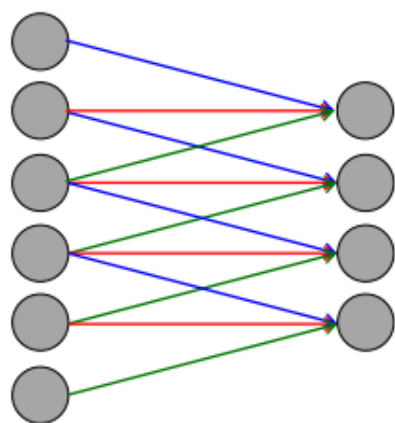


Figure 23.5 – Shared weights example

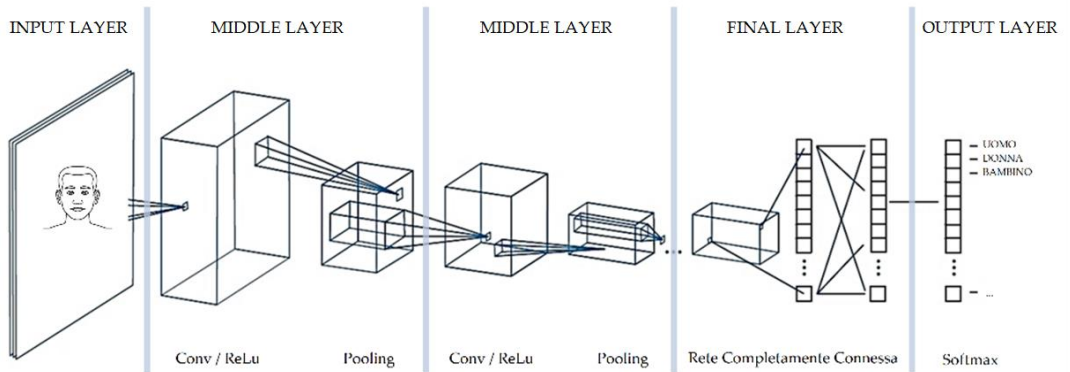


Figure 23.6 – CNN architecture

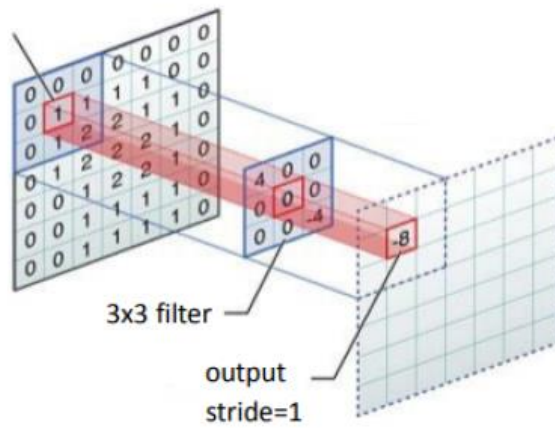


Figure 23.7 – Applying filter

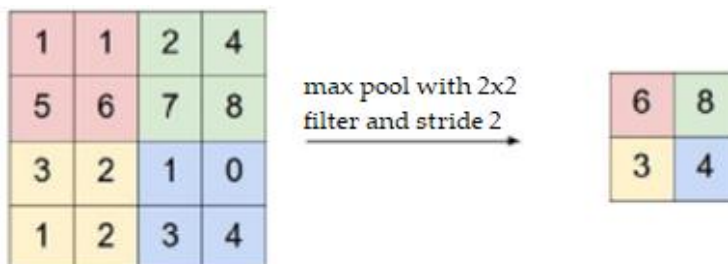


Figure 23.8 - Max pool

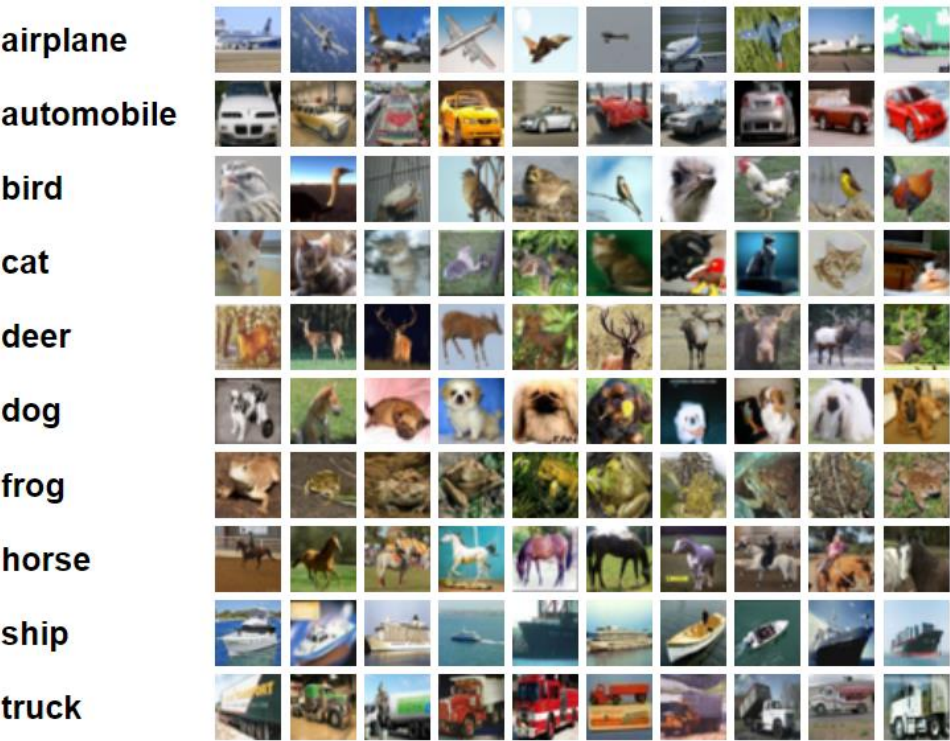


Figure 23.9 - The dataset classes, with some example images

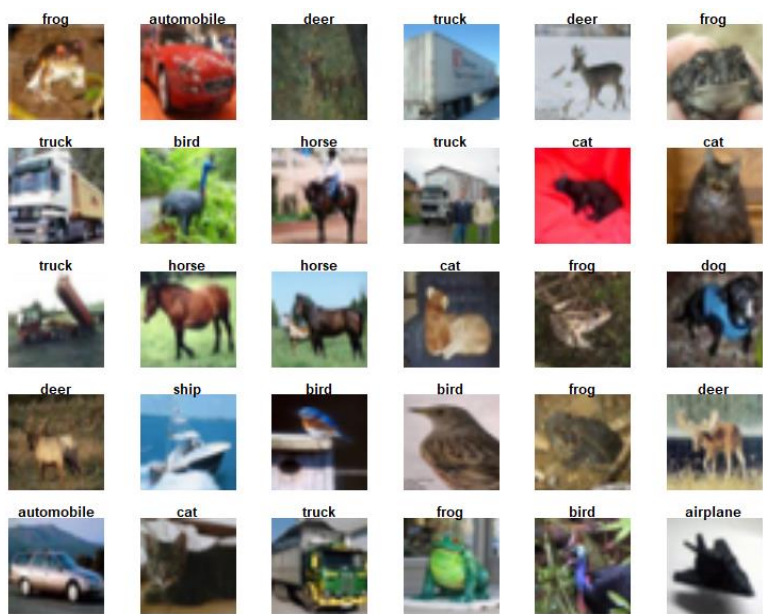


Figure 23.10 - Some dataset images

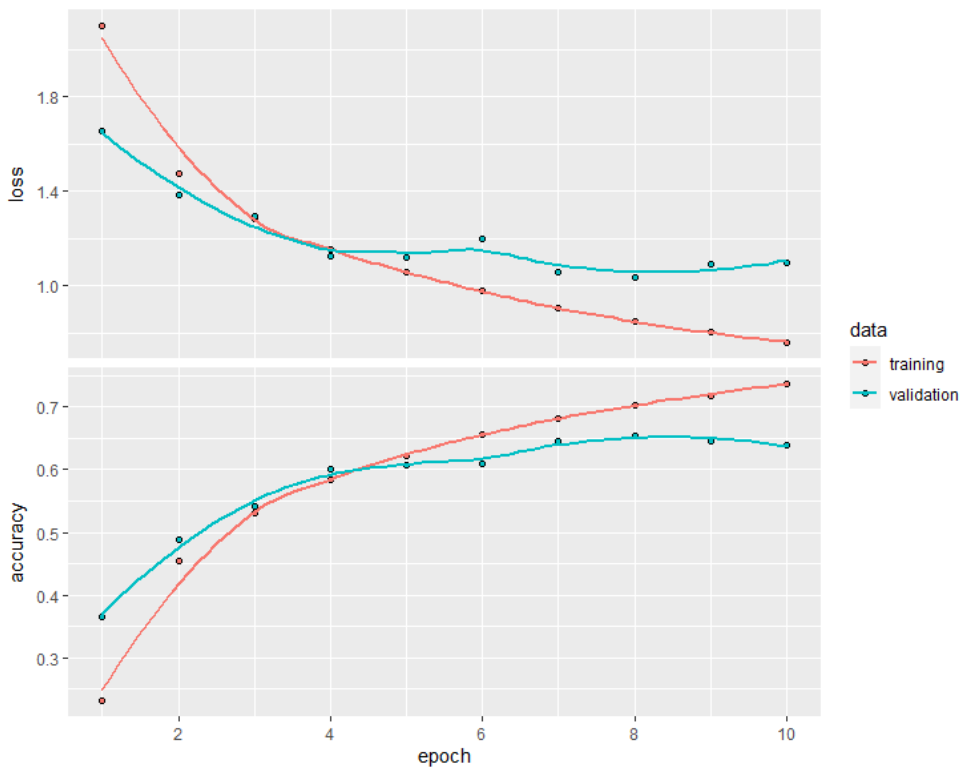


Figure 23.11 – Model's results

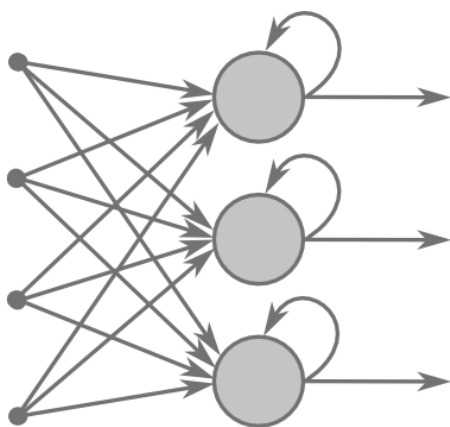


Figure 23.12 – Recurrent Neural Network

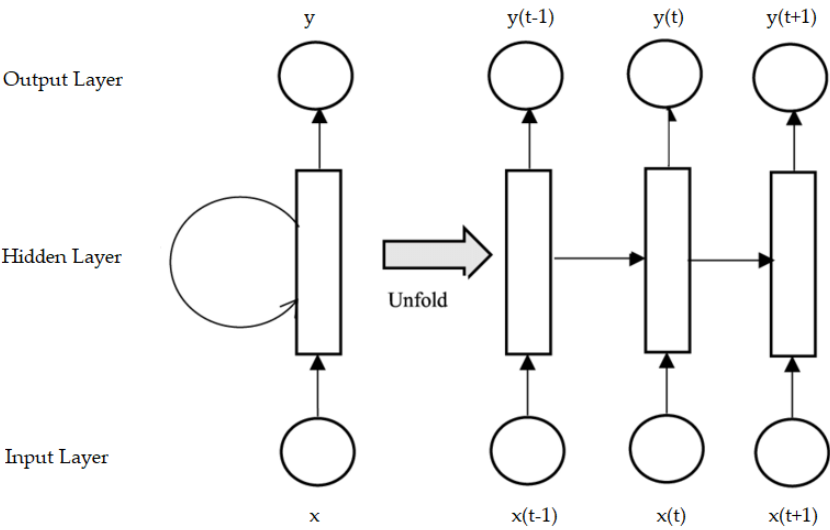


Figure 23.13 - Unfold of an RNN

	DATE	PRCP	TMAX	TMIN	RAIN
10	1948-01-10	0.74	43	34	TRUE
11	1948-01-11	0.01	42	32	TRUE
12	1948-01-12	0.00	41	26	FALSE
13	1948-01-13	0.00	45	29	FALSE
14	1948-01-14	0.00	38	26	FALSE
15	1948-01-15	0.00	34	31	FALSE
16	1948-01-16	0.00	34	28	FALSE

Figure 23.14 – Dataset slice

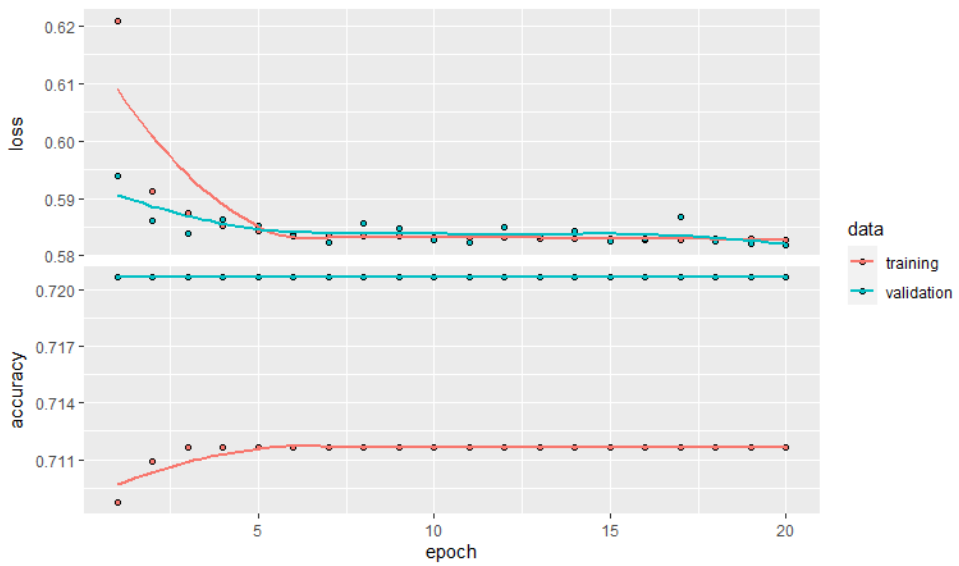


Figure 23.15 - Loss e Accuracy

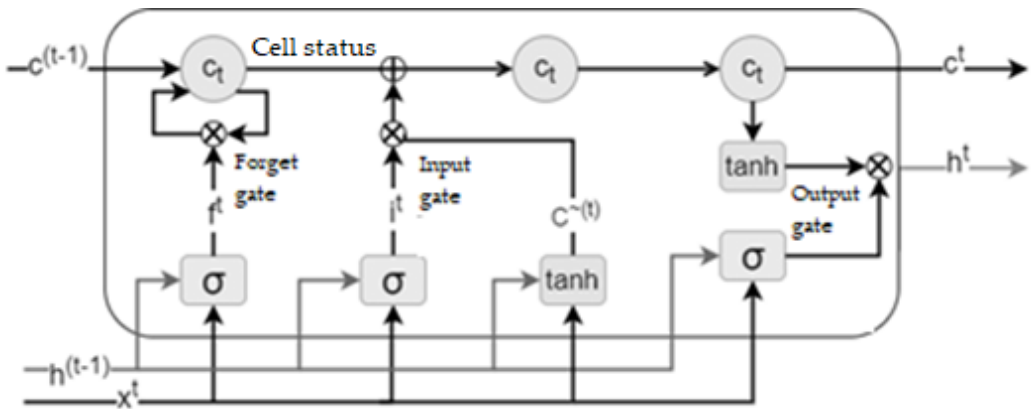


Figure 23.16 – LSTM architecture

review	sentiment
One of the other reviewers has mentioned that after watching just 1 Oz episode you'll be hooked. They are right, as this is exactly what happened with me. The first thing that struck me about Oz was its brutality and unflinching scenes of violence, which set in right from the word GO. Trust me, this is not a show for the faint hearted or timid. This show pulls no punches with regards to drugs, sex or violence. Its is hardcore, in the classic use of the word. It is called OZ as that is the nickname given to the Oswald Maximum Security State Penitentiary. It focuses mainly on Emerald City, an experimental section of the prison where all the cells have glass fronts and face inwards, so privacy is not high on the agenda. Em City is home to many..Aryans, Muslims, gangstas, Latinos, Christians, Italians, Irish and more....so scuffles, death stares, dodgy dealings and shady agreements are never far away. I would say the main appeal of the show is due to the fact that it goes where other shows wouldn't dare. Forget pretty pictures painted for mainstream audiences, forget charm, forget romance...OZ doesn't mess around. The first episode I ever saw struck me as so nasty it was surreal, I couldn't say I was ready for it, but as I watched more, I developed a taste for Oz, and got accustomed to the high levels of graphic violence. Not just violence, but injustice (crooked guards who'll be sold out for a nickel, inmates who'll kill on order and get away with it, well mannered, middle class inmates being turned into prison bitches due to their lack of street skills or prison experience) Watching Oz, you may become comfortable with what is uncomfortable viewing....thats if you can get in touch with your darker side.	positive
A wonderful little production. The filming technique is very unassuming- very old-time-BBC fashion and gives a comforting, and sometimes discomforting, sense of realism to the entire piece. The actors are extremely well chosen- Michael Sheen not only "has got all the polari" but he has all the voices down pat too! You can truly see the seamless editing guided by the references to Williams' diary entries, not only is it well worth the watching but it is a terrifically written and performed piece. A masterful production about one of the great master's of comedy and his life. The realism really comes home with the little things: the fantasy of the guard which, rather than use the traditional 'dream' techniques remains solid then disappears. It plays on our knowledge and our senses, particularly with the scenes concerning Orton and Halliwell and the sets (particularly of their flat with Halliwell's murals decorating every surface) are terribly well done.	positive
I thought this was a wonderful way to spend time on a too hot summer weekend, sitting in the air conditioned theater and watching a light-hearted comedy. The plot is simplistic, but the dialogue is witty and the characters are likable (even the well bread suspected serial killer). While some may be disappointed when they realize this is not Match Point 2: Risk Addiction, I thought it was proof that Woody Allen is still fully in control of the style many of us have grown to love. This was the most I'd laughed at one of Woody's comedies in years (dare I say a decade?). While I've never been impressed with Scarlet Johanson, in this she managed to tone down her "sexy" image and jumped right into a average, but spirited young woman. This may not be the crown jewel of his career, but it was wittier than "Devil Wears Prada" and more interesting than "Superman" a great comedy to go see with friends.	positive
Basically there's a family where a little boy (Jake) thinks there's a zombie in his closet & his parents are fighting all the time. This movie is slower than a soap opera... and suddenly, Jake decides to become Rambo and kill the zombie. OK, first of all when you're going to make a film you must Decide if it's a thriller or a drama! As a drama the movie is watchable. Parents are divorcing & arguing like in real life. And then we have Jake with his closet which totally ruins all the film! I expected to see a BOOGEYMAN similar movie, and instead I watched a drama with some meaningless thriller spots. 3 out of 10 just for the well playing parents & descent dialogs. As for the shots with Jake: just ignore them.	negative

Figure 23.17 – Some review in the IMDB dataset

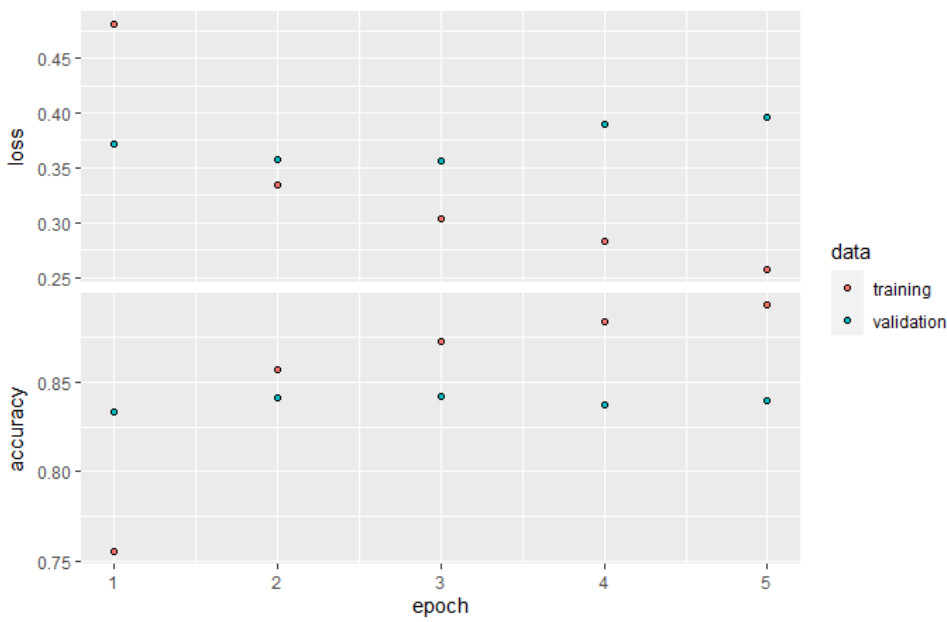


Figure 23.18 – Training results

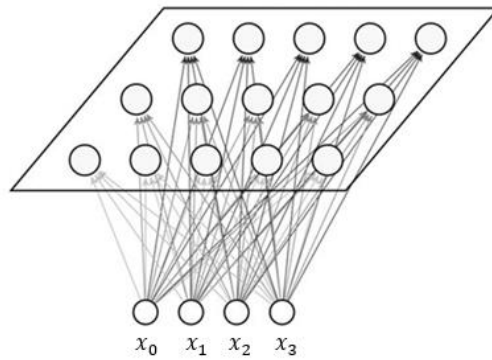


Figure 23.19 – SOM structure



Figure 23.20 - A representation of the BMU neuron and its neighbors

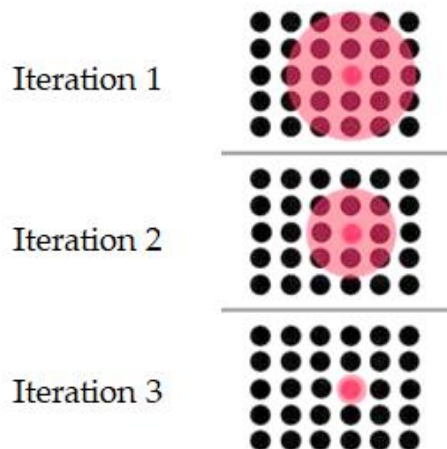


Figure 23.21 - Contraction effect of $h_{i,j}(n)$

	alcohol	malic acid	ash	ash alkalinity	magnesium	tot. phenols	flavonoids	non-flav. phenols	proanth	col. int.	col. hue	OD ratio	proline
1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.050	3.40	1050
2	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	2.81	5.68	1.030	3.17	1185
3	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	2.18	7.80	0.860	3.45	1480
4	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	1.82	4.32	1.040	2.93	735
5	14.20	1.76	2.45	15.2	112	3.27	3.39	0.34	1.97	6.75	1.050	2.85	1450
6	14.39	1.87	2.45	14.6	96	2.50	2.52	0.30	1.98	5.25	1.020	3.58	1290
7	14.06	2.15	2.61	17.6	121	2.60	2.51	0.31	1.25	5.05	1.060	3.58	1295
8	14.83	1.64	2.17	14.0	97	2.80	2.98	0.29	1.98	5.20	1.080	2.85	1045
9	13.86	1.35	2.27	16.0	98	2.98	3.15	0.22	1.85	7.22	1.010	3.55	1045
10	14.10	2.16	2.30	18.0	105	2.95	3.32	0.22	2.38	5.75	1.250	3.17	1510
11	14.12	1.48	2.32	16.8	95	2.20	2.43	0.26	1.57	5.00	1.170	2.82	1280
12	13.75	1.73	2.41	16.0	89	2.60	2.76	0.29	1.81	5.60	1.150	2.90	1320
13	14.75	1.73	2.39	11.4	91	3.10	3.69	0.43	2.81	5.40	1.250	2.73	1150

Figure 23.22 – Wine dataset

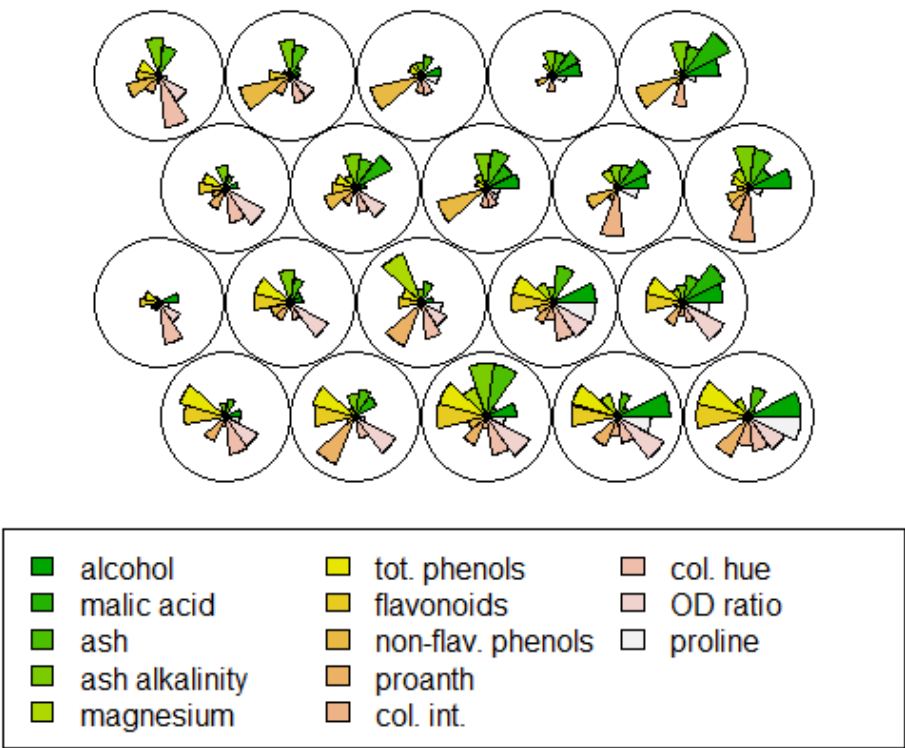


Figure 23.23 – SOM plot

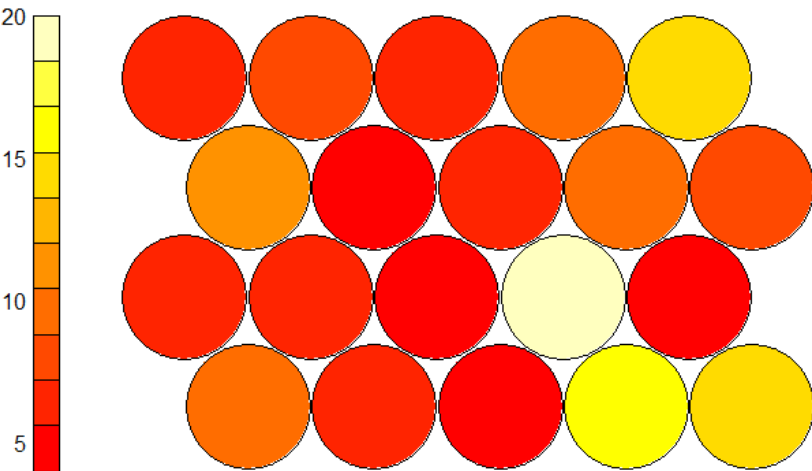


Figure 23.24 - Objects mapped for each unit

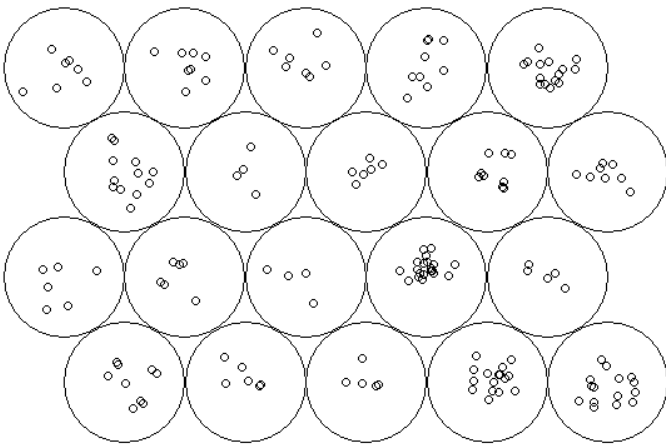


Figure 23.25 – Objects mapping

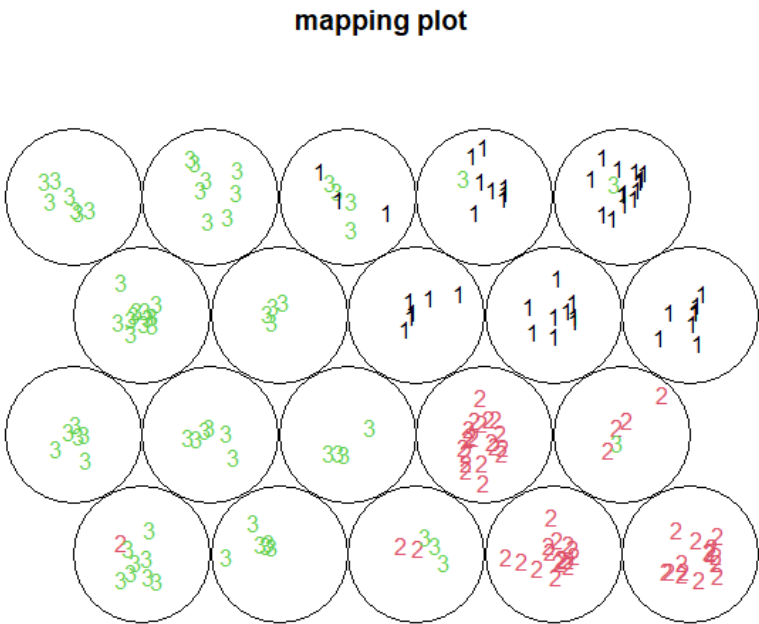


Figure 23.26 – Class mapping

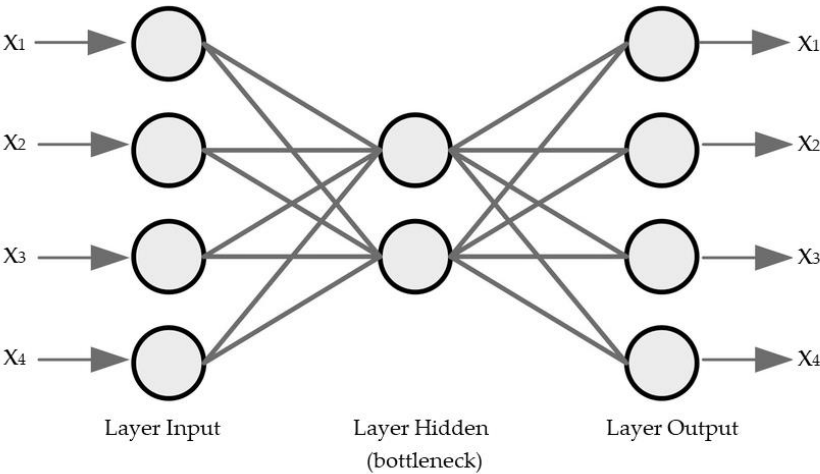


Figure 23.27 – Autoencoder structure

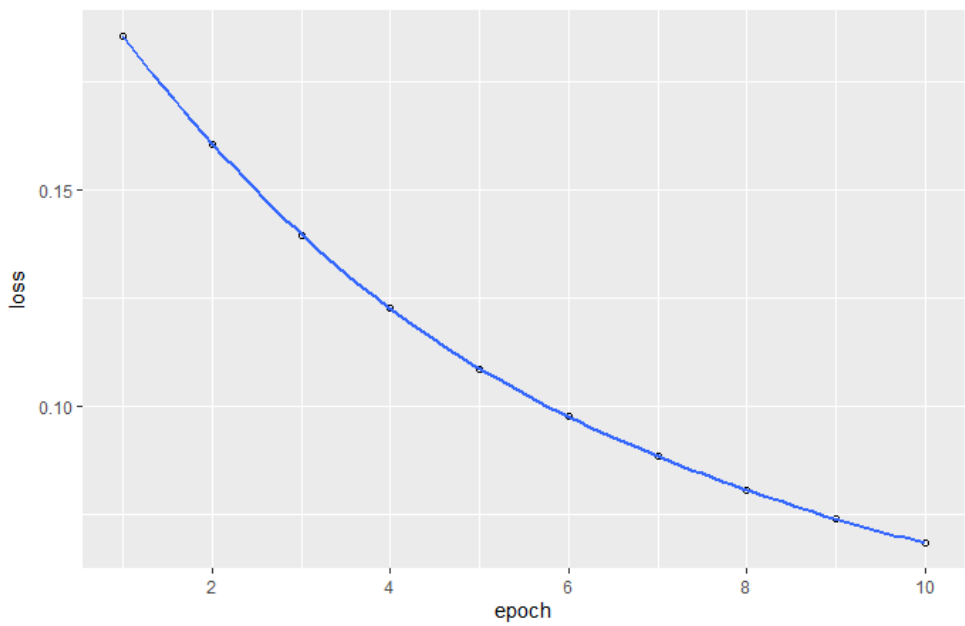


Figure 23.28 – Model Loss

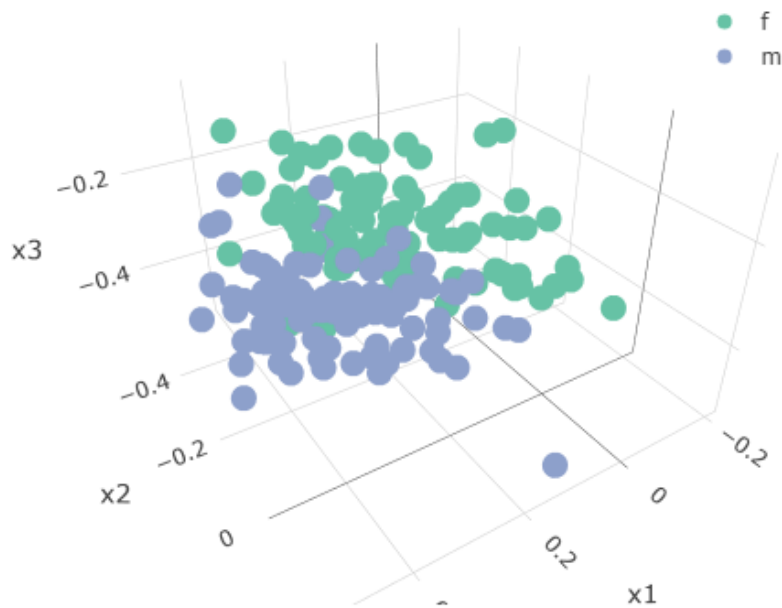


Figure 23.29 – Encoding results

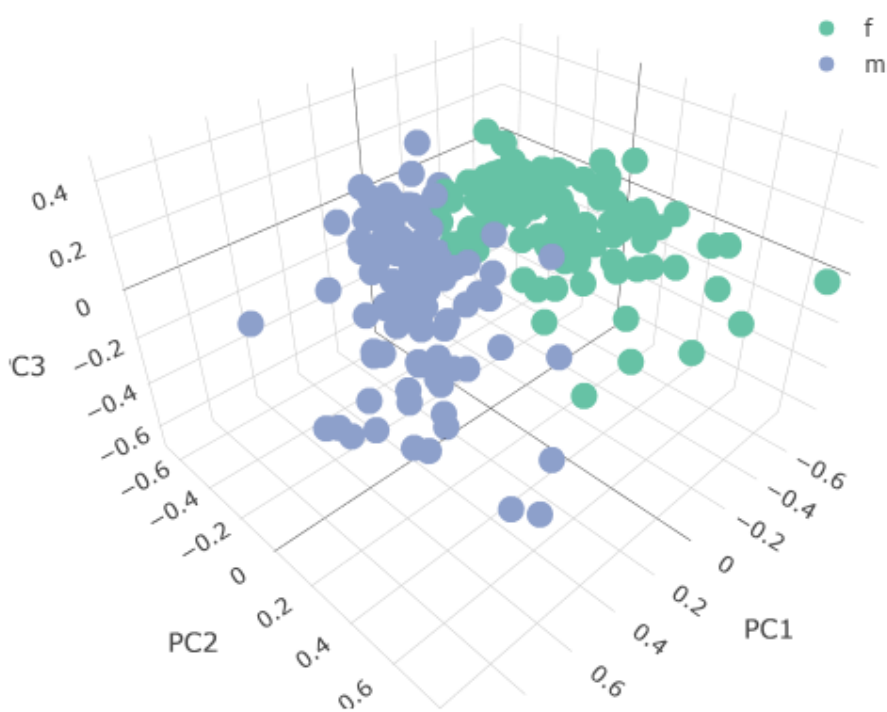


Figure 23.30 – PCA results

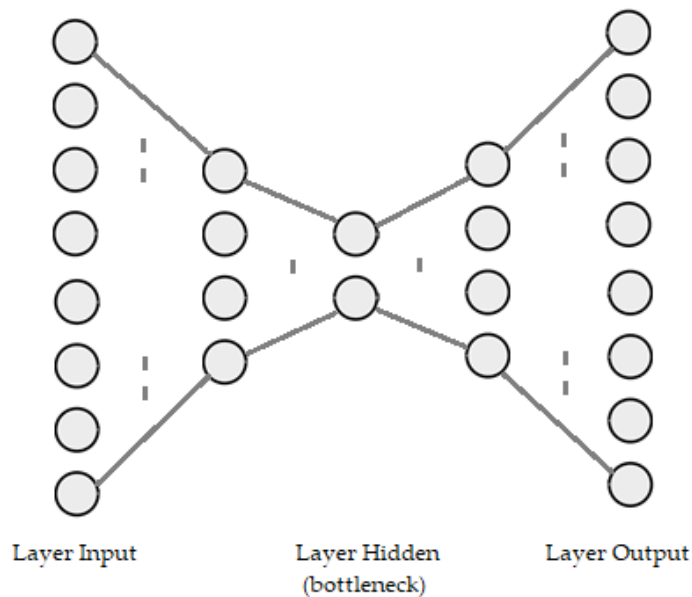


Figure 23.31 - Stacked Autoencoder example

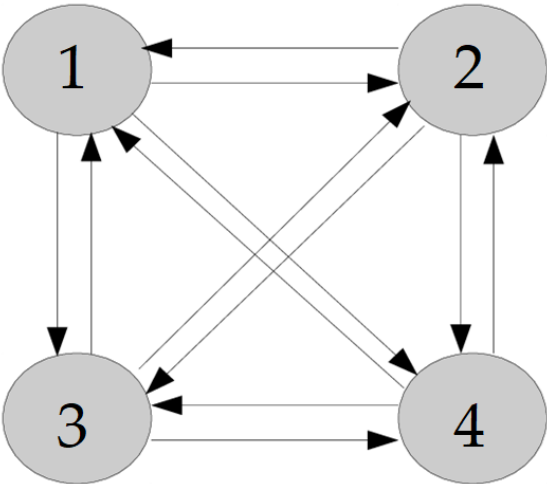


Figure 23.32 – Hopfield Network example

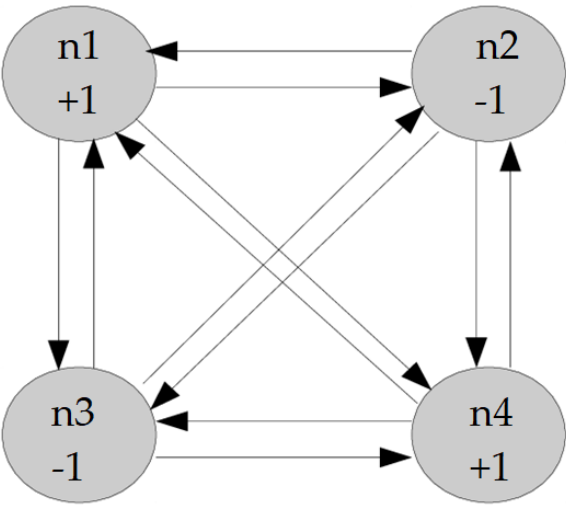


Figure 23.33 – The initial network status

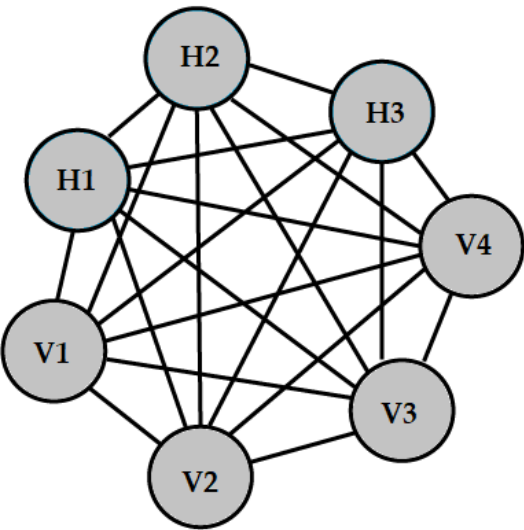


Figure 23.34 - Boltzmann Machine structure

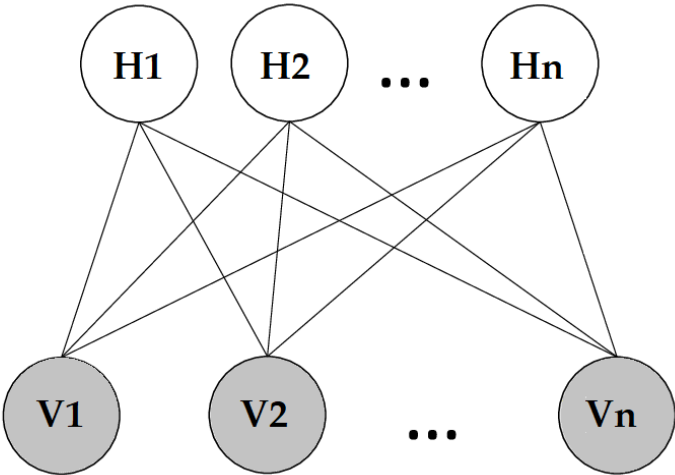


Figure 23.35 – RBM Structure

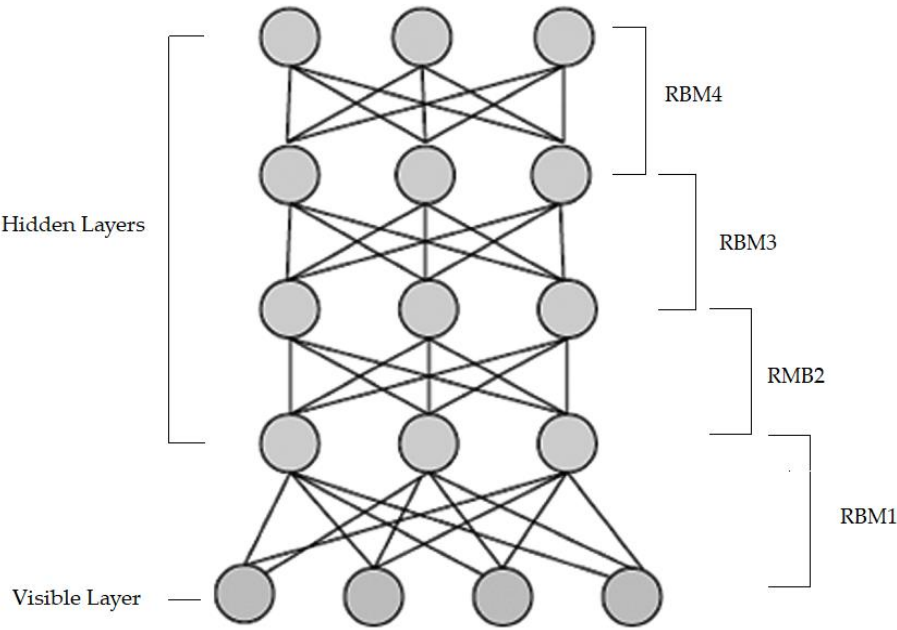


Figure 23.36 – DBN architecture

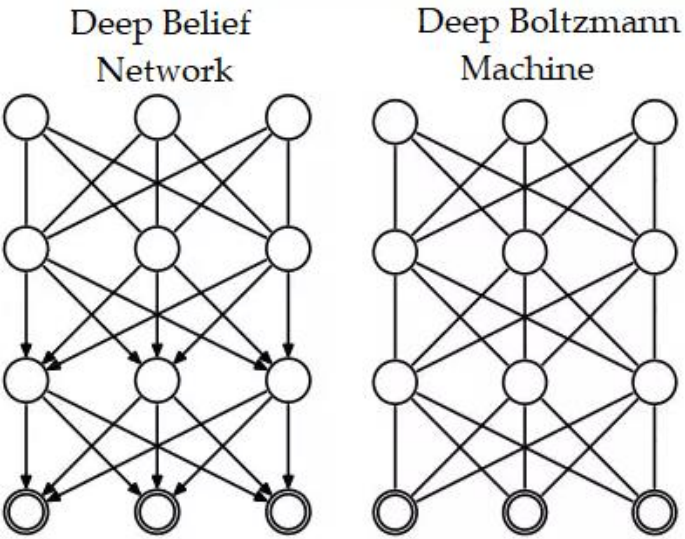


Figure 23.37 - DBN vs DBM

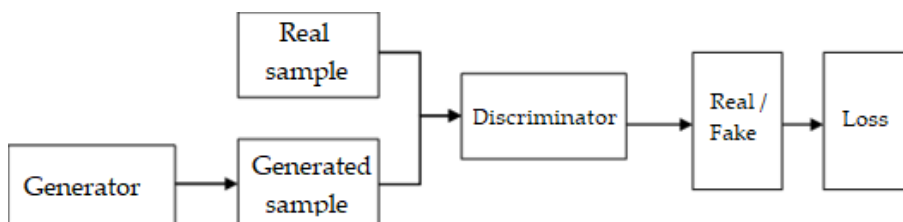


Figure 23.38 – GAN example



Figure 23.39 - A picture showing an African elephant

24. AUTOMATED MACHINE LEARNING

25. VISUALIZING DATA



Figure 25.1 – Horizontal and vertical bar chart

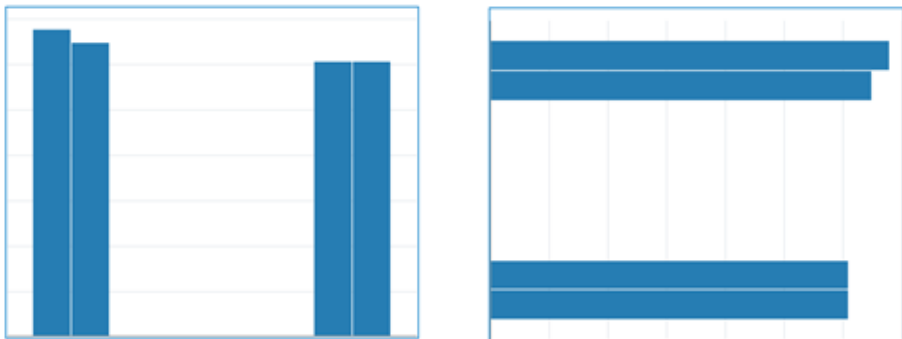


Figure 25.2 – Grouped bar charts

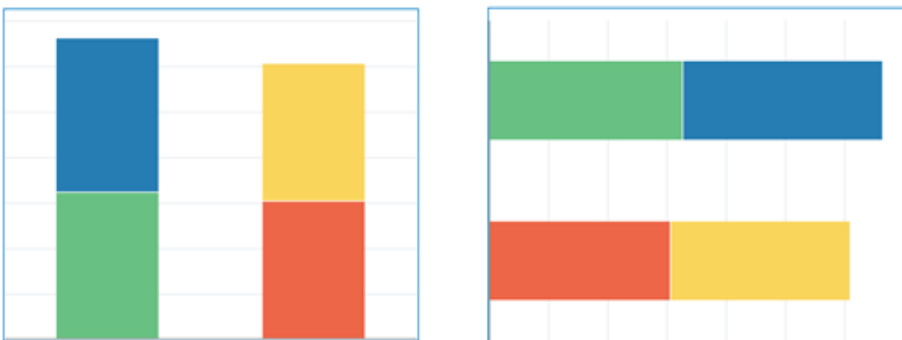


Figure 25.3 – Stacked bar charts

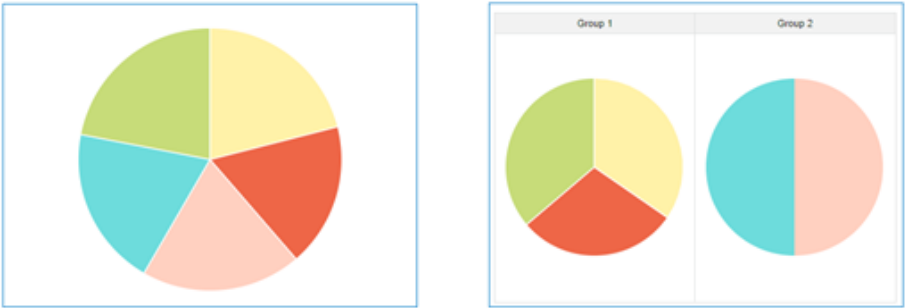


Figure 25.4 - Pie chart in single and multiple configuration

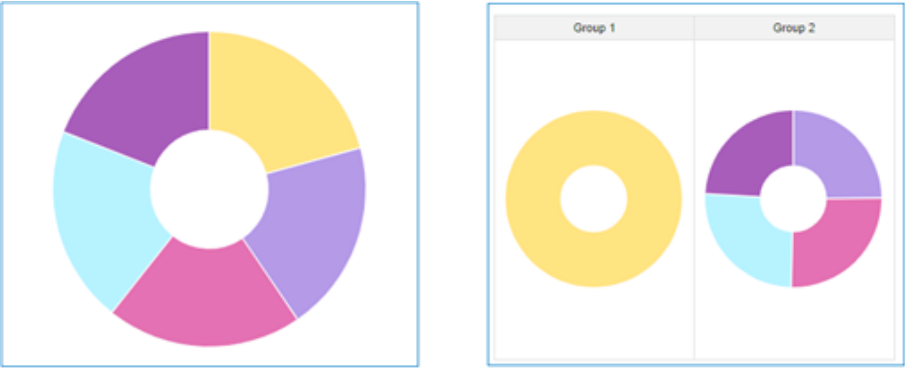


Figure 25.5 – Donut chart

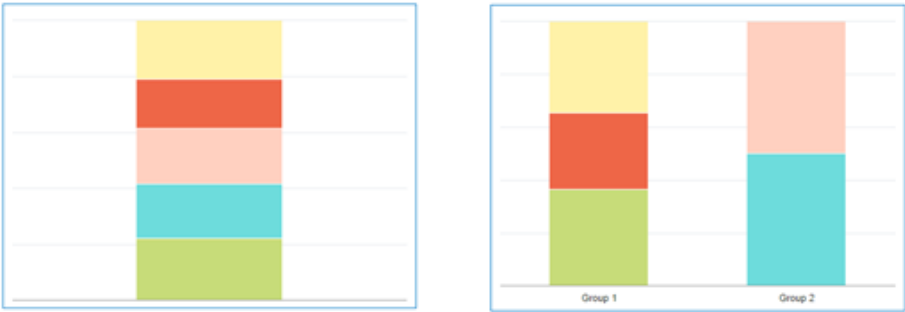


Figure 25.6 – Stacked 100% charts

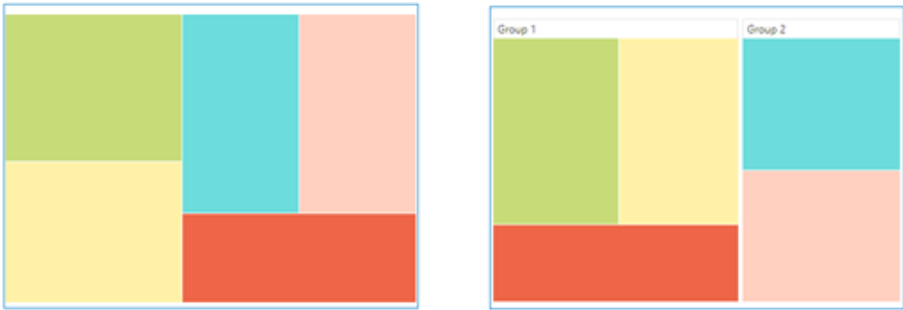


Figure 25.7 – Treemap chart



Figure 25.8 – Boxplot charts

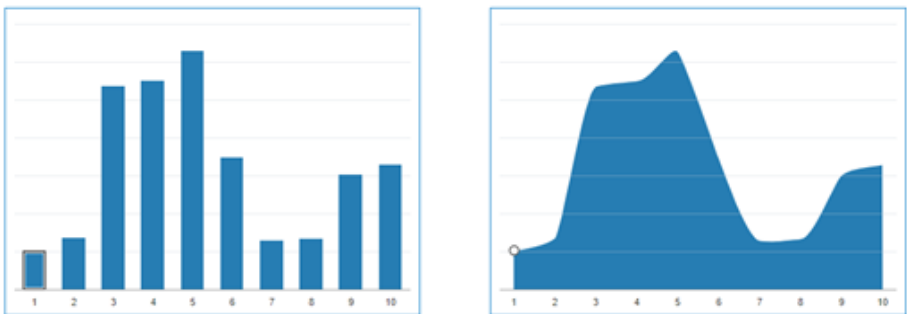


Figure 25.9 – Histogram and density plot

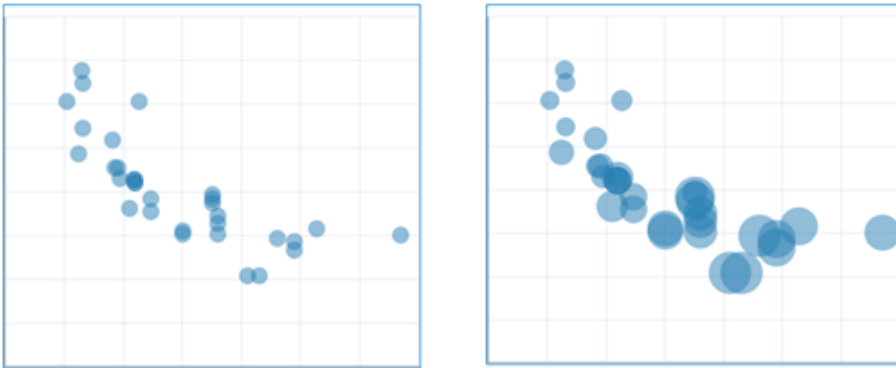


Figure 25.10 - Scatterplot and Bubble Chart



Figure 25.11 - Correlogram for four and seven variables

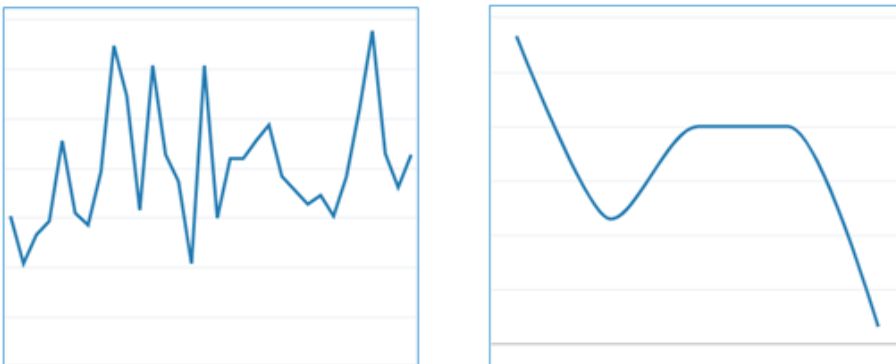


Figure 25.12 – Line charts

26. COMMUNICATING DATA

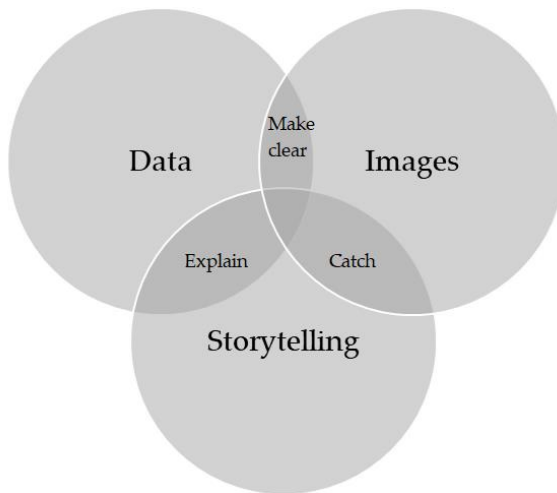


Figure 26.1 - Data Storytelling

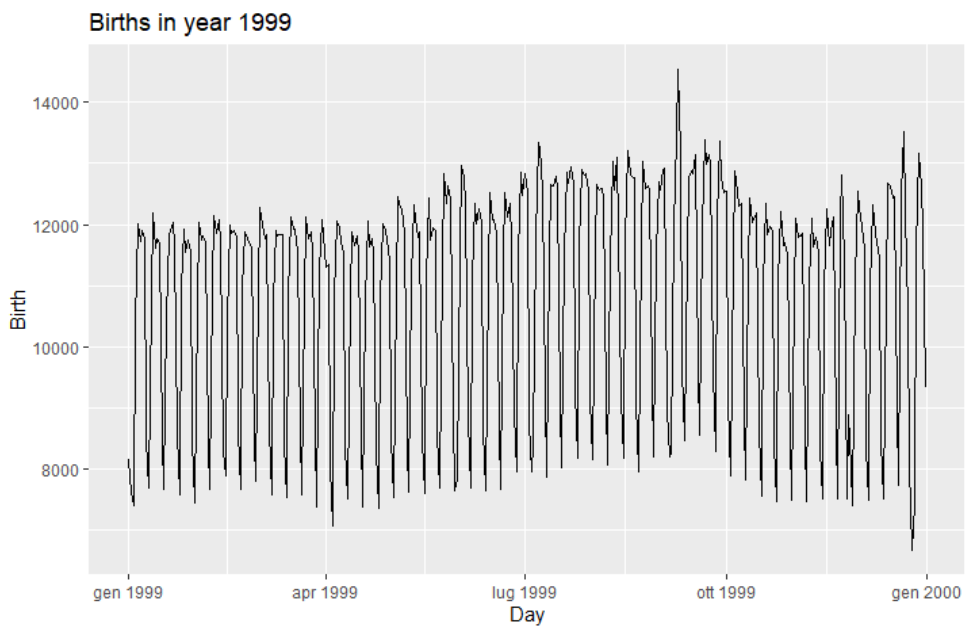


Figure 26.2 – US Birth in year 1999

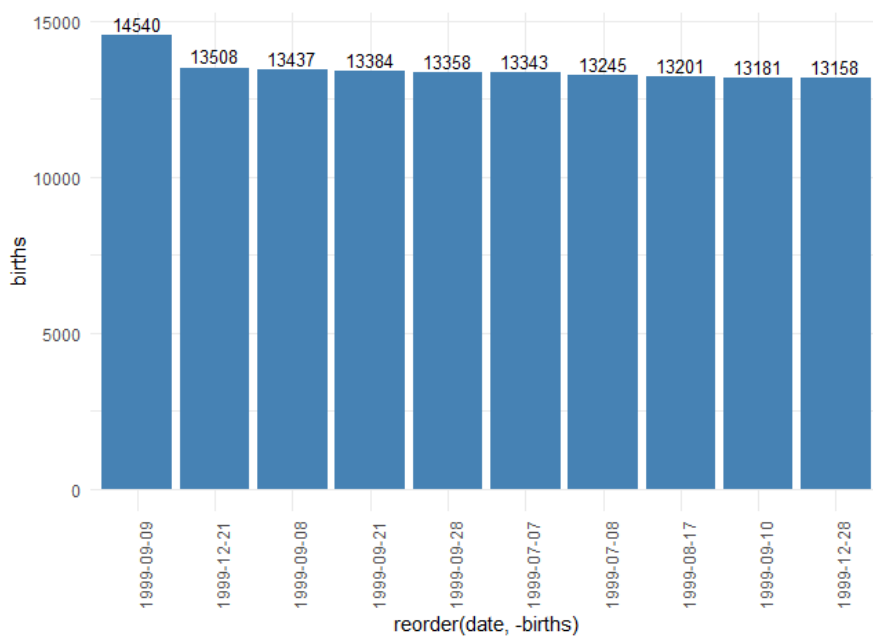


Figure 26.3 - The 10 days with most births

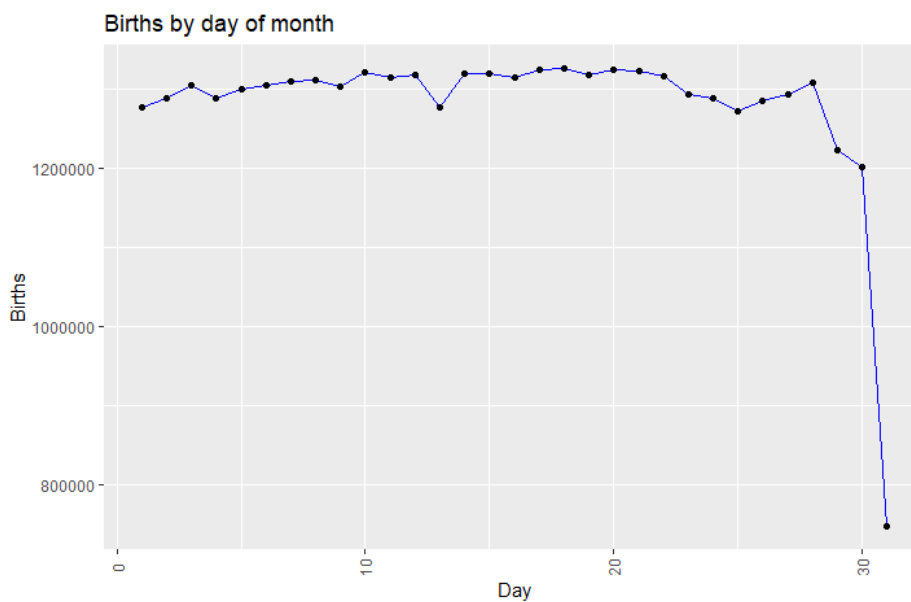


Figure 26.4 – Birth by day of month

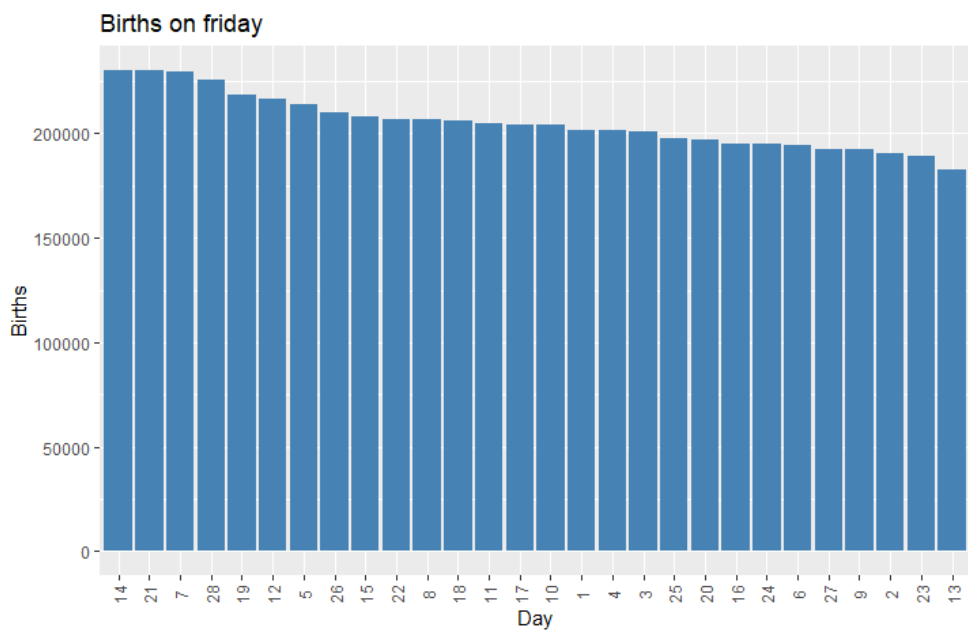


Figure 26.5 - Friday births by day of month

Contact

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